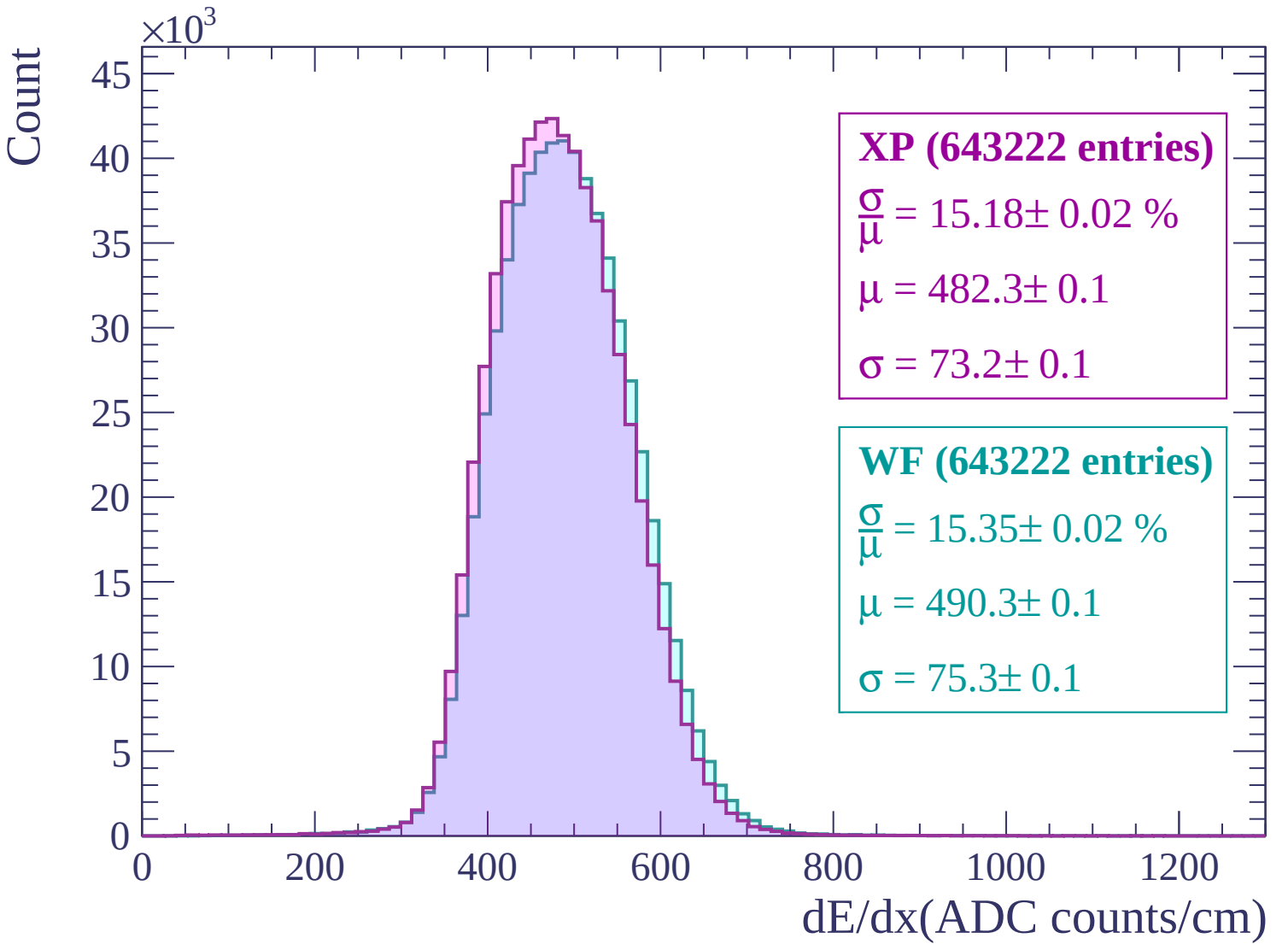
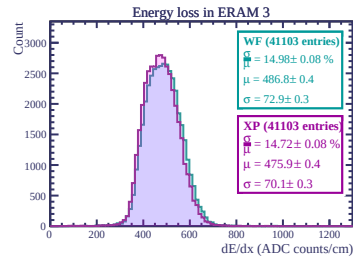
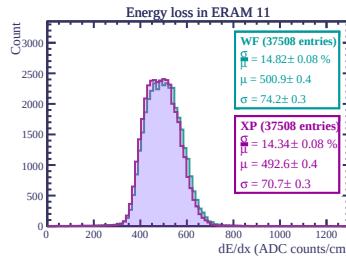
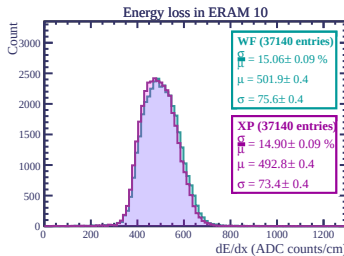
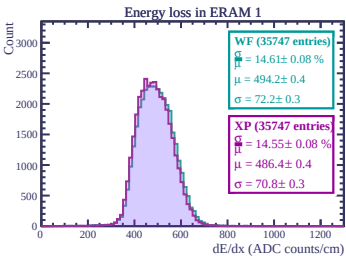
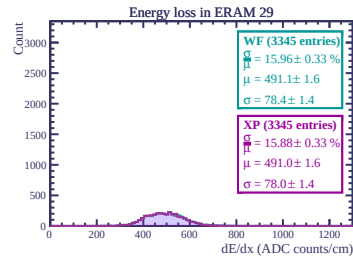
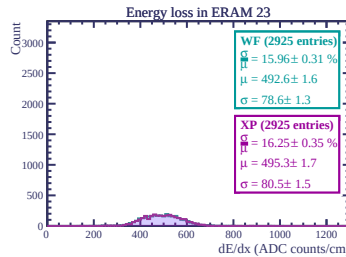
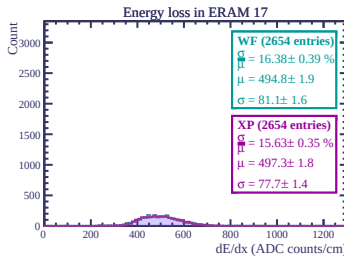
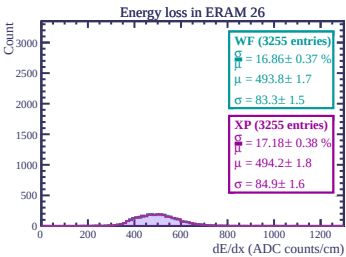
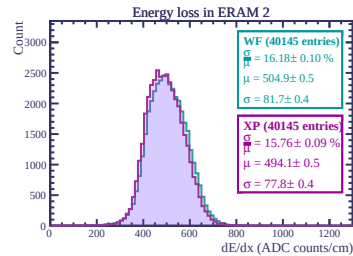
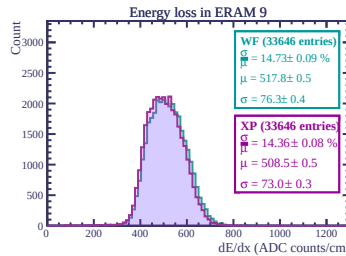
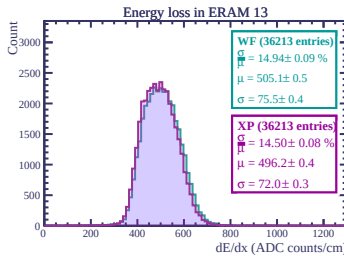
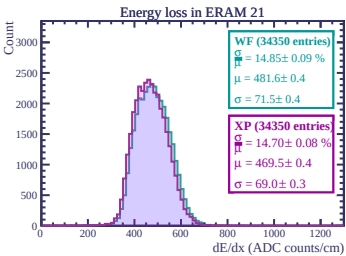
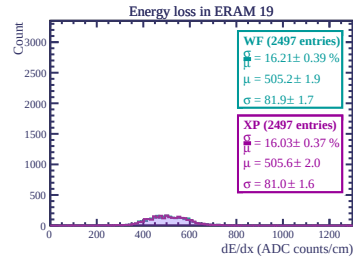
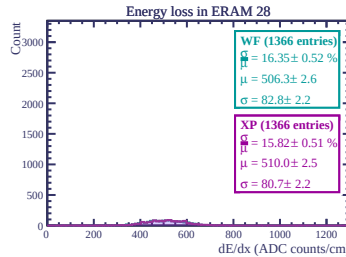
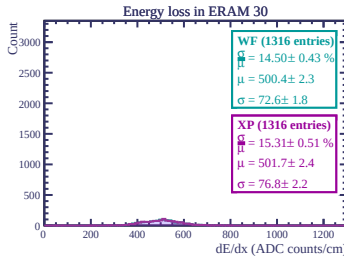
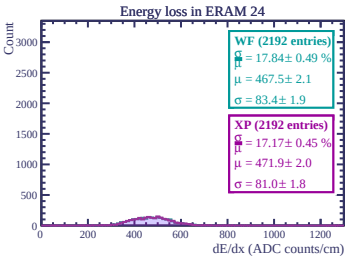
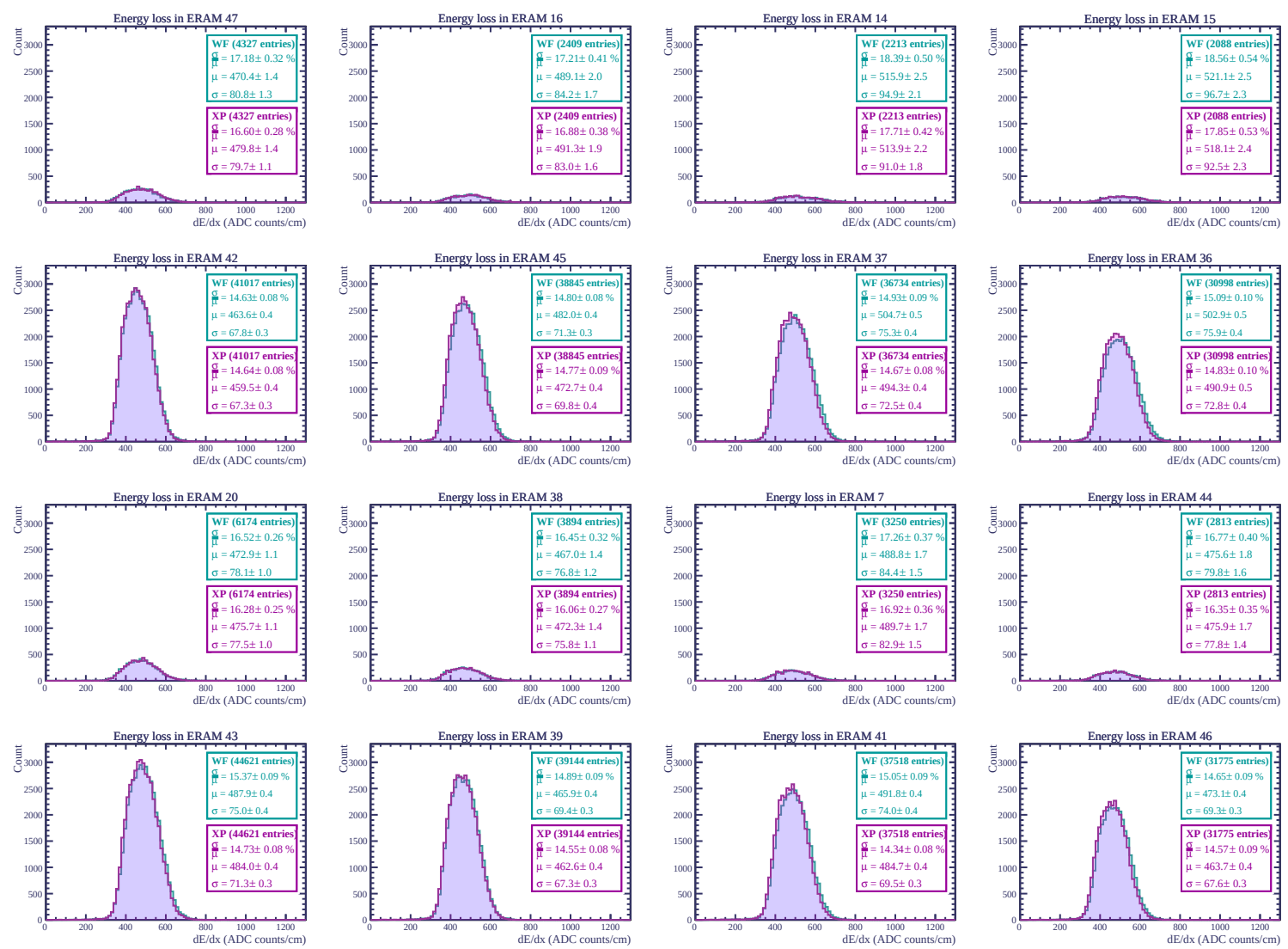
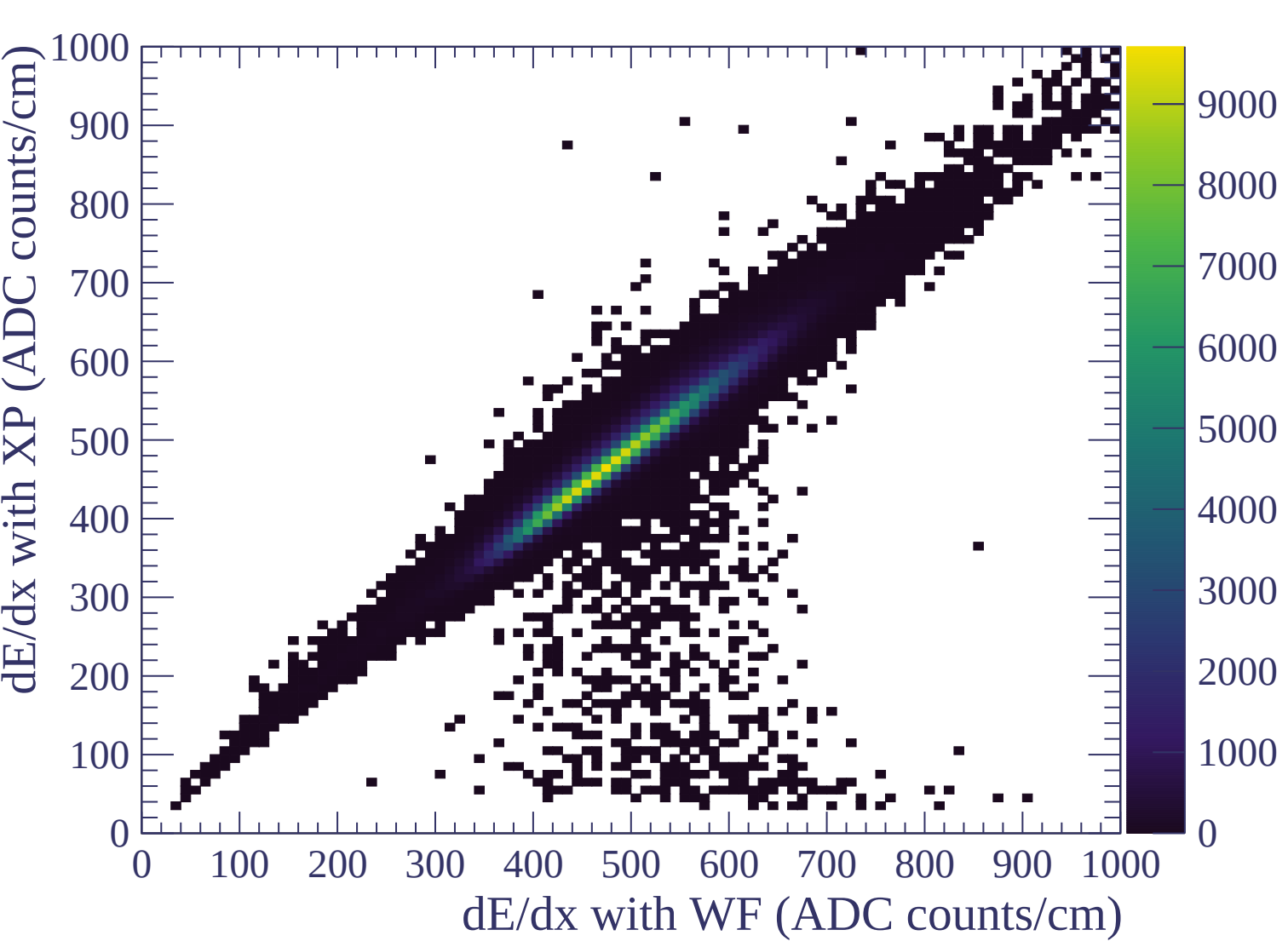
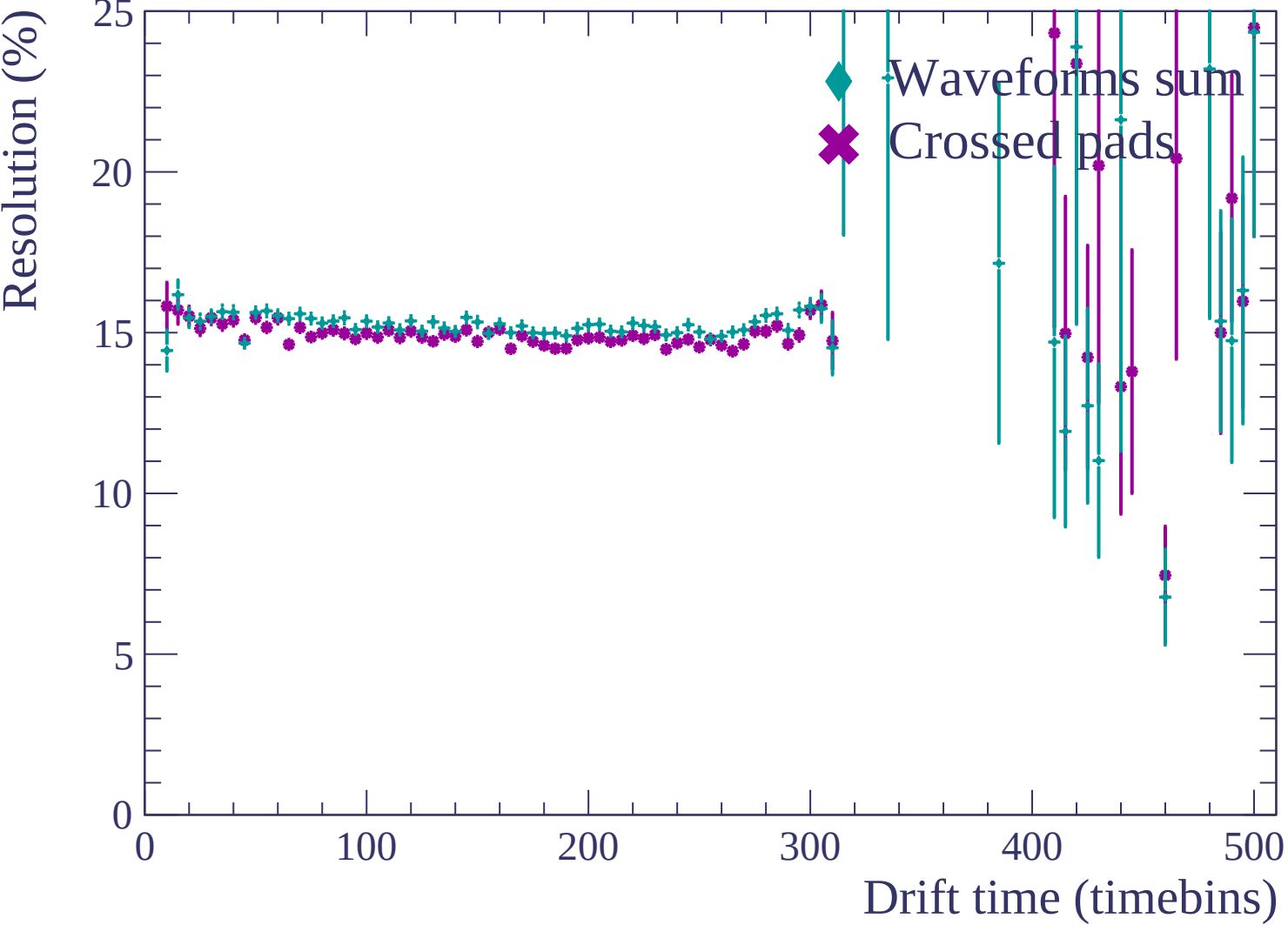


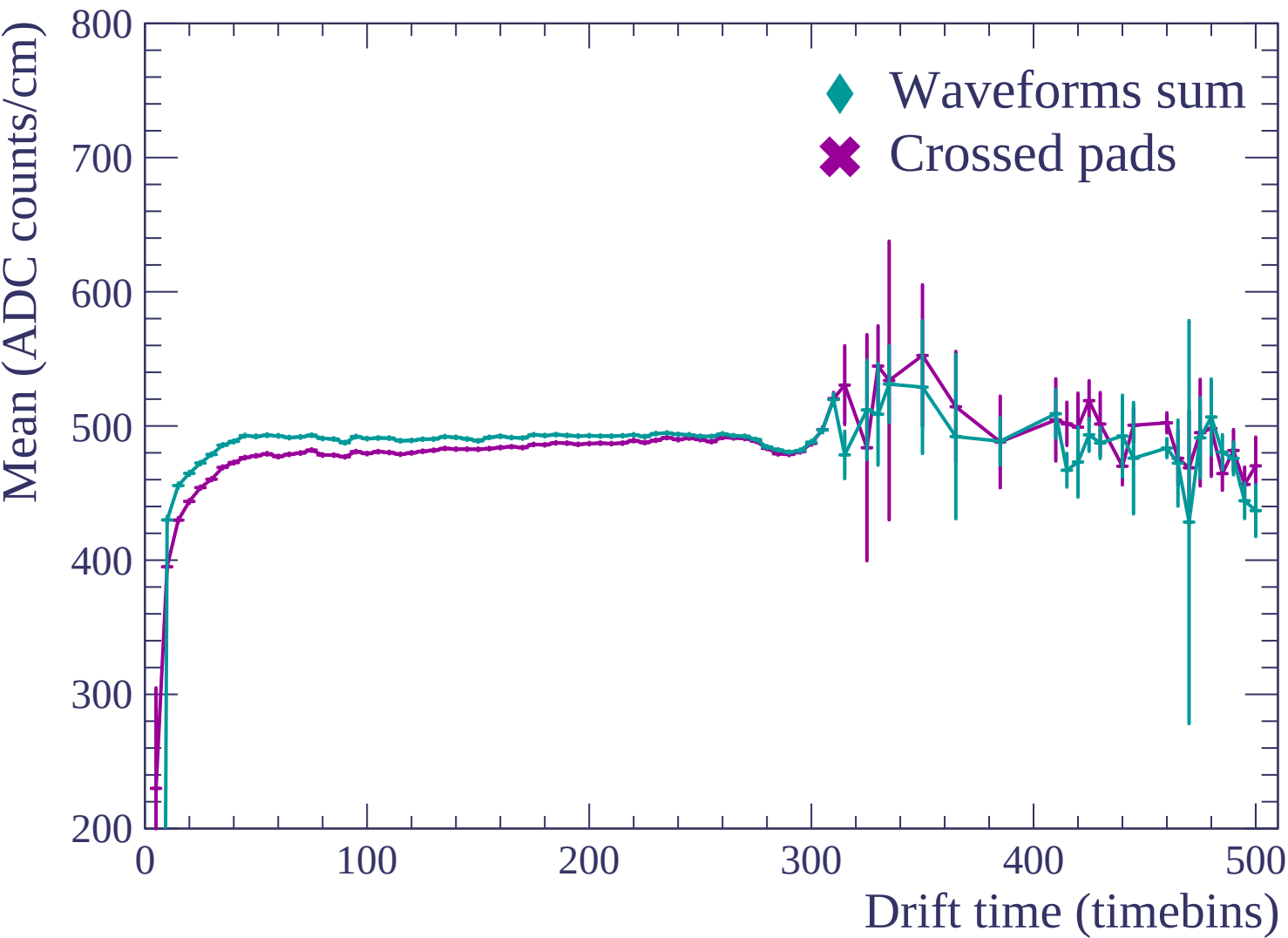


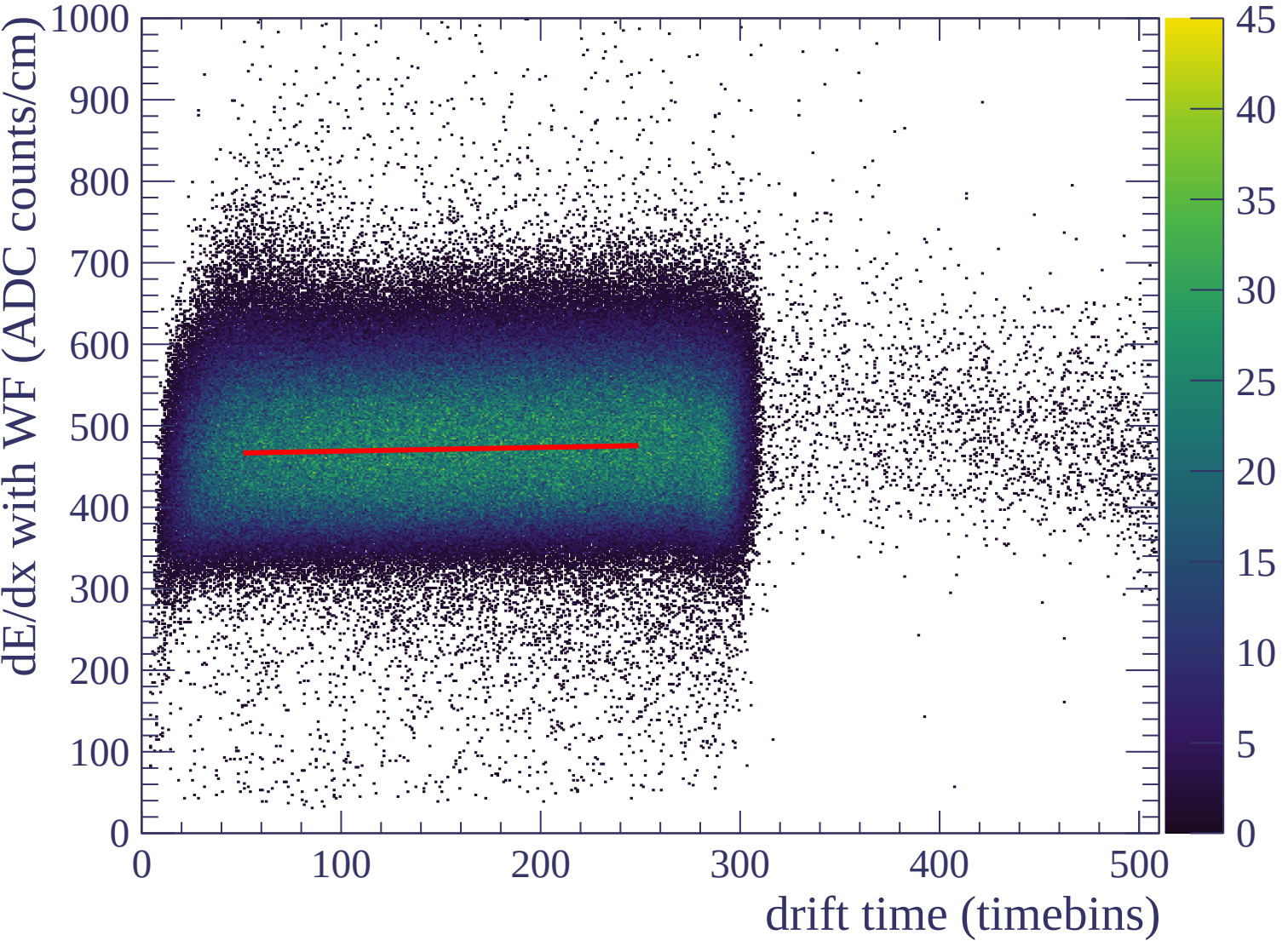


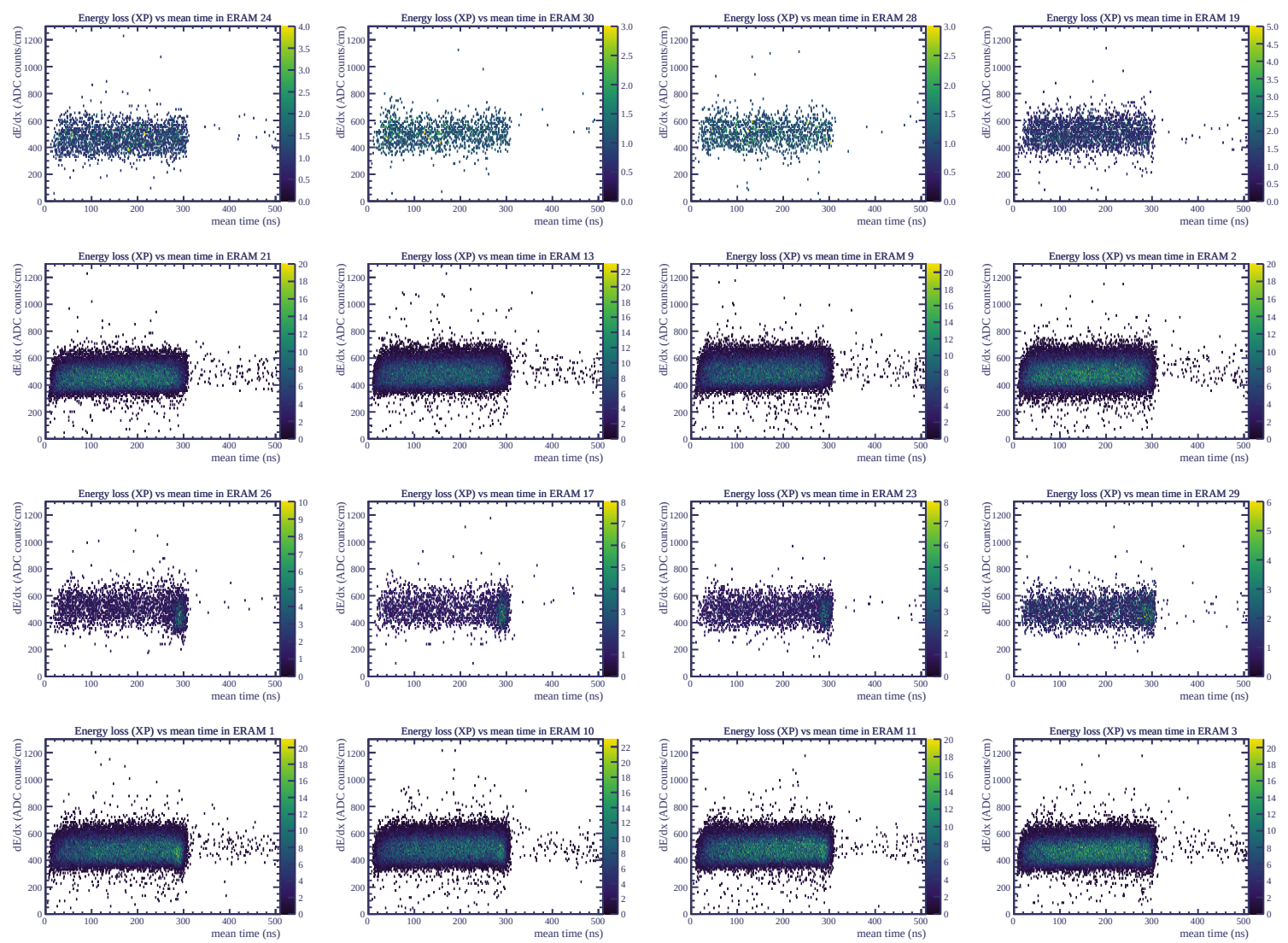


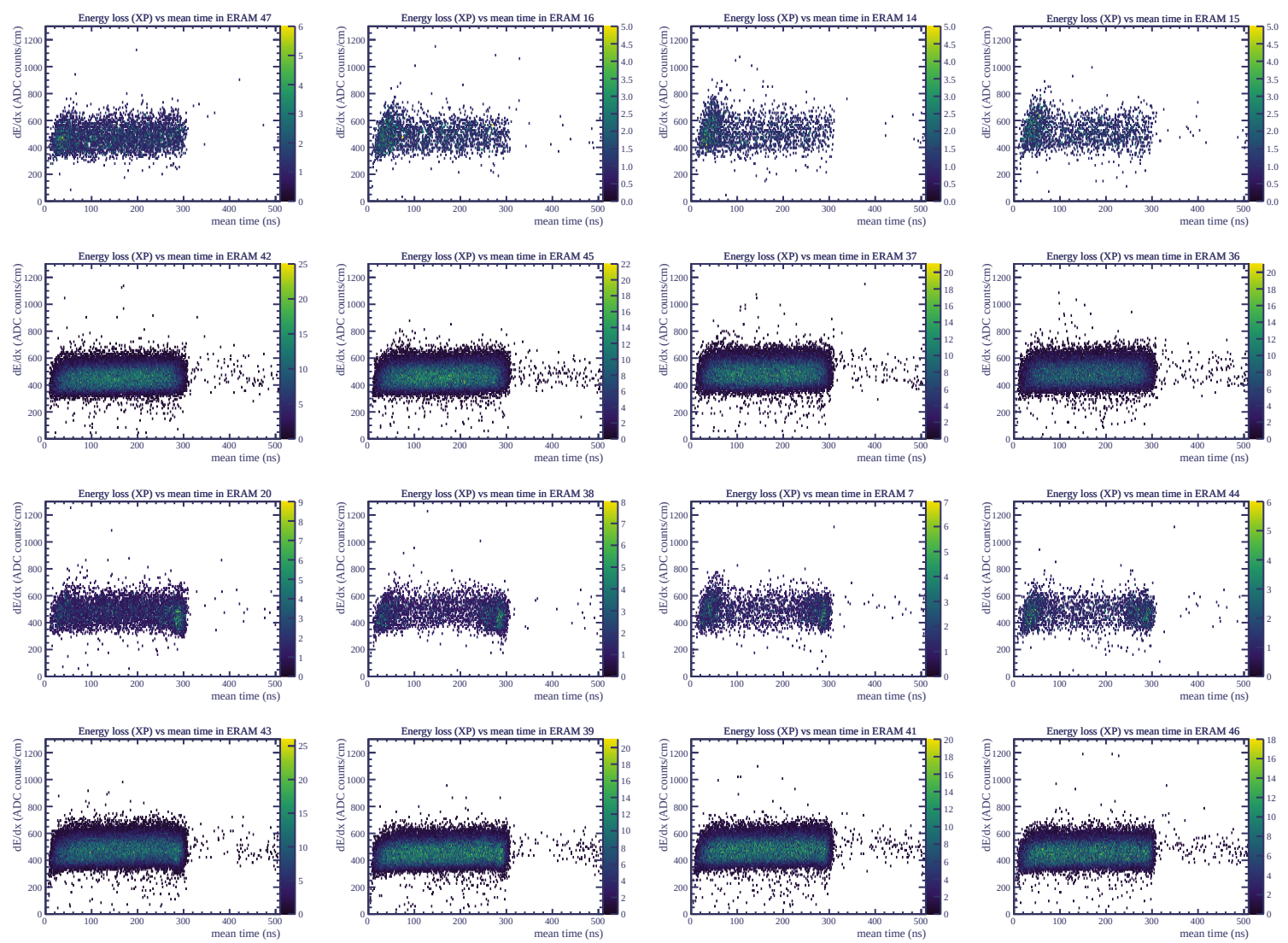


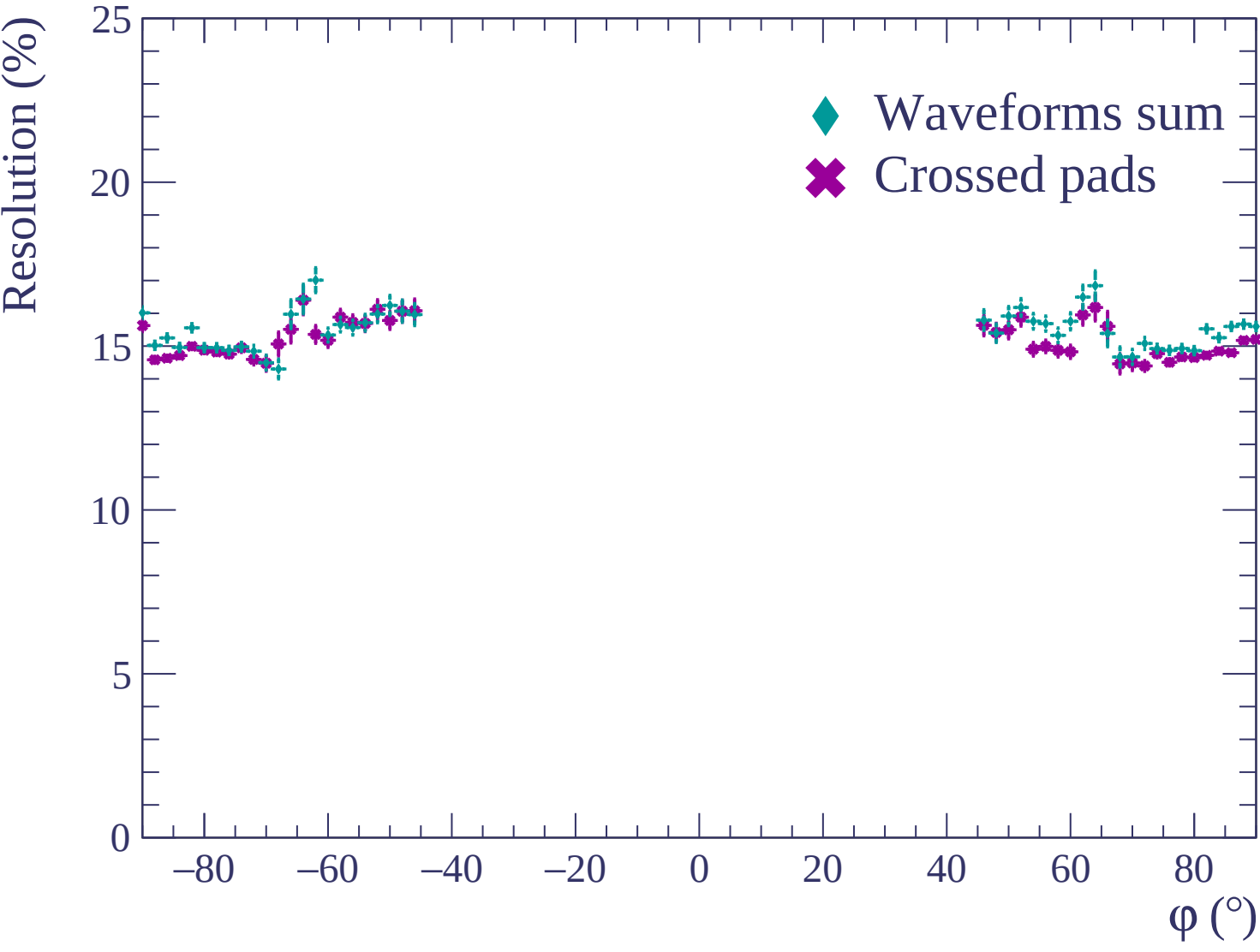


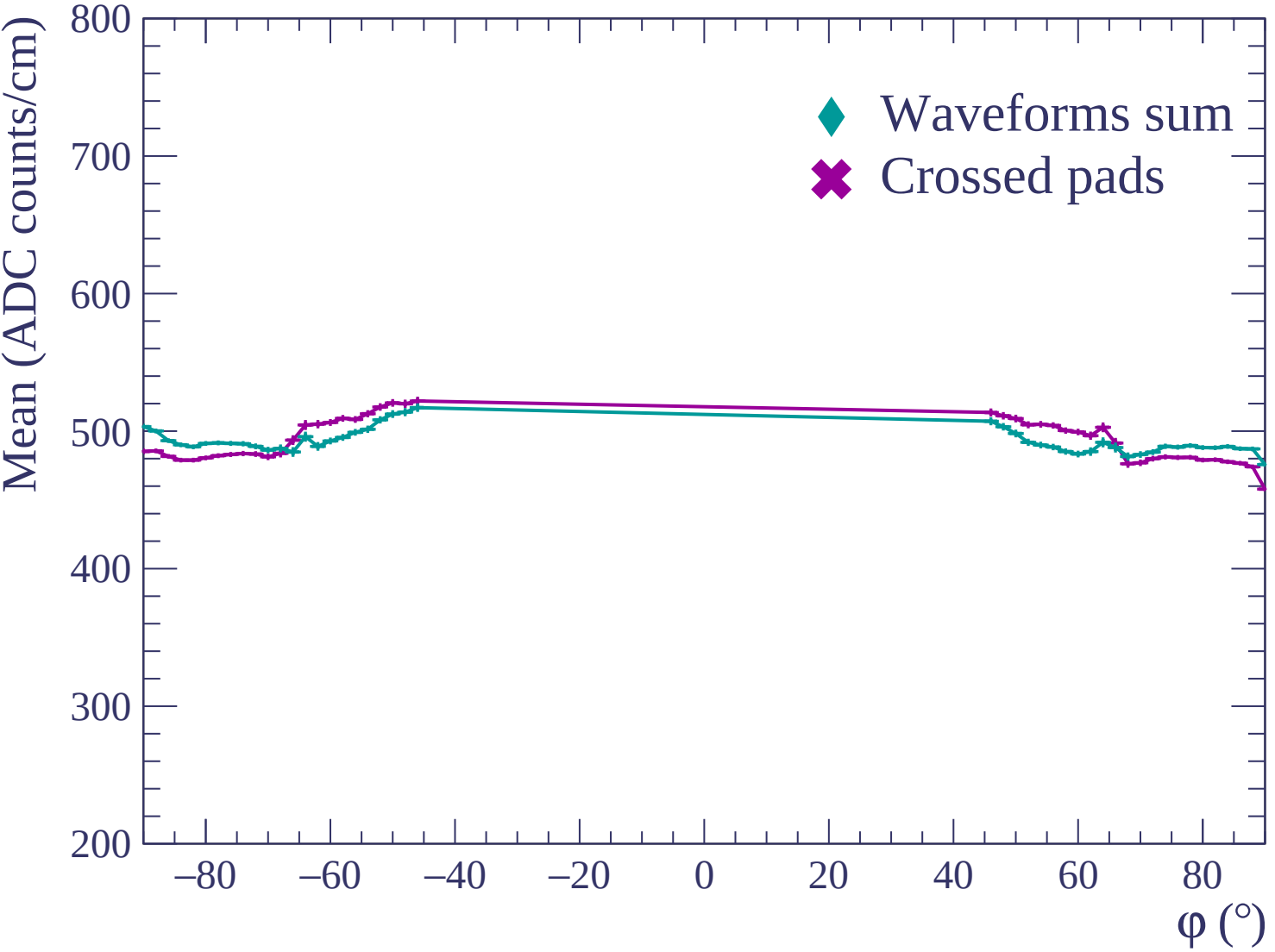


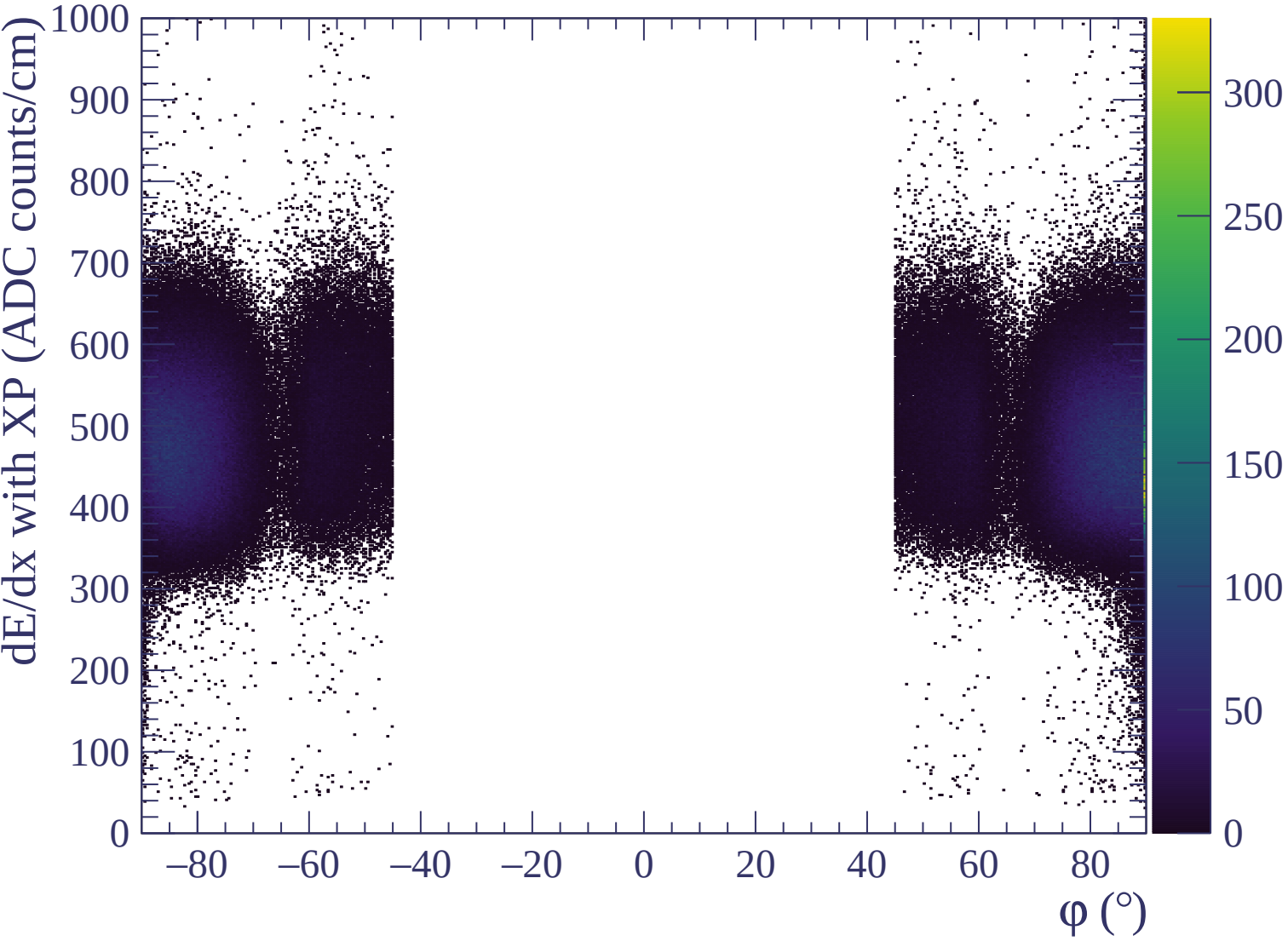


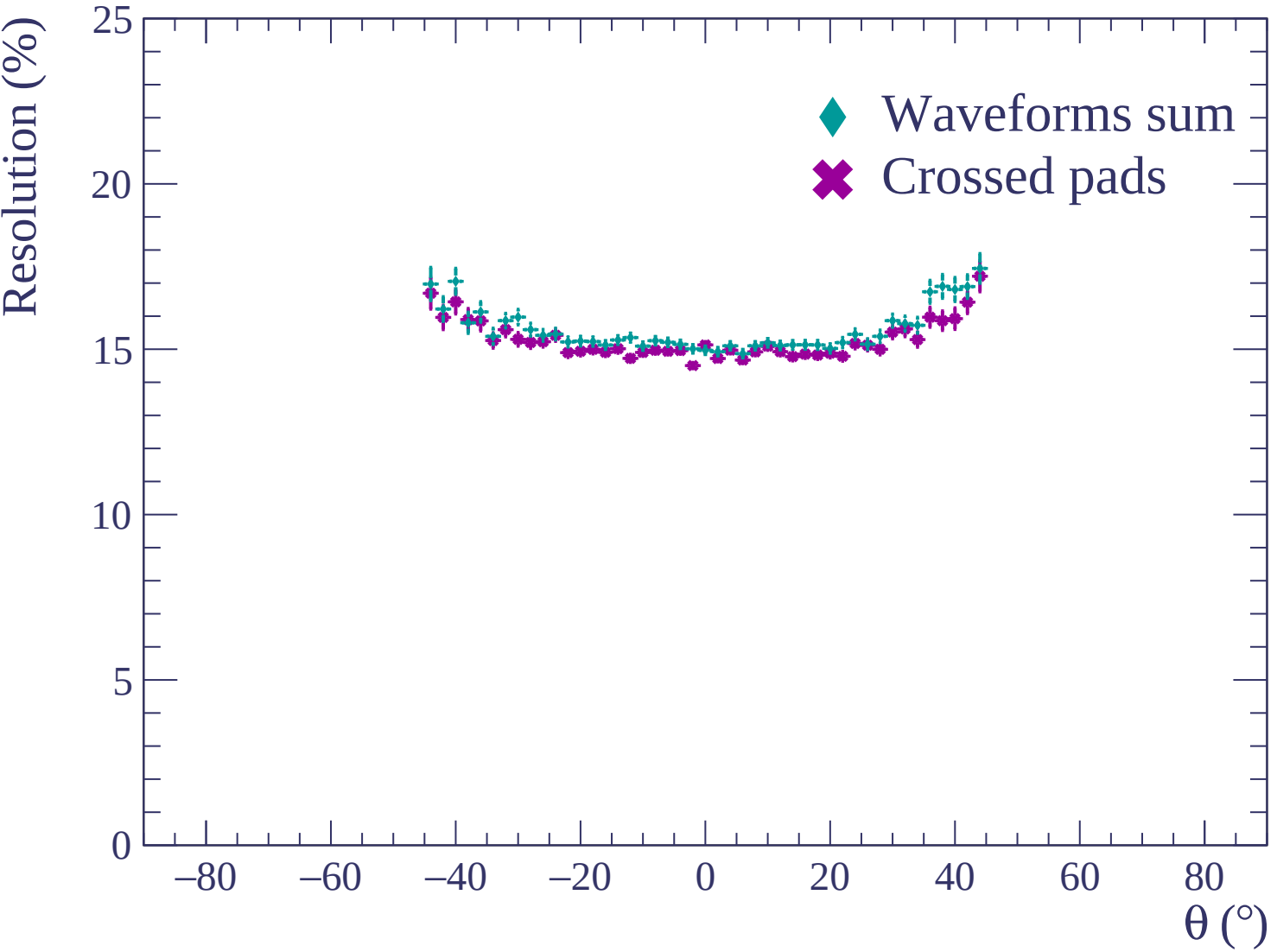


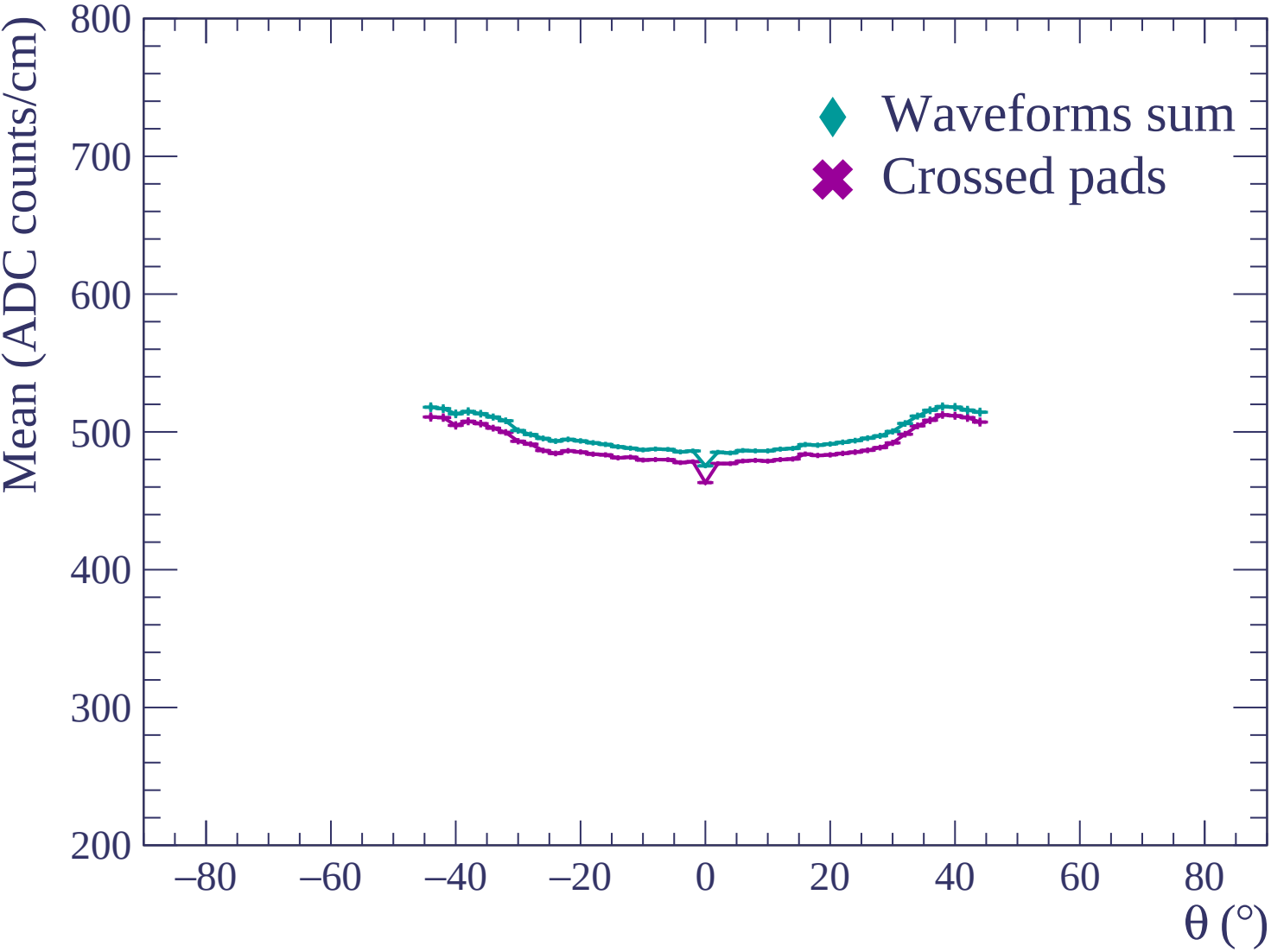


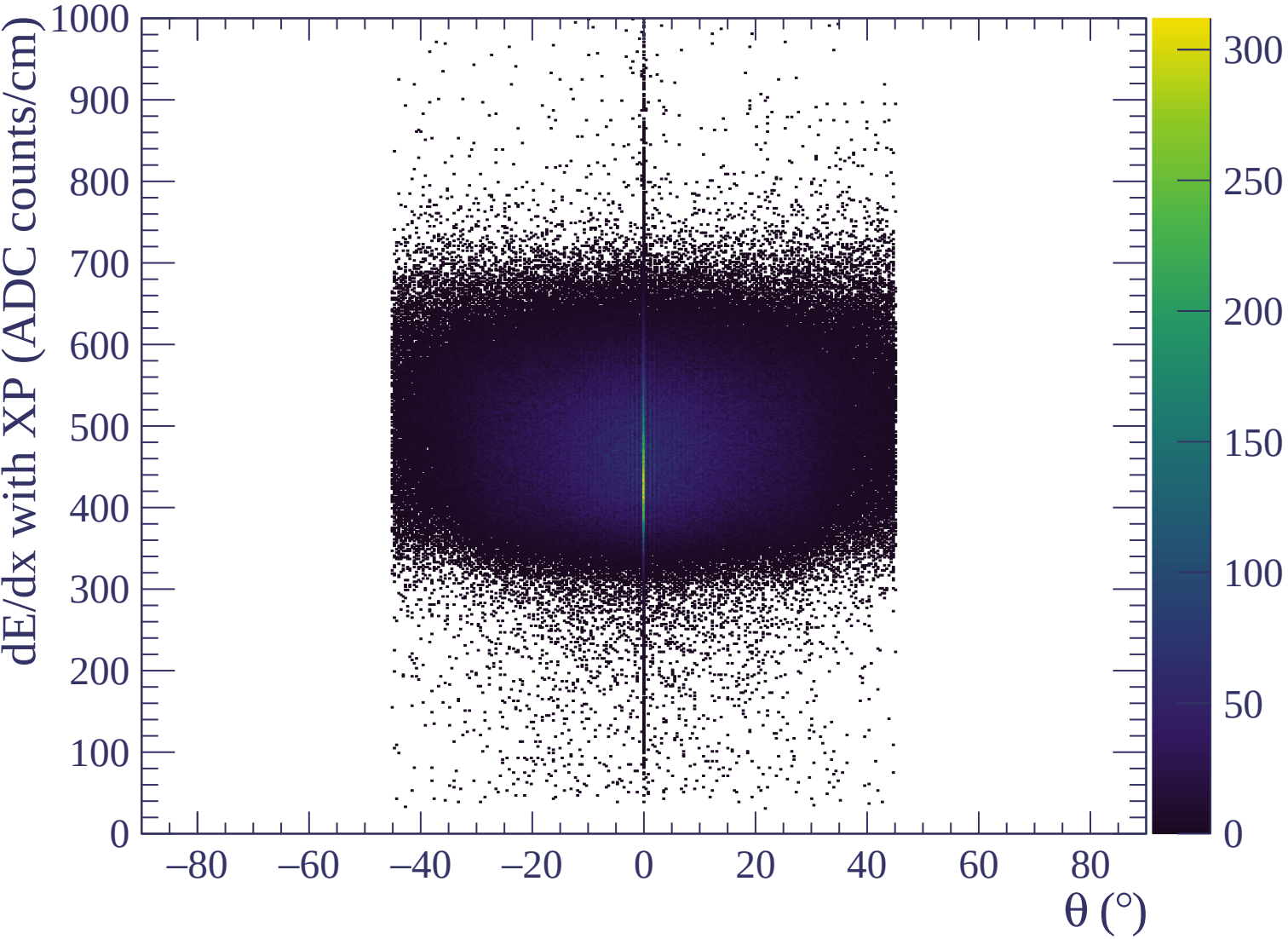


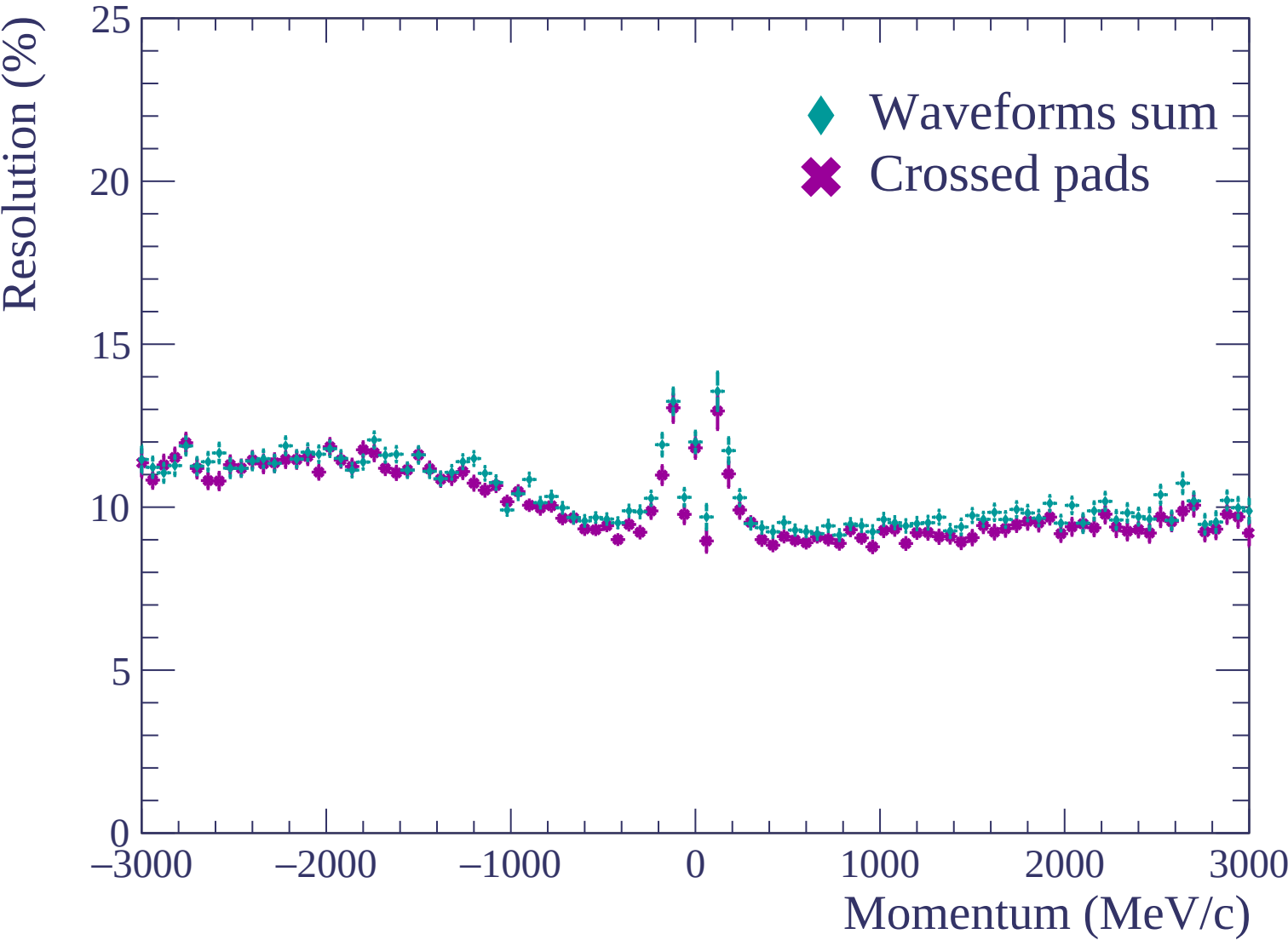


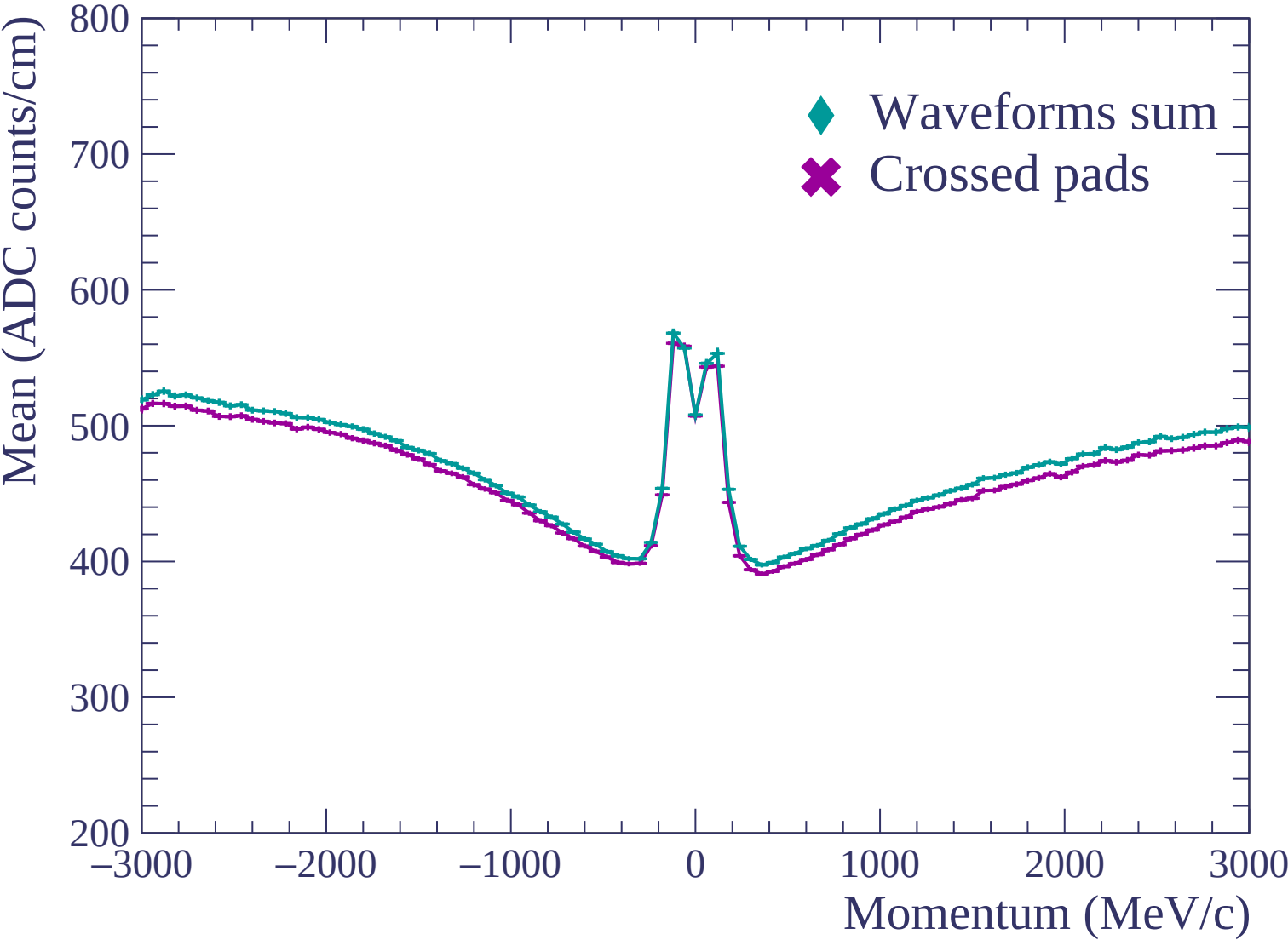


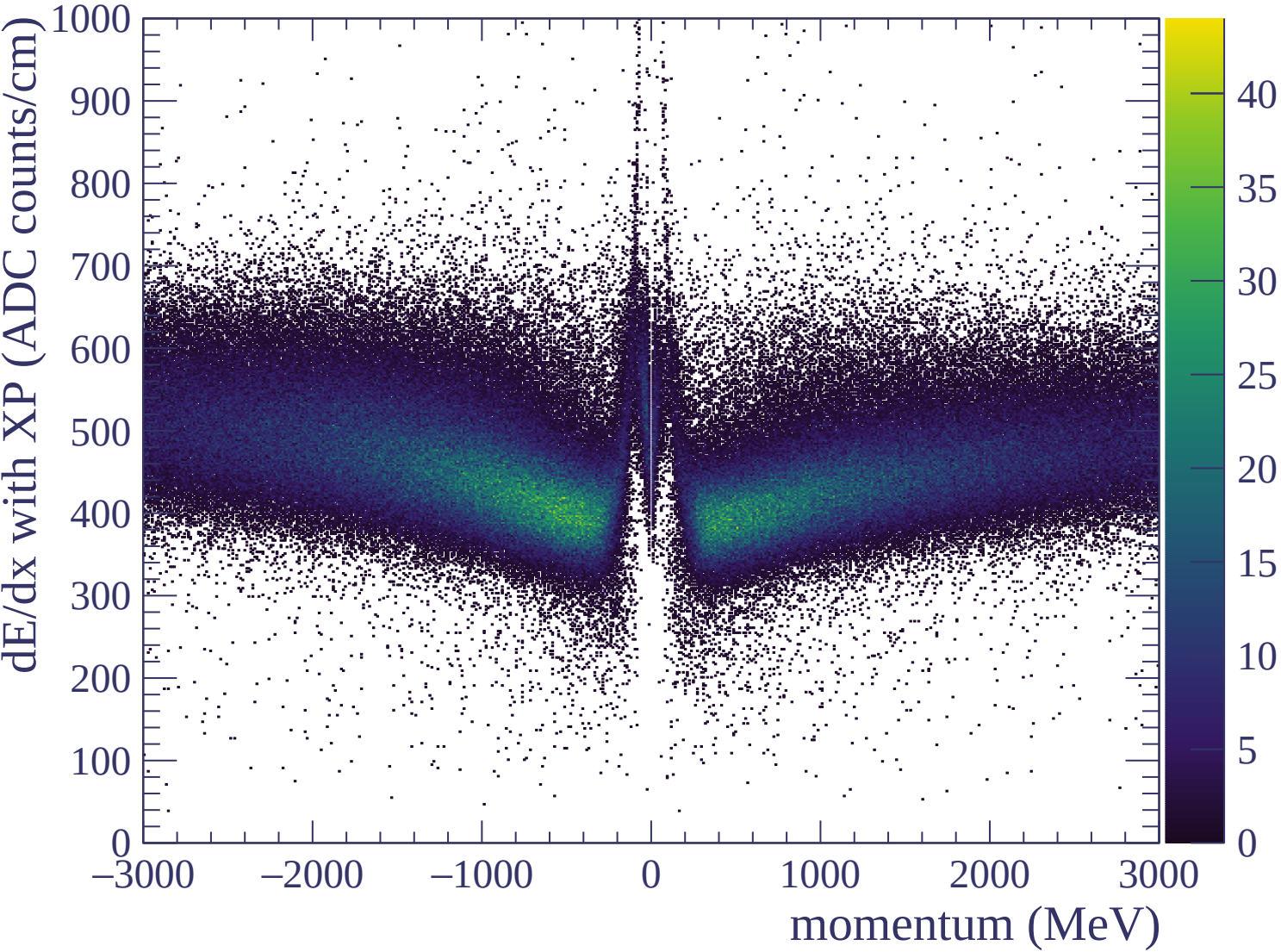


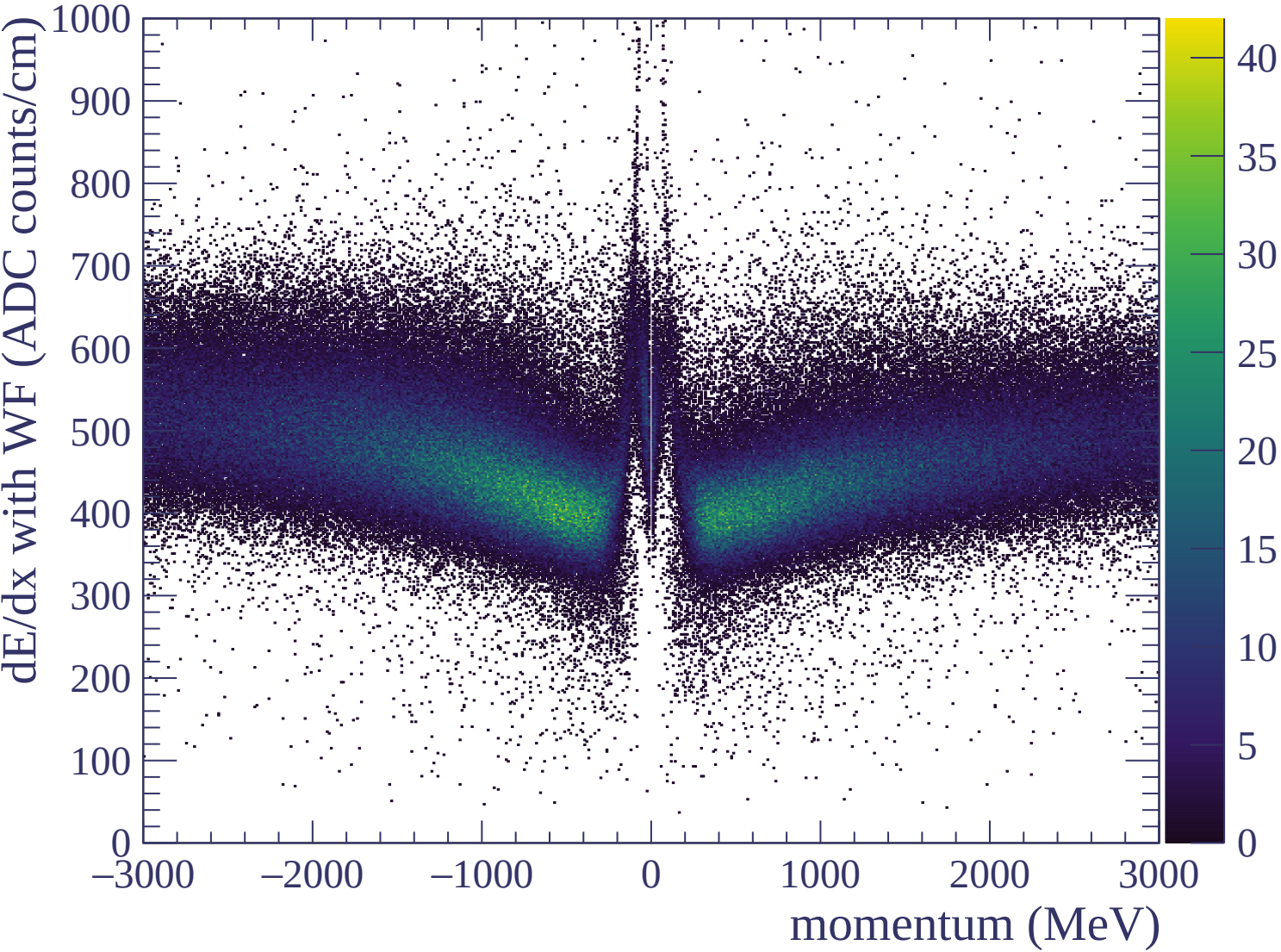


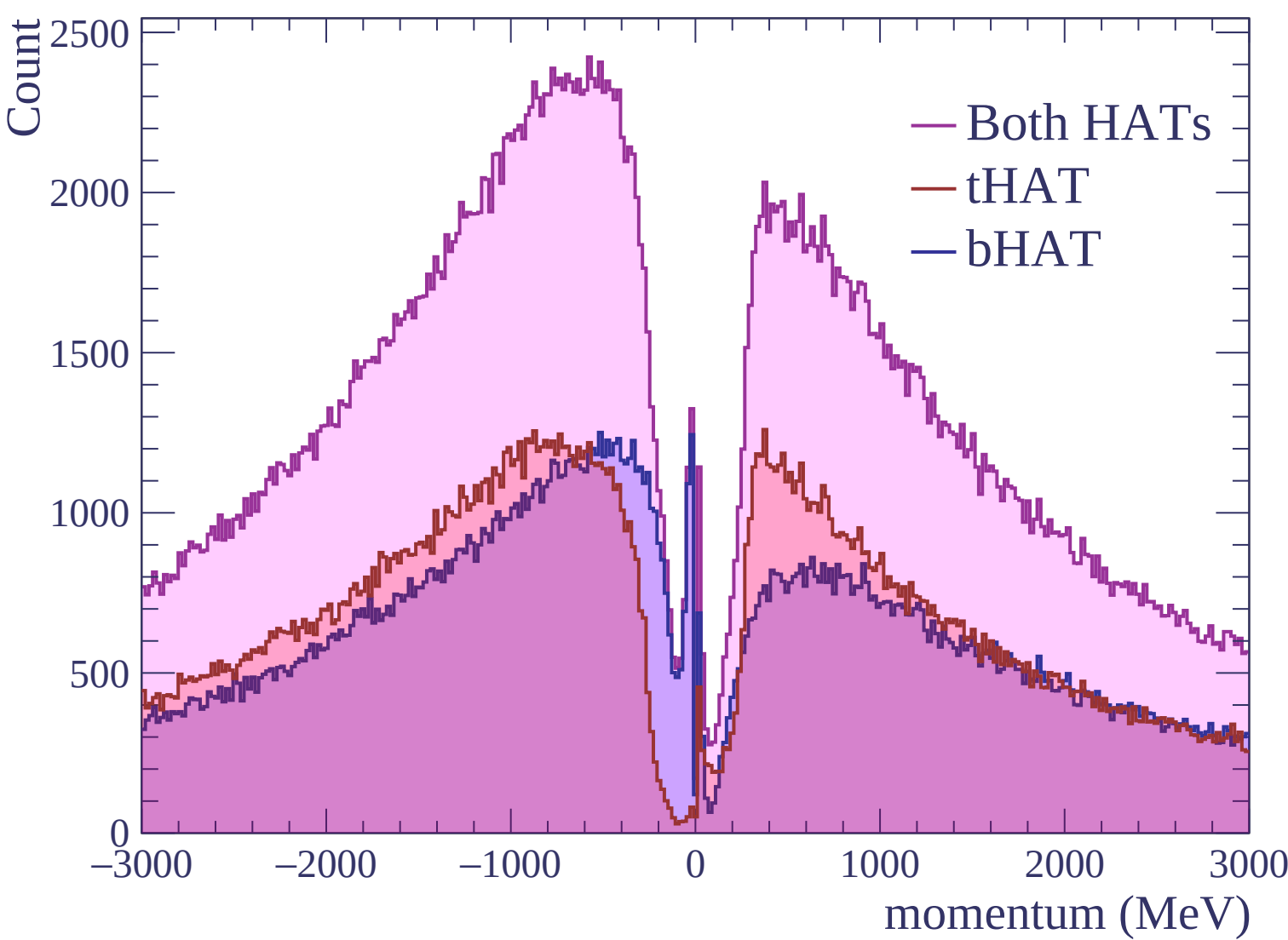


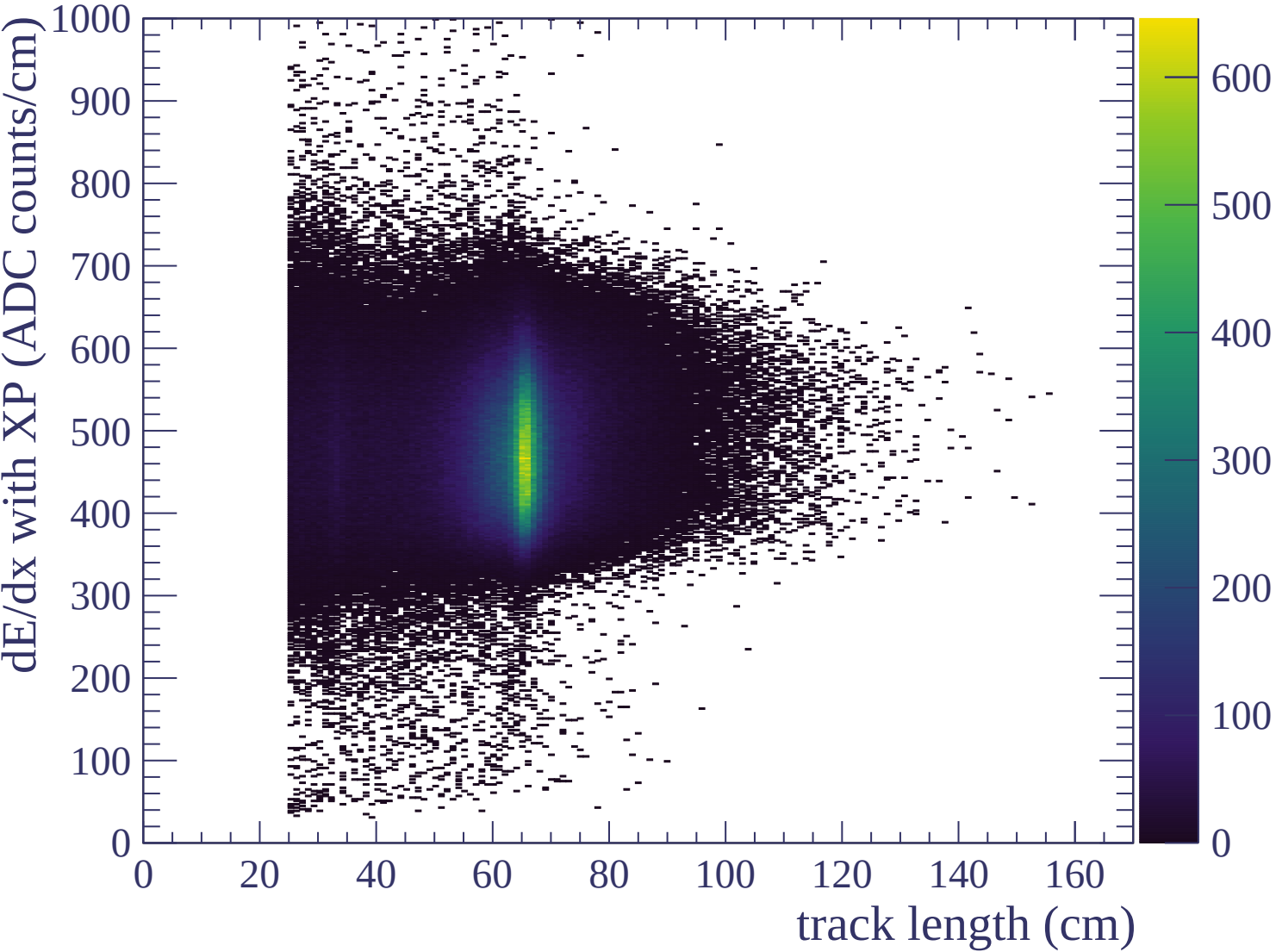


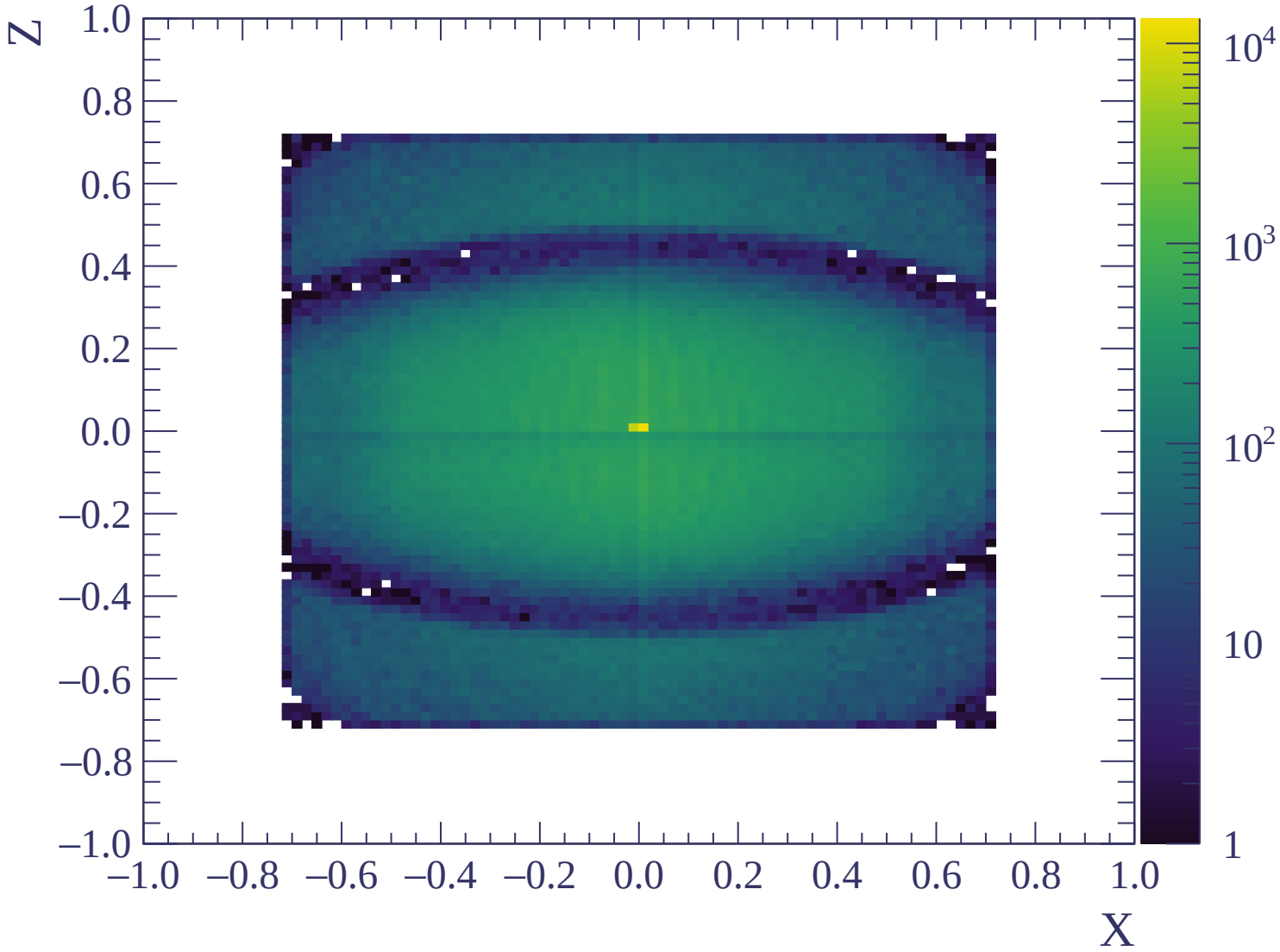


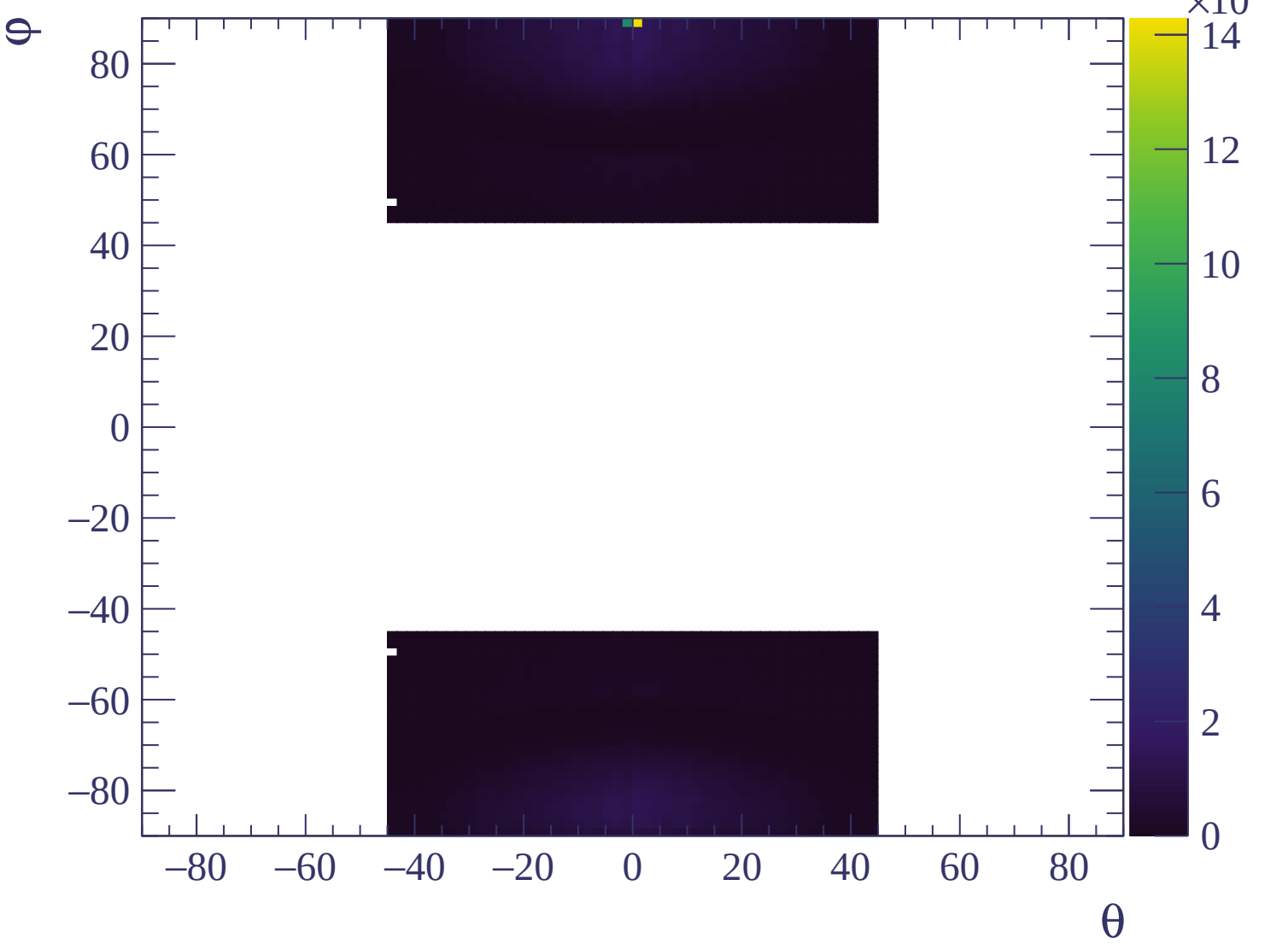


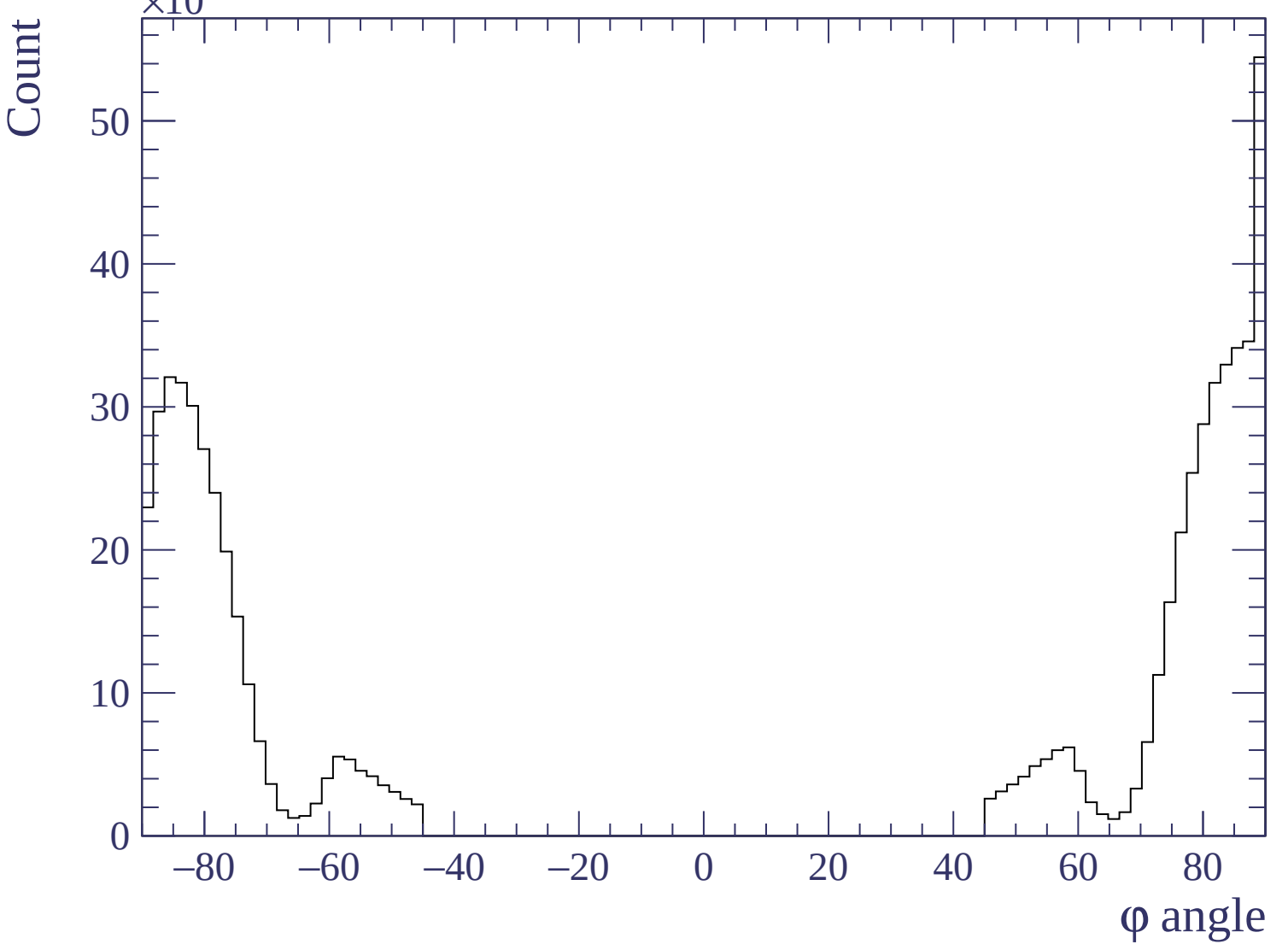


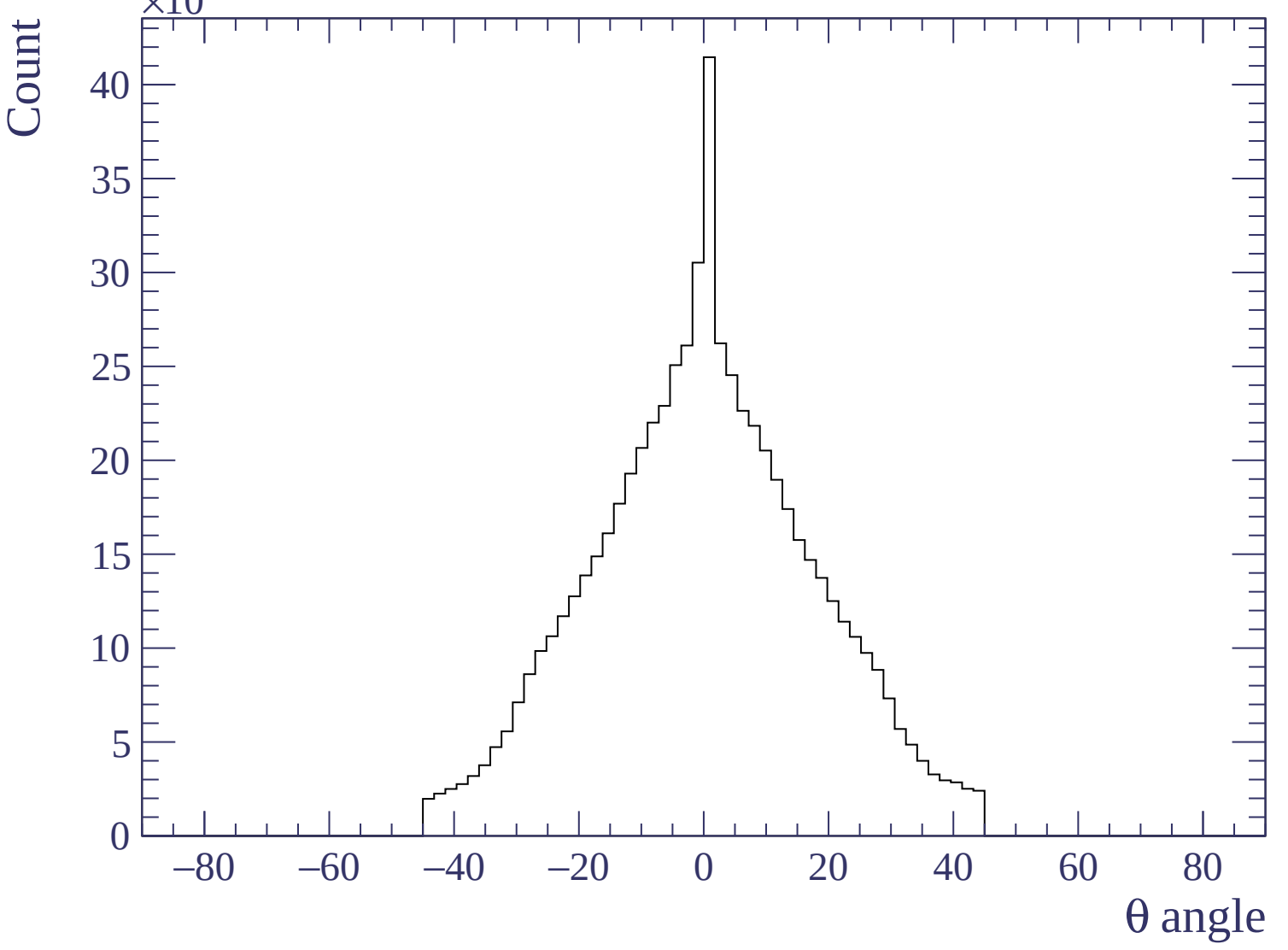


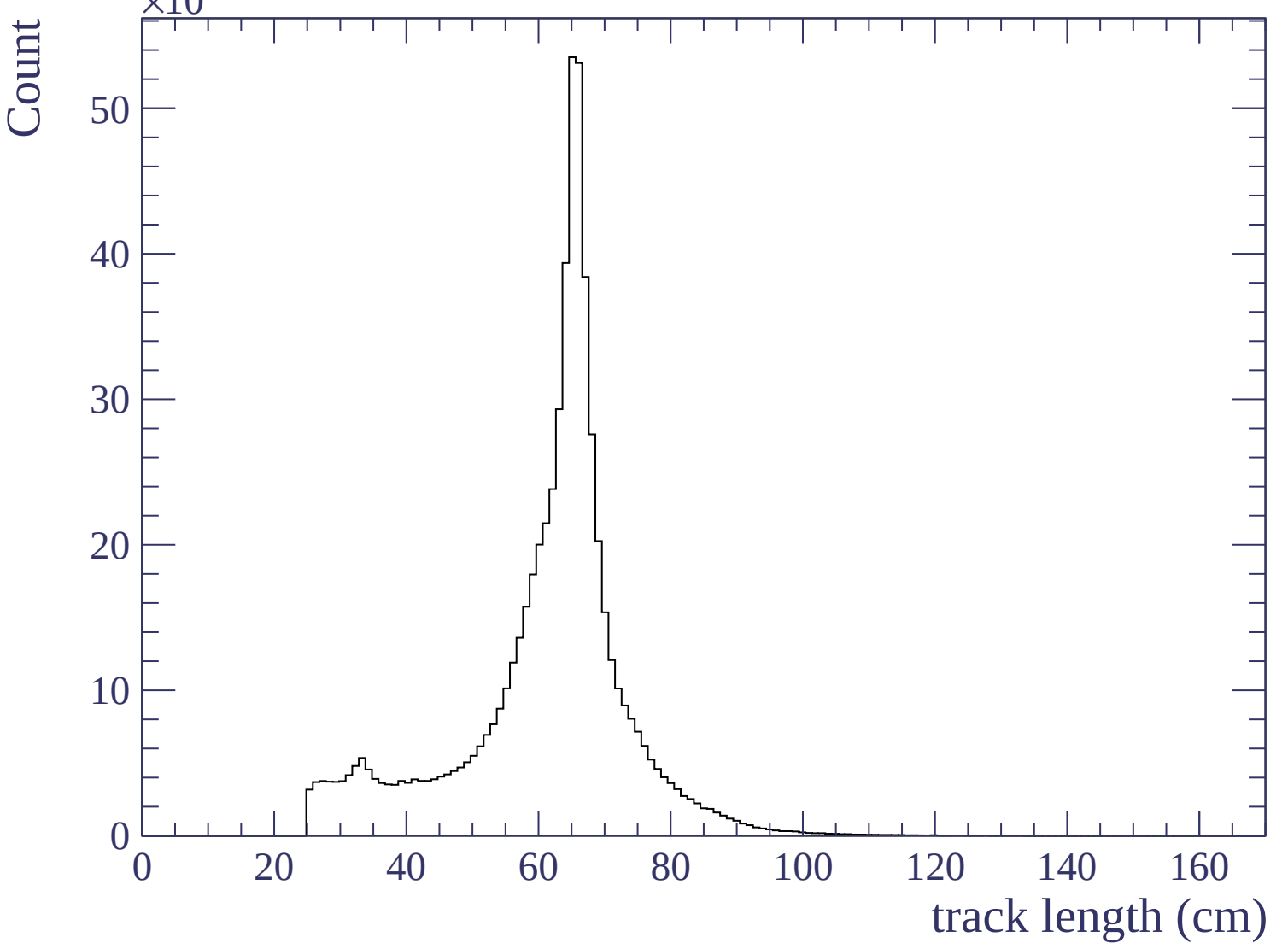


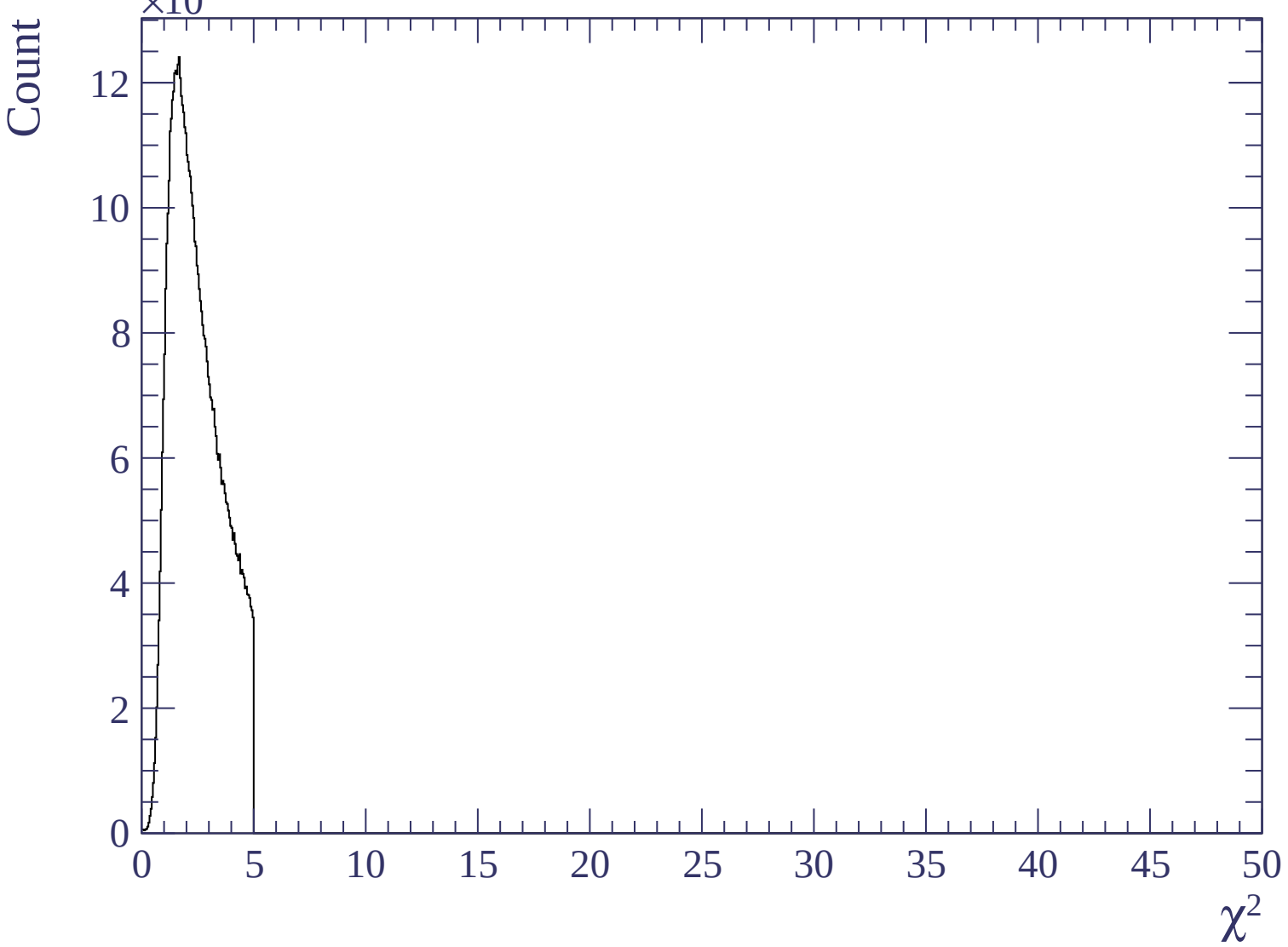


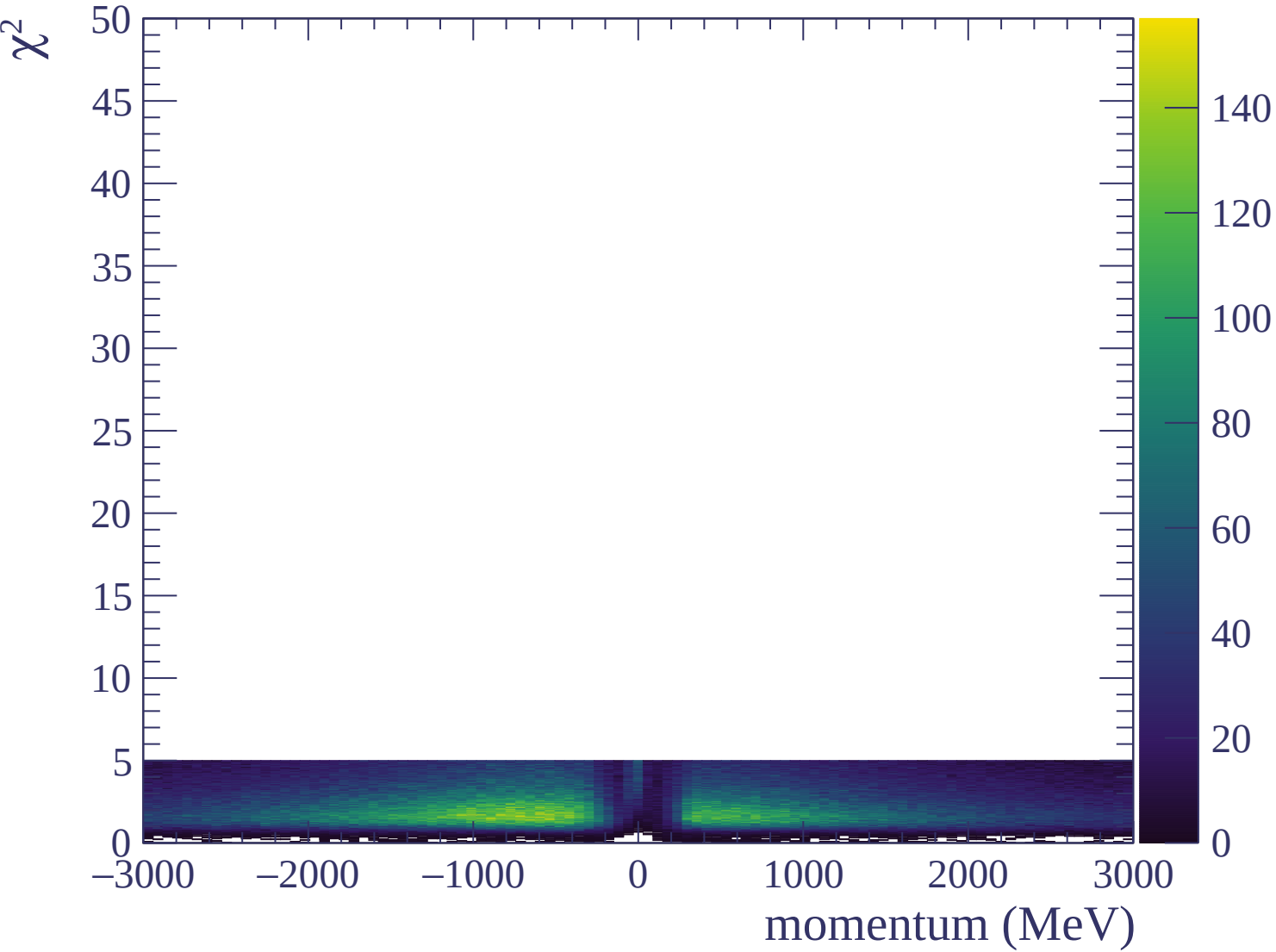


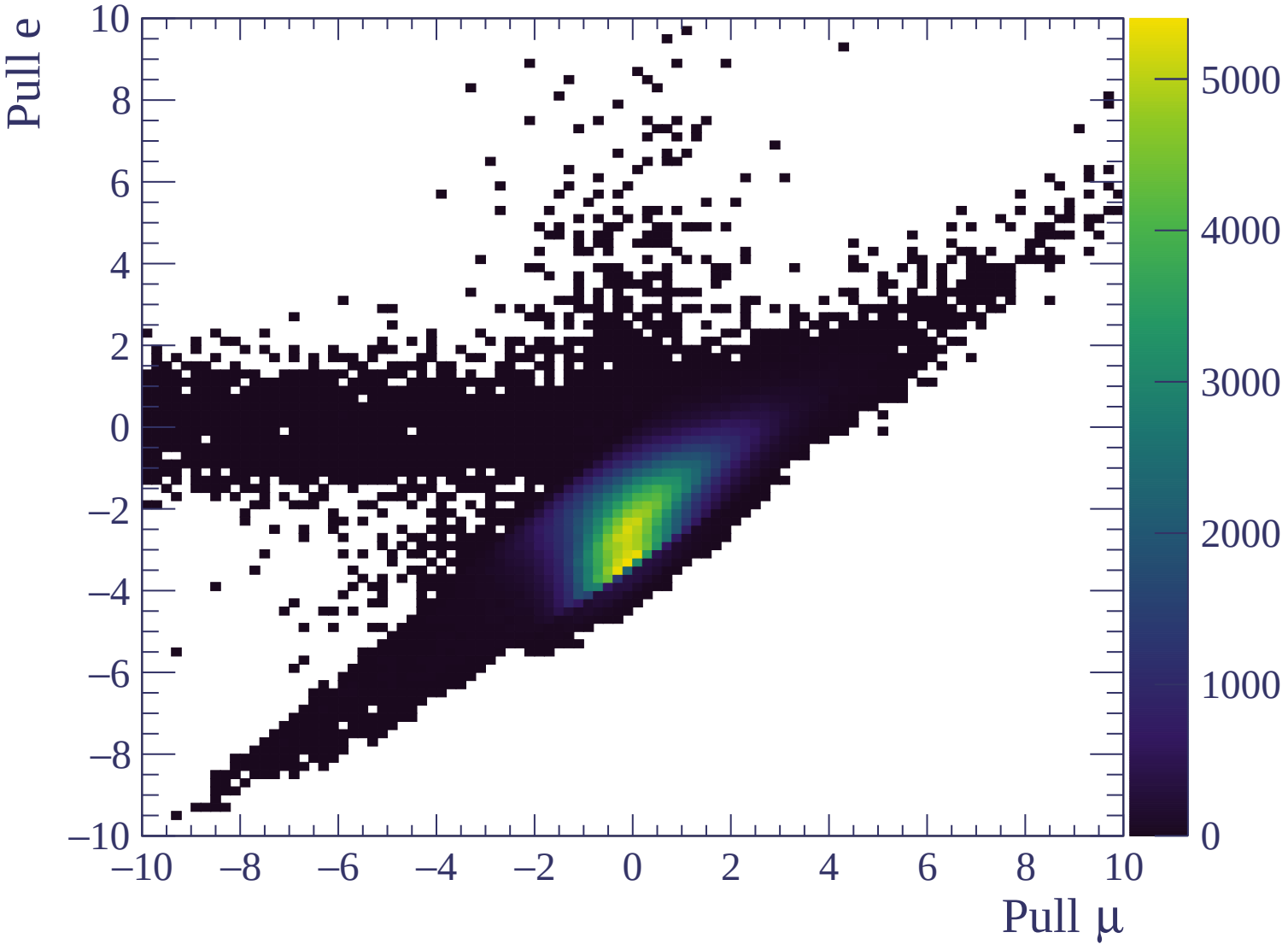


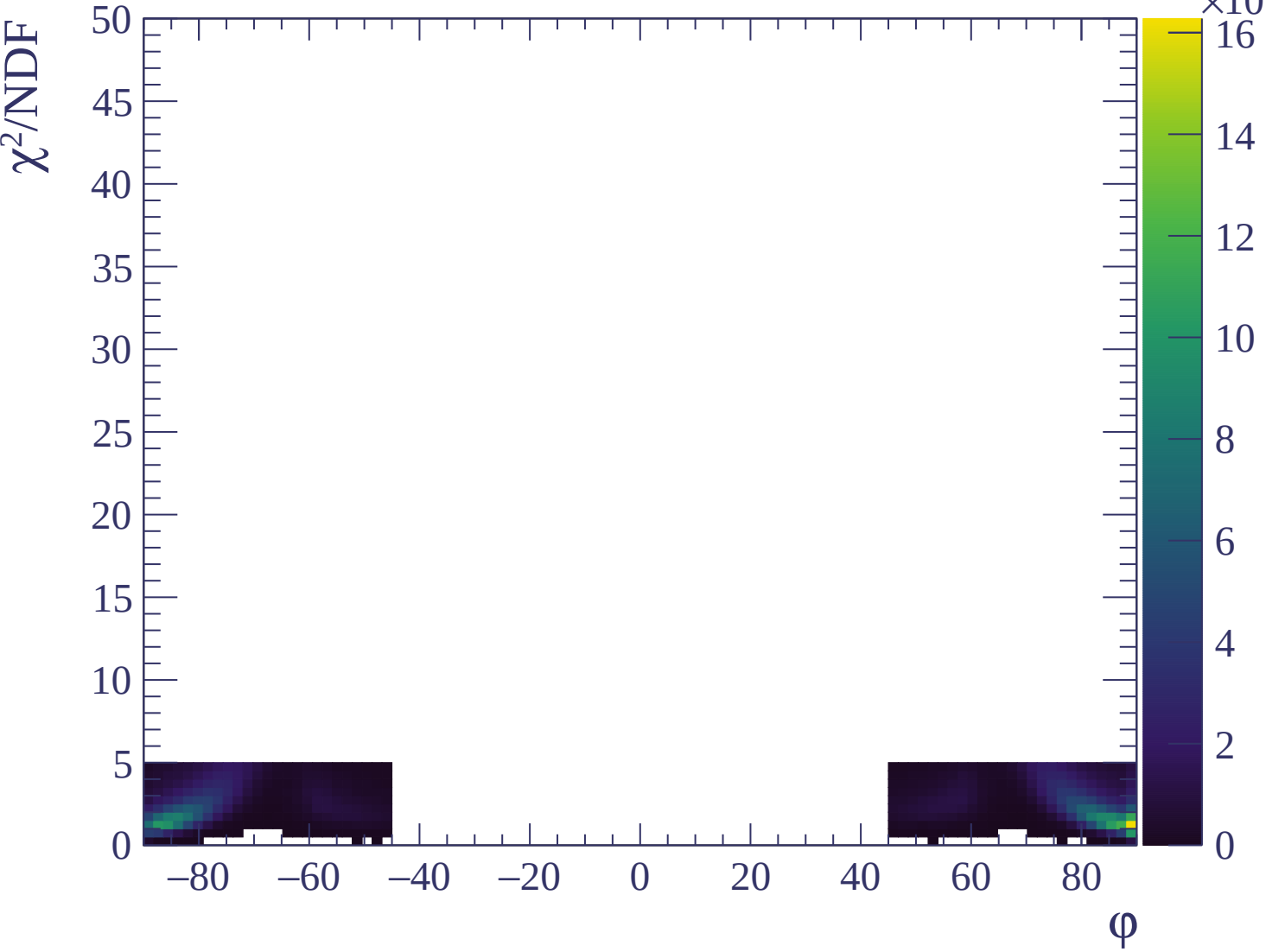


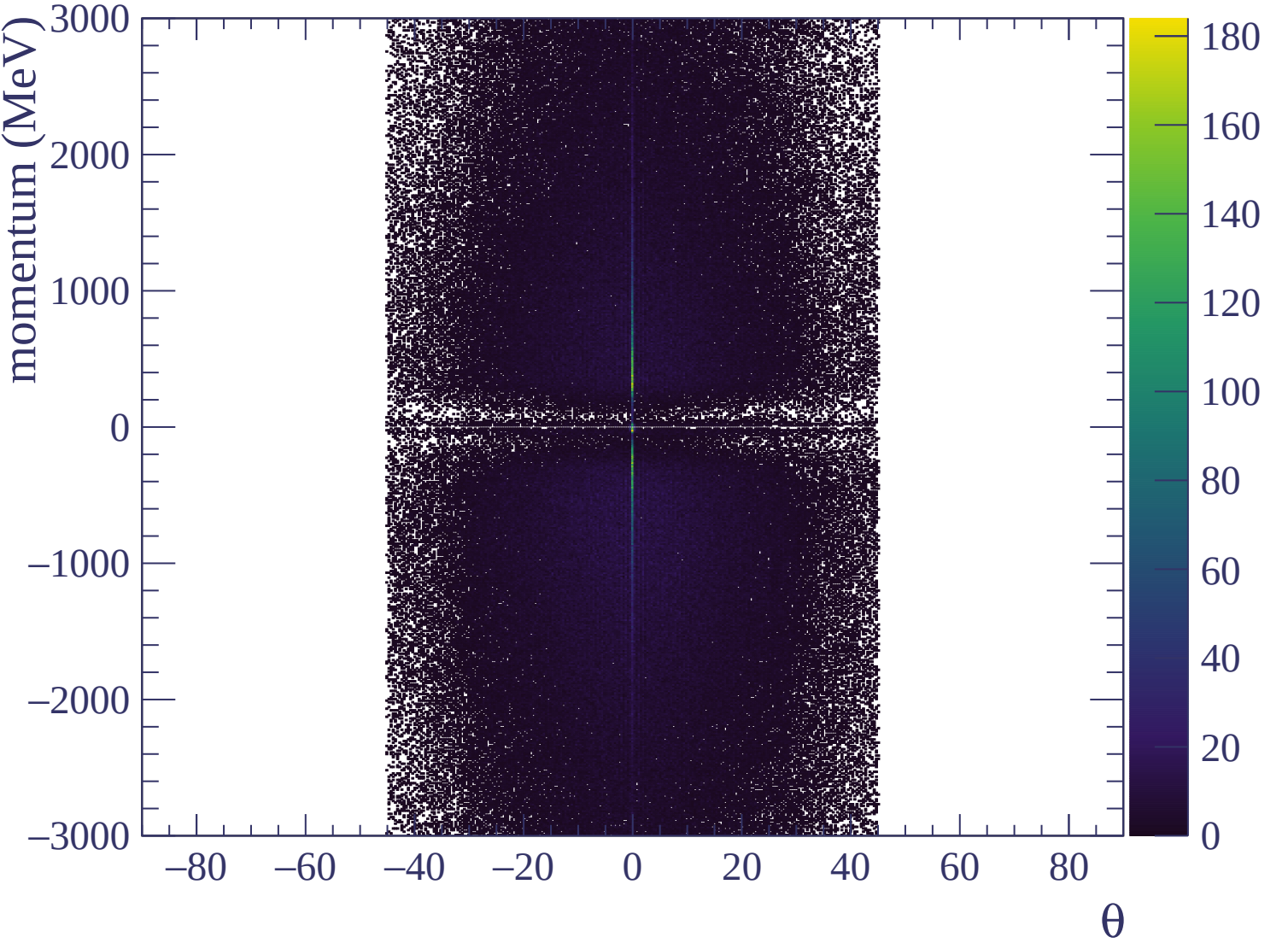


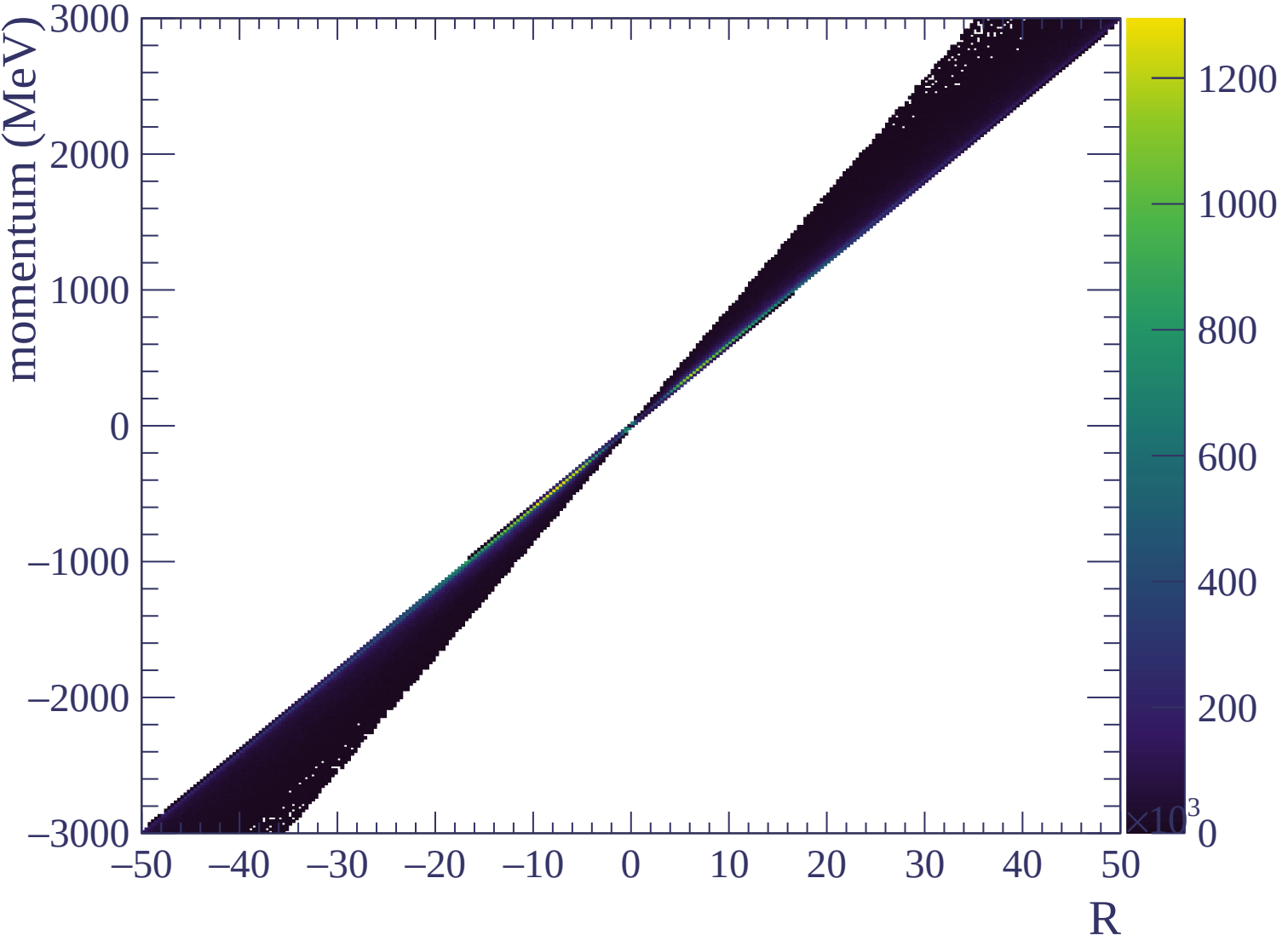


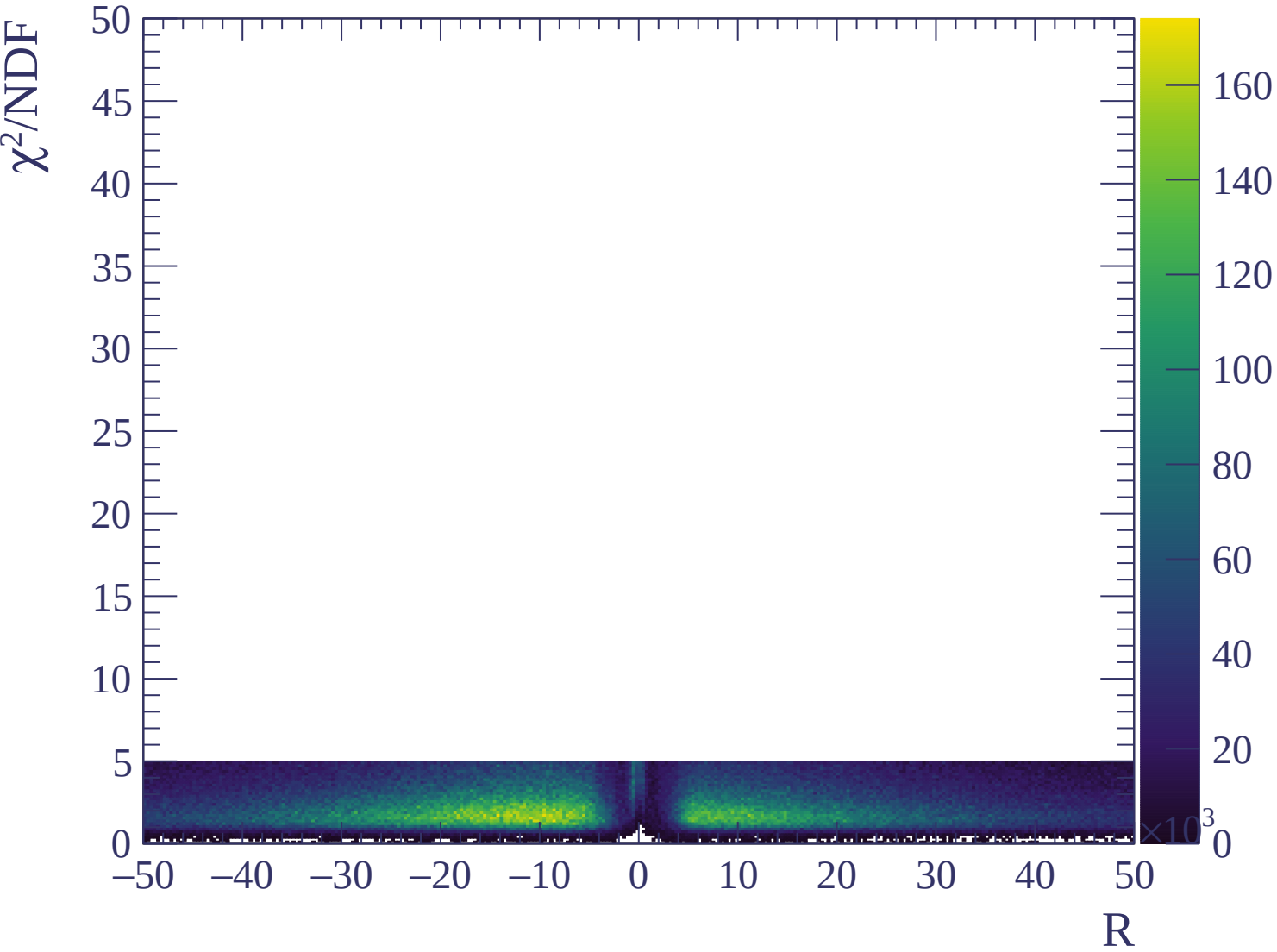


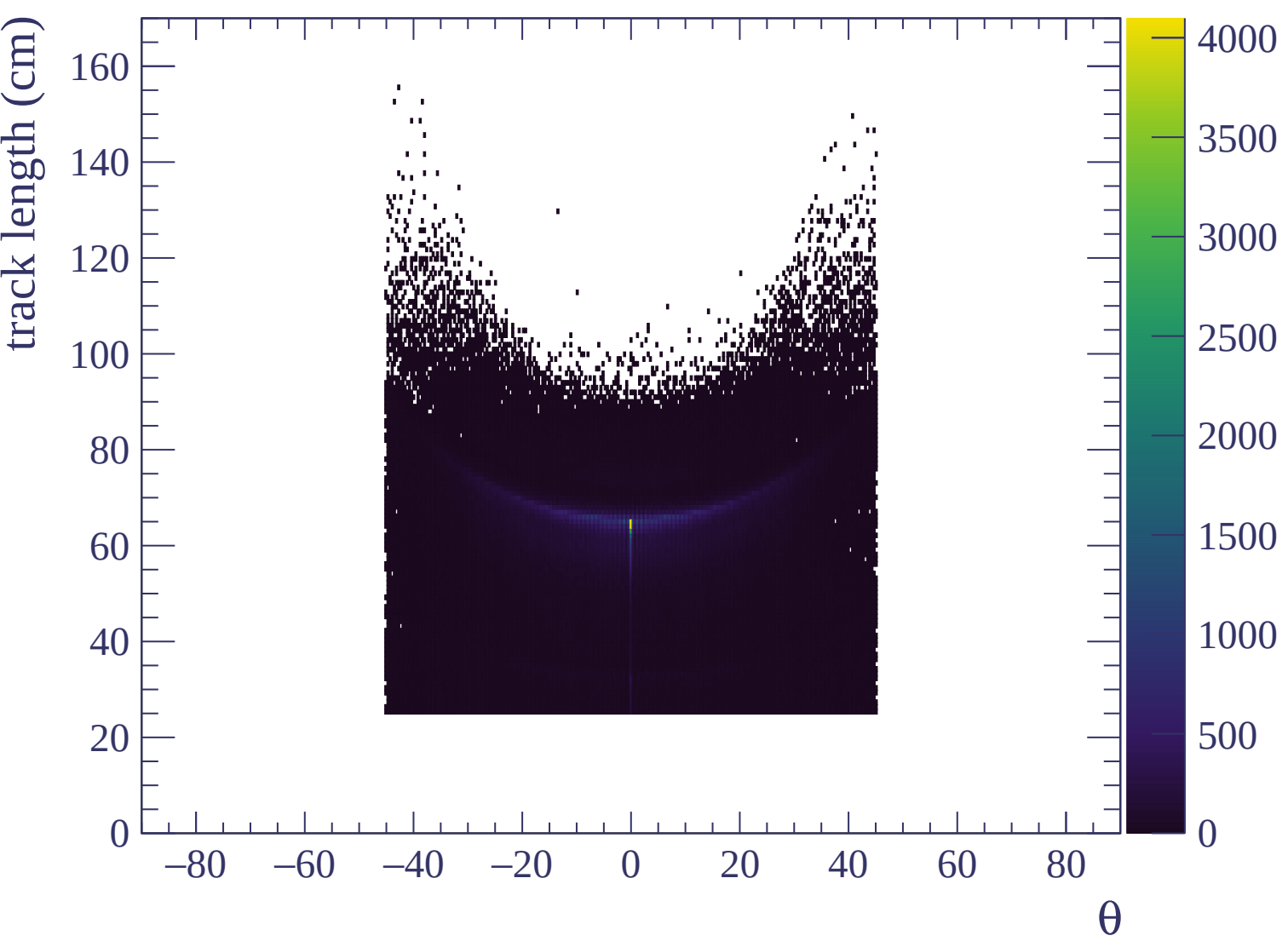


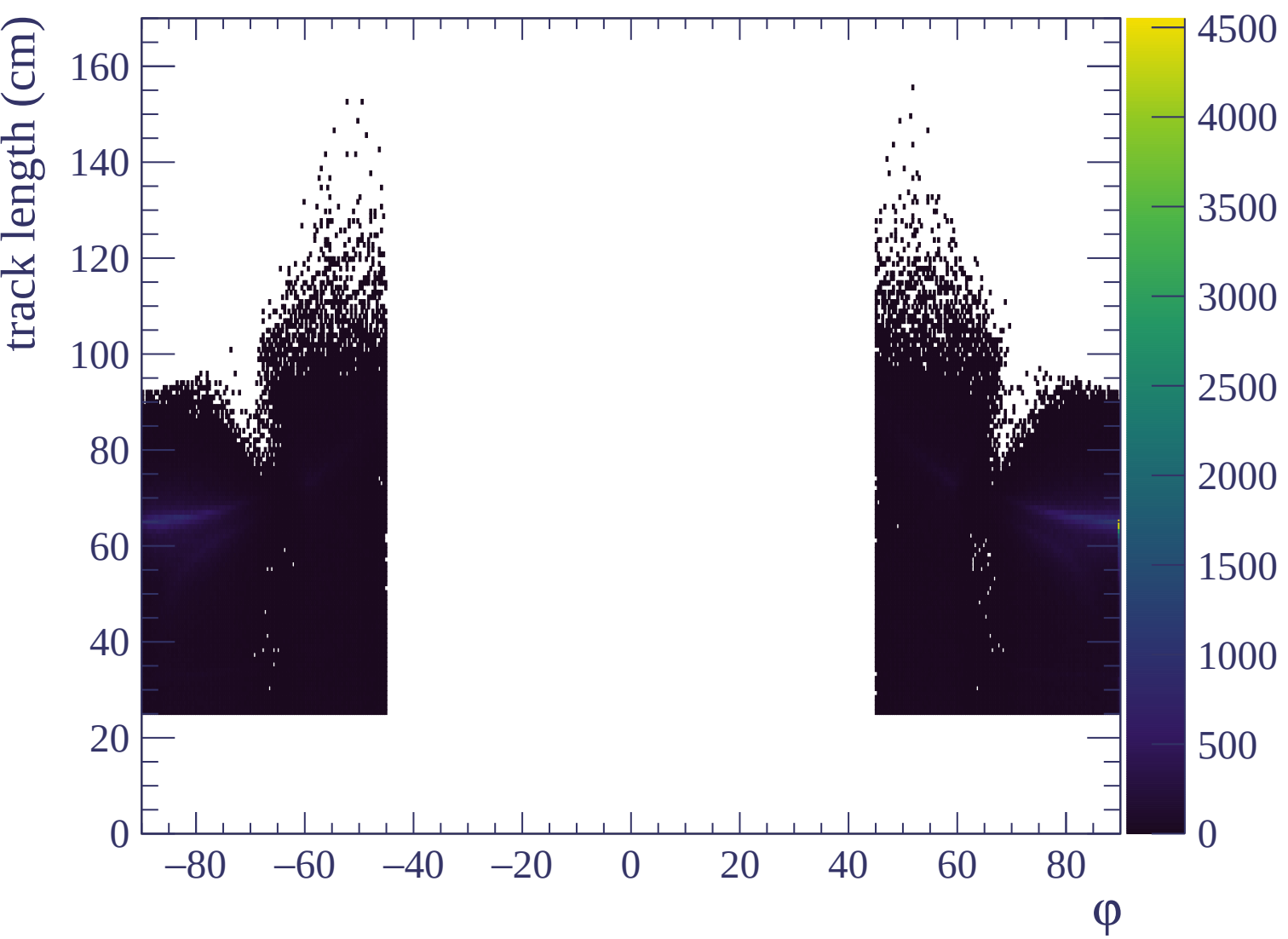






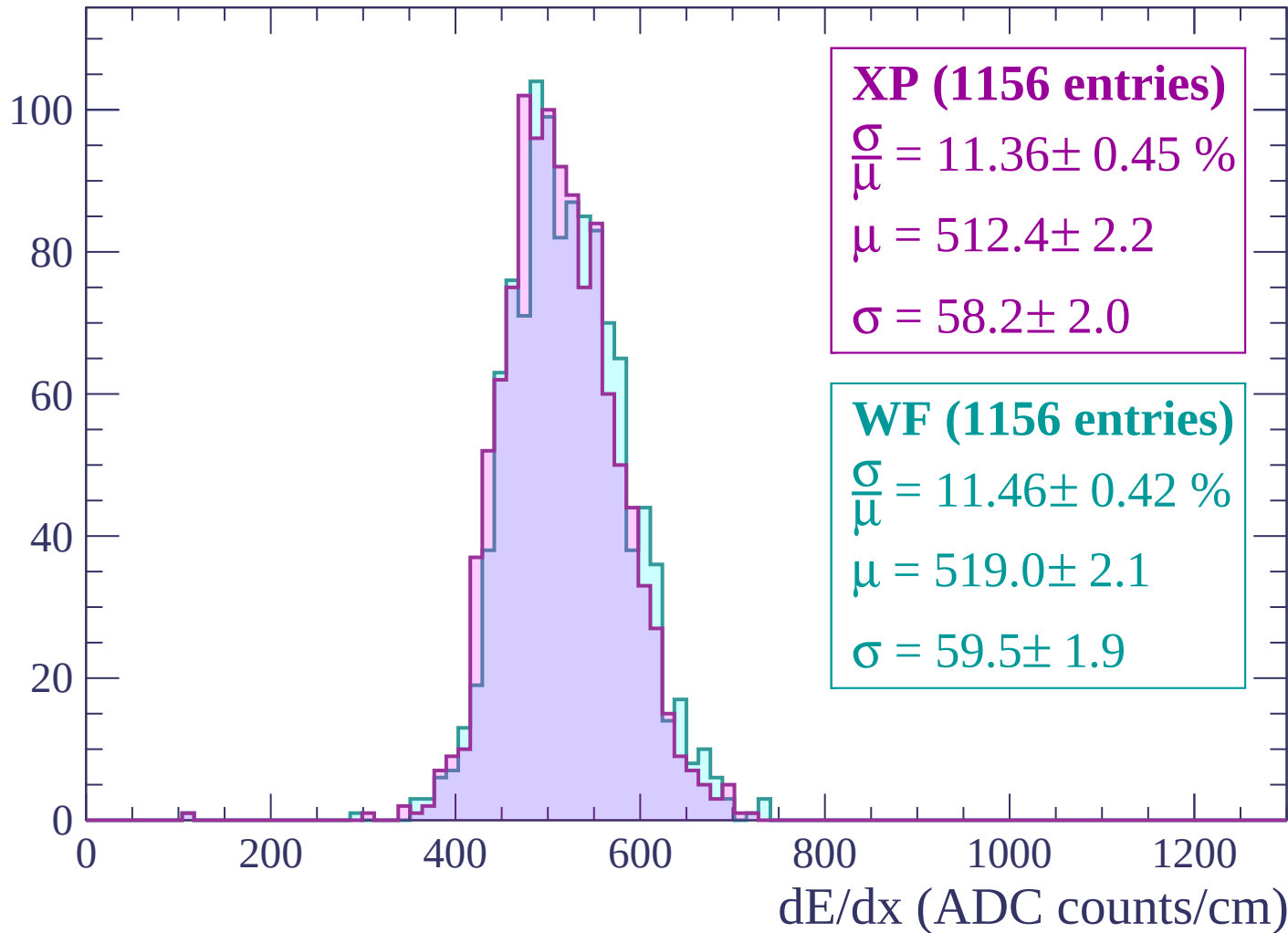






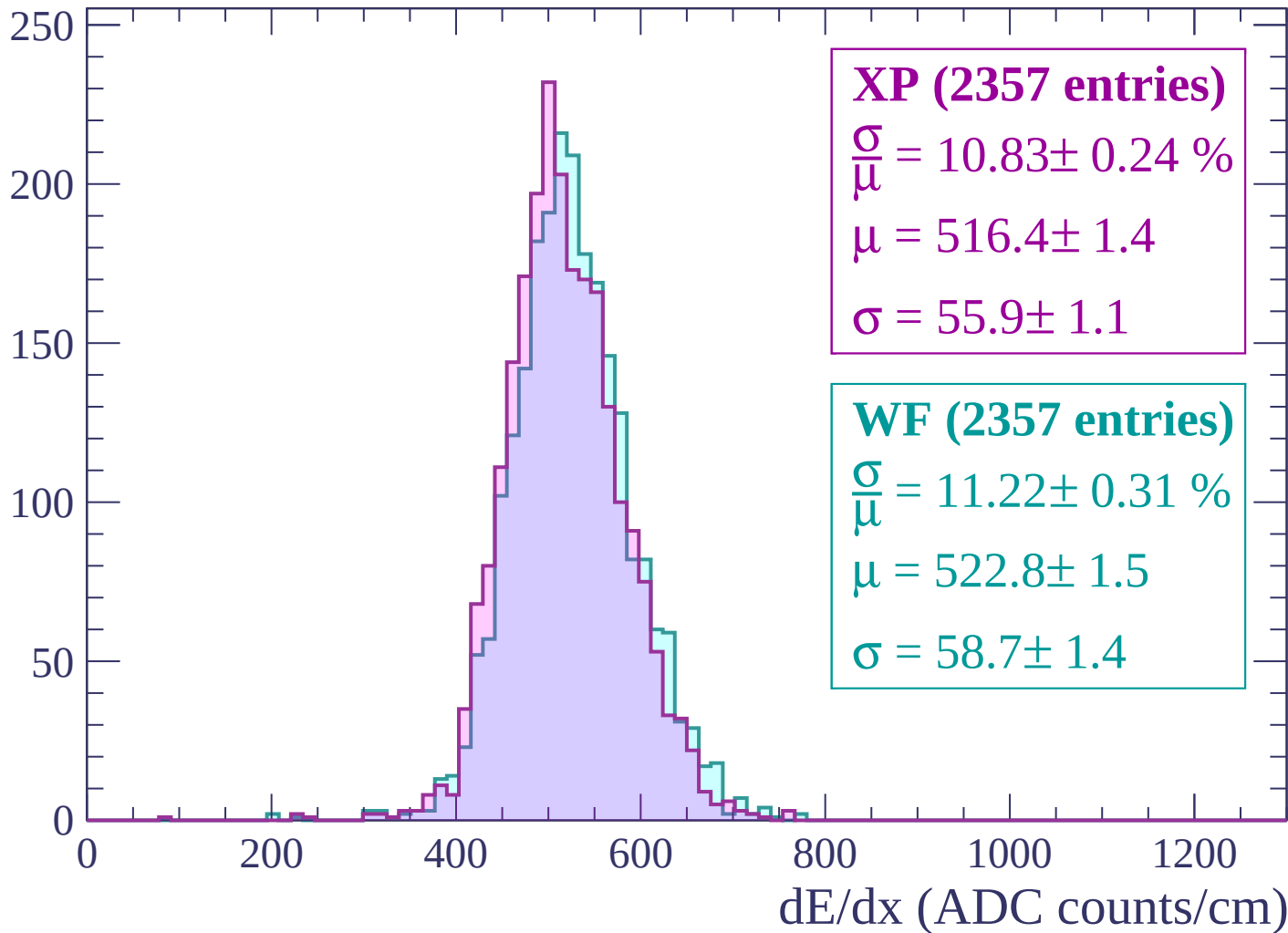
Energy loss | $-3000 < p < -2940$

Count



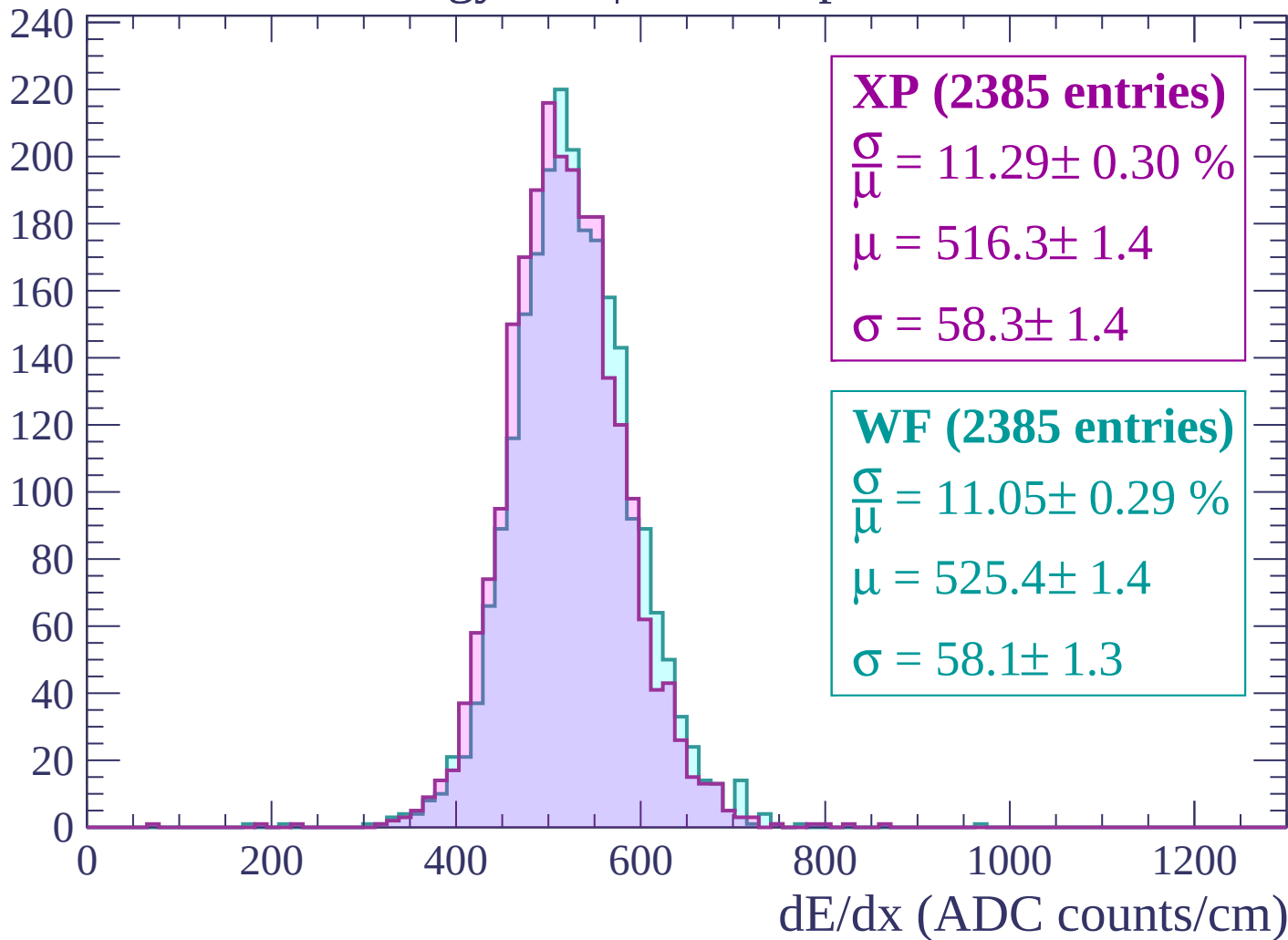
Energy loss | $-2940 < p < -2880$

Count



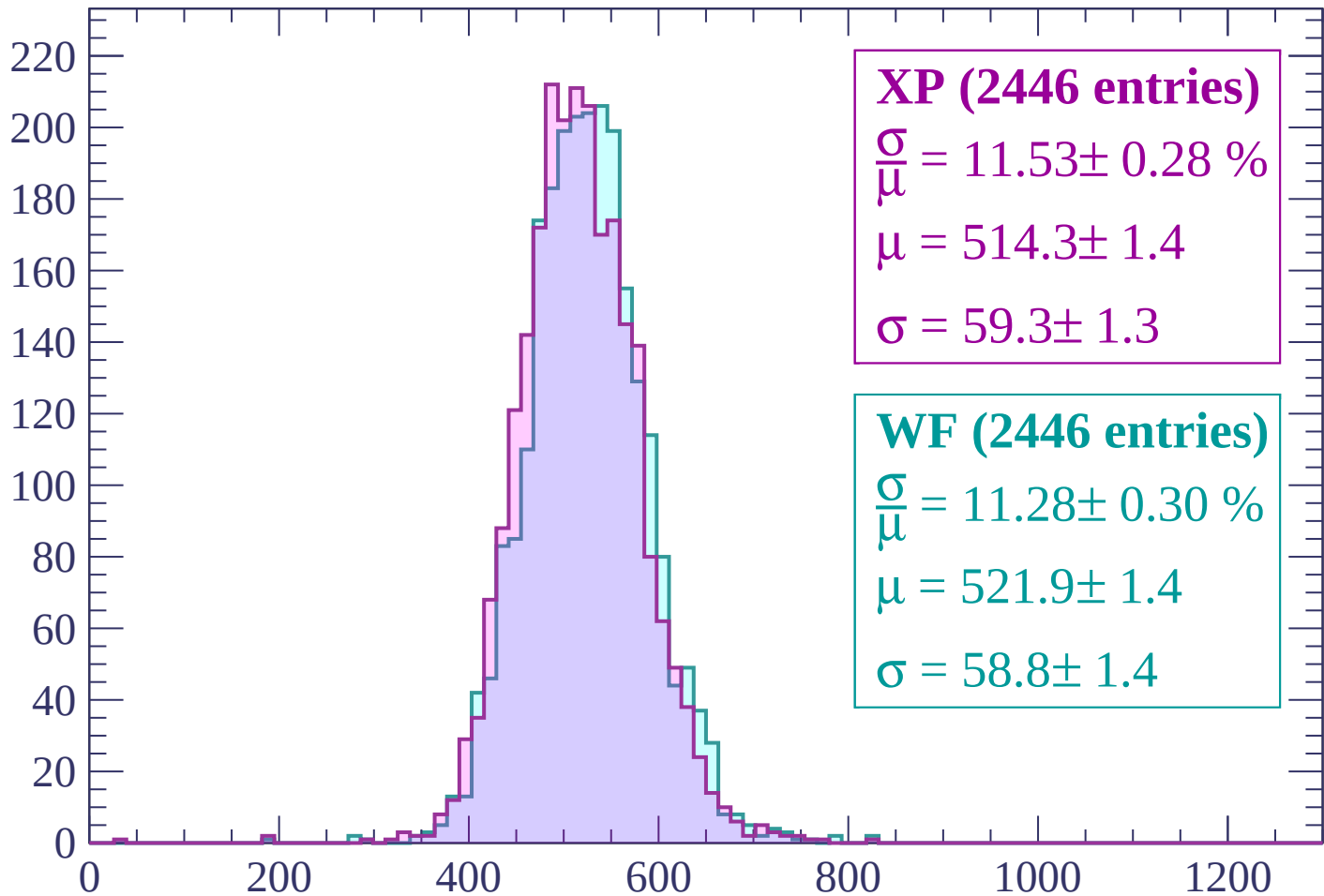
Energy loss | $-2880 < p < -2820$

Count



Energy loss | $-2820 < p < -2760$

Count



XP (2446 entries)

$$\frac{\sigma}{\mu} = 11.53 \pm 0.28 \%$$

$$\mu = 514.3 \pm 1.4$$

$$\sigma = 59.3 \pm 1.3$$

WF (2446 entries)

$$\frac{\sigma}{\mu} = 11.28 \pm 0.30 \%$$

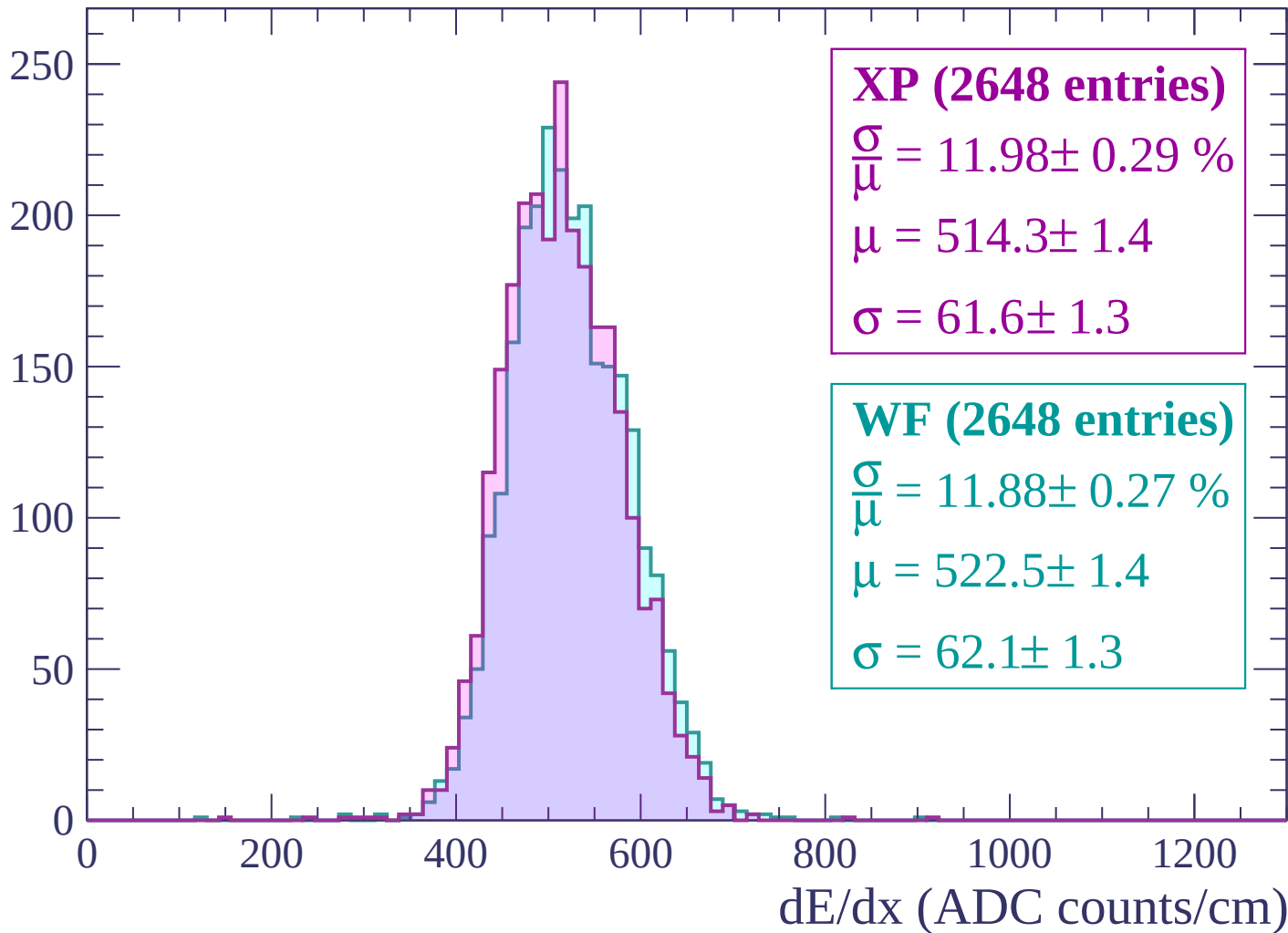
$$\mu = 521.9 \pm 1.4$$

$$\sigma = 58.8 \pm 1.4$$

dE/dx (ADC counts/cm)

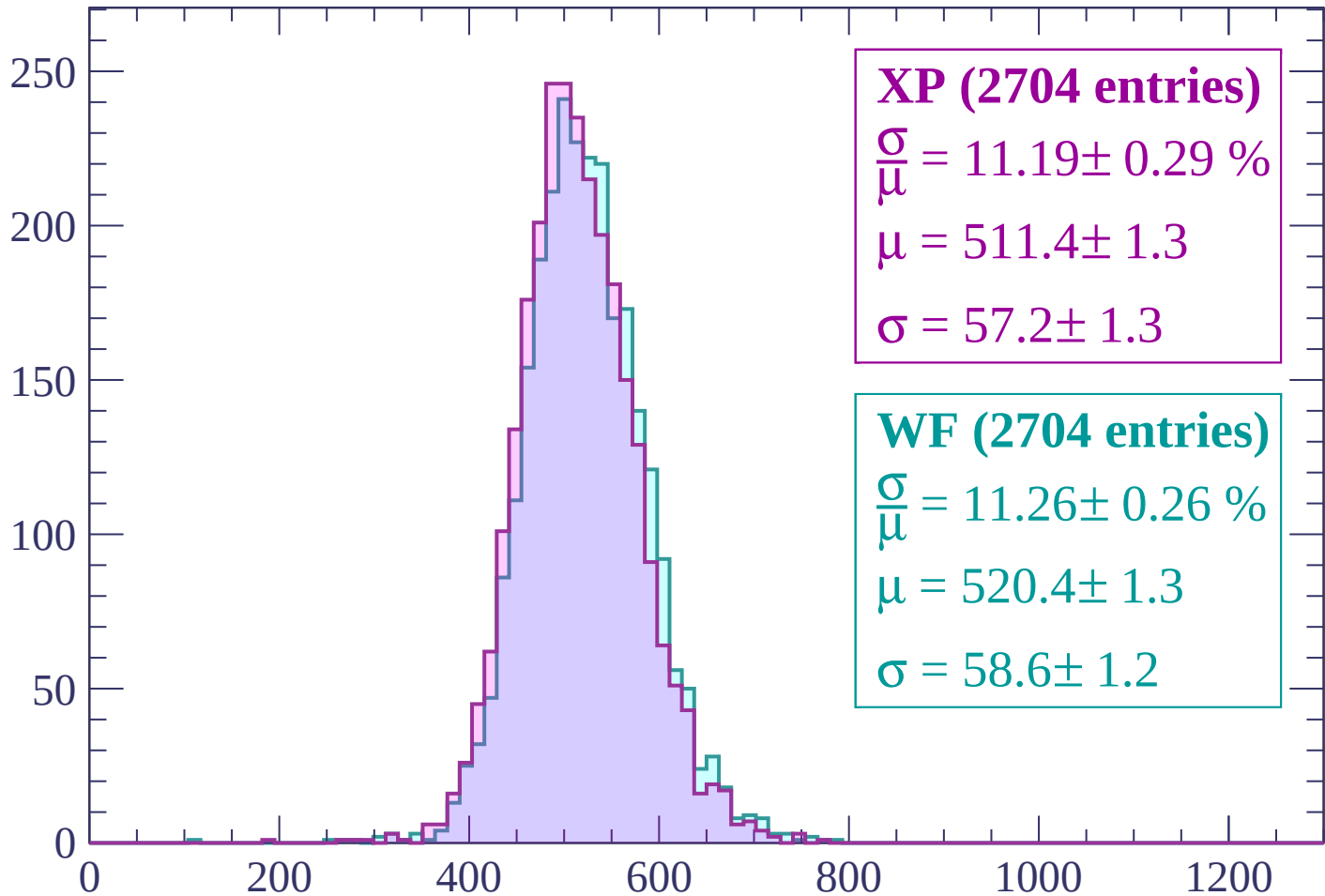
Energy loss | -2760 < p < -2700

Count



Energy loss | -2700 < p < -2640

Count



XP (2704 entries)

$\frac{\sigma}{\mu} = 11.19 \pm 0.29 \%$

$\mu = 511.4 \pm 1.3$

$\sigma = 57.2 \pm 1.3$

WF (2704 entries)

$\frac{\sigma}{\mu} = 11.26 \pm 0.26 \%$

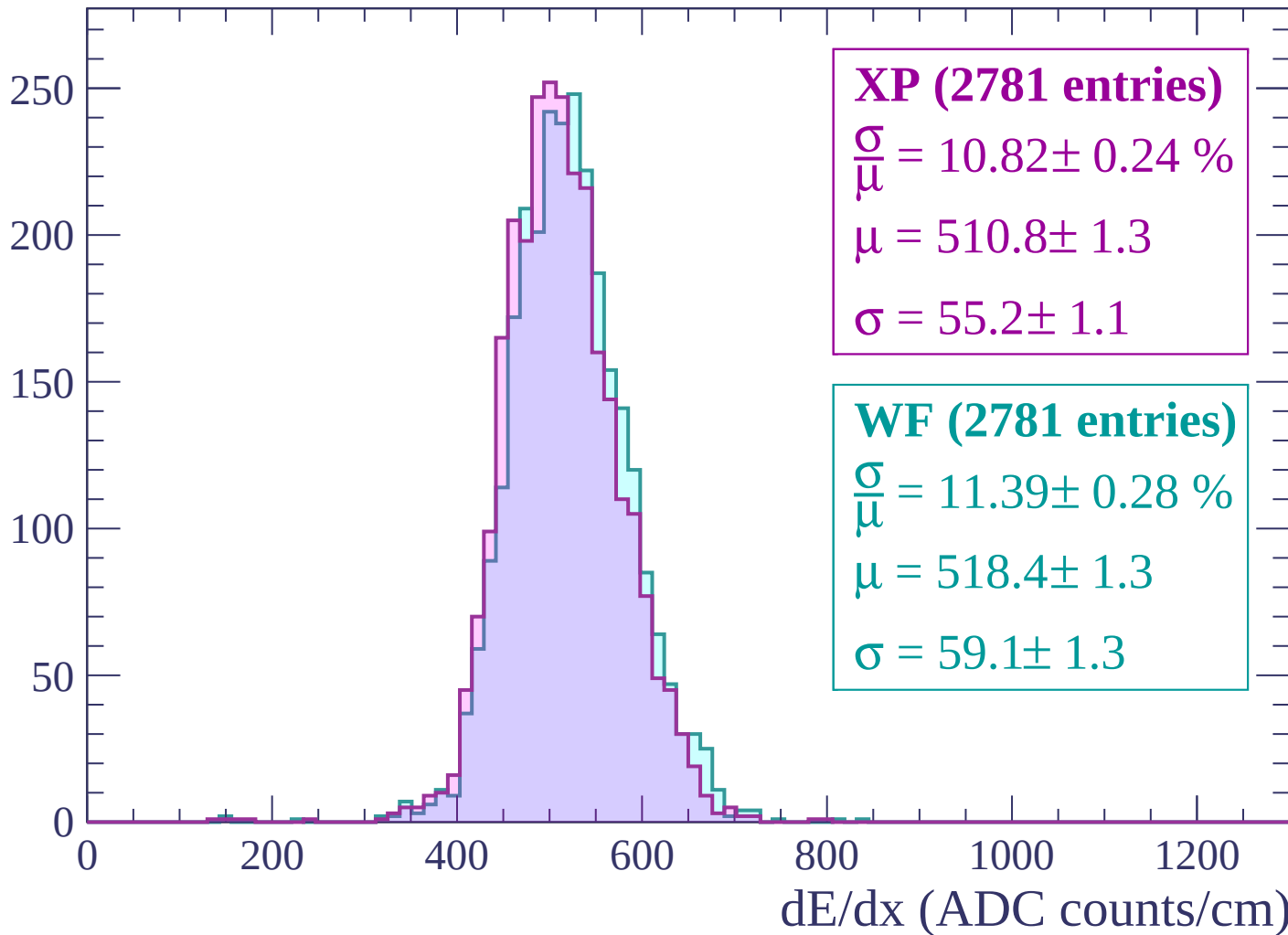
$\mu = 520.4 \pm 1.3$

$\sigma = 58.6 \pm 1.2$

dE/dx (ADC counts/cm)

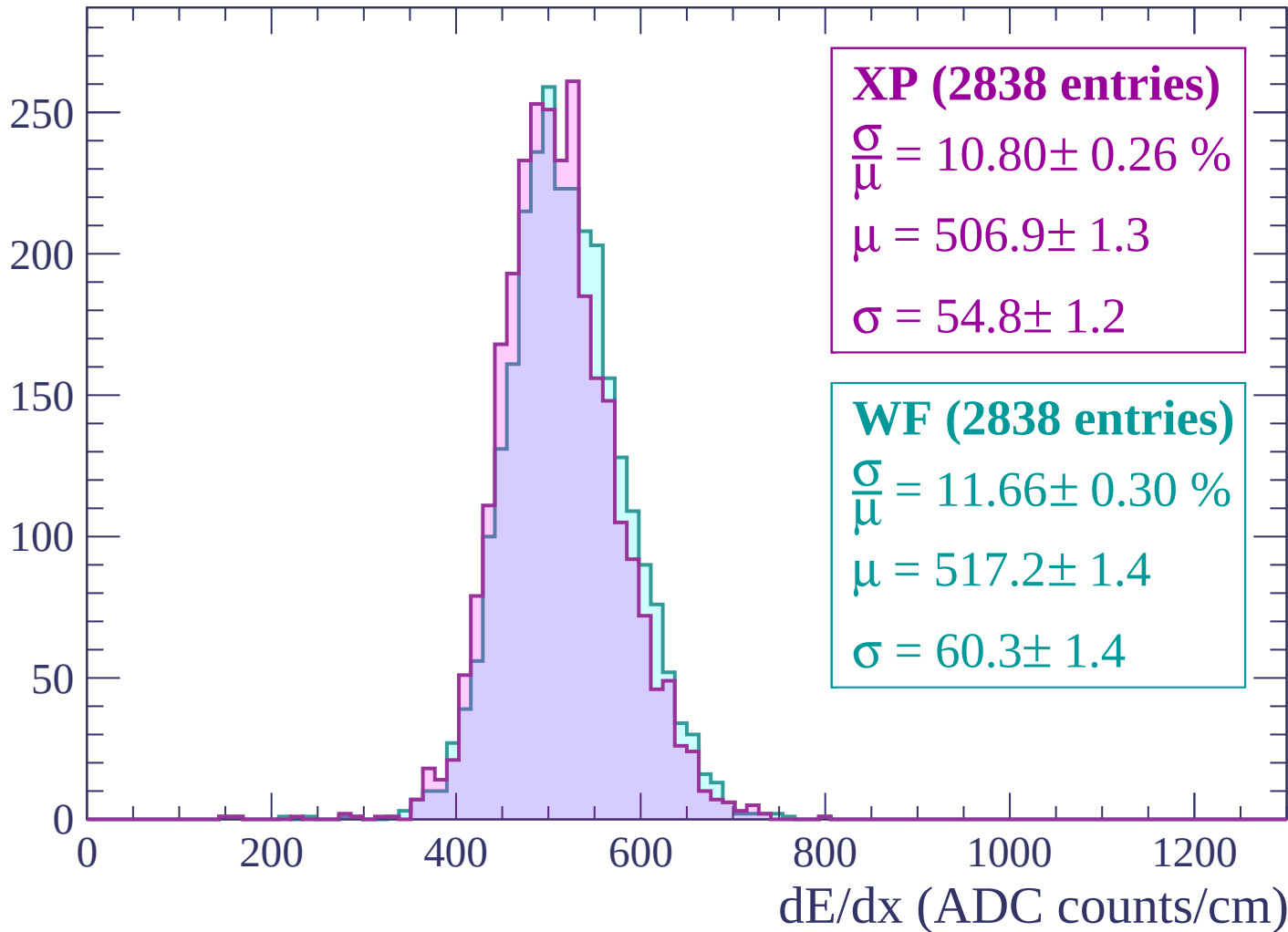
Energy loss | $-2640 < p < -2580$

Count



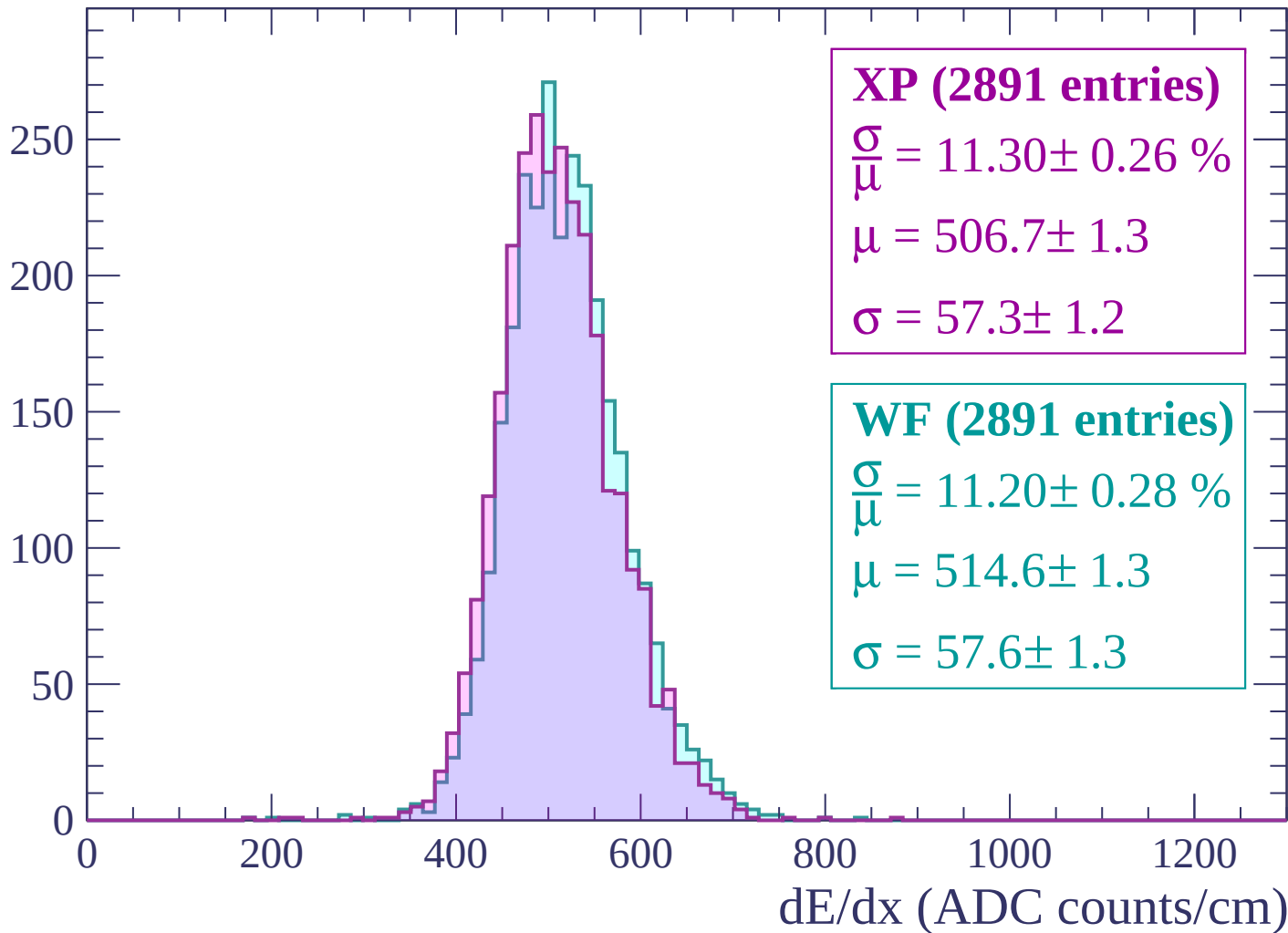
Energy loss | $-2580 < p < -2520$

Count



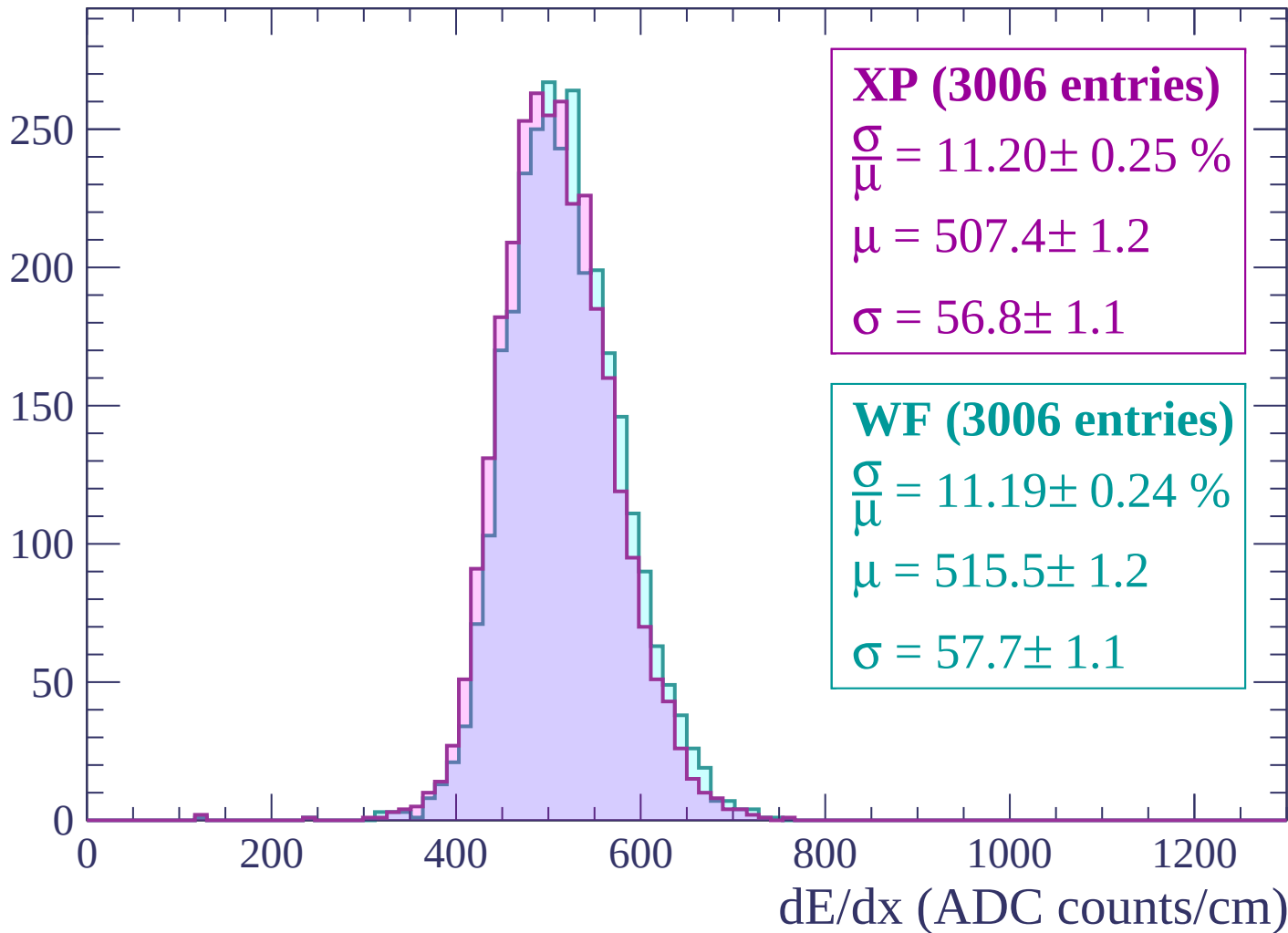
Energy loss | -2520 < p < -2460

Count



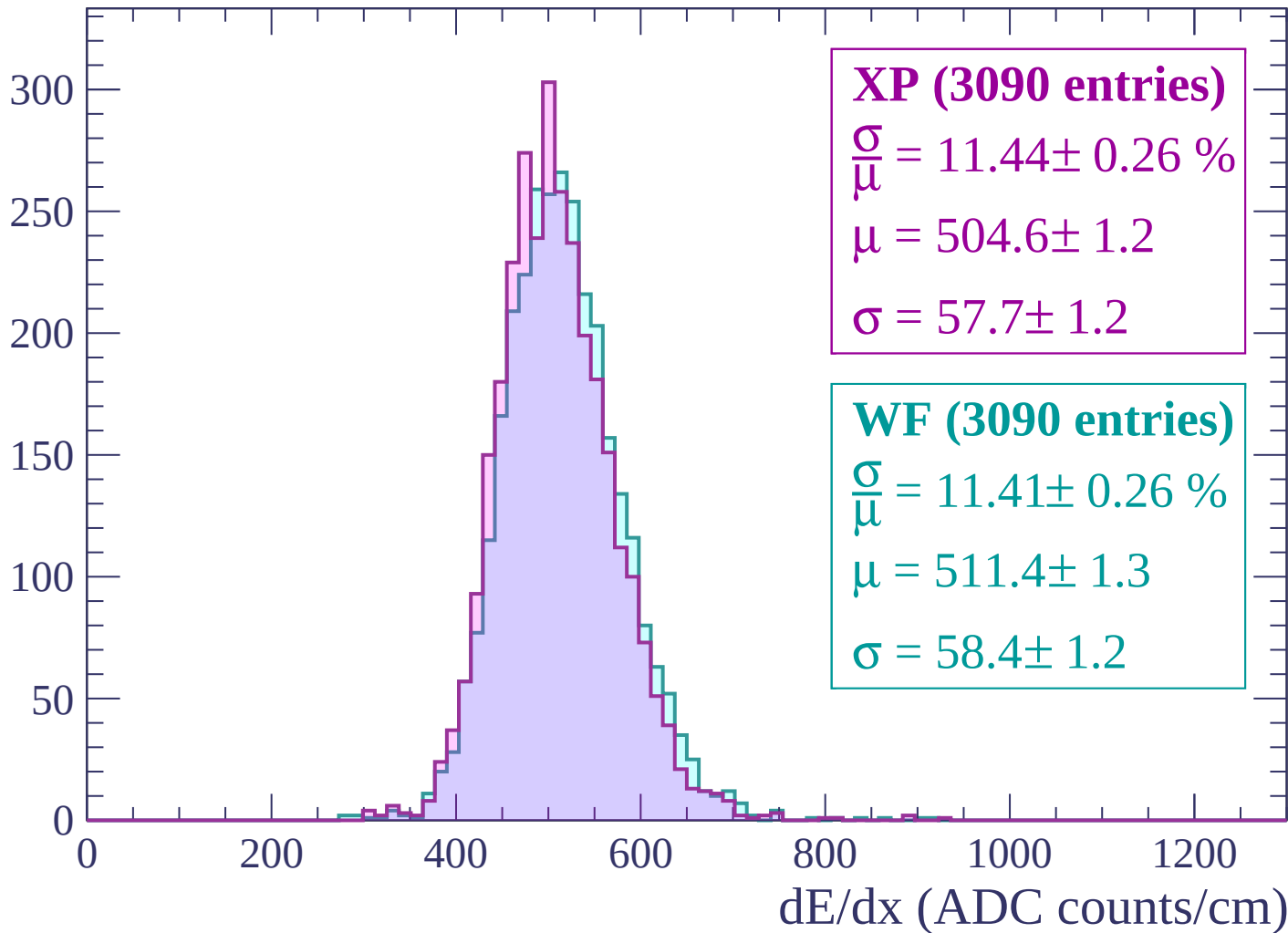
Energy loss | $-2460 < p < -2400$

Count



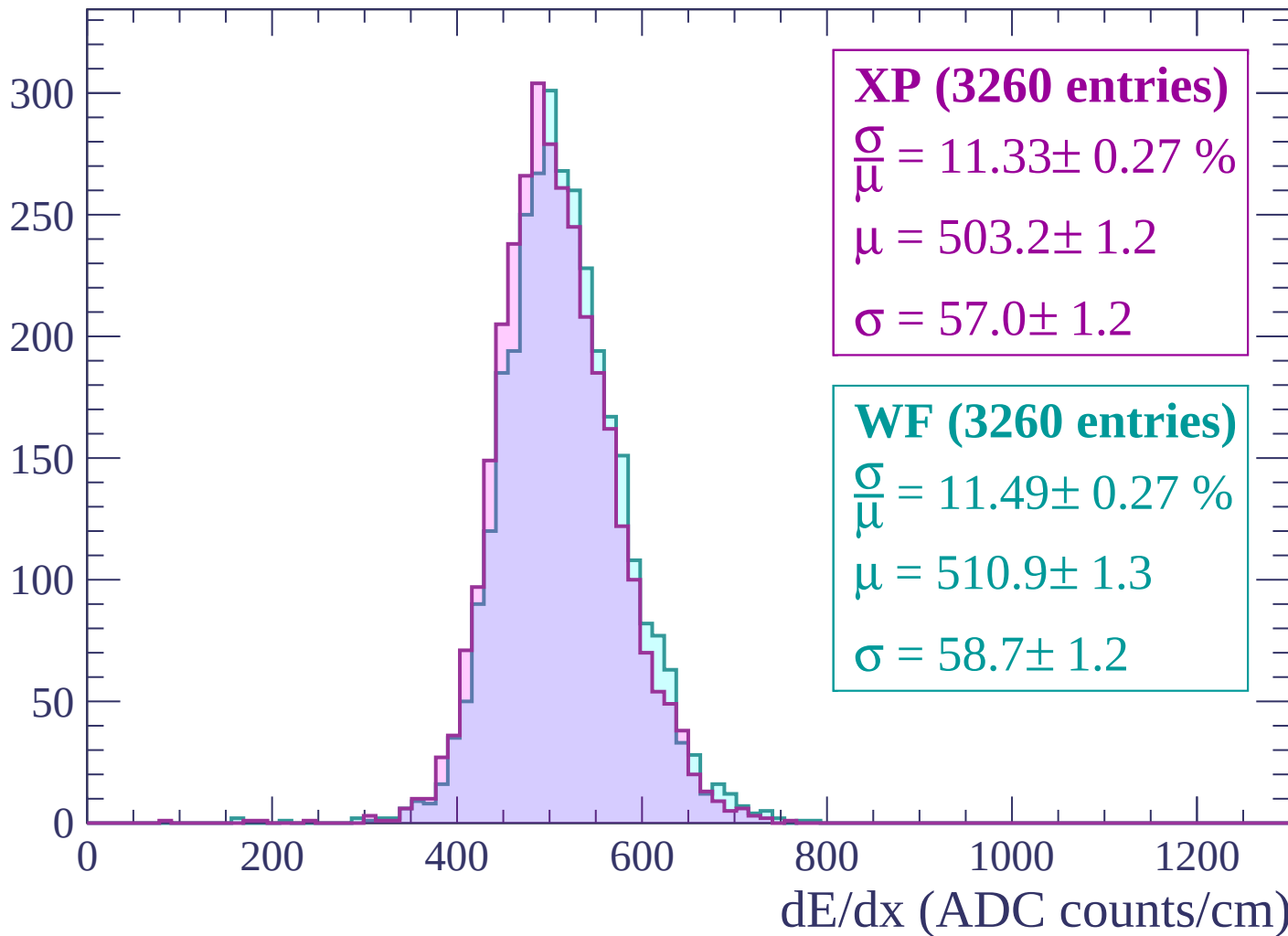
Energy loss | -2400 < p < -2340

Count



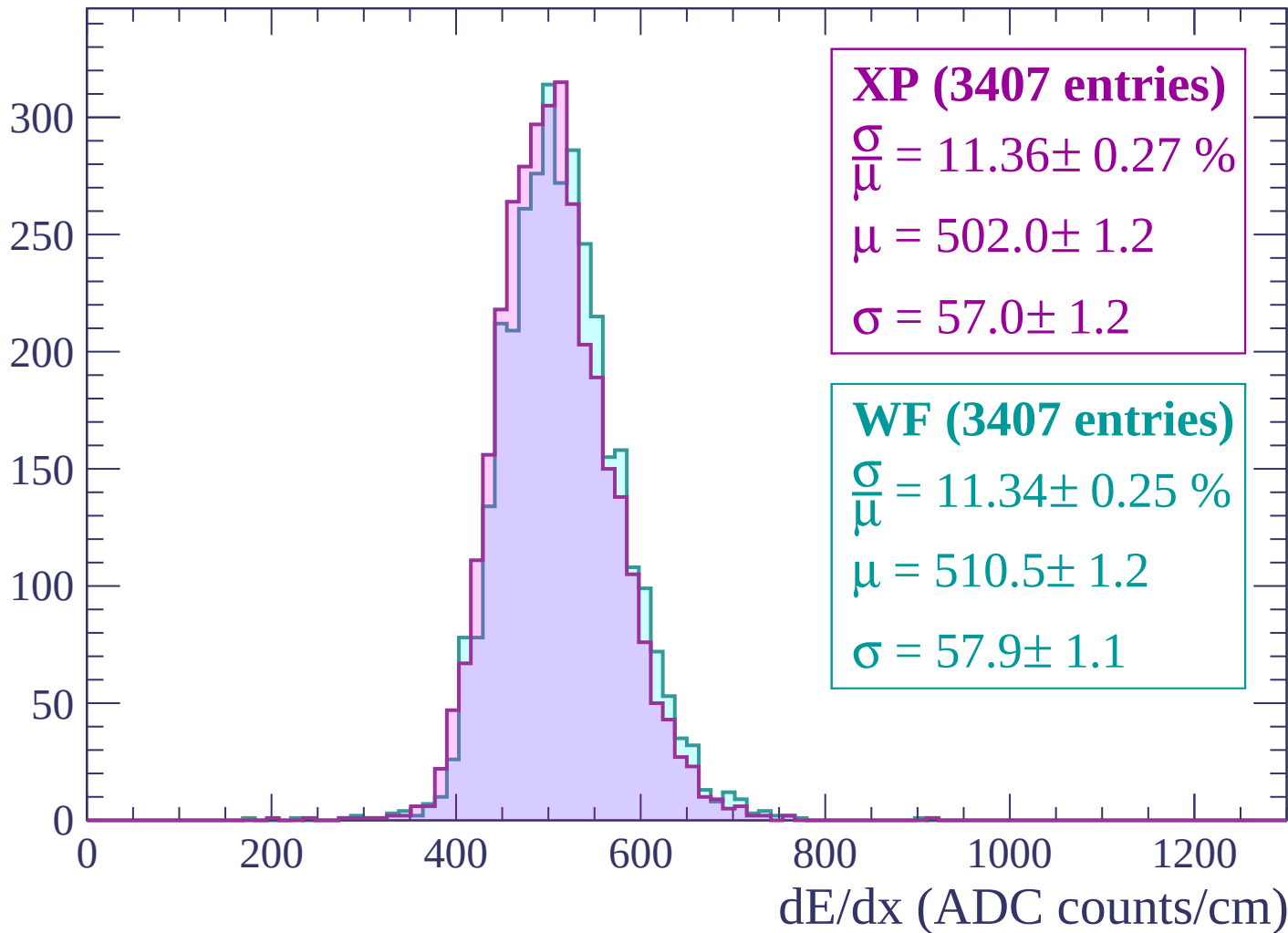
Energy loss | $-2340 < p < -2280$

Count



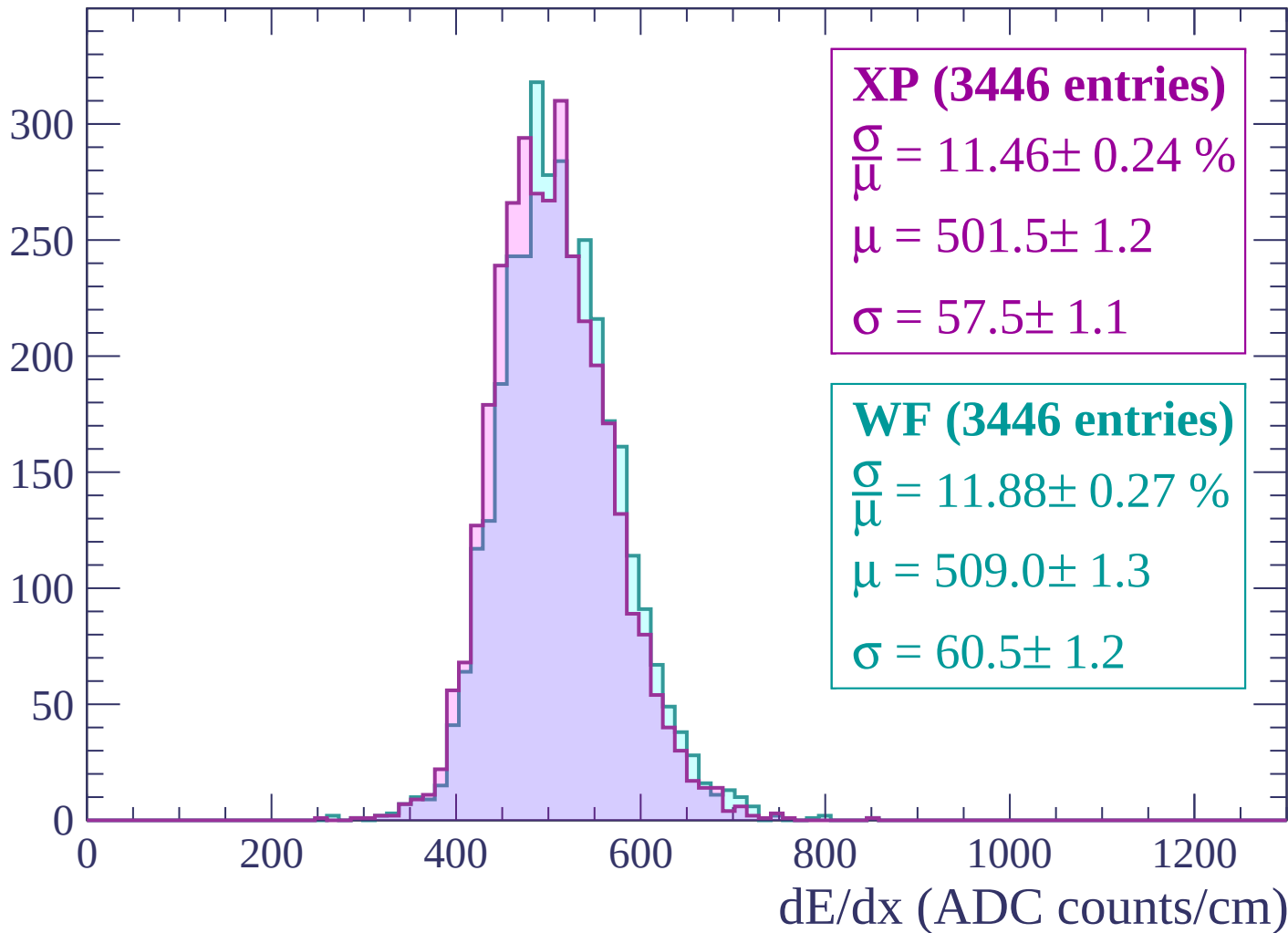
Energy loss | -2280 < p < -2220

Count



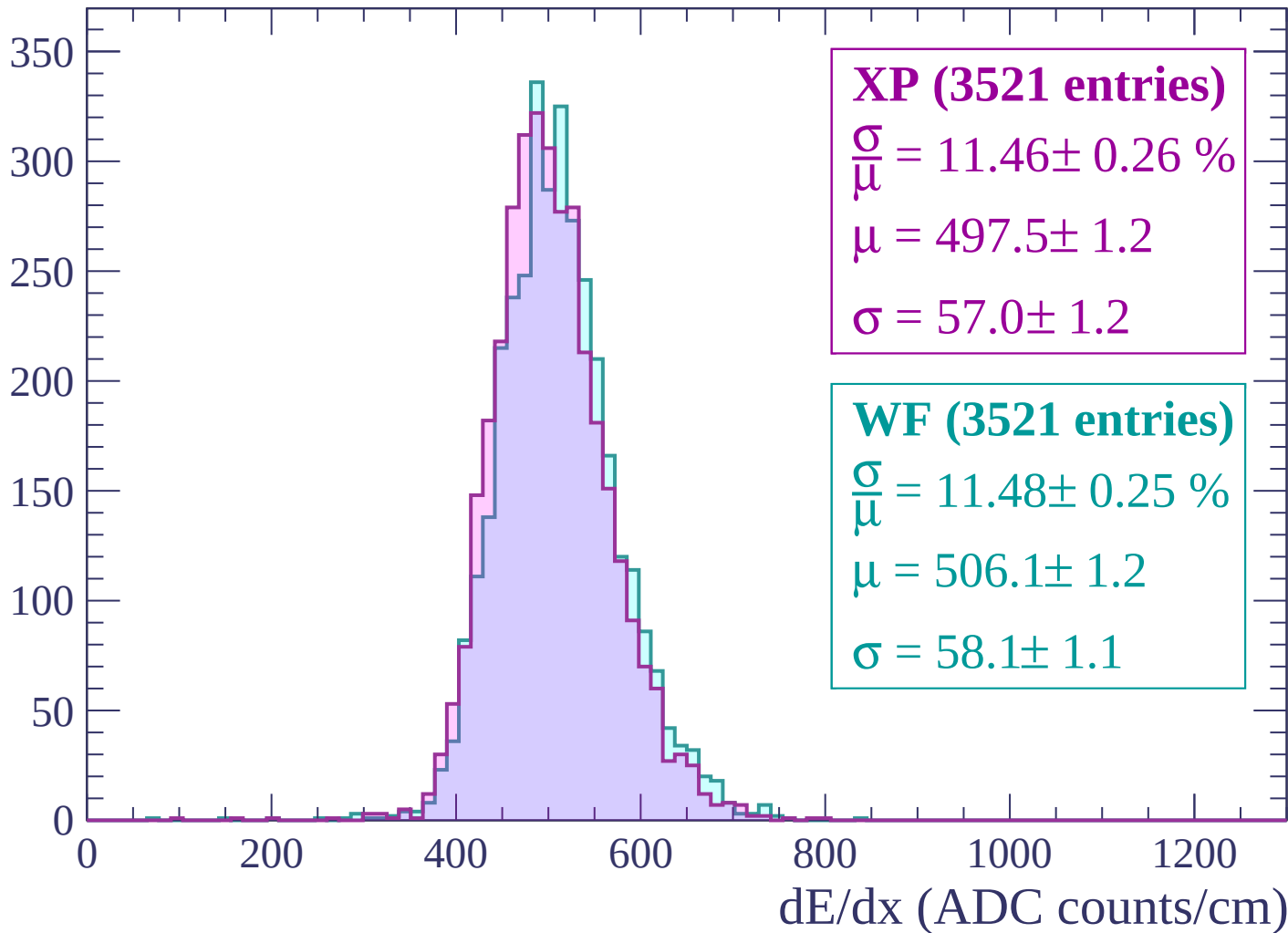
Energy loss | -2220 < p < -2160

Count



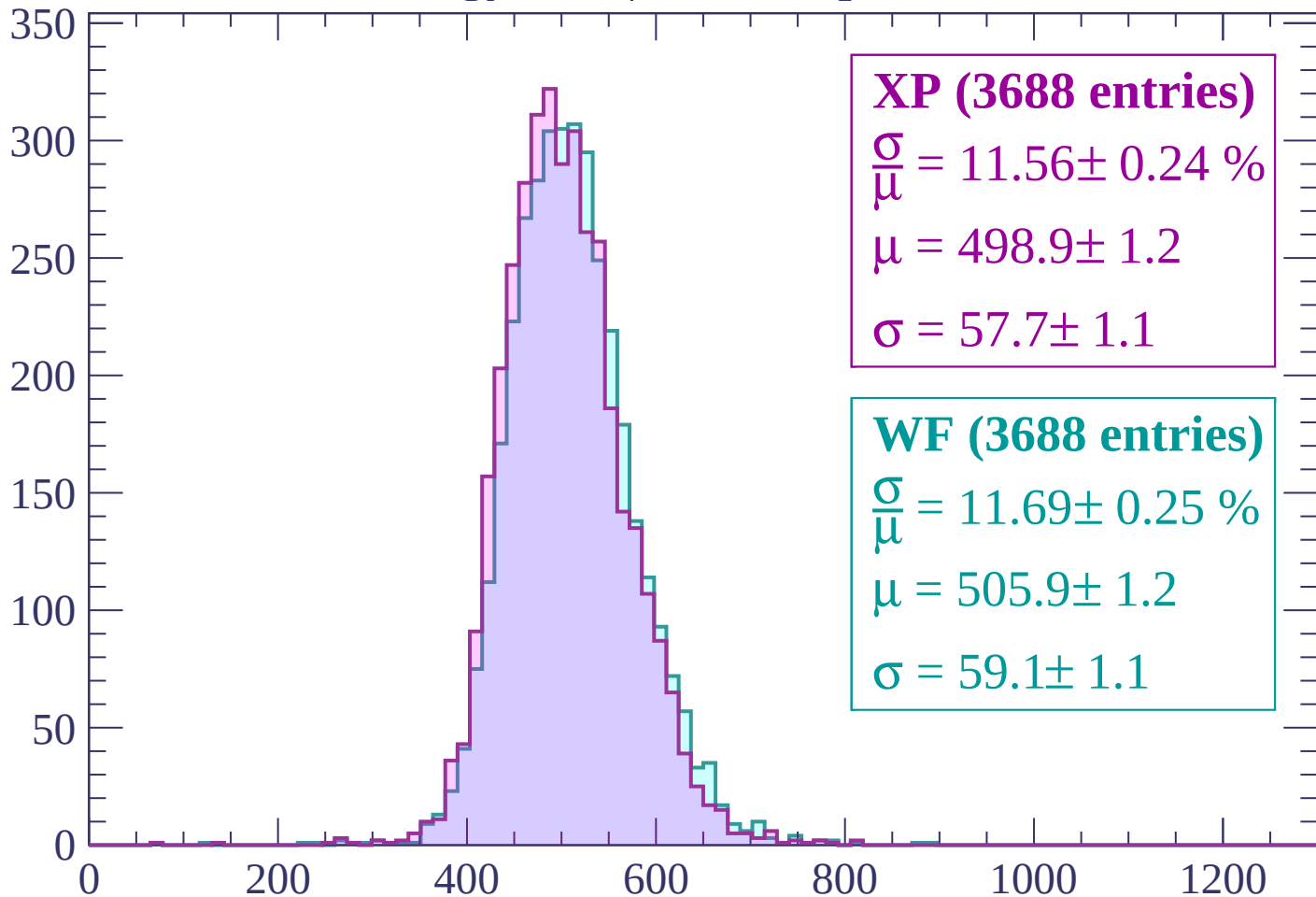
Energy loss | -2160 < p < -2100

Count



Energy loss | -2100 < p < -2040

Count



XP (3688 entries)

$$\frac{\sigma}{\mu} = 11.56 \pm 0.24 \%$$

$$\mu = 498.9 \pm 1.2$$

$$\sigma = 57.7 \pm 1.1$$

WF (3688 entries)

$$\frac{\sigma}{\mu} = 11.69 \pm 0.25 \%$$

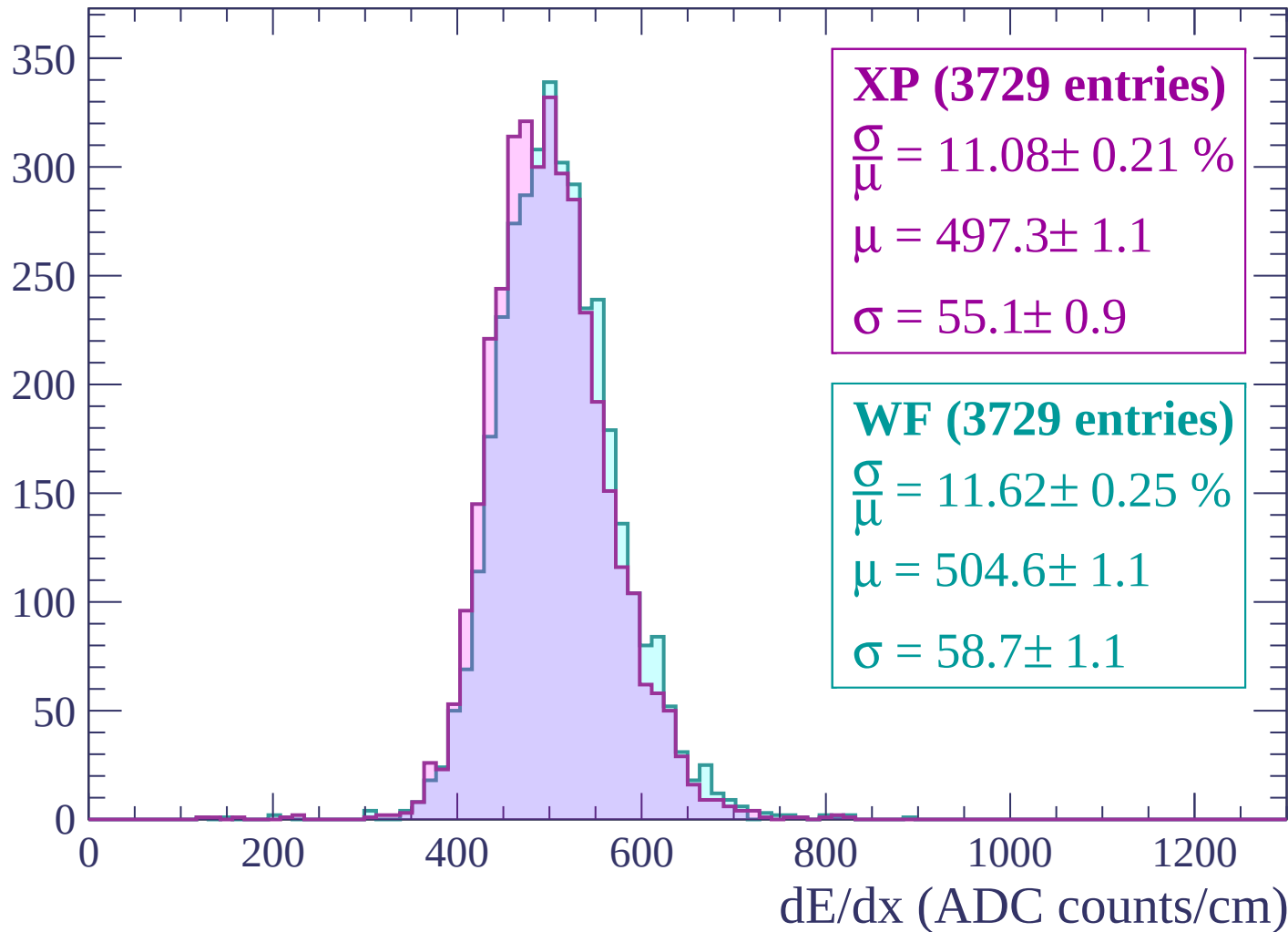
$$\mu = 505.9 \pm 1.2$$

$$\sigma = 59.1 \pm 1.1$$

dE/dx (ADC counts/cm)

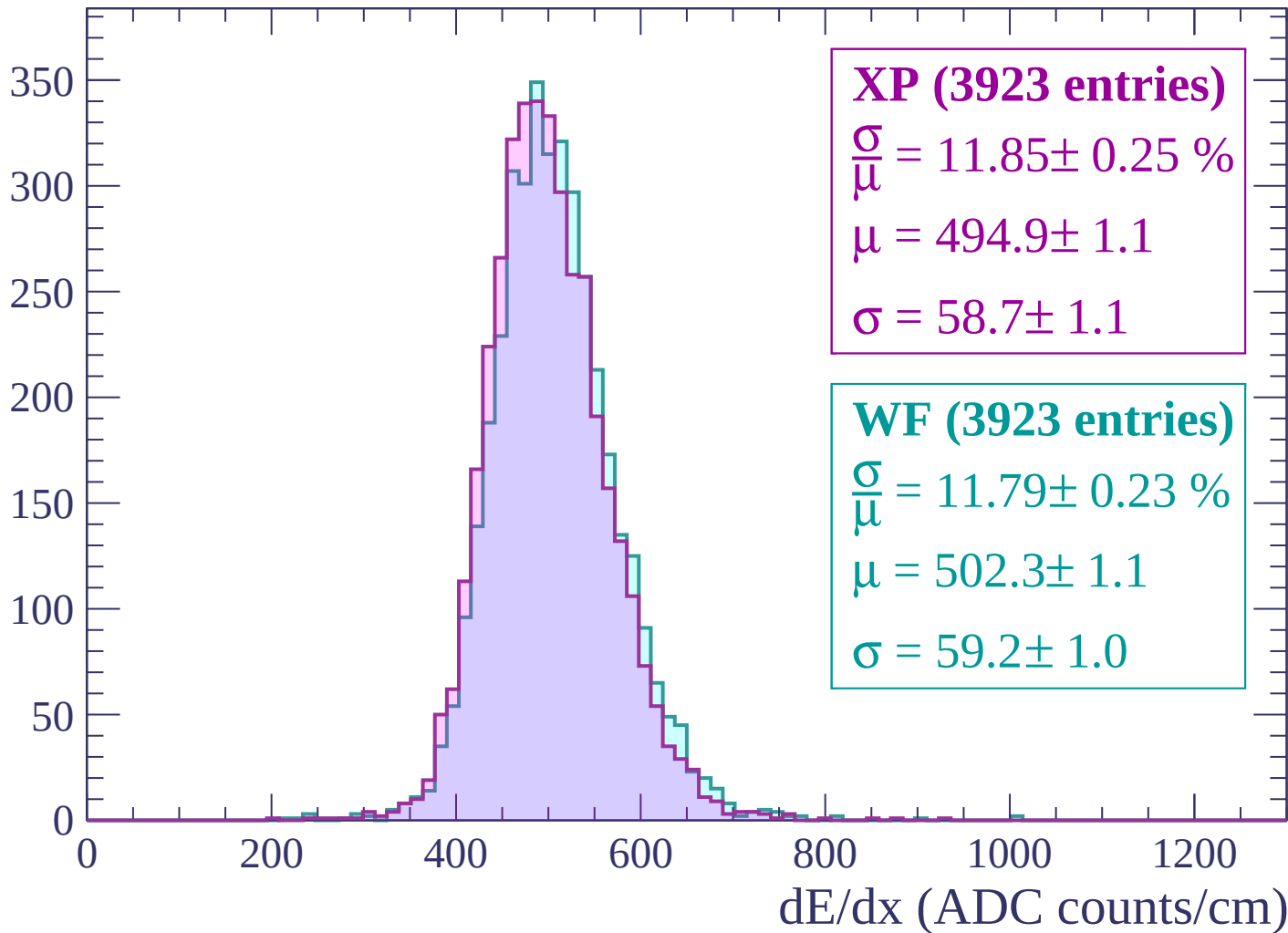
Energy loss | -2040 < p < -1980

Count



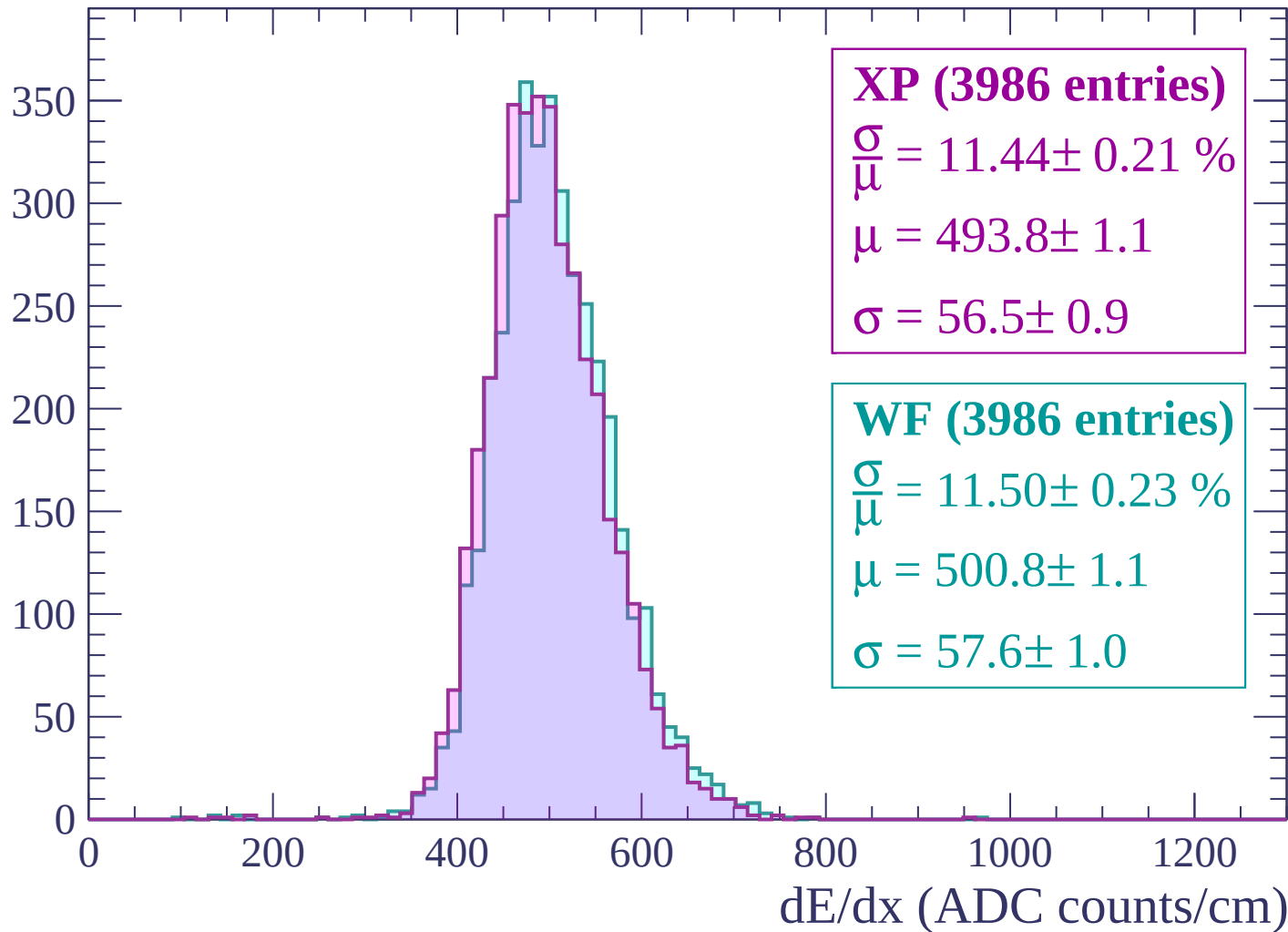
Energy loss | $-1980 < p < -1920$

Count



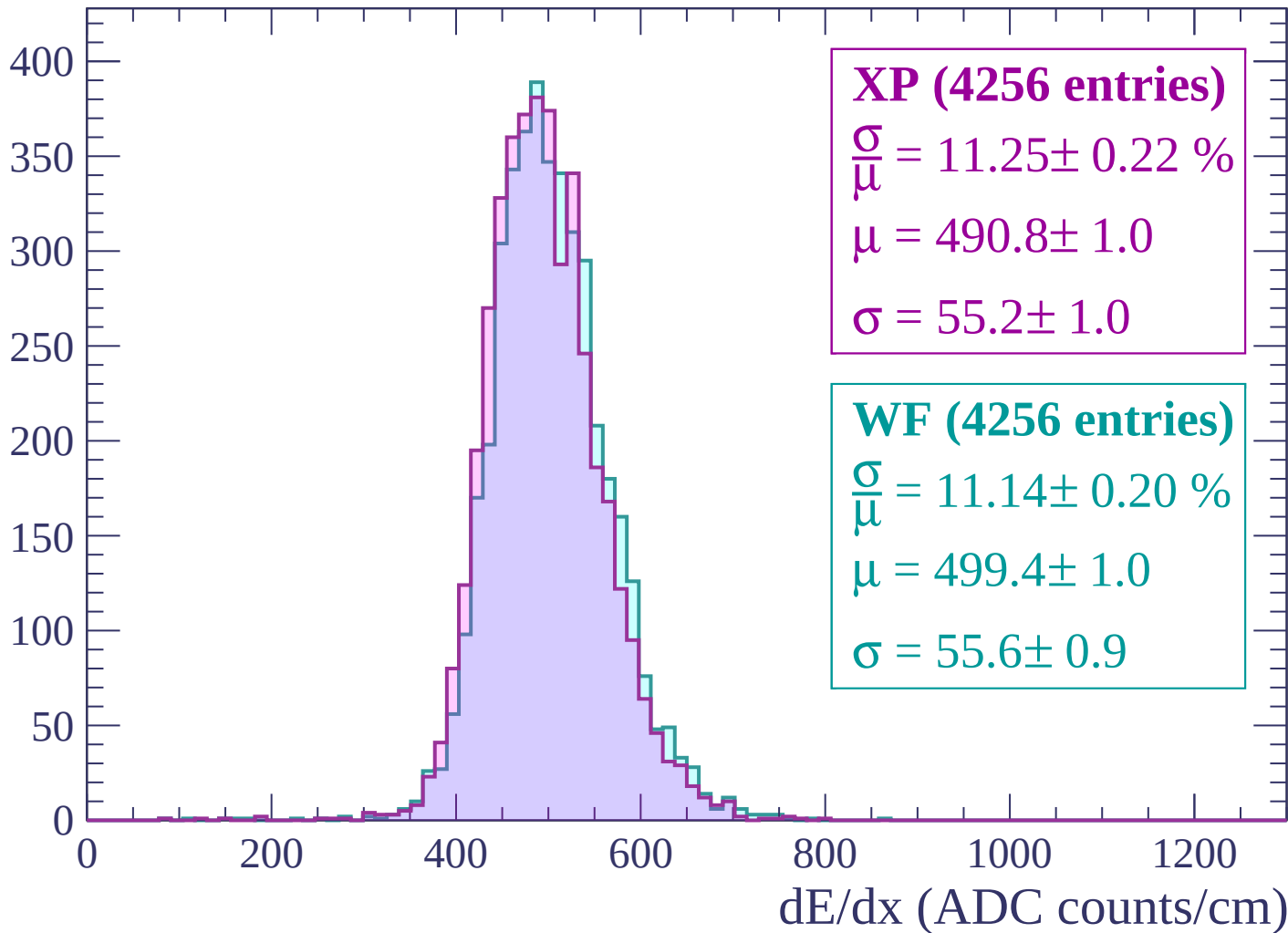
Energy loss | -1920 < p < -1860

Count



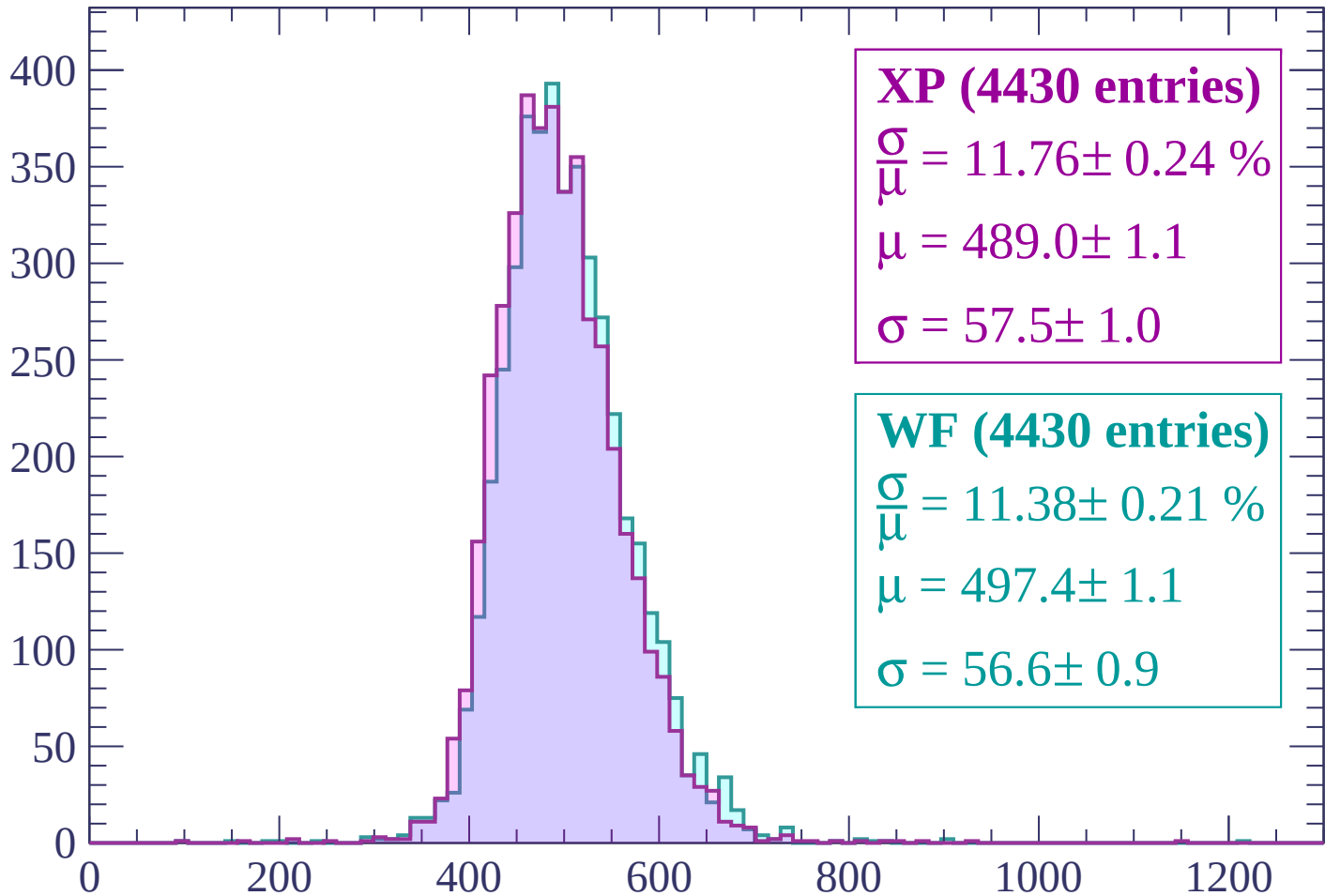
Energy loss | -1860 < p < -1800

Count



Energy loss | $-1800 < p < -1740$

Count



XP (4430 entries)

$\frac{\sigma}{\mu} = 11.76 \pm 0.24 \%$

$\mu = 489.0 \pm 1.1$

$\sigma = 57.5 \pm 1.0$

WF (4430 entries)

$\frac{\sigma}{\mu} = 11.38 \pm 0.21 \%$

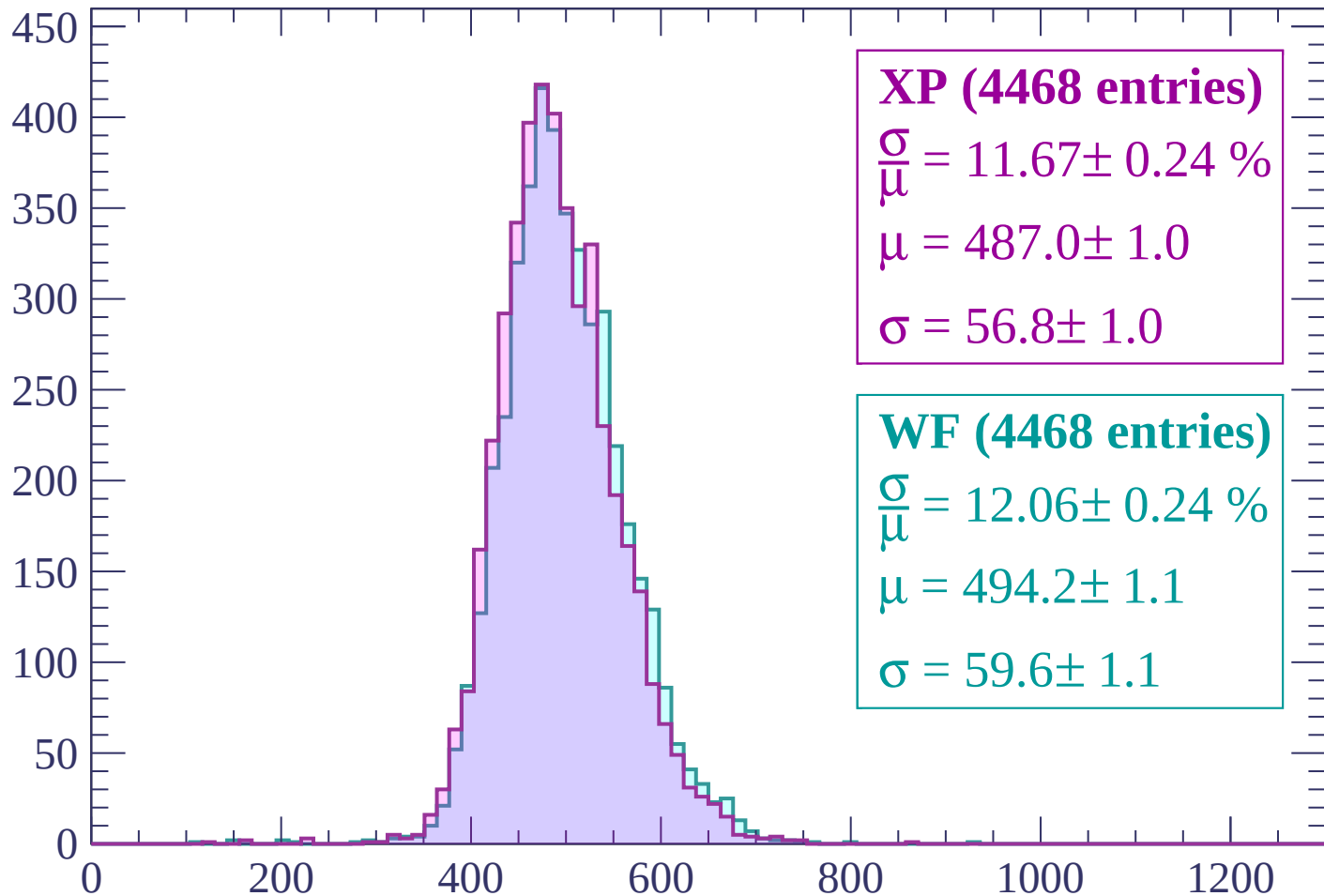
$\mu = 497.4 \pm 1.1$

$\sigma = 56.6 \pm 0.9$

dE/dx (ADC counts/cm)

Energy loss | $-1740 < p < -1680$

Count



XP (4468 entries)

$$\frac{\sigma}{\mu} = 11.67 \pm 0.24 \%$$

$$\mu = 487.0 \pm 1.0$$

$$\sigma = 56.8 \pm 1.0$$

WF (4468 entries)

$$\frac{\sigma}{\mu} = 12.06 \pm 0.24 \%$$

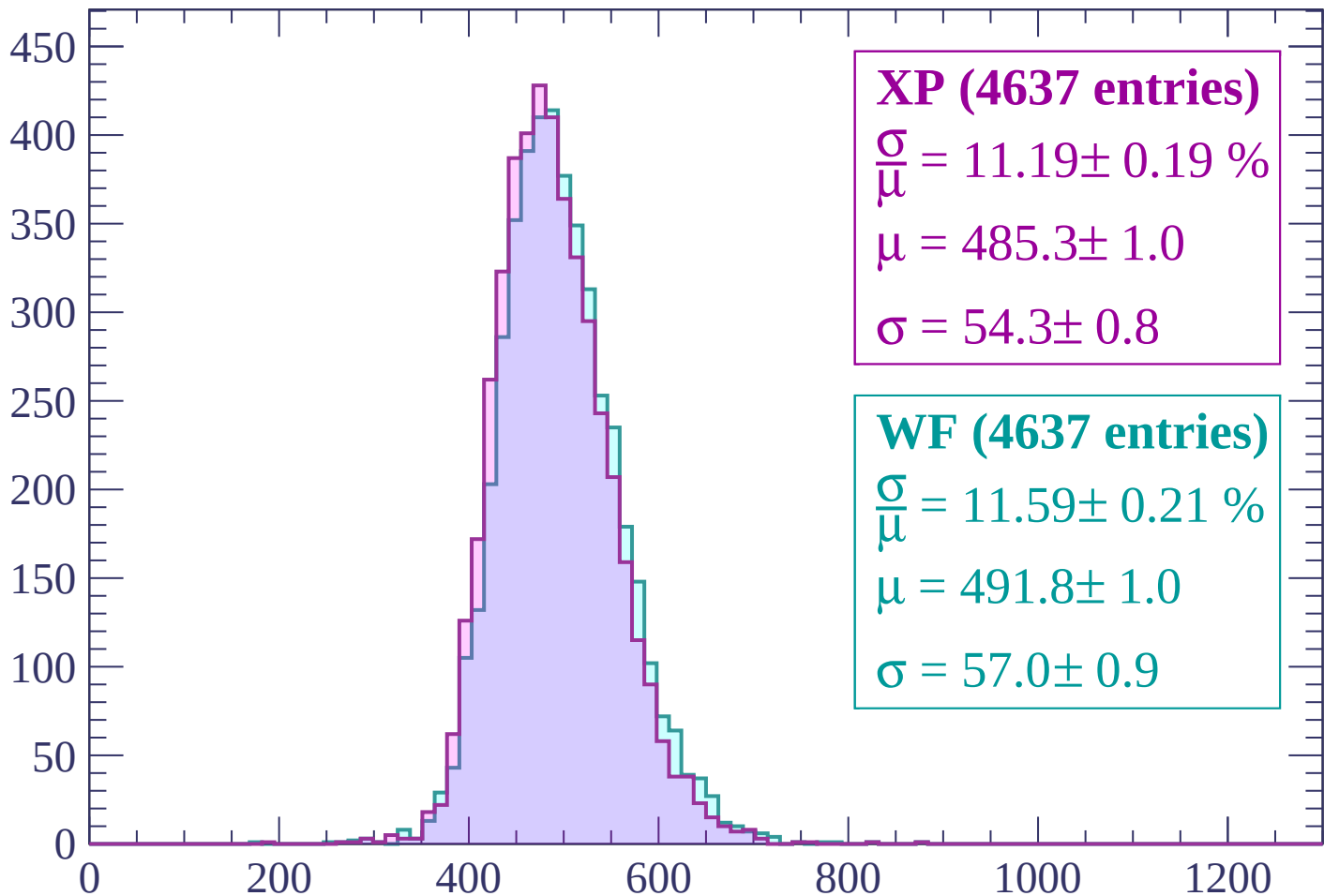
$$\mu = 494.2 \pm 1.1$$

$$\sigma = 59.6 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | -1680 < p < -1620

Count



XP (4637 entries)

$\frac{\sigma}{\mu} = 11.19 \pm 0.19 \%$

$\mu = 485.3 \pm 1.0$

$\sigma = 54.3 \pm 0.8$

WF (4637 entries)

$\frac{\sigma}{\mu} = 11.59 \pm 0.21 \%$

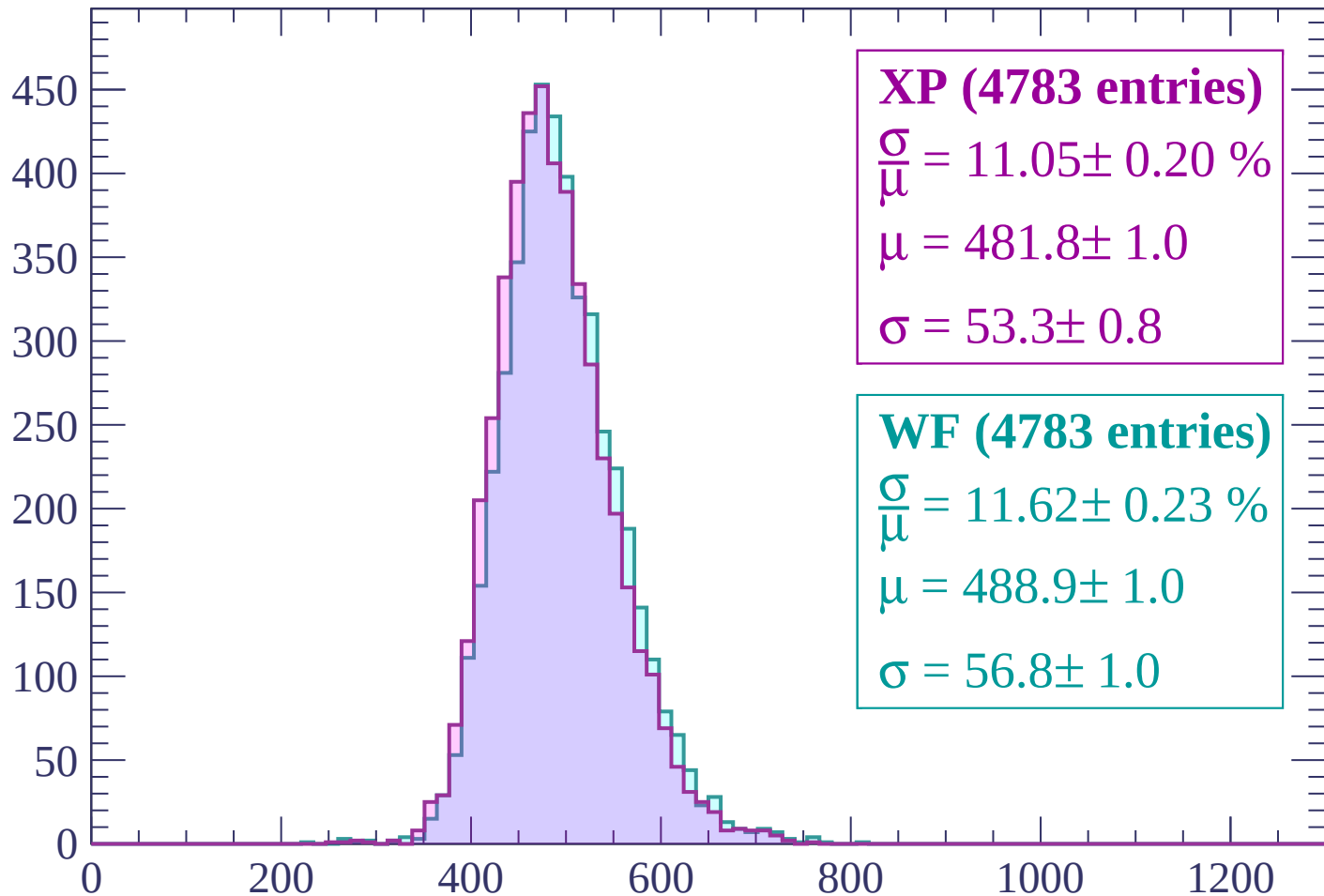
$\mu = 491.8 \pm 1.0$

$\sigma = 57.0 \pm 0.9$

dE/dx (ADC counts/cm)

Energy loss | $-1620 < p < -1560$

Count



XP (4783 entries)

$\frac{\sigma}{\mu} = 11.05 \pm 0.20 \%$

$\mu = 481.8 \pm 1.0$

$\sigma = 53.3 \pm 0.8$

WF (4783 entries)

$\frac{\sigma}{\mu} = 11.62 \pm 0.23 \%$

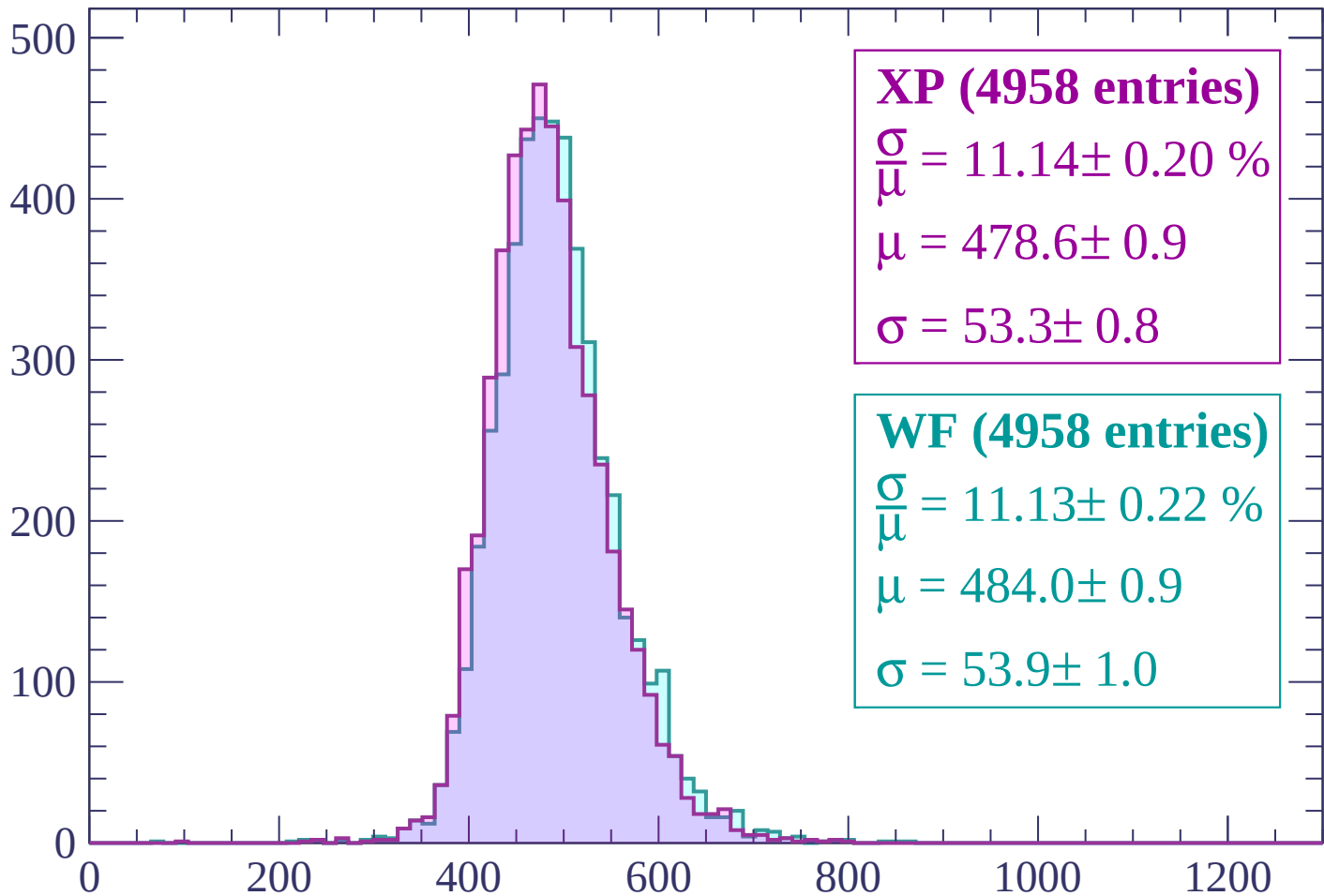
$\mu = 488.9 \pm 1.0$

$\sigma = 56.8 \pm 1.0$

dE/dx (ADC counts/cm)

Energy loss | -1560 < p < -1500

Count



XP (4958 entries)

$$\frac{\sigma}{\mu} = 11.14 \pm 0.20 \%$$

$$\mu = 478.6 \pm 0.9$$

$$\sigma = 53.3 \pm 0.8$$

WF (4958 entries)

$$\frac{\sigma}{\mu} = 11.13 \pm 0.22 \%$$

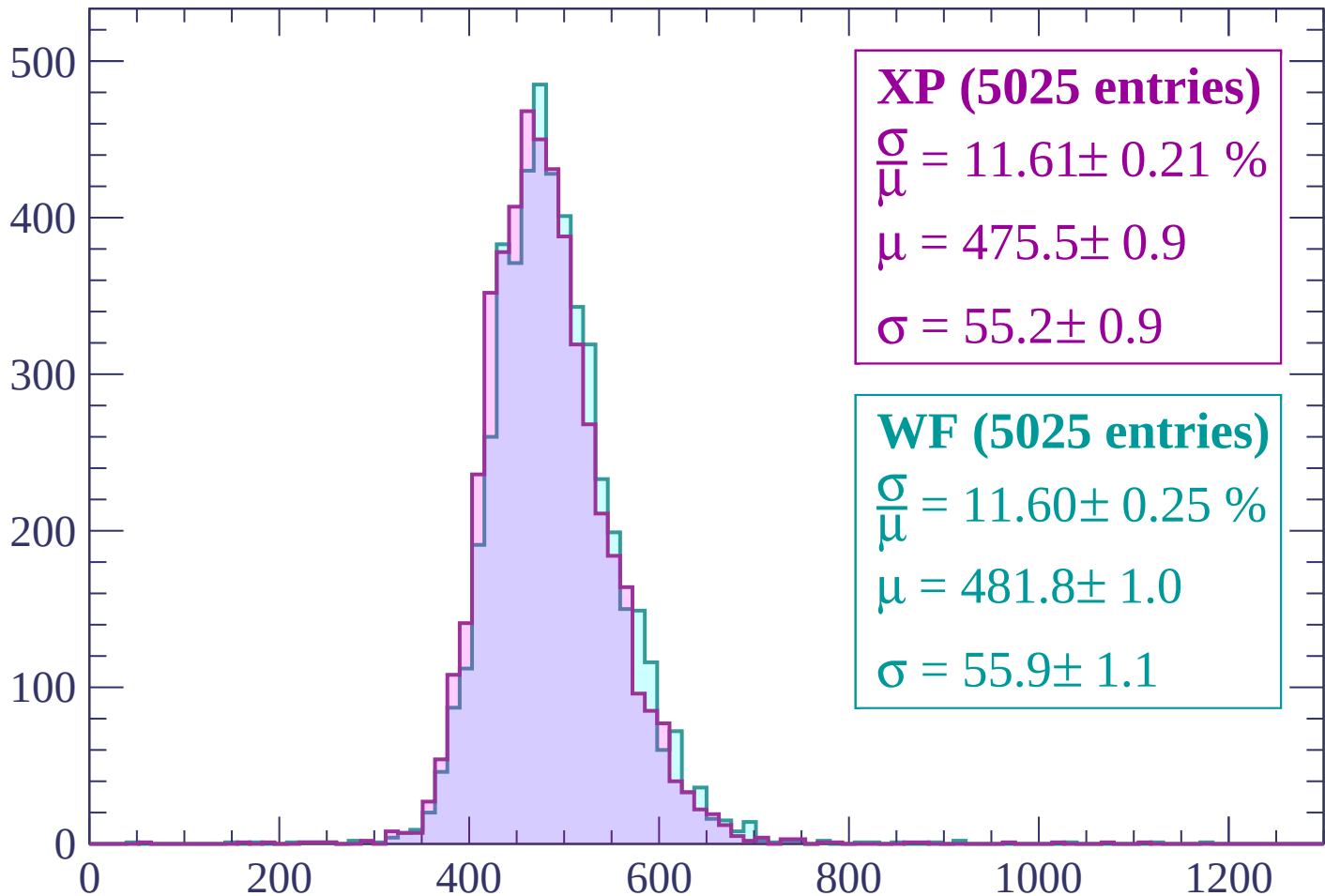
$$\mu = 484.0 \pm 0.9$$

$$\sigma = 53.9 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | -1500 < p < -1440

Count



XP (5025 entries)

$\frac{\sigma}{\mu} = 11.61 \pm 0.21 \%$

$\mu = 475.5 \pm 0.9$

$\sigma = 55.2 \pm 0.9$

WF (5025 entries)

$\frac{\sigma}{\mu} = 11.60 \pm 0.25 \%$

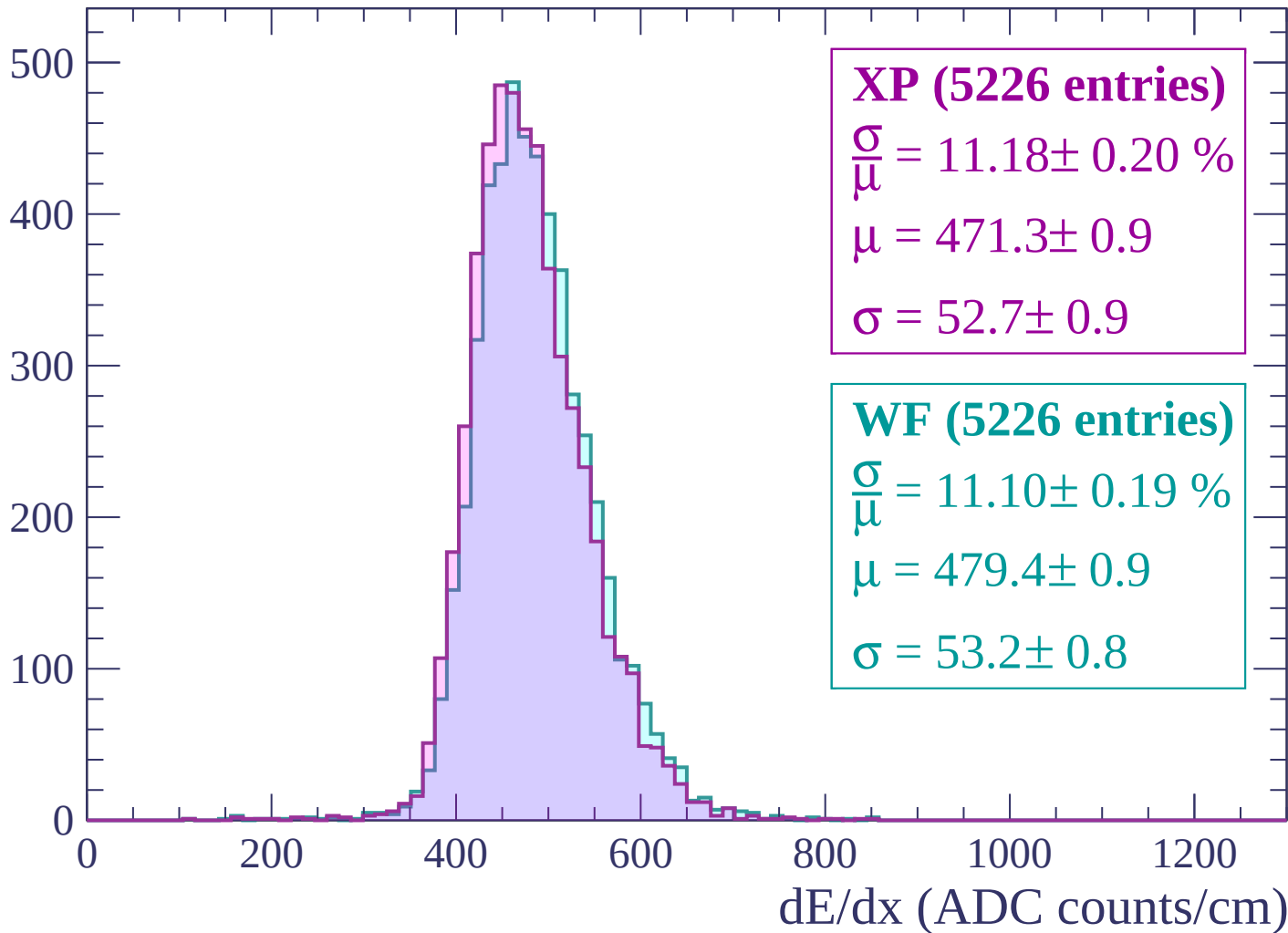
$\mu = 481.8 \pm 1.0$

$\sigma = 55.9 \pm 1.1$

dE/dx (ADC counts/cm)

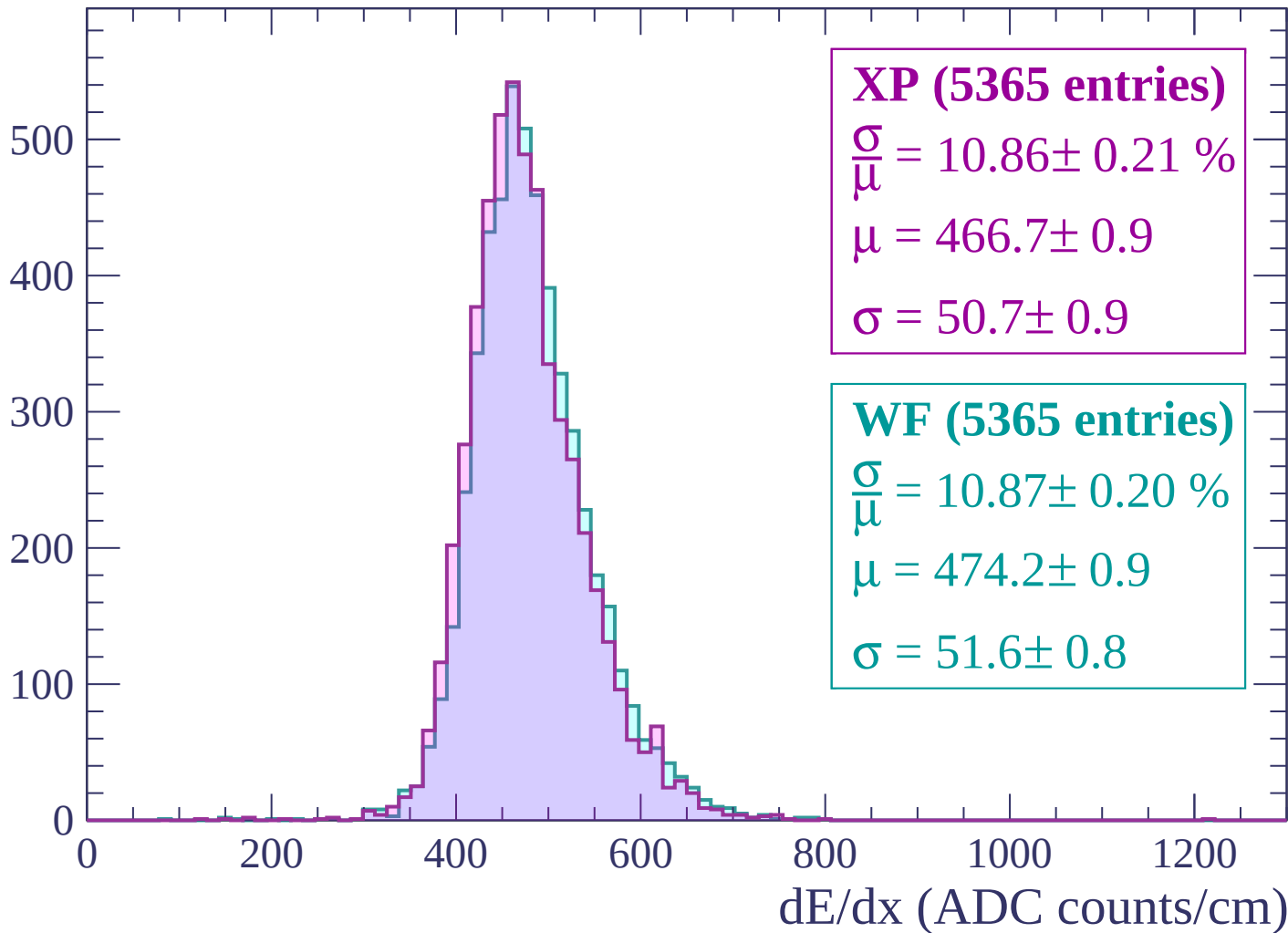
Energy loss | -1440 < p < -1380

Count



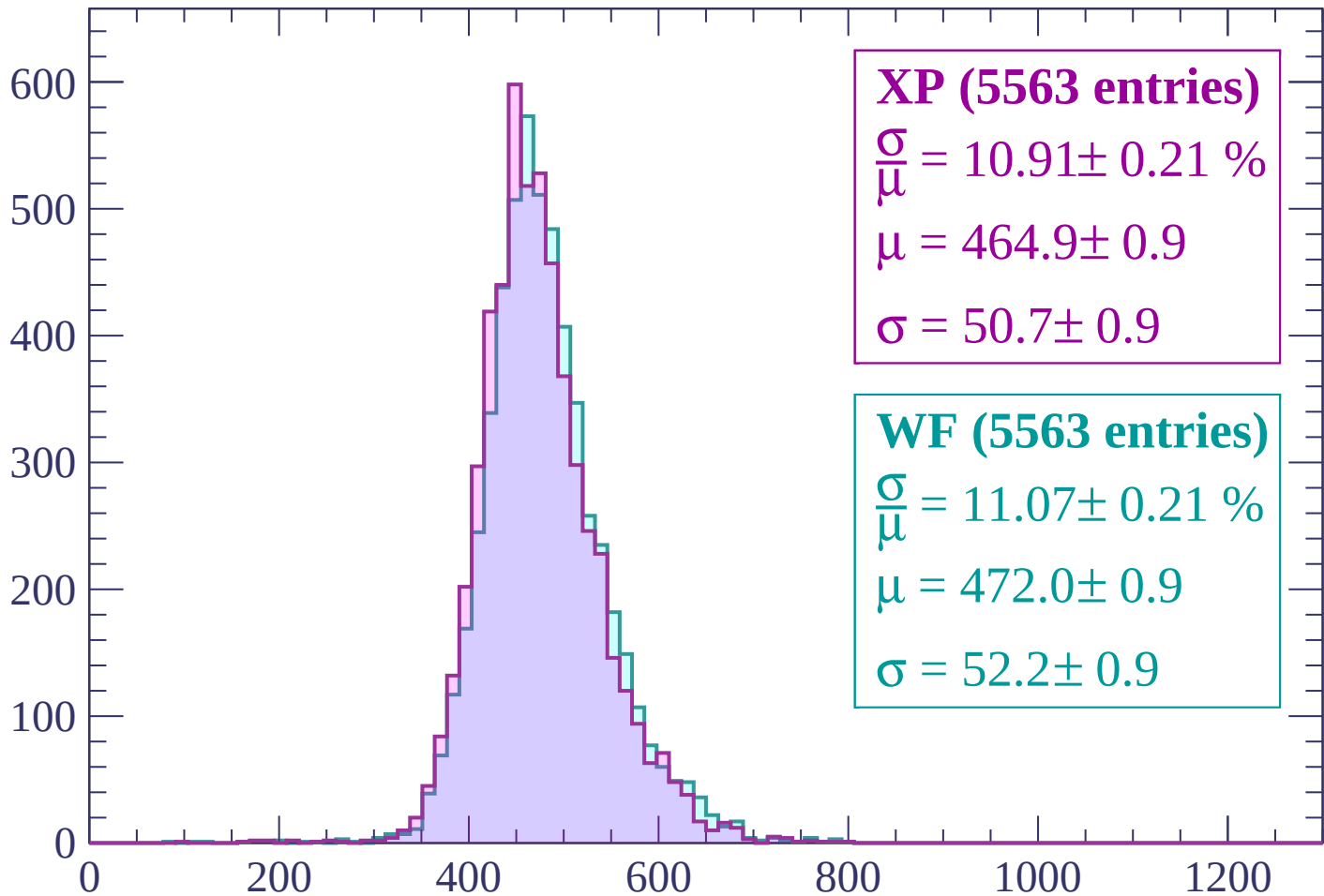
Energy loss | -1380 < p < -1320

Count



Energy loss | $-1320 < p < -1260$

Count



XP (5563 entries)

$\frac{\sigma}{\mu} = 10.91 \pm 0.21 \%$

$\mu = 464.9 \pm 0.9$

$\sigma = 50.7 \pm 0.9$

WF (5563 entries)

$\frac{\sigma}{\mu} = 11.07 \pm 0.21 \%$

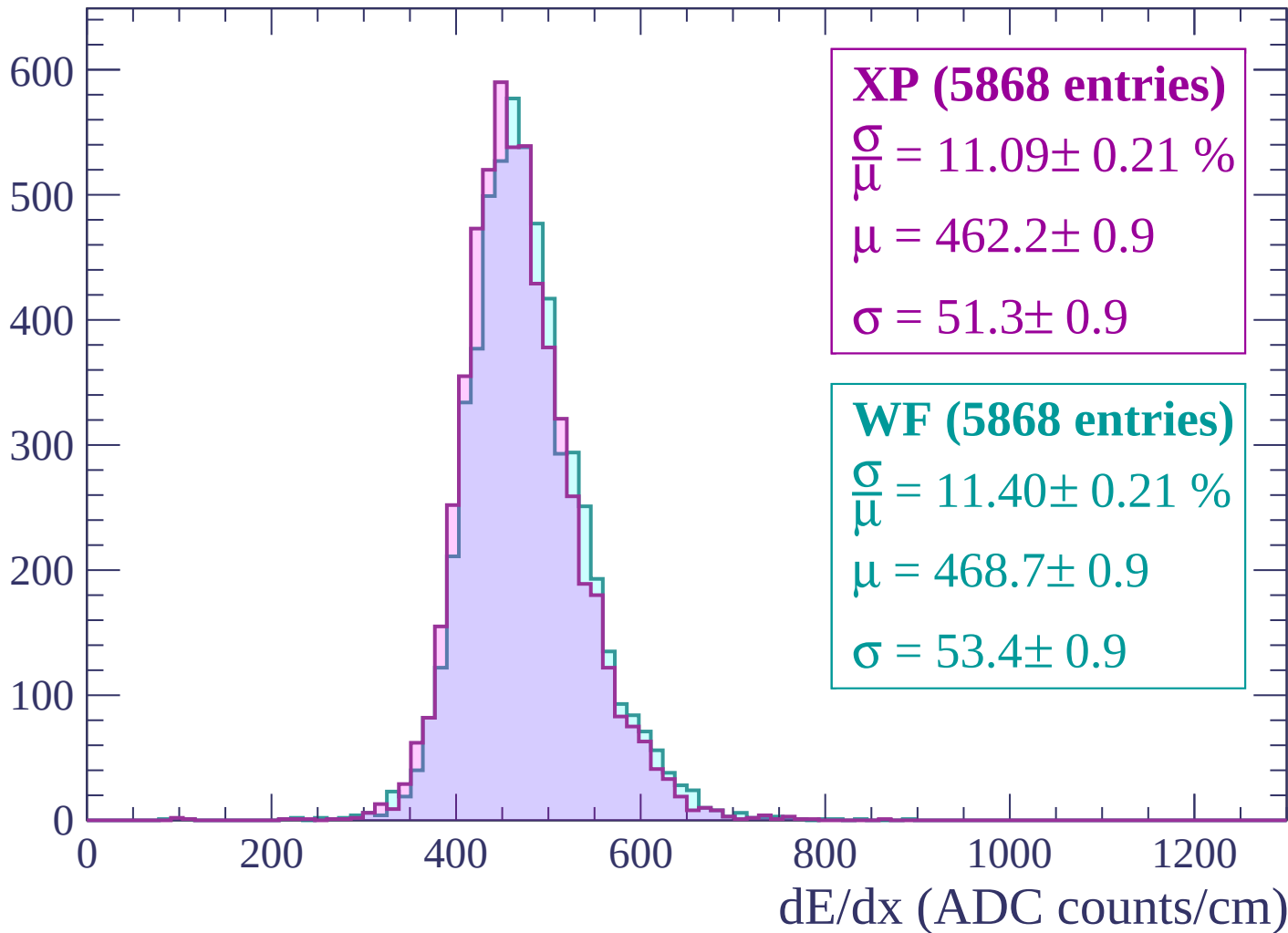
$\mu = 472.0 \pm 0.9$

$\sigma = 52.2 \pm 0.9$

dE/dx (ADC counts/cm)

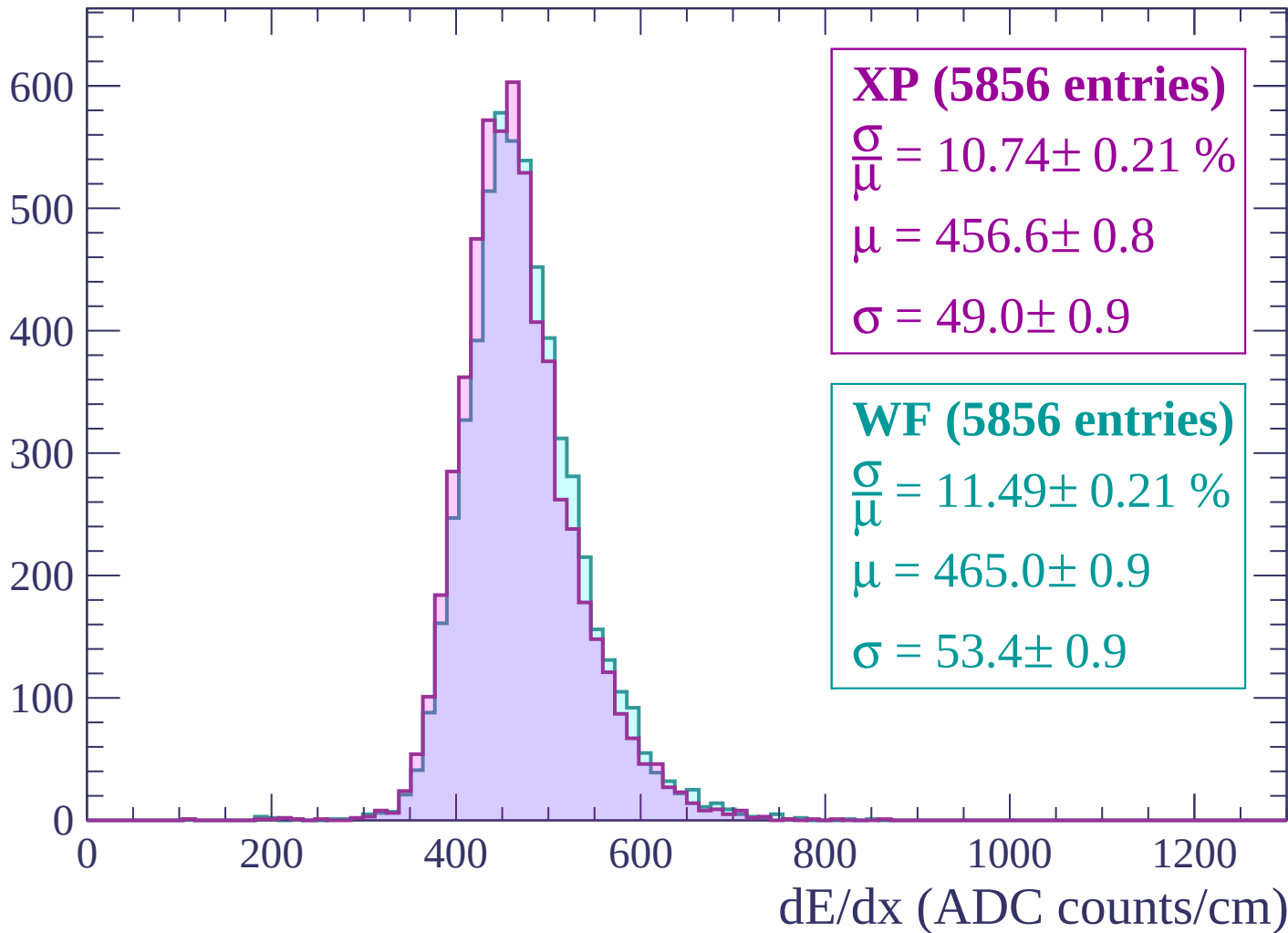
Energy loss | $-1260 < p < -1200$

Count



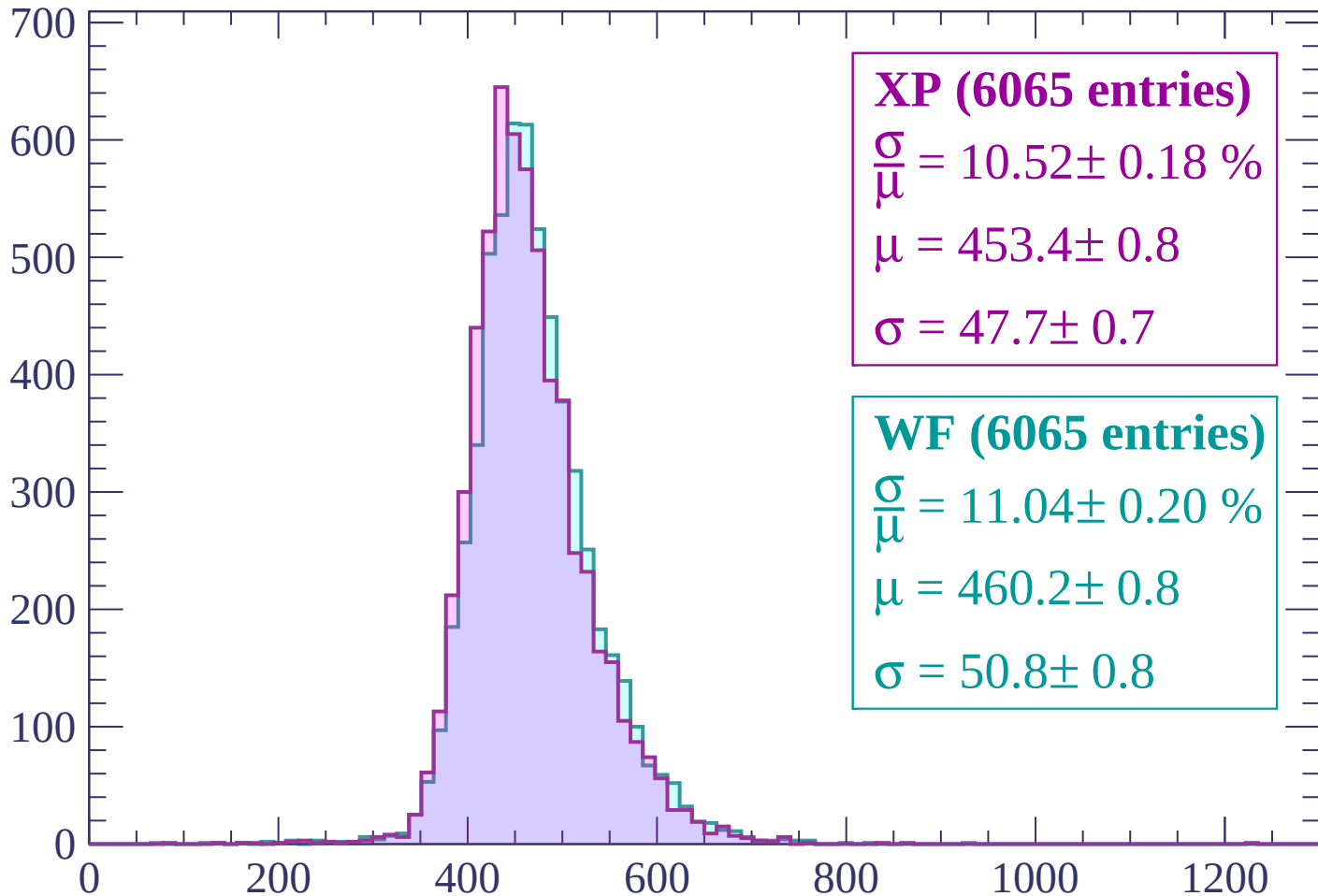
Energy loss | $-1200 < p < -1140$

Count



Energy loss | $-1140 < p < -1080$

Count



XP (6065 entries)

$$\frac{\sigma}{\mu} = 10.52 \pm 0.18 \%$$

$$\mu = 453.4 \pm 0.8$$

$$\sigma = 47.7 \pm 0.7$$

WF (6065 entries)

$$\frac{\sigma}{\mu} = 11.04 \pm 0.20 \%$$

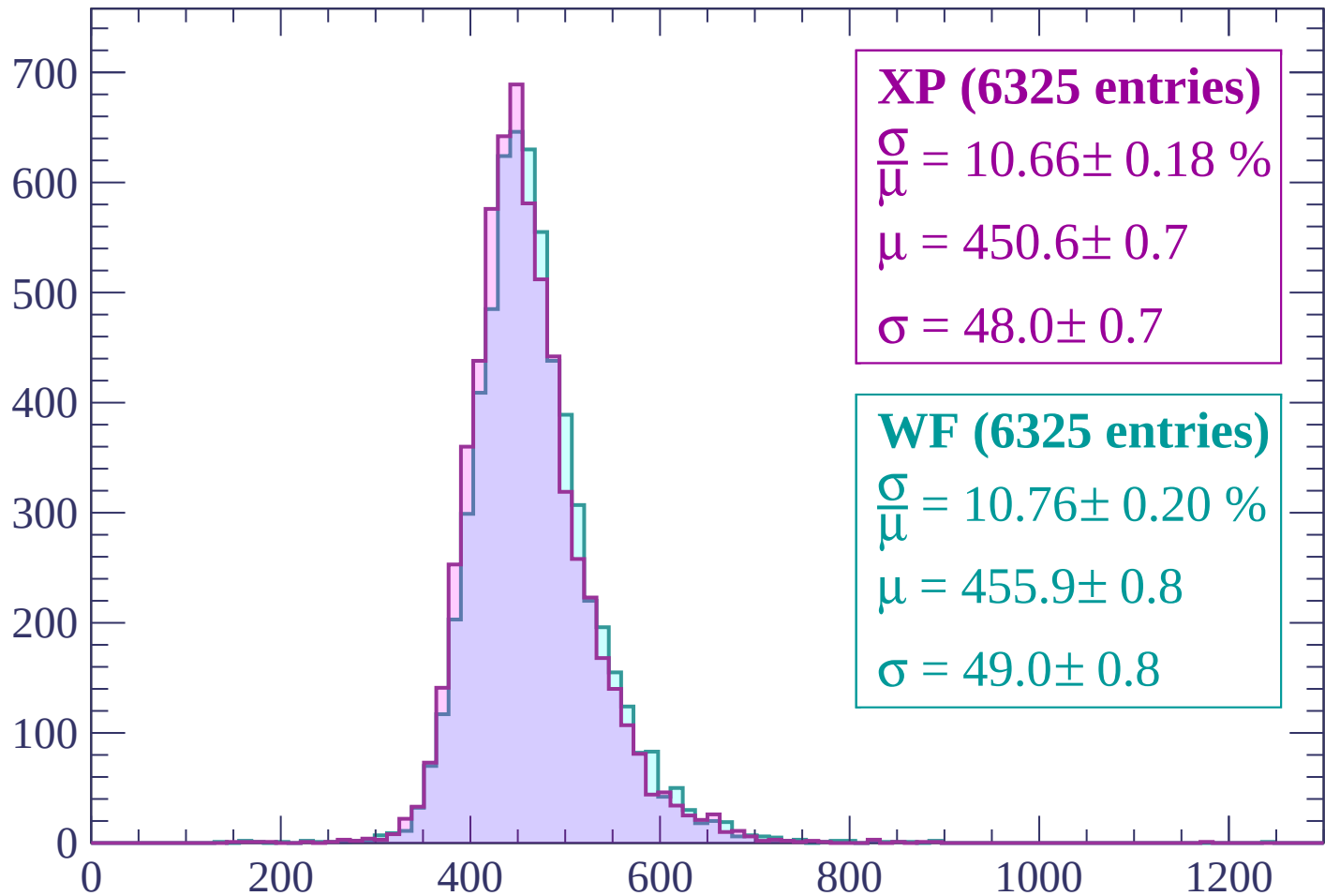
$$\mu = 460.2 \pm 0.8$$

$$\sigma = 50.8 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-1080 < p < -1020$

Count



XP (6325 entries)

$$\frac{\sigma}{\mu} = 10.66 \pm 0.18 \%$$

$$\mu = 450.6 \pm 0.7$$

$$\sigma = 48.0 \pm 0.7$$

WF (6325 entries)

$$\frac{\sigma}{\mu} = 10.76 \pm 0.20 \%$$

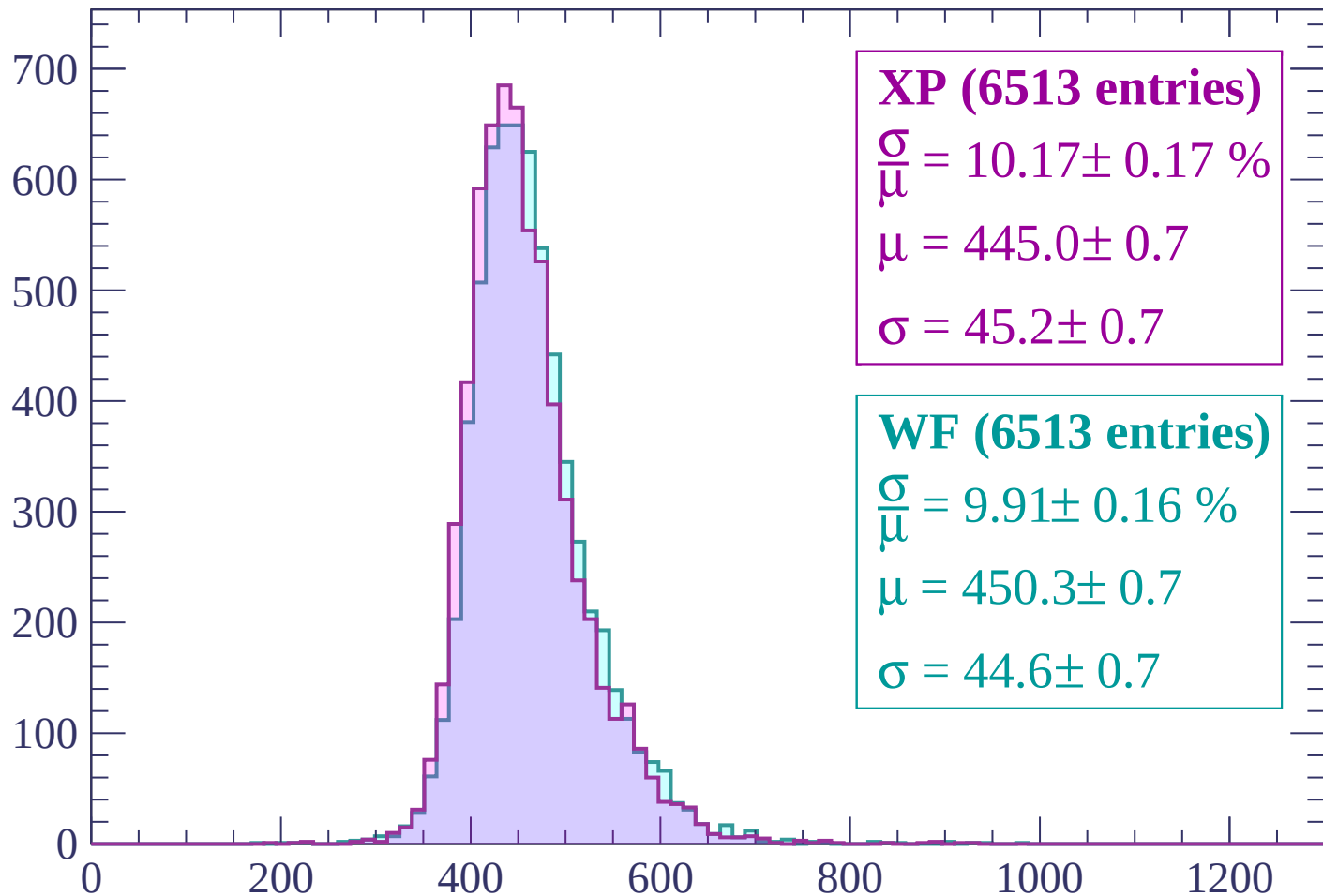
$$\mu = 455.9 \pm 0.8$$

$$\sigma = 49.0 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-1020 < p < -960$

Count



XP (6513 entries)

$$\frac{\sigma}{\mu} = 10.17 \pm 0.17 \%$$

$$\mu = 445.0 \pm 0.7$$

$$\sigma = 45.2 \pm 0.7$$

WF (6513 entries)

$$\frac{\sigma}{\mu} = 9.91 \pm 0.16 \%$$

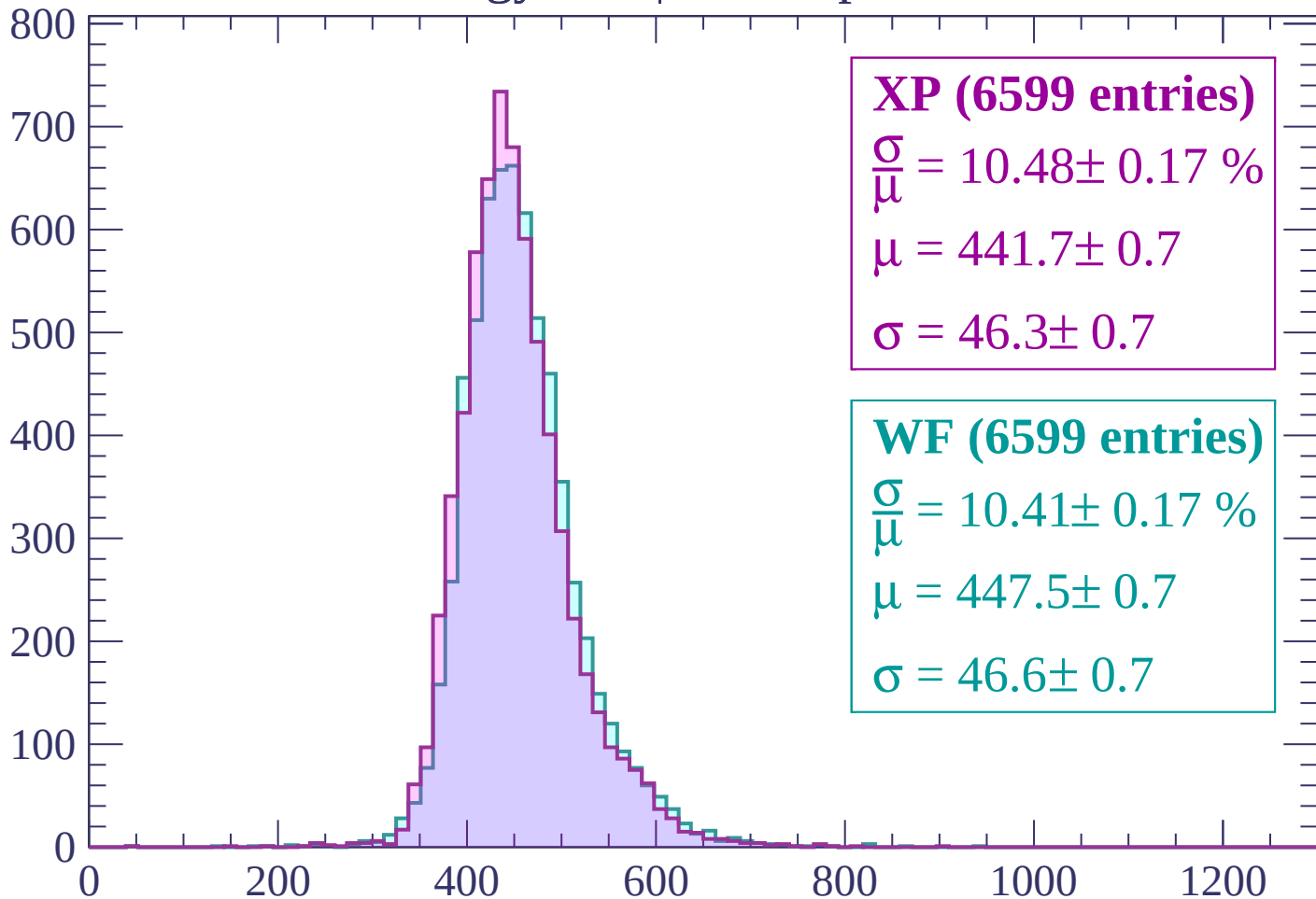
$$\mu = 450.3 \pm 0.7$$

$$\sigma = 44.6 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-960 < p < -900$

Count



XP (6599 entries)

$\frac{\sigma}{\mu} = 10.48 \pm 0.17 \%$

$\mu = 441.7 \pm 0.7$

$\sigma = 46.3 \pm 0.7$

WF (6599 entries)

$\frac{\sigma}{\mu} = 10.41 \pm 0.17 \%$

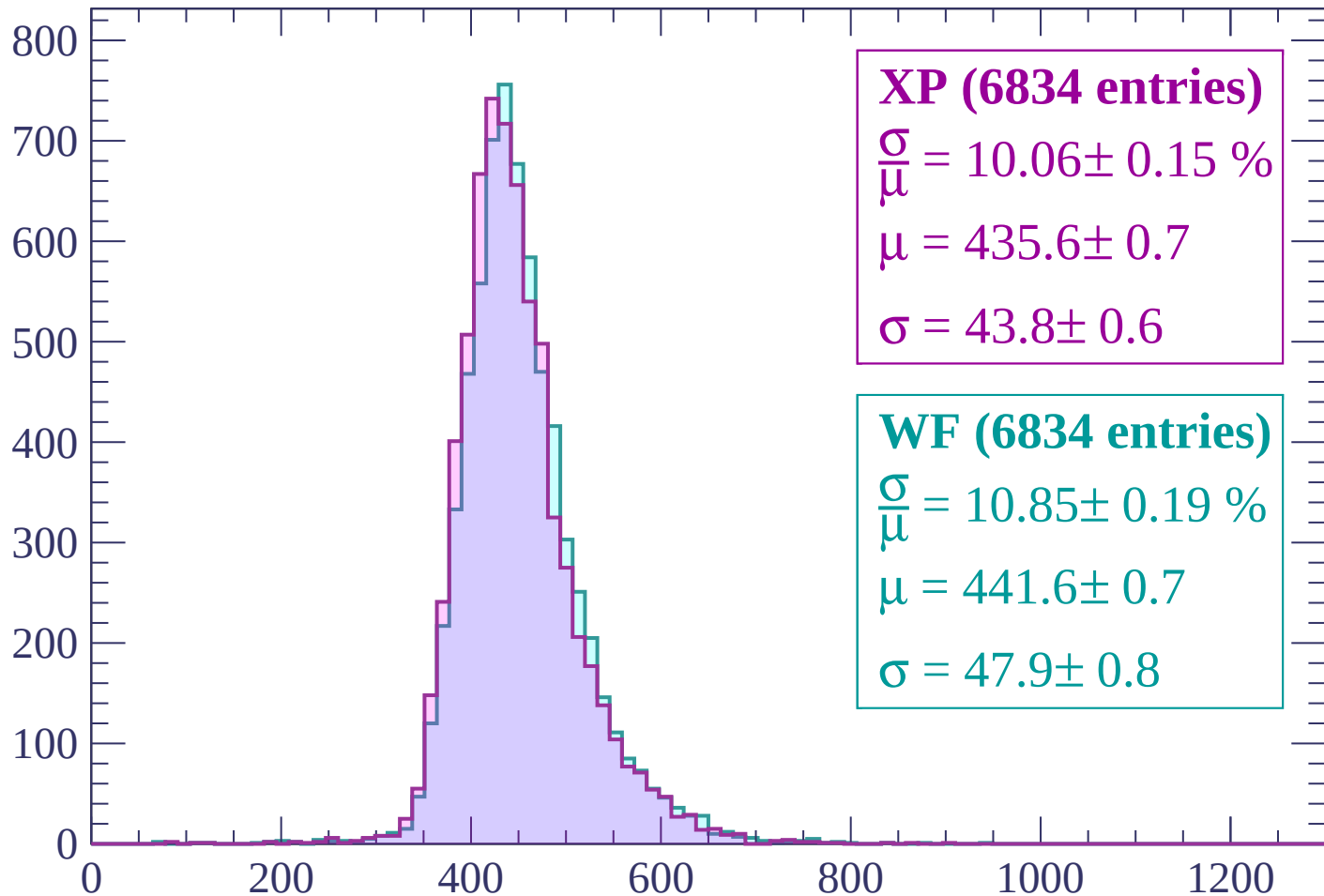
$\mu = 447.5 \pm 0.7$

$\sigma = 46.6 \pm 0.7$

dE/dx (ADC counts/cm)

Energy loss | $-900 < p < -840$

Count



XP (6834 entries)

$$\frac{\sigma}{\mu} = 10.06 \pm 0.15 \%$$

$$\mu = 435.6 \pm 0.7$$

$$\sigma = 43.8 \pm 0.6$$

WF (6834 entries)

$$\frac{\sigma}{\mu} = 10.85 \pm 0.19 \%$$

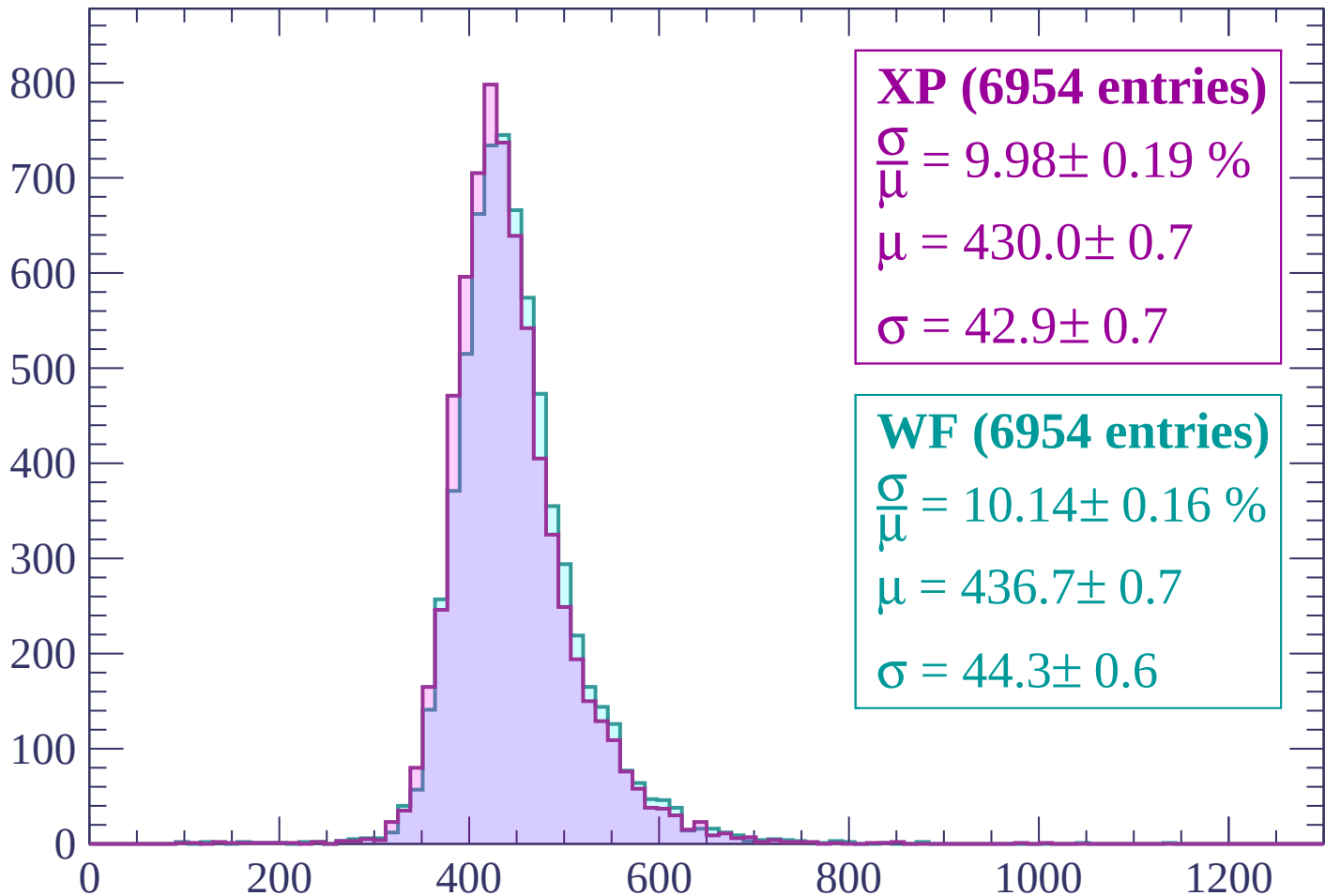
$$\mu = 441.6 \pm 0.7$$

$$\sigma = 47.9 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-840 < p < -780$

Count



XP (6954 entries)

$$\frac{\sigma}{\mu} = 9.98 \pm 0.19 \%$$

$$\mu = 430.0 \pm 0.7$$

$$\sigma = 42.9 \pm 0.7$$

WF (6954 entries)

$$\frac{\sigma}{\mu} = 10.14 \pm 0.16 \%$$

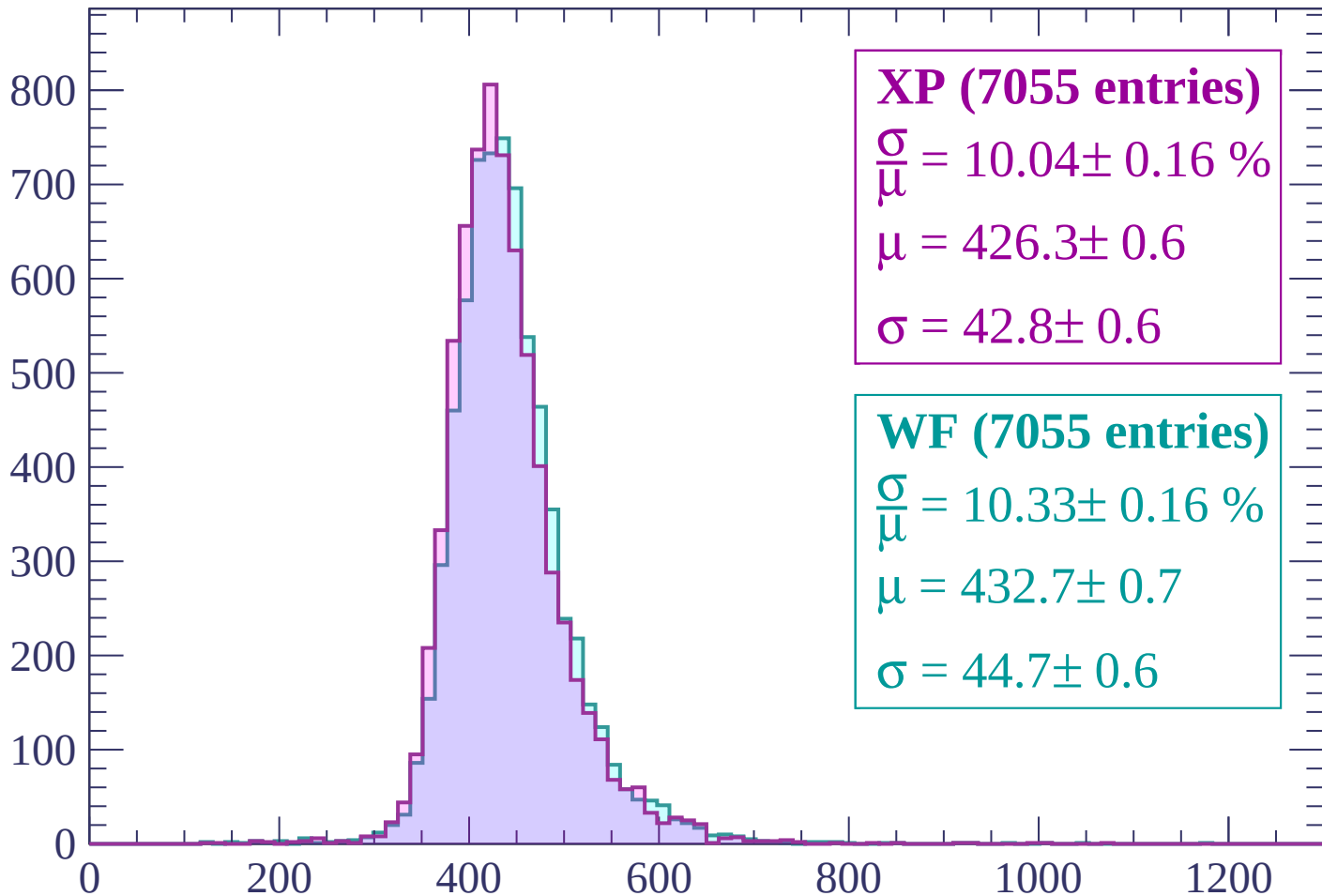
$$\mu = 436.7 \pm 0.7$$

$$\sigma = 44.3 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-780 < p < -720$

Count



XP (7055 entries)

$$\frac{\sigma}{\mu} = 10.04 \pm 0.16 \%$$

$$\mu = 426.3 \pm 0.6$$

$$\sigma = 42.8 \pm 0.6$$

WF (7055 entries)

$$\frac{\sigma}{\mu} = 10.33 \pm 0.16 \%$$

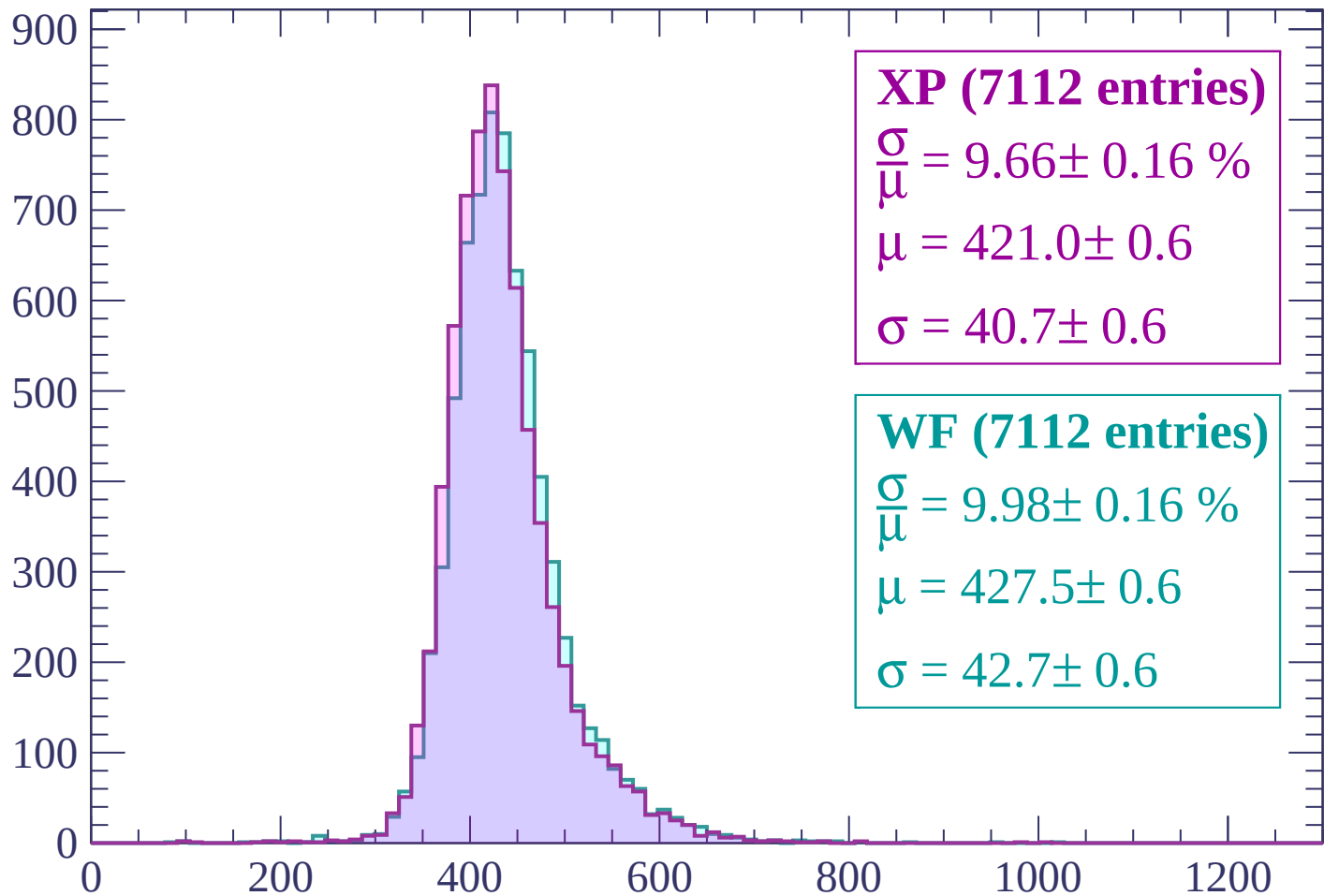
$$\mu = 432.7 \pm 0.7$$

$$\sigma = 44.7 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-720 < p < -660$

Count



XP (7112 entries)

$$\frac{\sigma}{\mu} = 9.66 \pm 0.16 \%$$

$$\mu = 421.0 \pm 0.6$$

$$\sigma = 40.7 \pm 0.6$$

WF (7112 entries)

$$\frac{\sigma}{\mu} = 9.98 \pm 0.16 \%$$

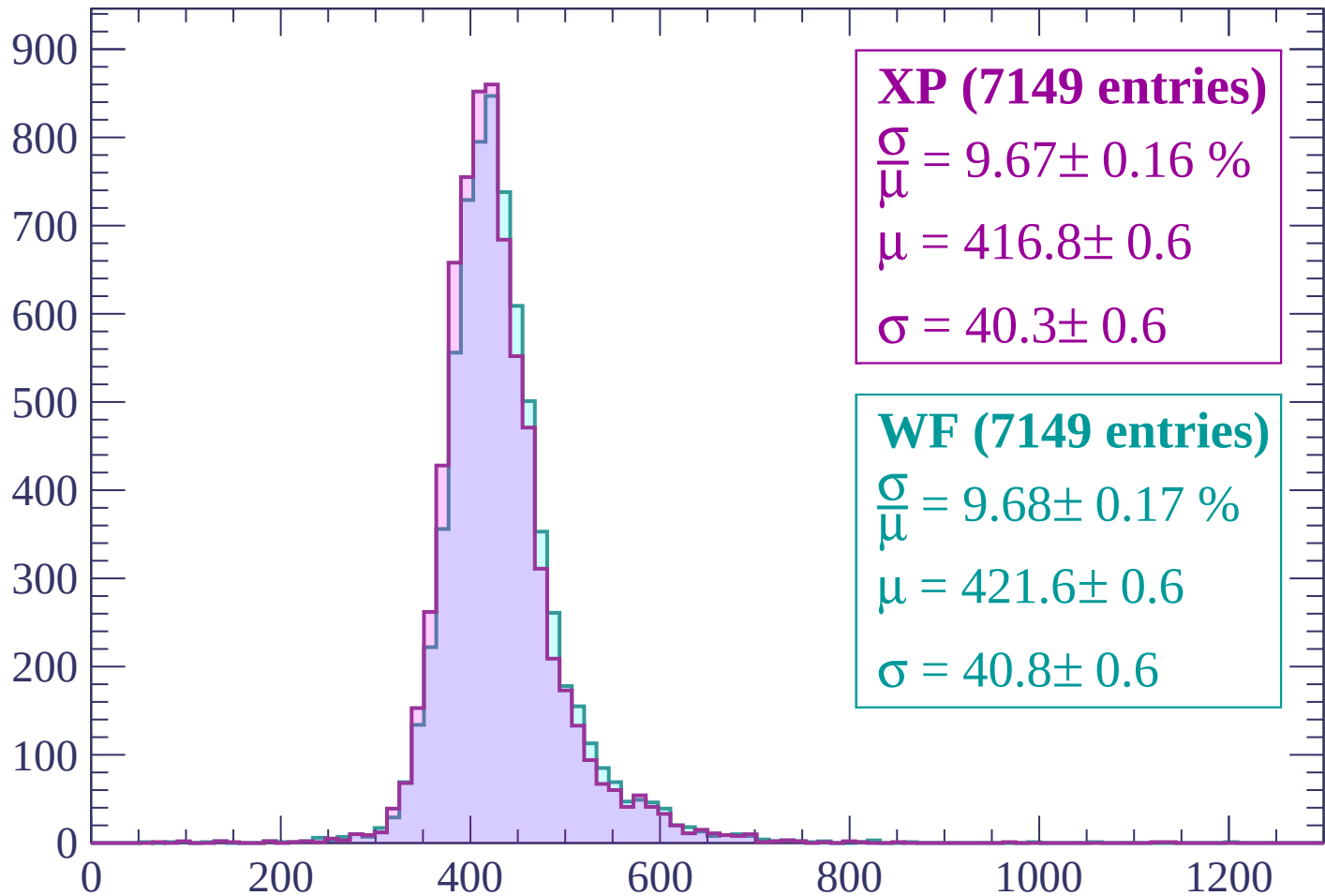
$$\mu = 427.5 \pm 0.6$$

$$\sigma = 42.7 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-660 < p < -600$

Count



XP (7149 entries)

$$\frac{\sigma}{\mu} = 9.67 \pm 0.16 \%$$

$$\mu = 416.8 \pm 0.6$$

$$\sigma = 40.3 \pm 0.6$$

WF (7149 entries)

$$\frac{\sigma}{\mu} = 9.68 \pm 0.17 \%$$

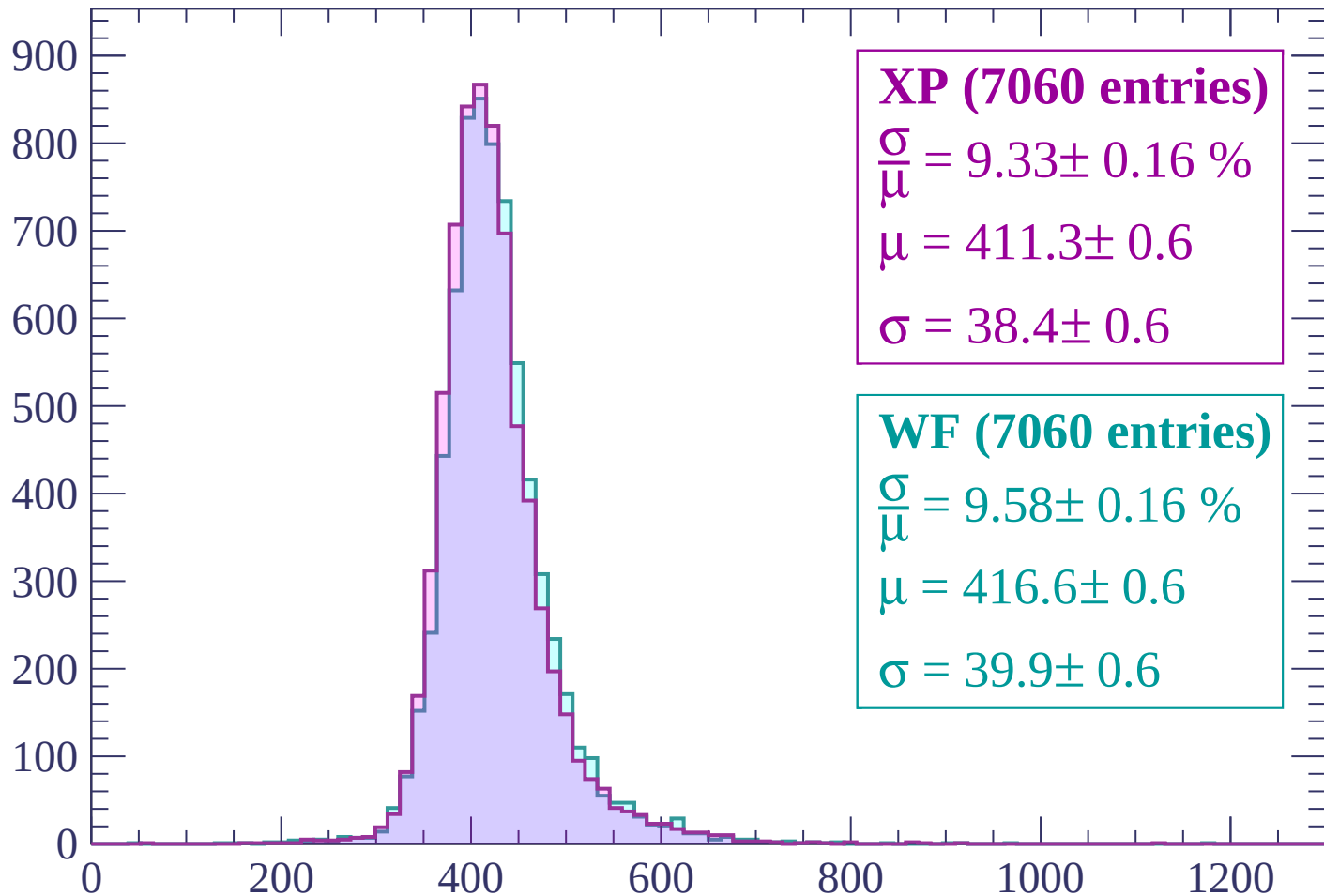
$$\mu = 421.6 \pm 0.6$$

$$\sigma = 40.8 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-600 < p < -540$

Count



XP (7060 entries)

$$\frac{\sigma}{\mu} = 9.33 \pm 0.16 \%$$

$$\mu = 411.3 \pm 0.6$$

$$\sigma = 38.4 \pm 0.6$$

WF (7060 entries)

$$\frac{\sigma}{\mu} = 9.58 \pm 0.16 \%$$

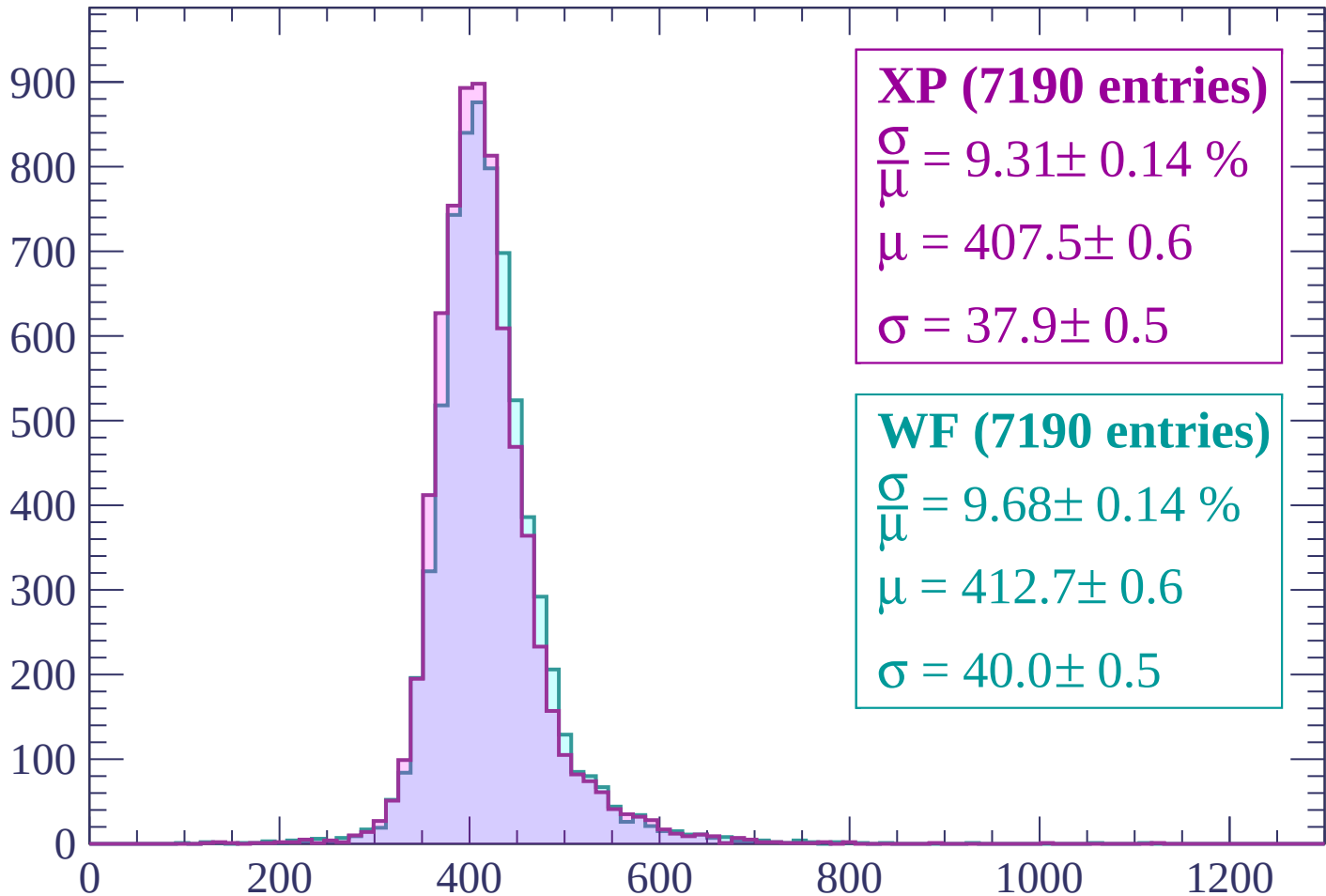
$$\mu = 416.6 \pm 0.6$$

$$\sigma = 39.9 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-540 < p < -480$

Count



XP (7190 entries)

$$\frac{\sigma}{\mu} = 9.31 \pm 0.14 \%$$

$$\mu = 407.5 \pm 0.6$$

$$\sigma = 37.9 \pm 0.5$$

WF (7190 entries)

$$\frac{\sigma}{\mu} = 9.68 \pm 0.14 \%$$

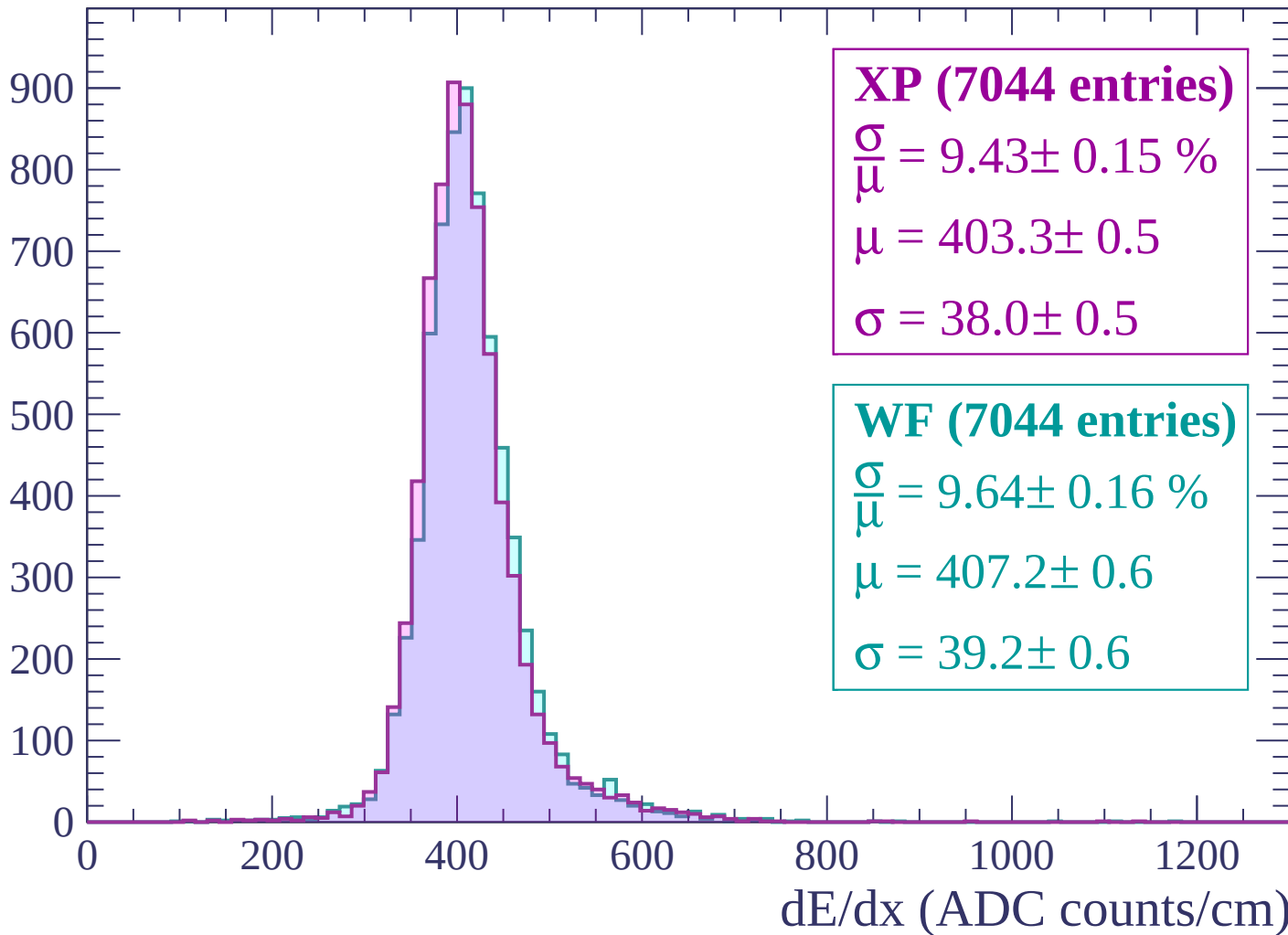
$$\mu = 412.7 \pm 0.6$$

$$\sigma = 40.0 \pm 0.5$$

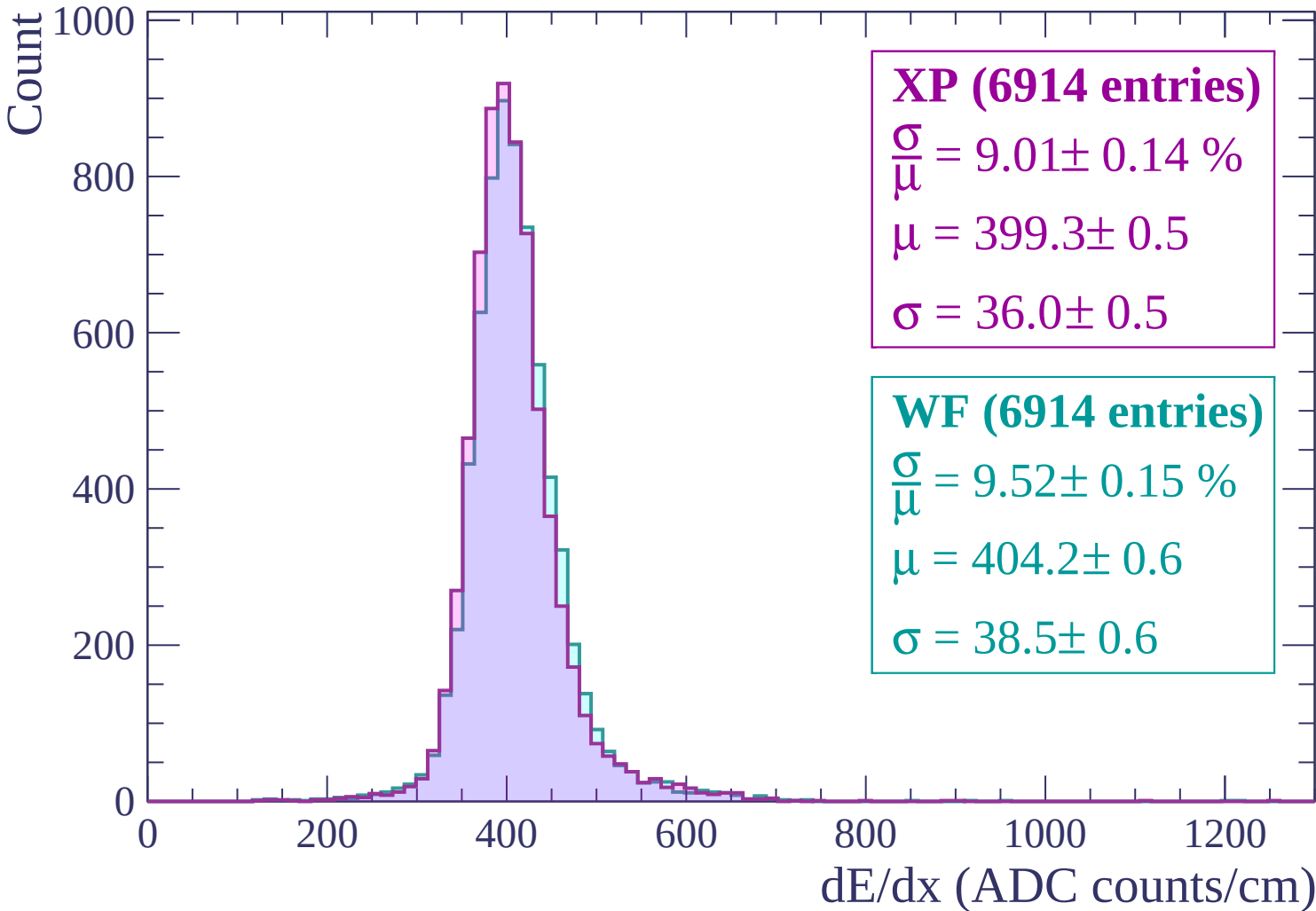
dE/dx (ADC counts/cm)

Energy loss | $-480 < p < -420$

Count

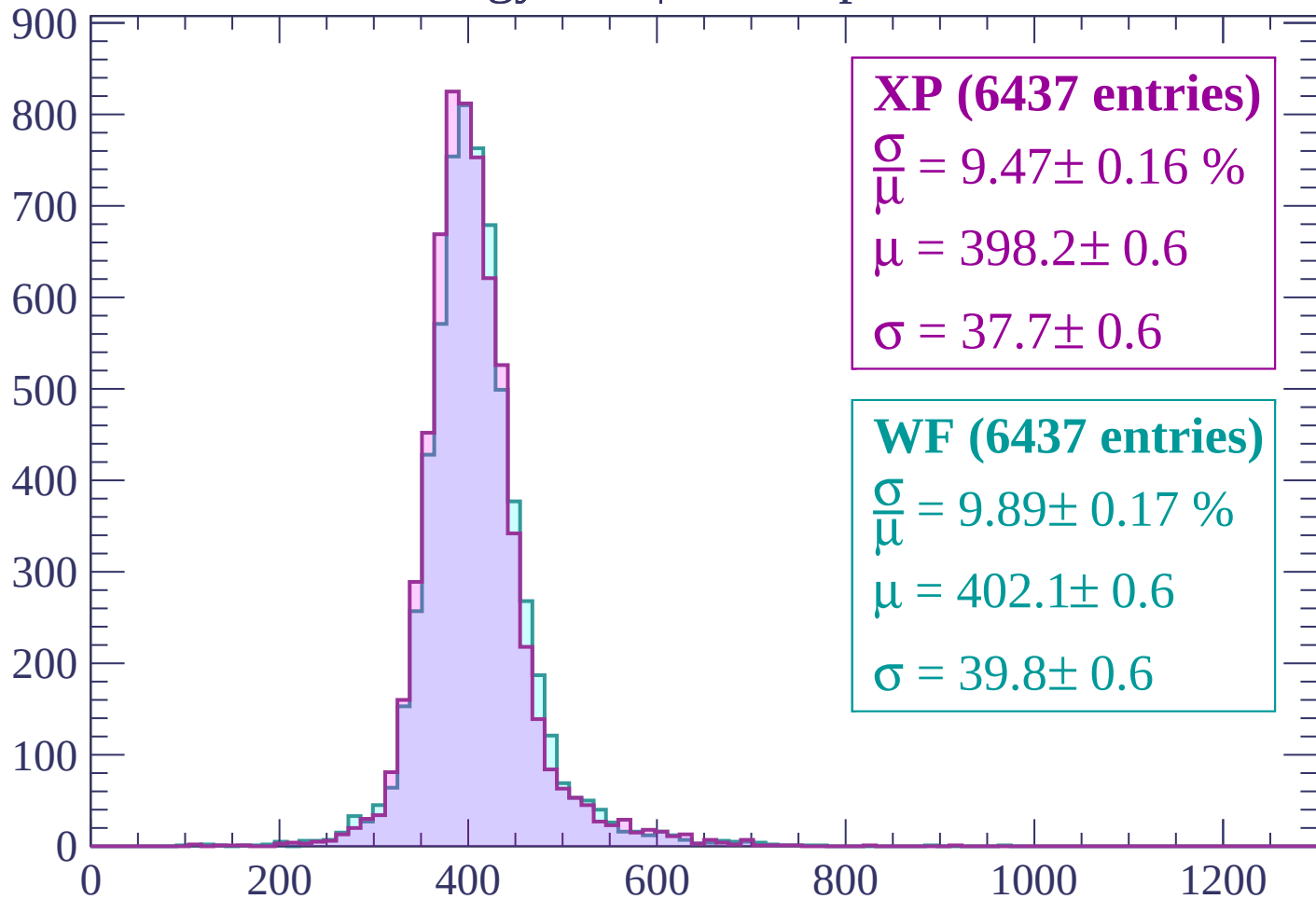


Energy loss | $-420 < p < -360$



Energy loss | $-360 < p < -300$

Count



XP (6437 entries)

$$\frac{\sigma}{\mu} = 9.47 \pm 0.16 \%$$

$$\mu = 398.2 \pm 0.6$$

$$\sigma = 37.7 \pm 0.6$$

WF (6437 entries)

$$\frac{\sigma}{\mu} = 9.89 \pm 0.17 \%$$

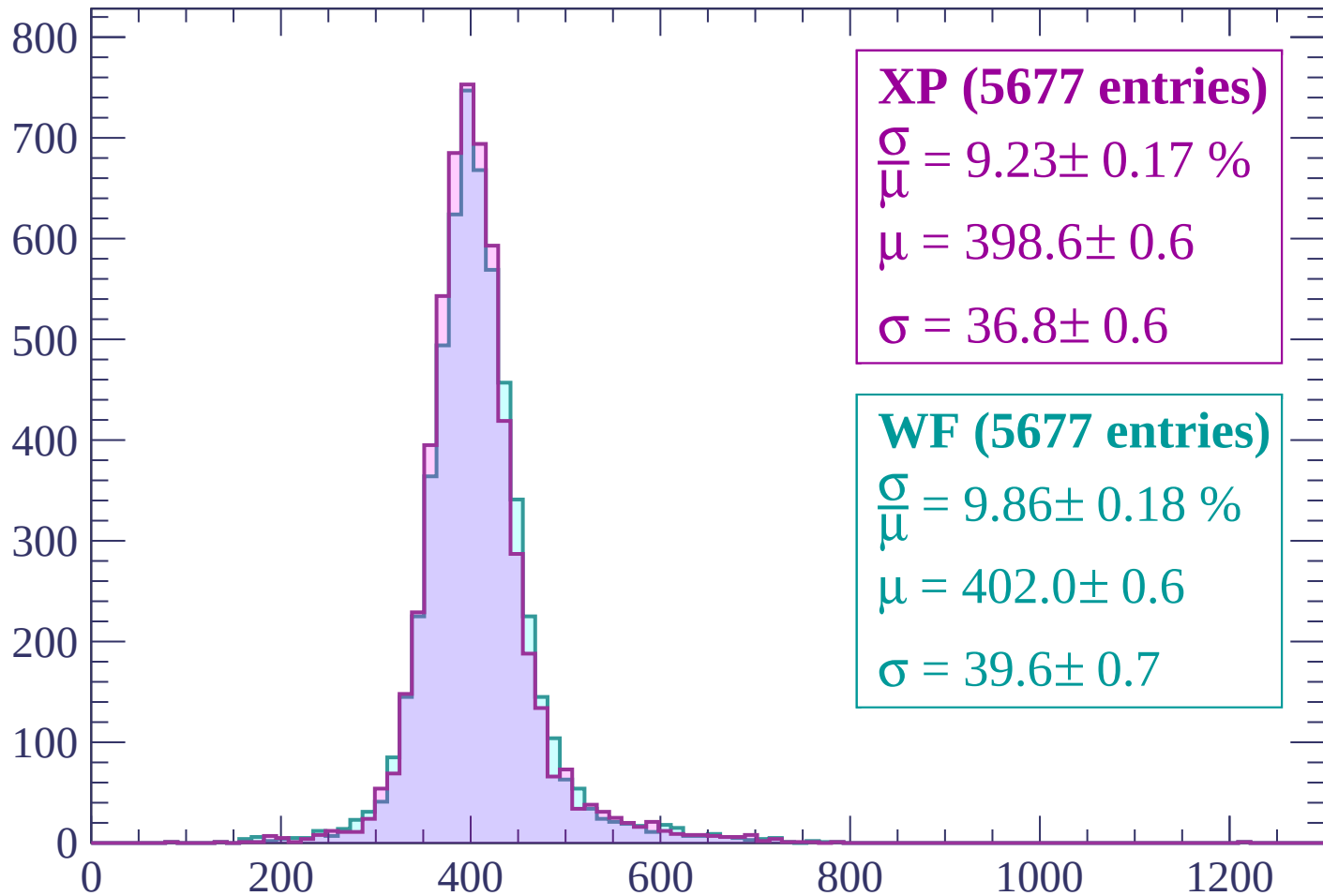
$$\mu = 402.1 \pm 0.6$$

$$\sigma = 39.8 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-300 < p < -240$

Count



XP (5677 entries)

$$\frac{\sigma}{\mu} = 9.23 \pm 0.17 \%$$

$$\mu = 398.6 \pm 0.6$$

$$\sigma = 36.8 \pm 0.6$$

WF (5677 entries)

$$\frac{\sigma}{\mu} = 9.86 \pm 0.18 \%$$

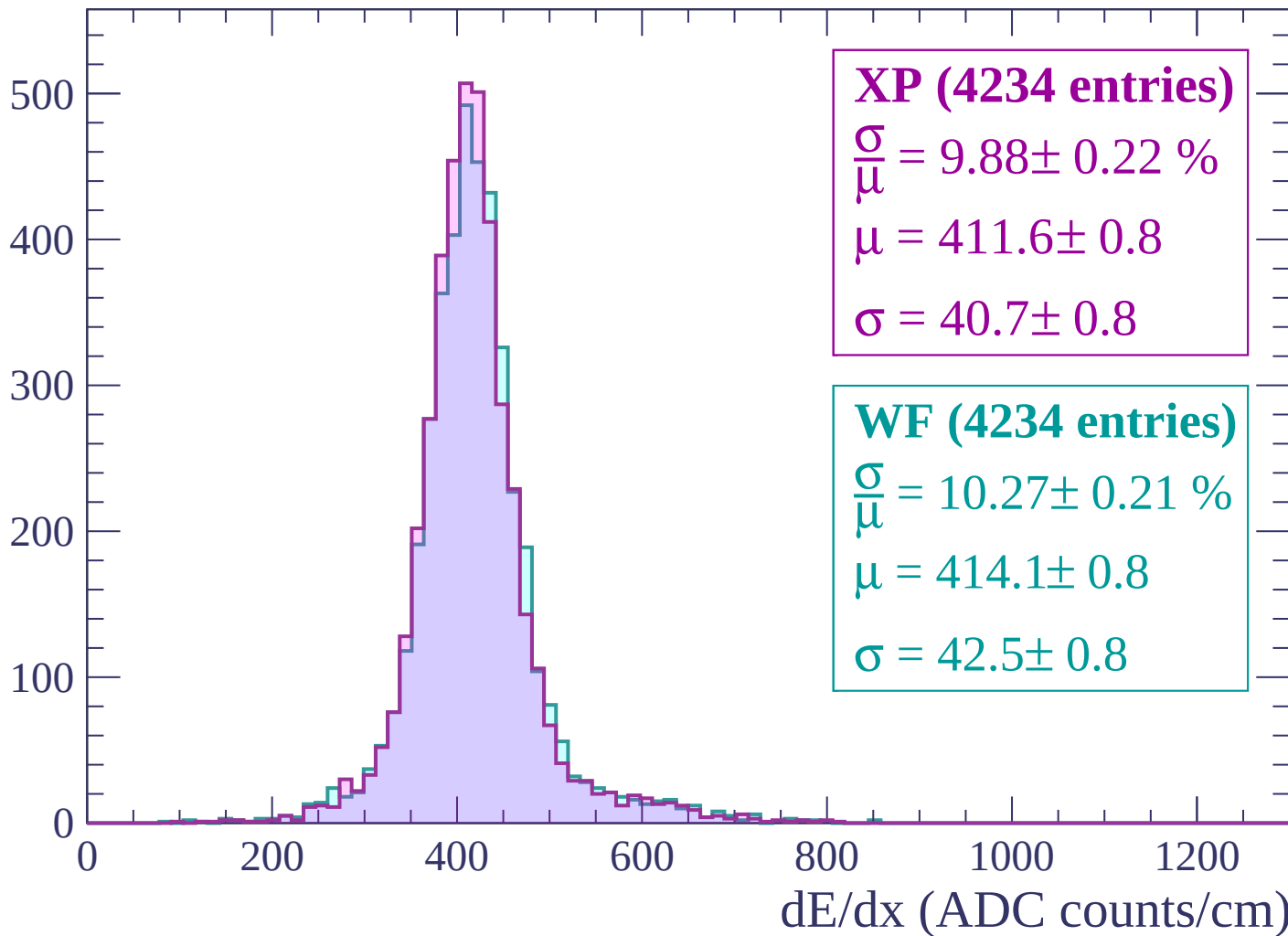
$$\mu = 402.0 \pm 0.6$$

$$\sigma = 39.6 \pm 0.7$$

dE/dx (ADC counts/cm)

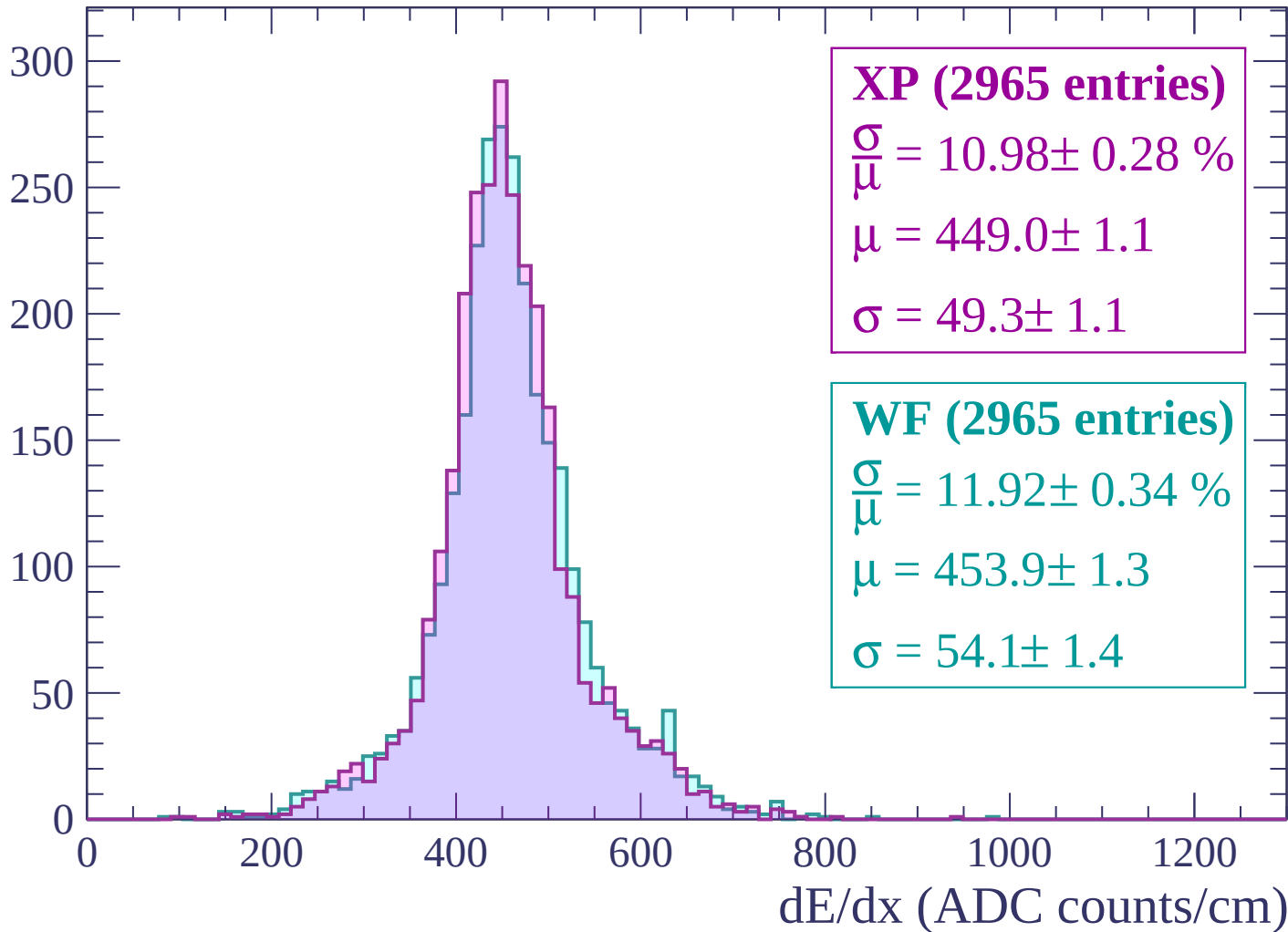
Energy loss | $-240 < p < -180$

Count



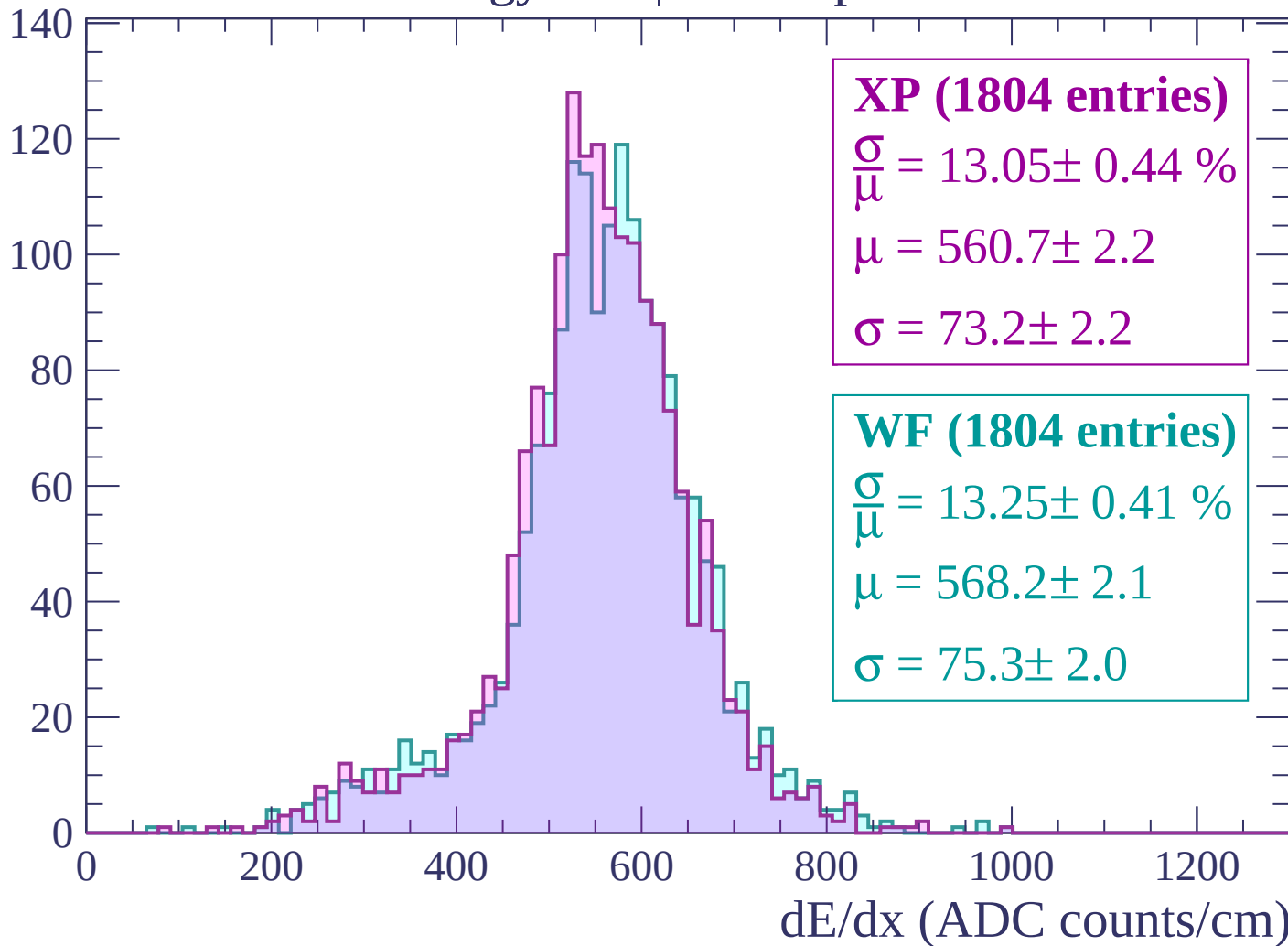
Energy loss | $-180 < p < -120$

Count



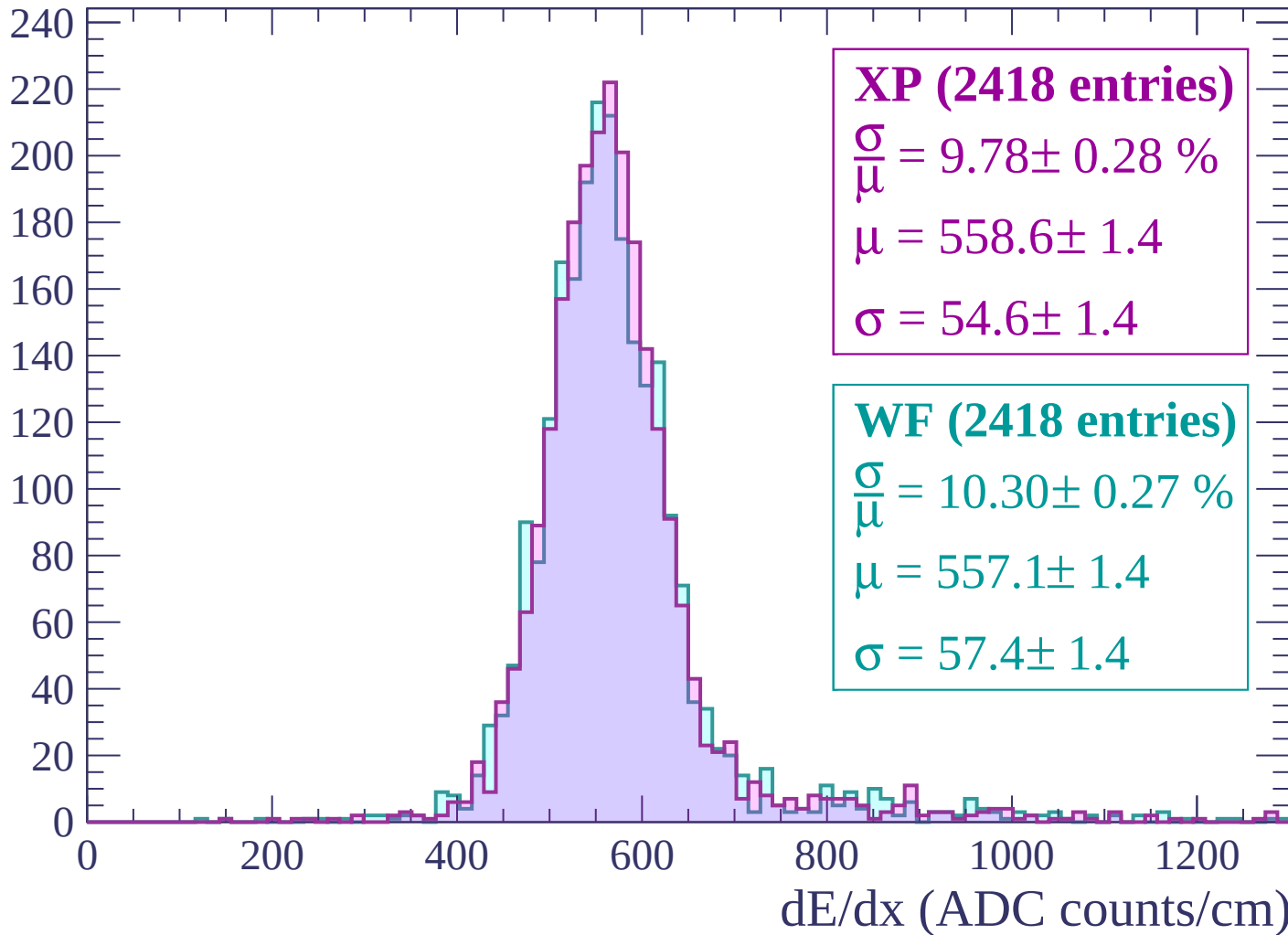
Energy loss | $-120 < p < -60$

Count



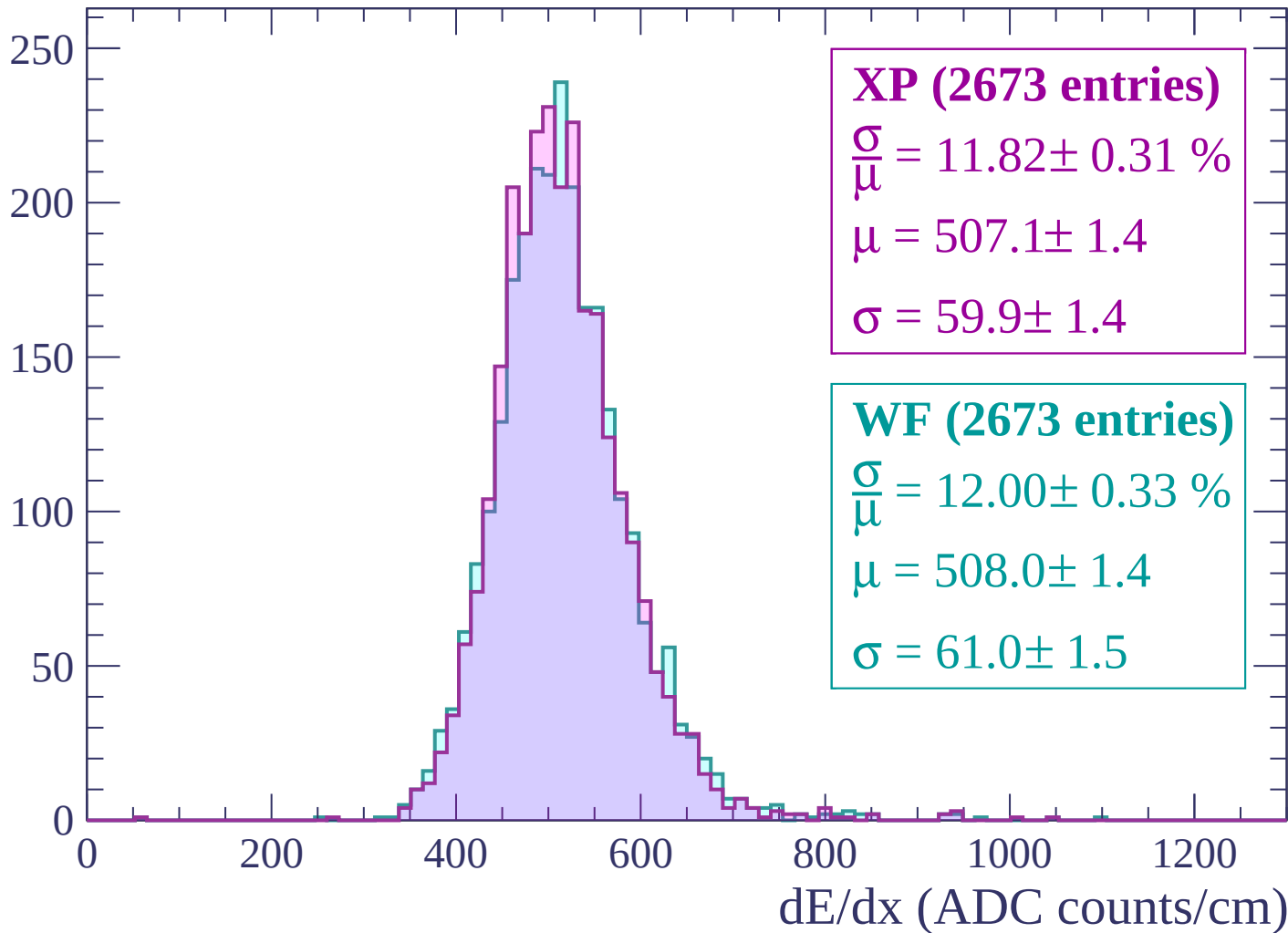
Energy loss | $-60 < p < 0$

Count



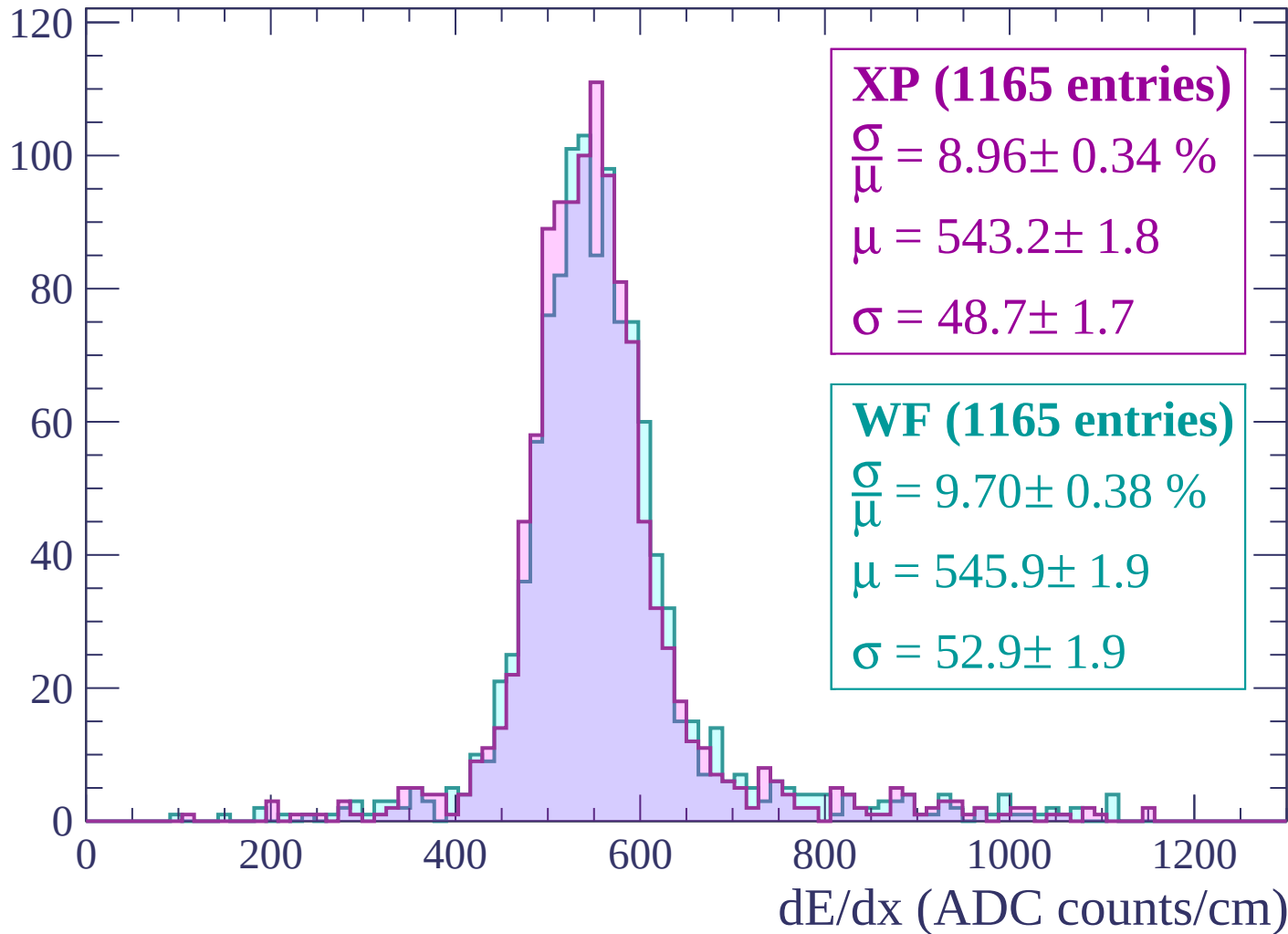
Energy loss | $0 < p < 60$

Count



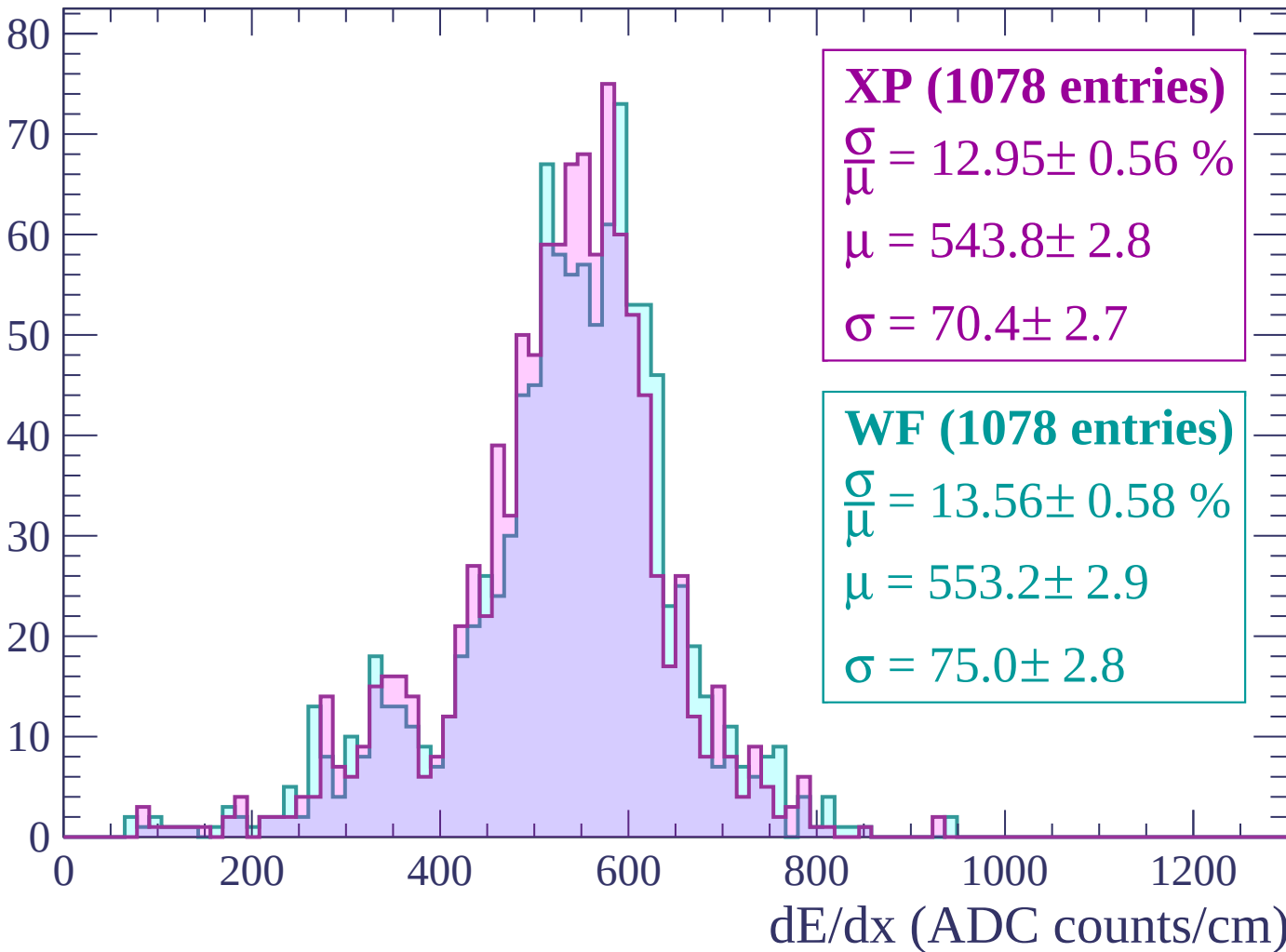
Energy loss | $60 < p < 120$

Count



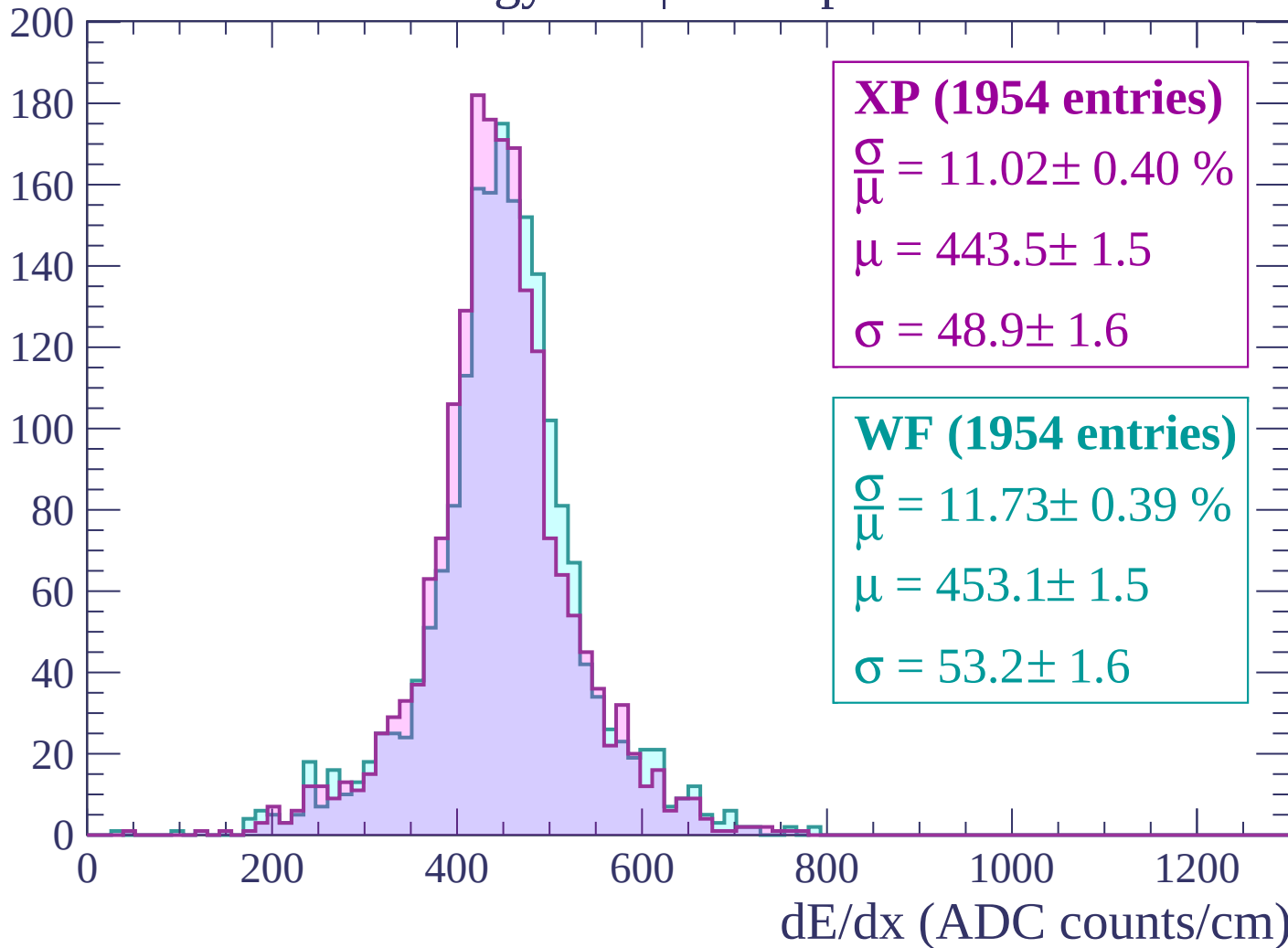
Energy loss | $120 < p < 180$

Count



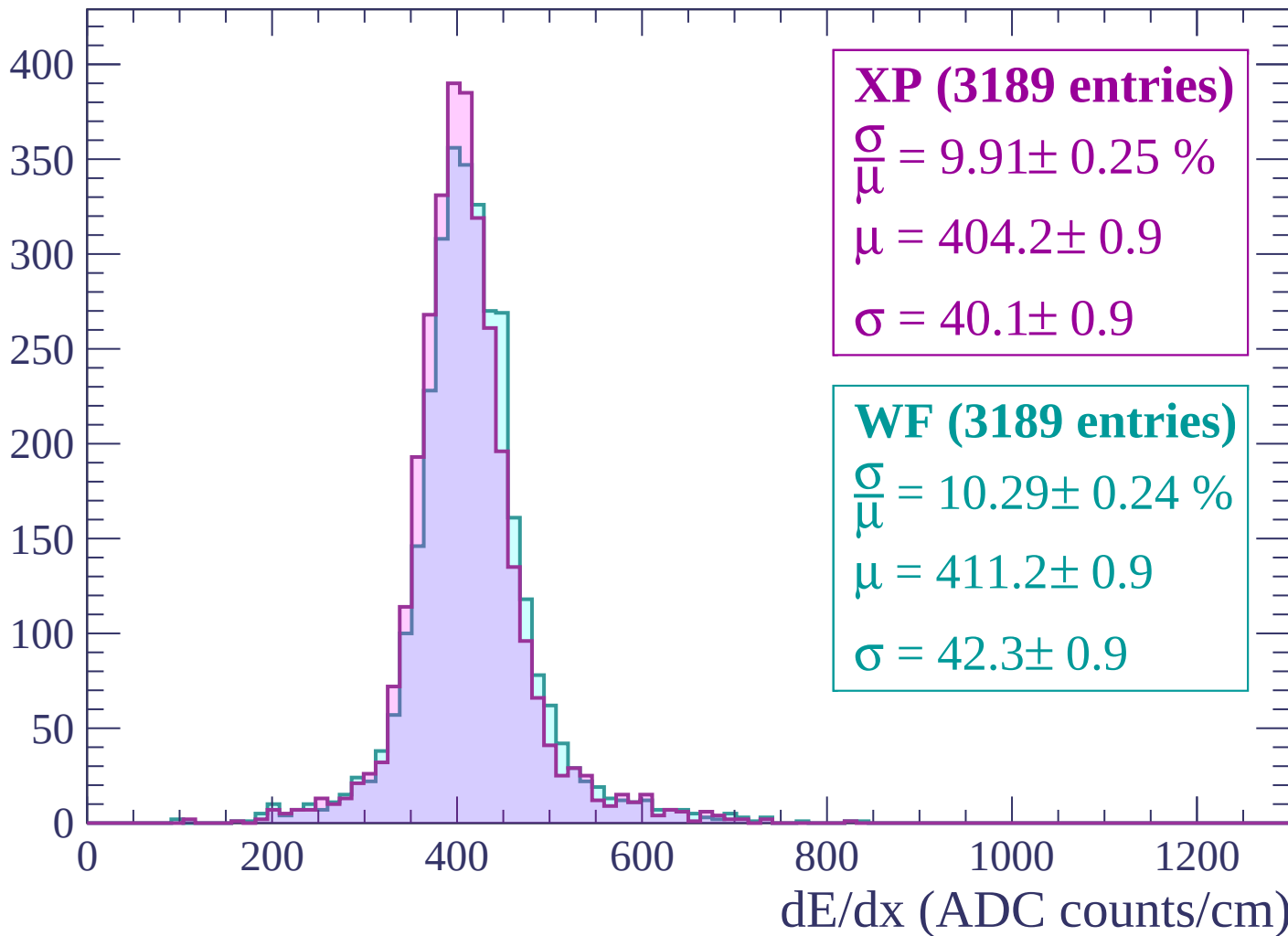
Energy loss | $180 < p < 240$

Count



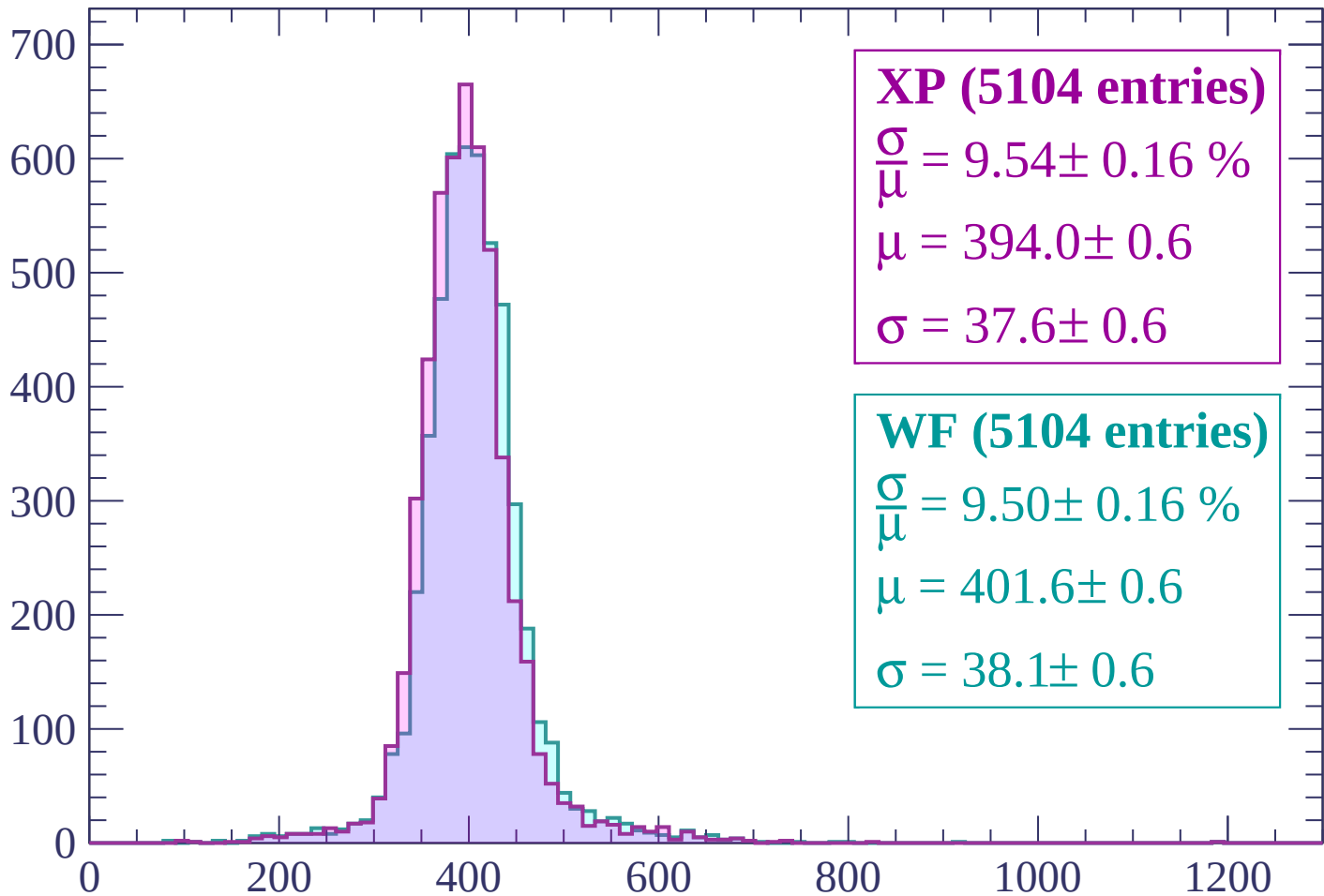
Energy loss | $240 < p < 300$

Count



Energy loss | $300 < p < 360$

Count



XP (5104 entries)

$\frac{\sigma}{\mu} = 9.54 \pm 0.16 \%$

$\mu = 394.0 \pm 0.6$

$\sigma = 37.6 \pm 0.6$

WF (5104 entries)

$\frac{\sigma}{\mu} = 9.50 \pm 0.16 \%$

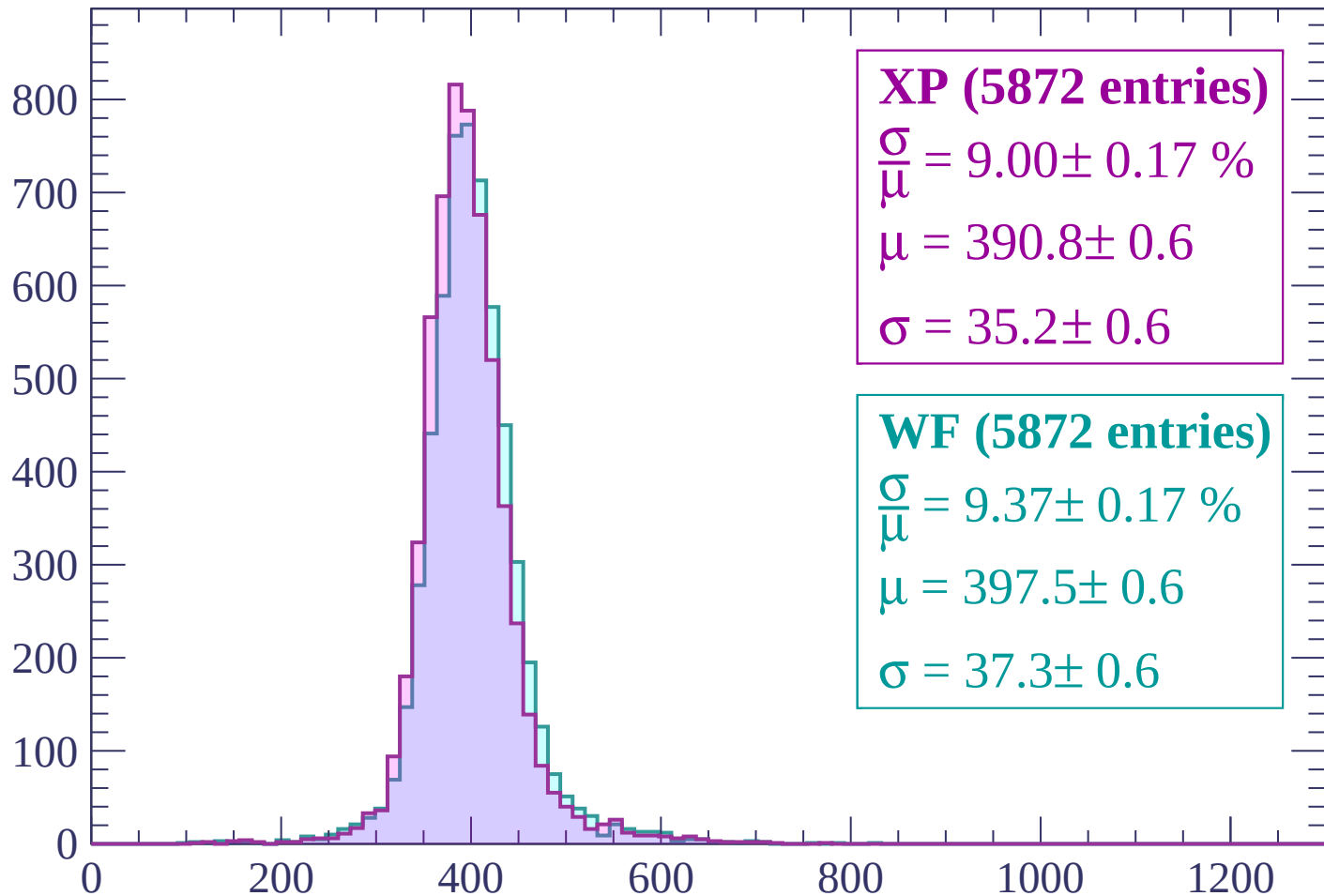
$\mu = 401.6 \pm 0.6$

$\sigma = 38.1 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $360 < p < 420$

Count



XP (5872 entries)

$\frac{\sigma}{\mu} = 9.00 \pm 0.17 \%$

$\mu = 390.8 \pm 0.6$

$\sigma = 35.2 \pm 0.6$

WF (5872 entries)

$\frac{\sigma}{\mu} = 9.37 \pm 0.17 \%$

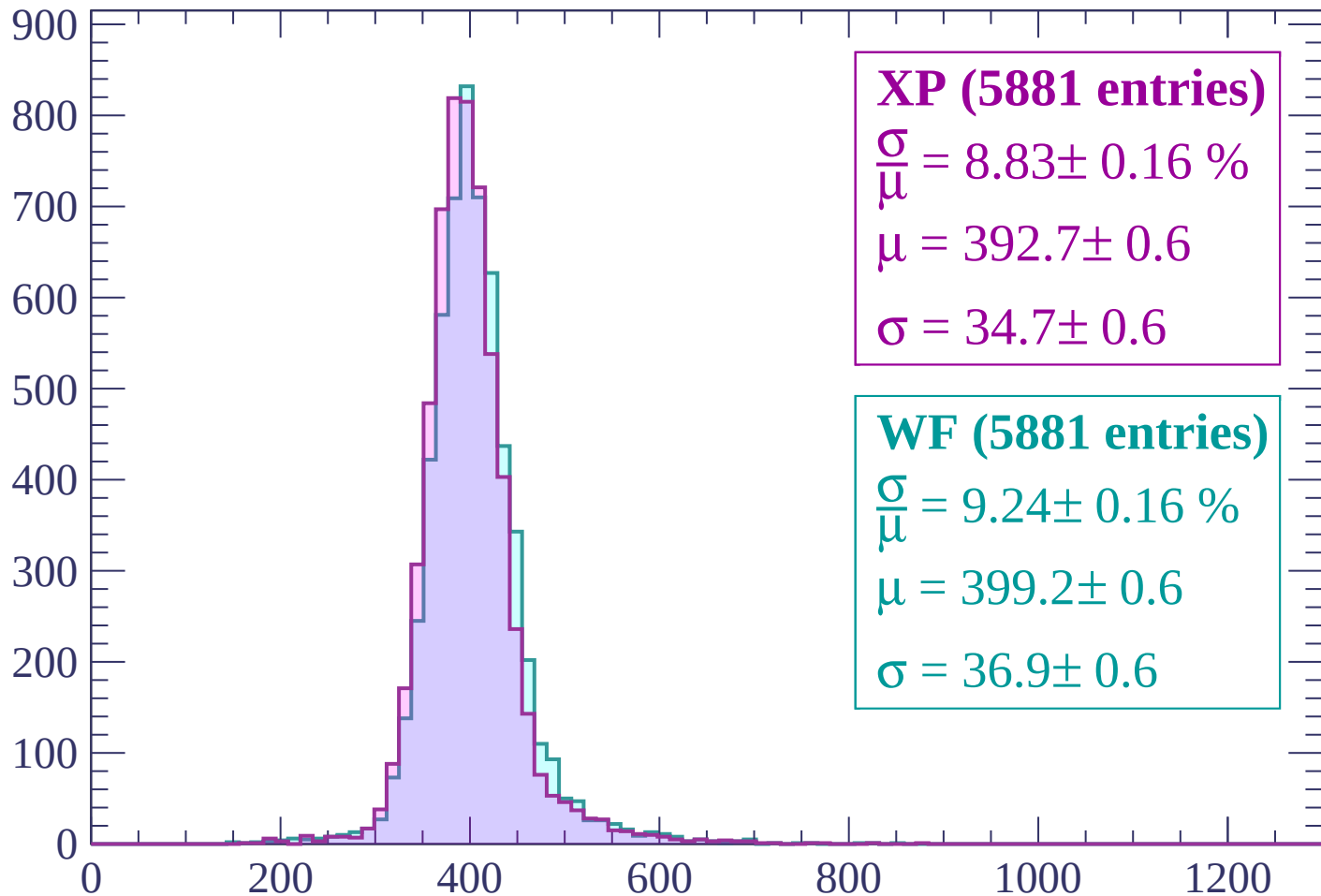
$\mu = 397.5 \pm 0.6$

$\sigma = 37.3 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $420 < p < 480$

Count



XP (5881 entries)

$$\frac{\sigma}{\mu} = 8.83 \pm 0.16 \%$$

$$\mu = 392.7 \pm 0.6$$

$$\sigma = 34.7 \pm 0.6$$

WF (5881 entries)

$$\frac{\sigma}{\mu} = 9.24 \pm 0.16 \%$$

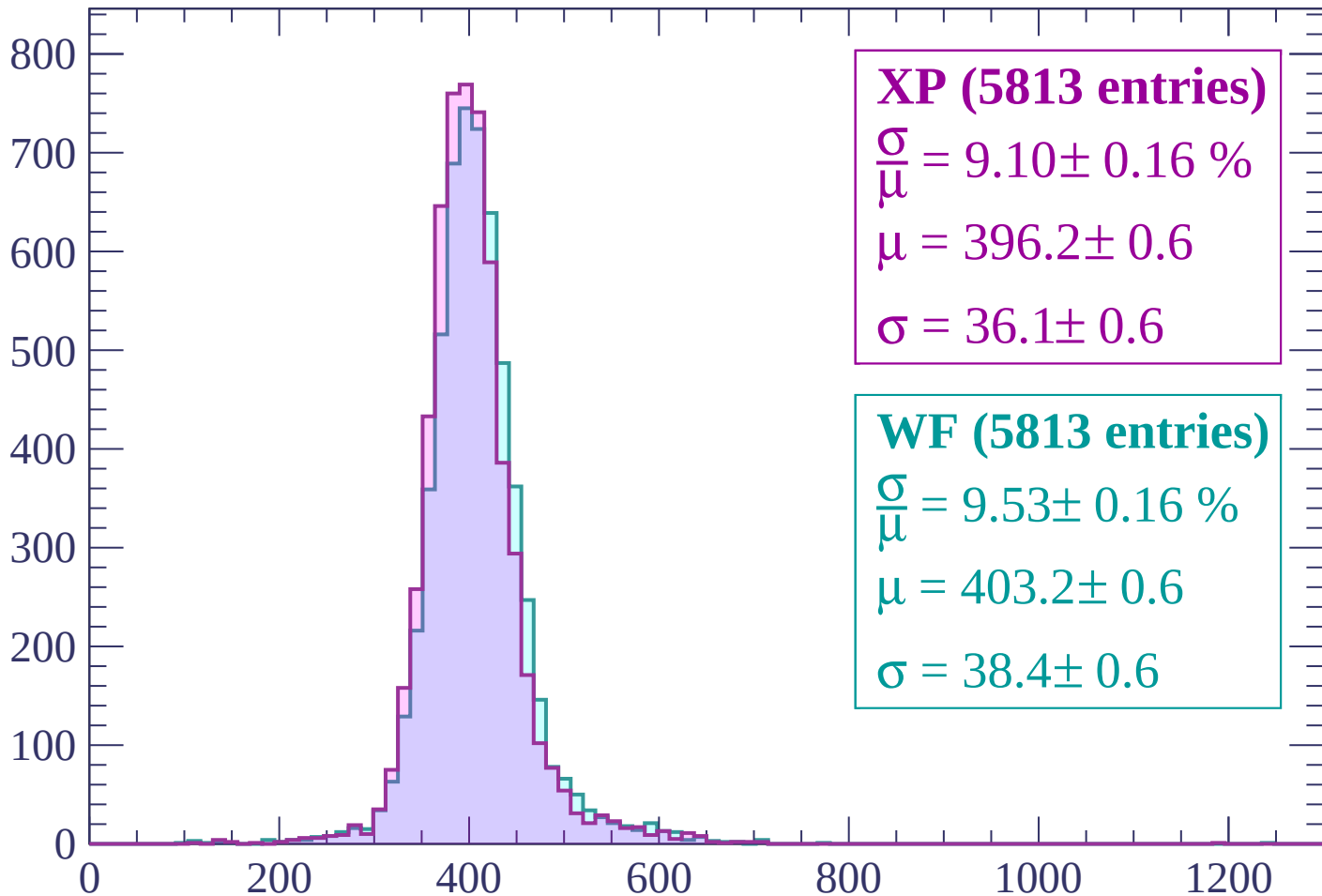
$$\mu = 399.2 \pm 0.6$$

$$\sigma = 36.9 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $480 < p < 540$

Count



XP (5813 entries)

$$\frac{\sigma}{\mu} = 9.10 \pm 0.16 \%$$

$$\mu = 396.2 \pm 0.6$$

$$\sigma = 36.1 \pm 0.6$$

WF (5813 entries)

$$\frac{\sigma}{\mu} = 9.53 \pm 0.16 \%$$

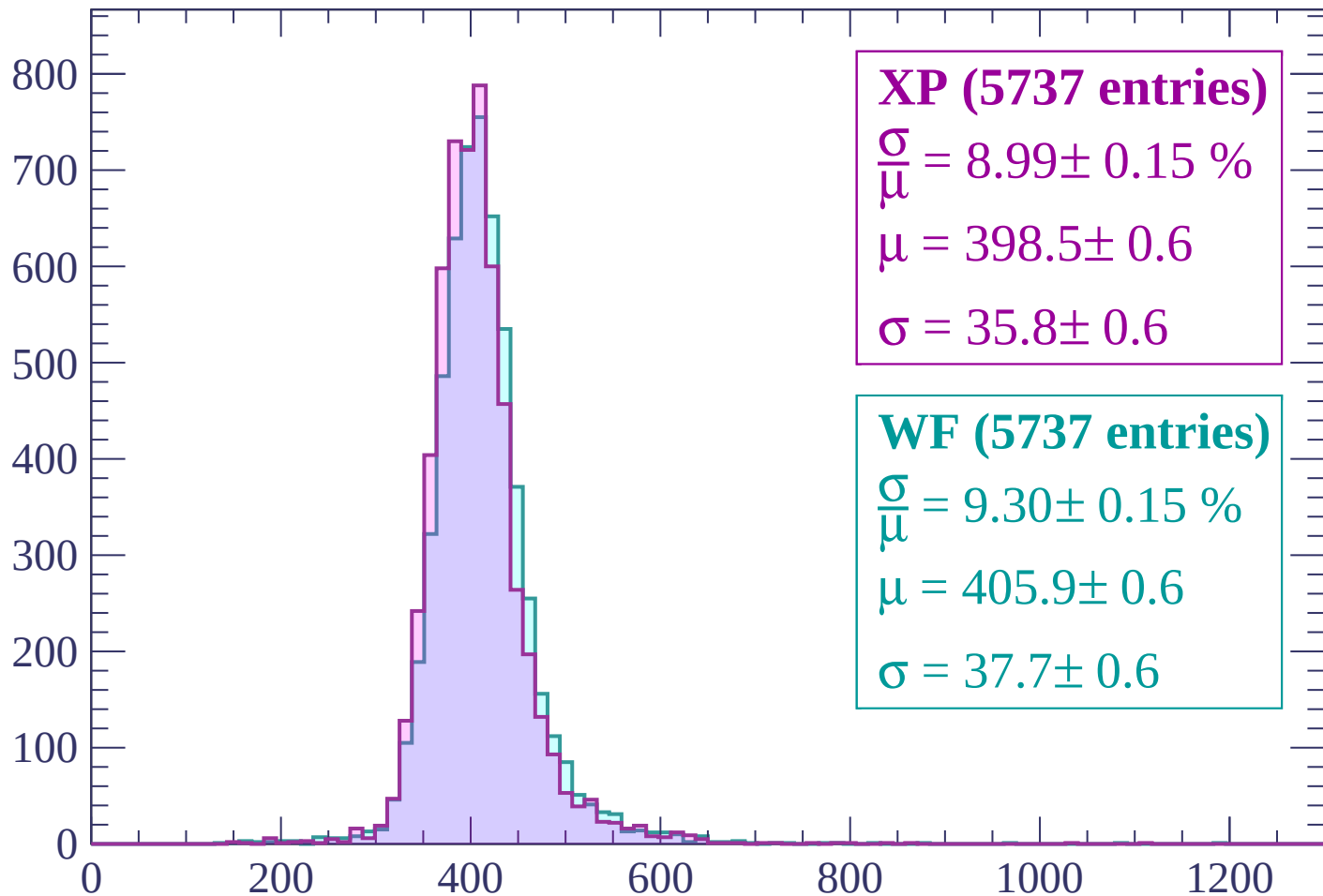
$$\mu = 403.2 \pm 0.6$$

$$\sigma = 38.4 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $540 < p < 600$

Count



XP (5737 entries)

$$\frac{\sigma}{\mu} = 8.99 \pm 0.15 \%$$

$$\mu = 398.5 \pm 0.6$$

$$\sigma = 35.8 \pm 0.6$$

WF (5737 entries)

$$\frac{\sigma}{\mu} = 9.30 \pm 0.15 \%$$

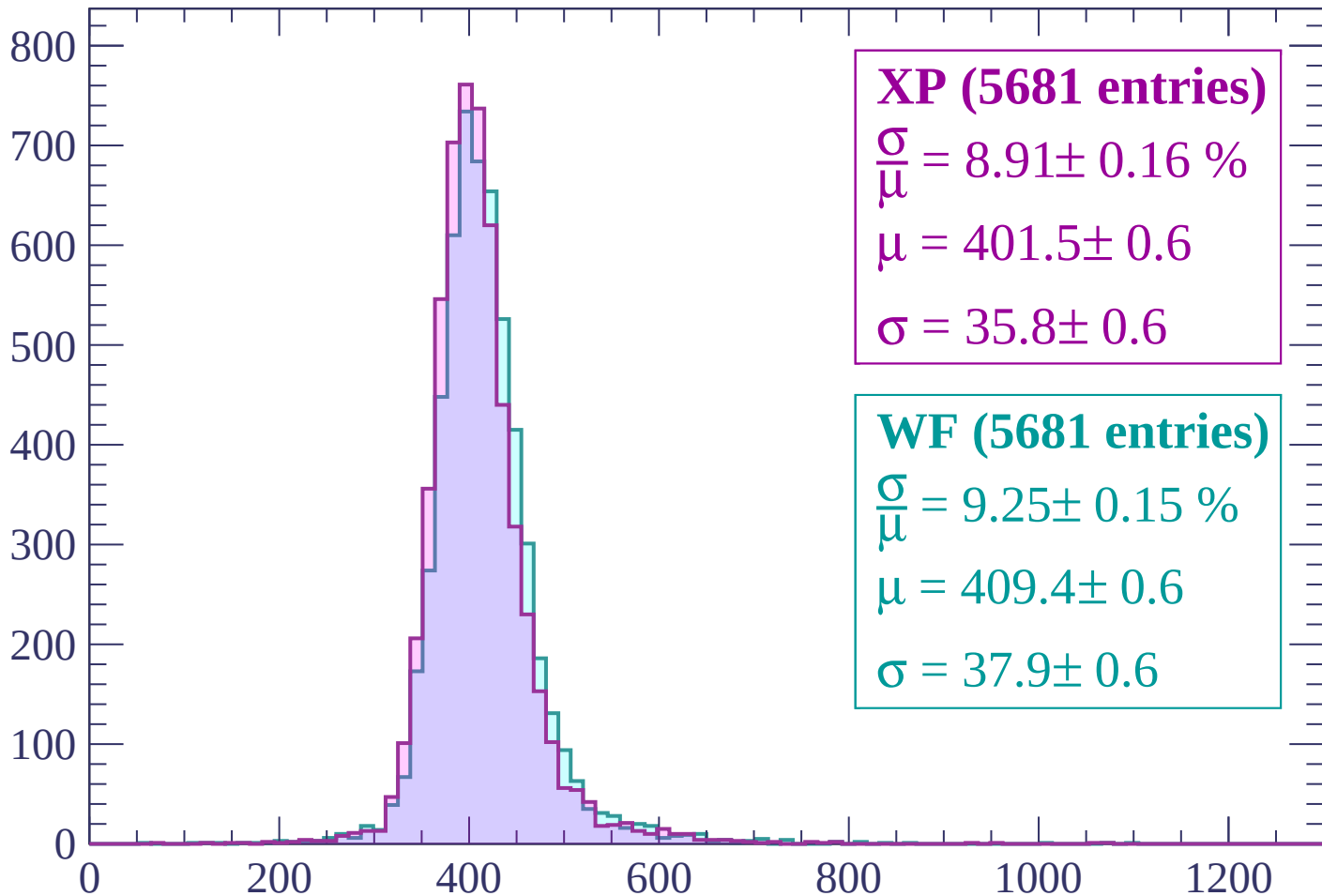
$$\mu = 405.9 \pm 0.6$$

$$\sigma = 37.7 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $600 < p < 660$

Count



XP (5681 entries)

$\frac{\sigma}{\mu} = 8.91 \pm 0.16 \%$

$\mu = 401.5 \pm 0.6$

$\sigma = 35.8 \pm 0.6$

WF (5681 entries)

$\frac{\sigma}{\mu} = 9.25 \pm 0.15 \%$

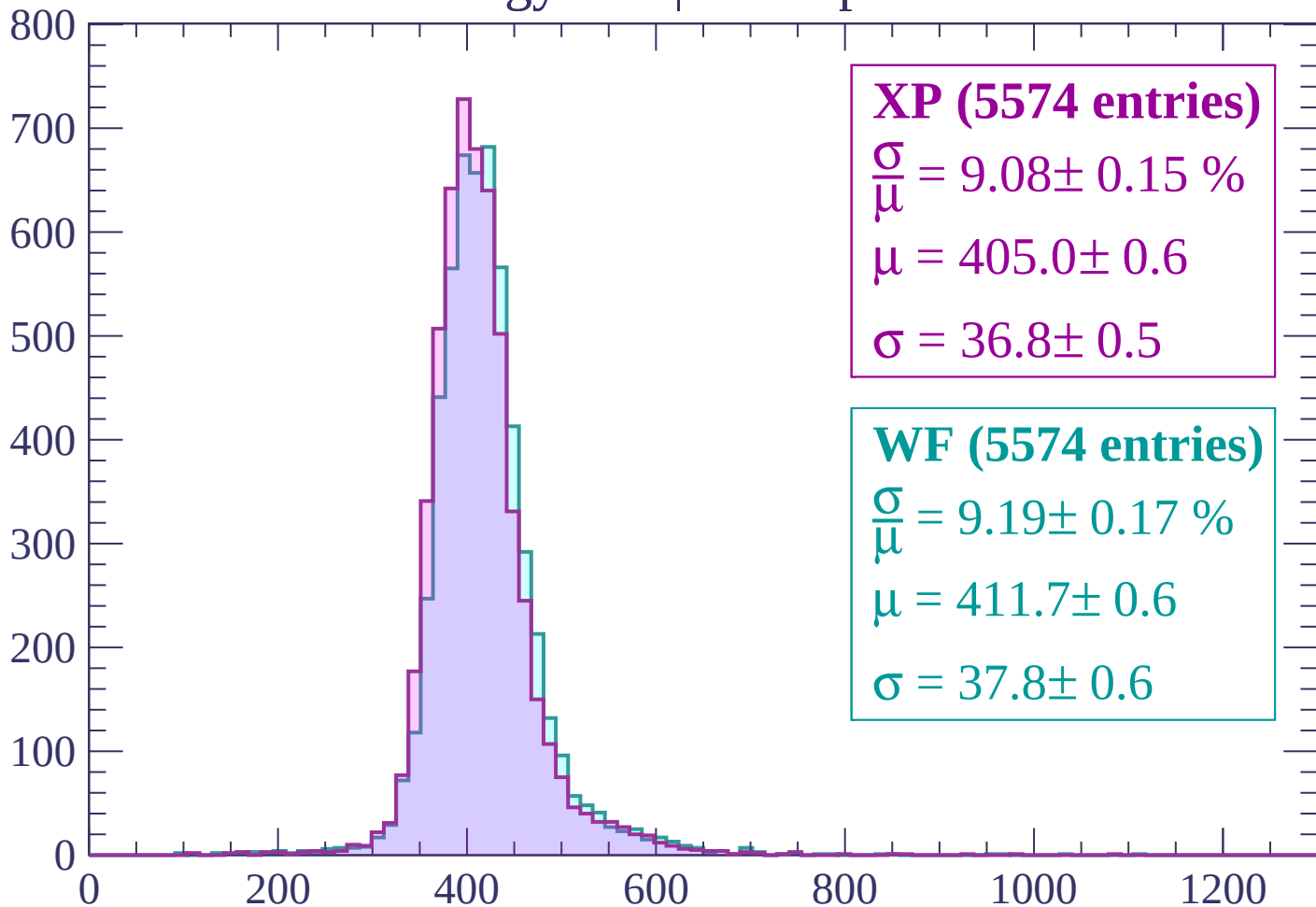
$\mu = 409.4 \pm 0.6$

$\sigma = 37.9 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $660 < p < 720$

Count



XP (5574 entries)

$\frac{\sigma}{\mu} = 9.08 \pm 0.15 \%$

$\mu = 405.0 \pm 0.6$

$\sigma = 36.8 \pm 0.5$

WF (5574 entries)

$\frac{\sigma}{\mu} = 9.19 \pm 0.17 \%$

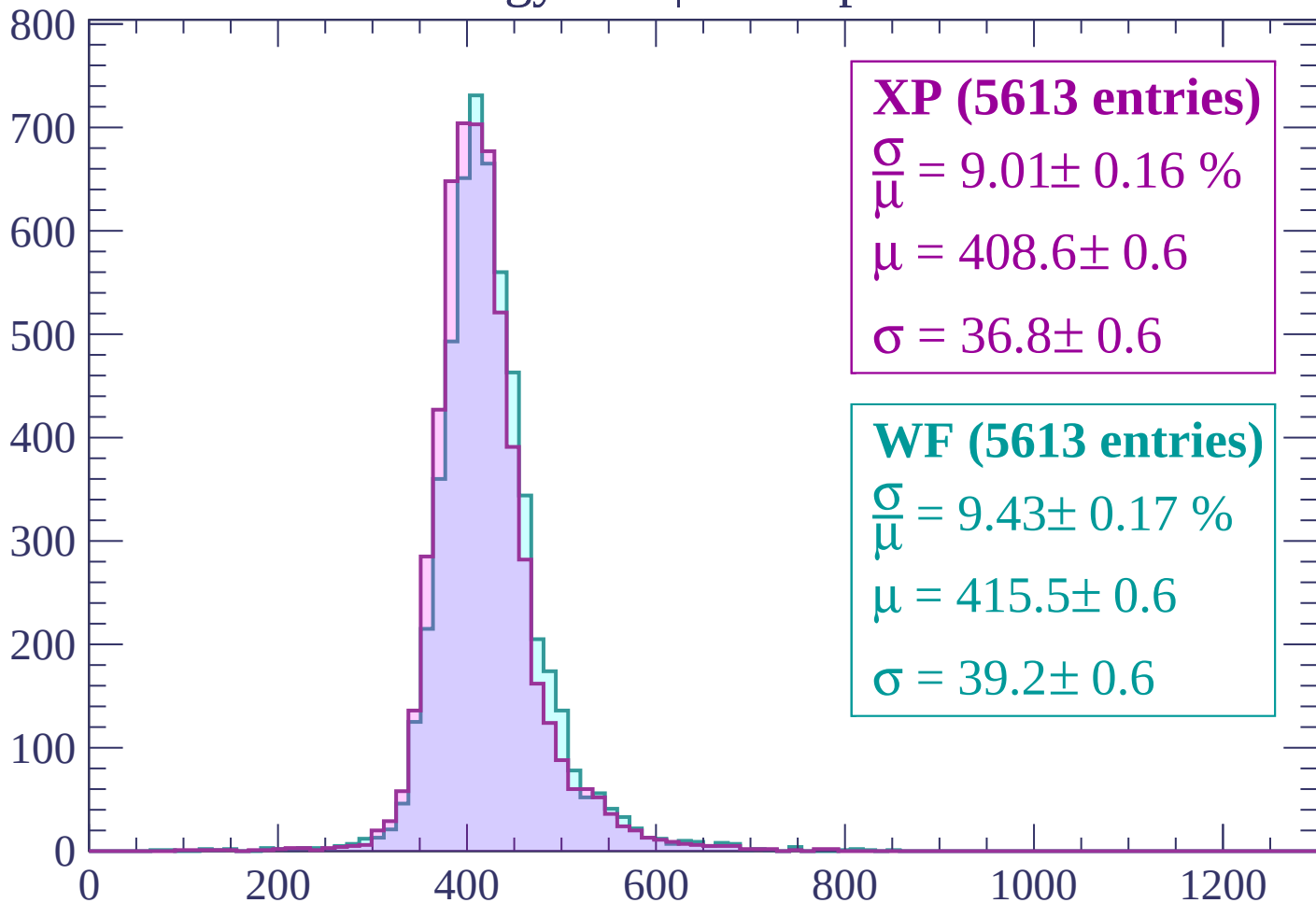
$\mu = 411.7 \pm 0.6$

$\sigma = 37.8 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $720 < p < 780$

Count



XP (5613 entries)

$$\frac{\sigma}{\mu} = 9.01 \pm 0.16 \%$$

$$\mu = 408.6 \pm 0.6$$

$$\sigma = 36.8 \pm 0.6$$

WF (5613 entries)

$$\frac{\sigma}{\mu} = 9.43 \pm 0.17 \%$$

$$\mu = 415.5 \pm 0.6$$

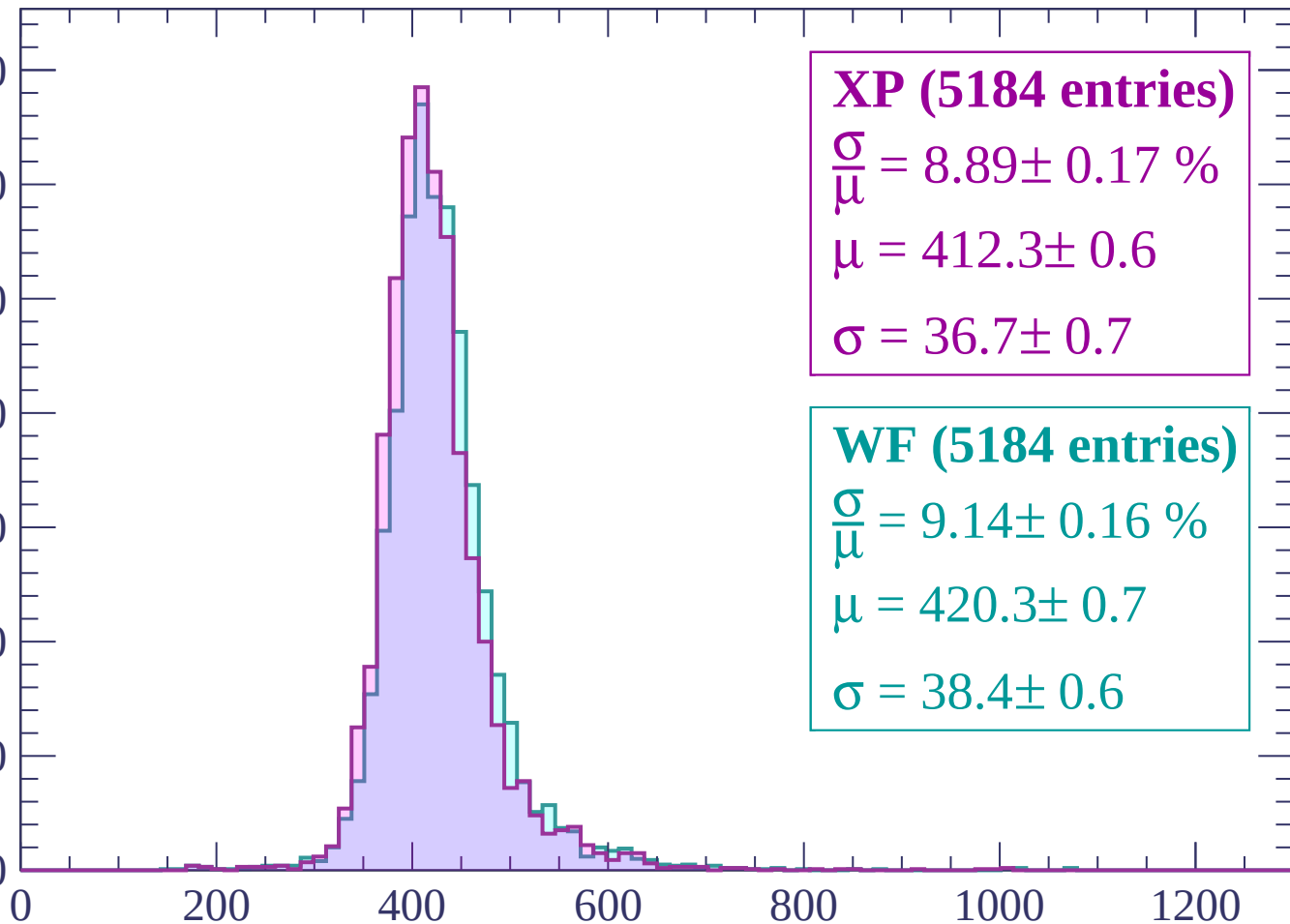
$$\sigma = 39.2 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $780 < p < 840$

Count

700
600
500
400
300
200
100
0



XP (5184 entries)

$\frac{\sigma}{\mu} = 8.89 \pm 0.17 \%$

$\mu = 412.3 \pm 0.6$

$\sigma = 36.7 \pm 0.7$

WF (5184 entries)

$\frac{\sigma}{\mu} = 9.14 \pm 0.16 \%$

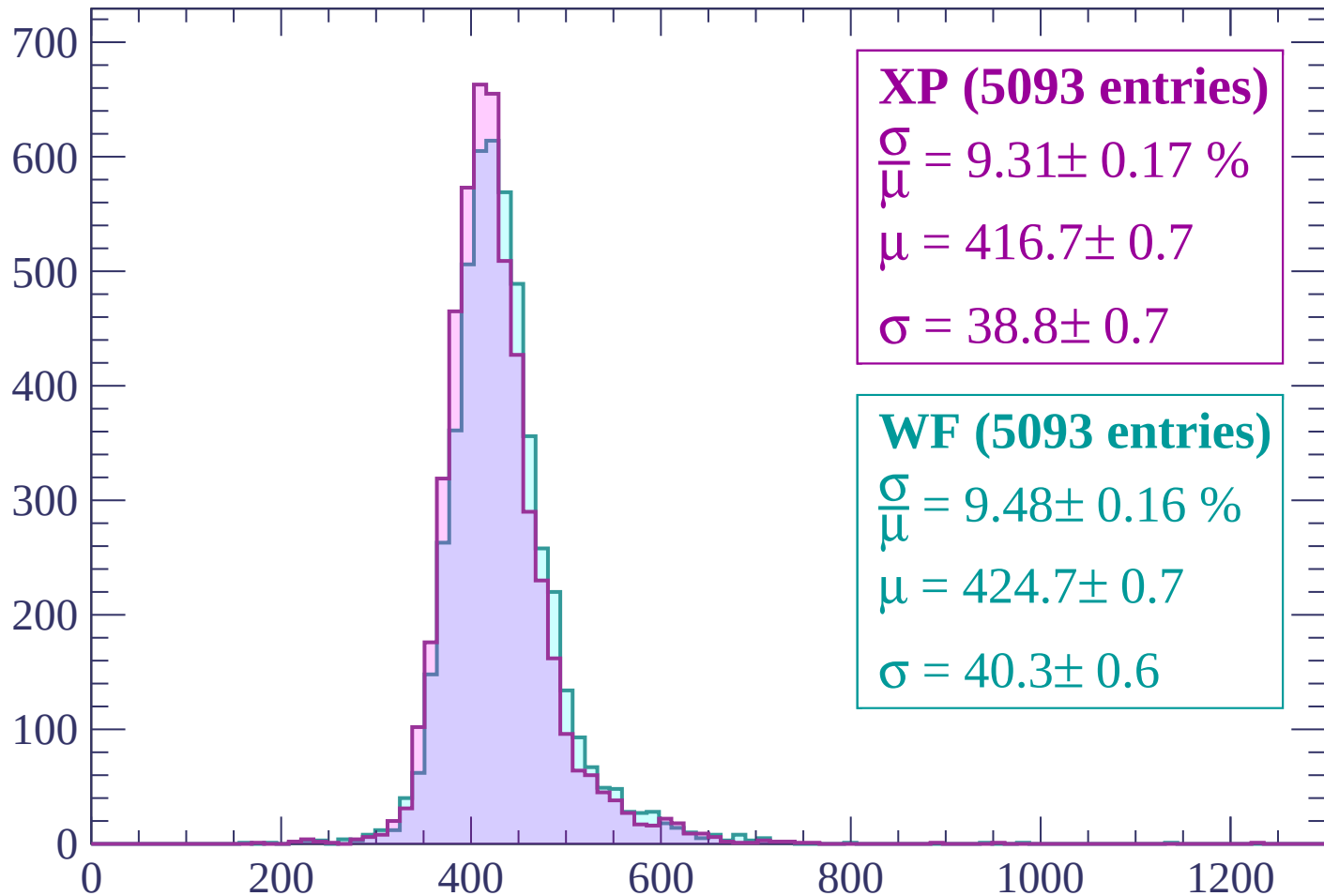
$\mu = 420.3 \pm 0.7$

$\sigma = 38.4 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $840 < p < 900$

Count



XP (5093 entries)

$$\frac{\sigma}{\mu} = 9.31 \pm 0.17 \%$$

$$\mu = 416.7 \pm 0.7$$

$$\sigma = 38.8 \pm 0.7$$

WF (5093 entries)

$$\frac{\sigma}{\mu} = 9.48 \pm 0.16 \%$$

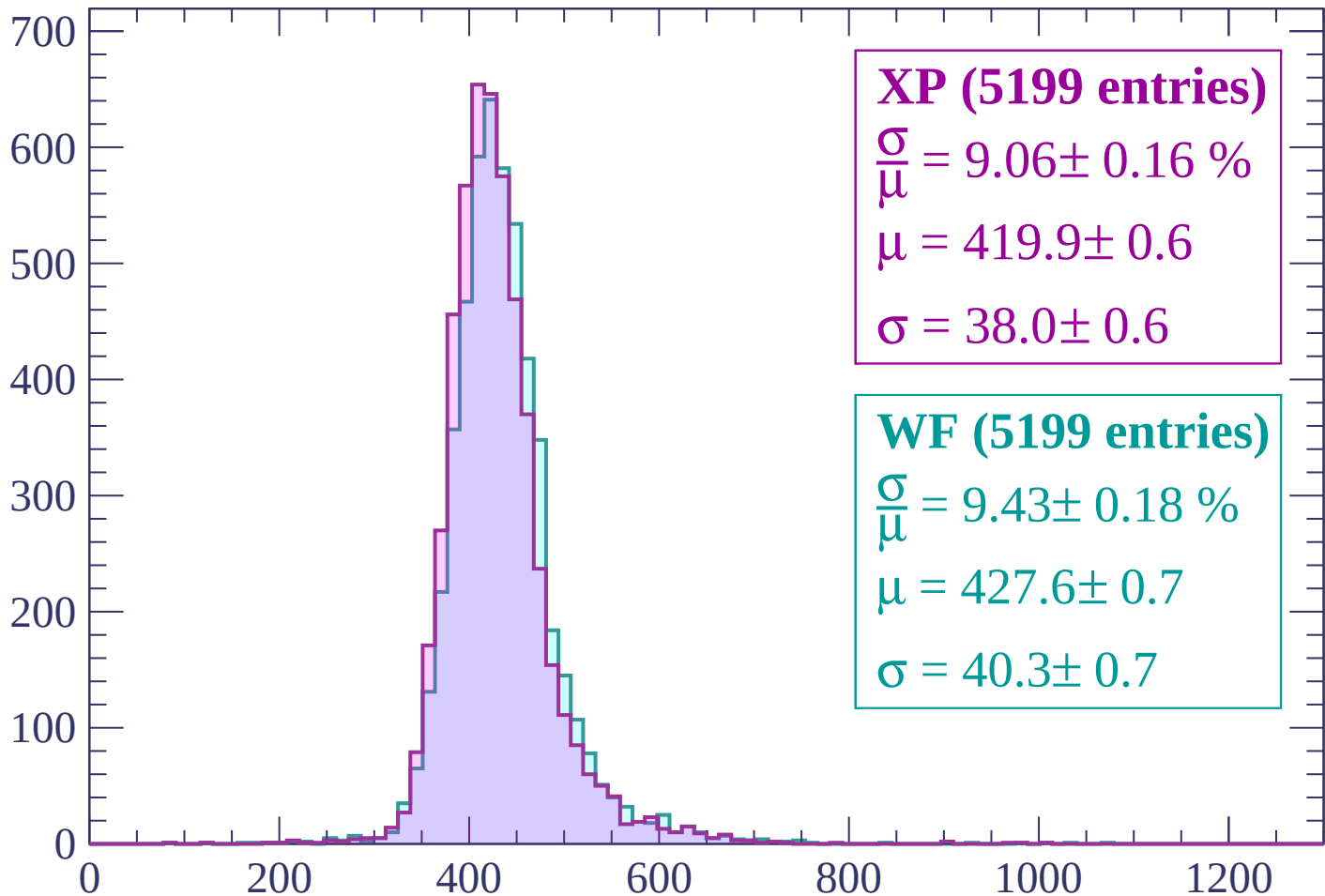
$$\mu = 424.7 \pm 0.7$$

$$\sigma = 40.3 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $900 < p < 960$

Count



XP (5199 entries)

$$\frac{\sigma}{\mu} = 9.06 \pm 0.16 \%$$

$$\mu = 419.9 \pm 0.6$$

$$\sigma = 38.0 \pm 0.6$$

WF (5199 entries)

$$\frac{\sigma}{\mu} = 9.43 \pm 0.18 \%$$

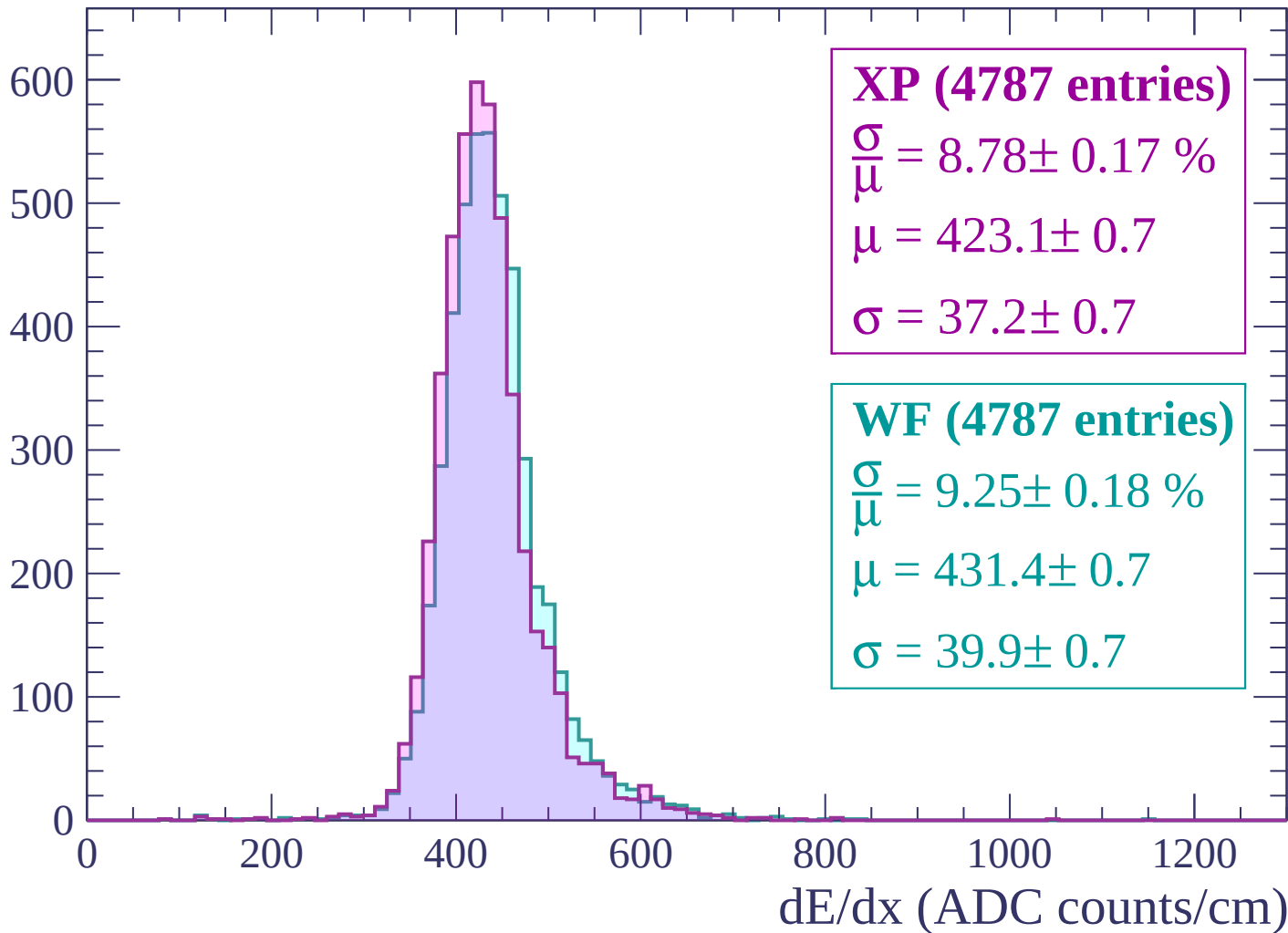
$$\mu = 427.6 \pm 0.7$$

$$\sigma = 40.3 \pm 0.7$$

dE/dx (ADC counts/cm)

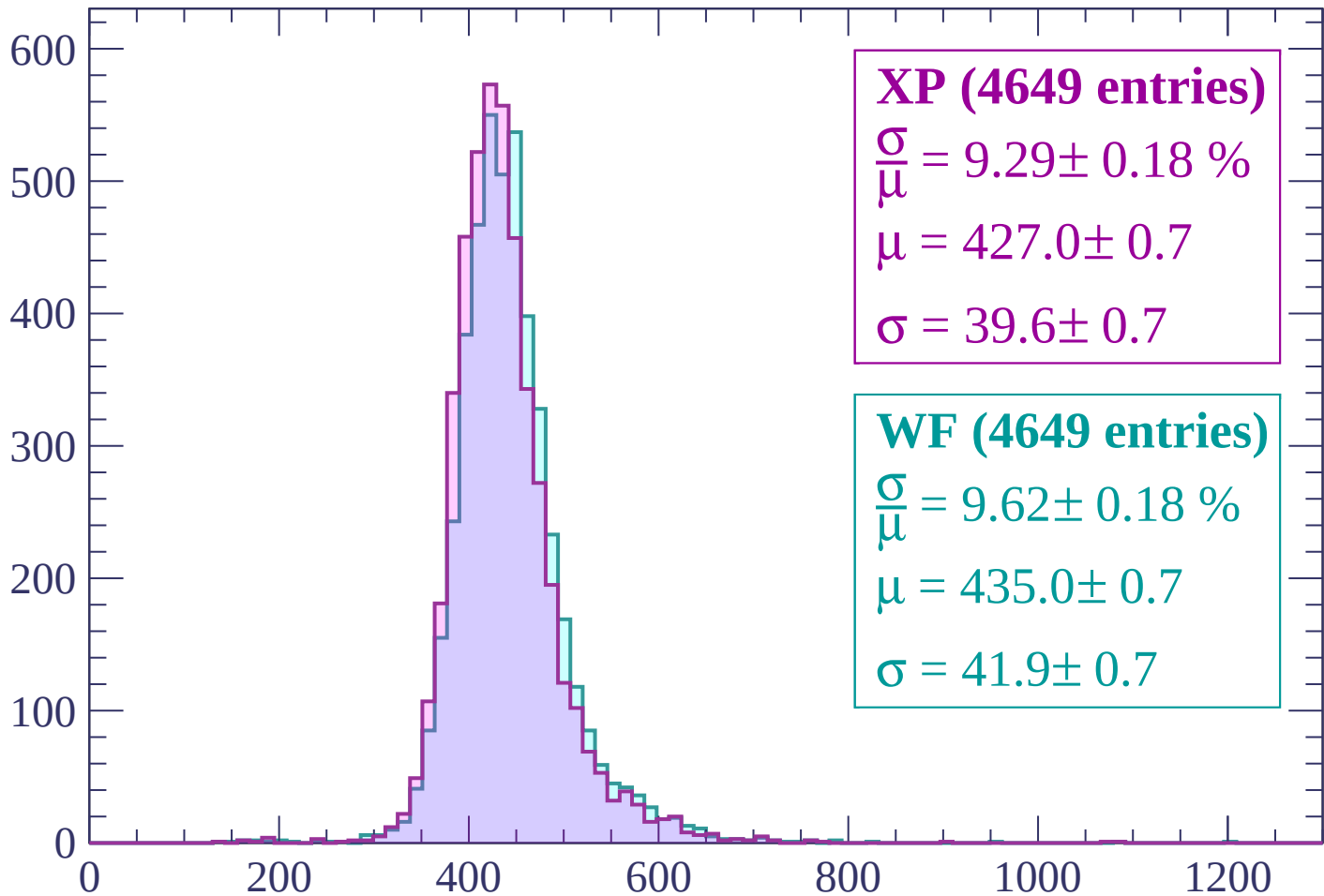
Energy loss | $960 < p < 1020$

Count



Energy loss | $1020 < p < 1080$

Count



XP (4649 entries)

$$\frac{\sigma}{\mu} = 9.29 \pm 0.18 \%$$

$$\mu = 427.0 \pm 0.7$$

$$\sigma = 39.6 \pm 0.7$$

WF (4649 entries)

$$\frac{\sigma}{\mu} = 9.62 \pm 0.18 \%$$

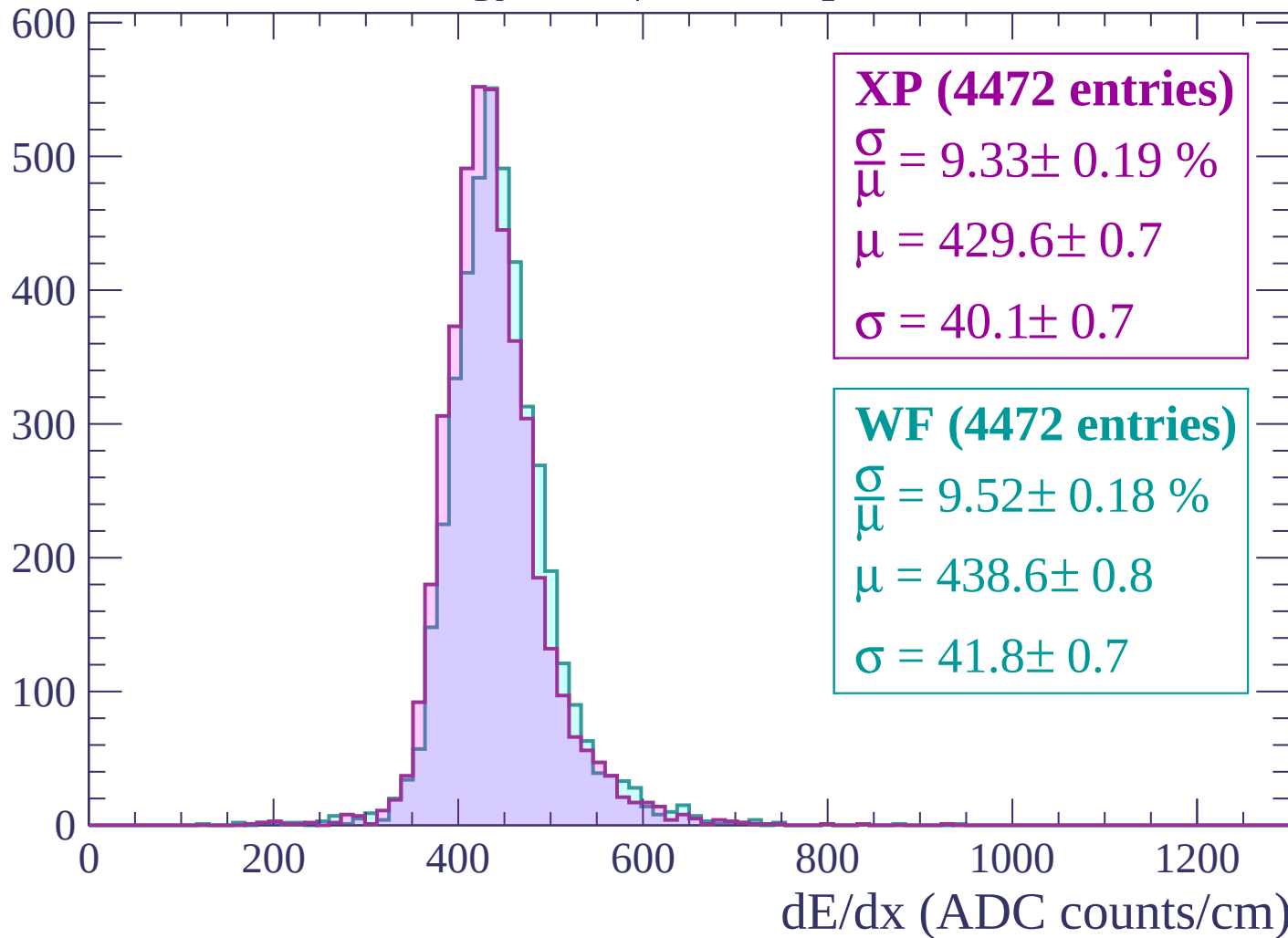
$$\mu = 435.0 \pm 0.7$$

$$\sigma = 41.9 \pm 0.7$$

dE/dx (ADC counts/cm)

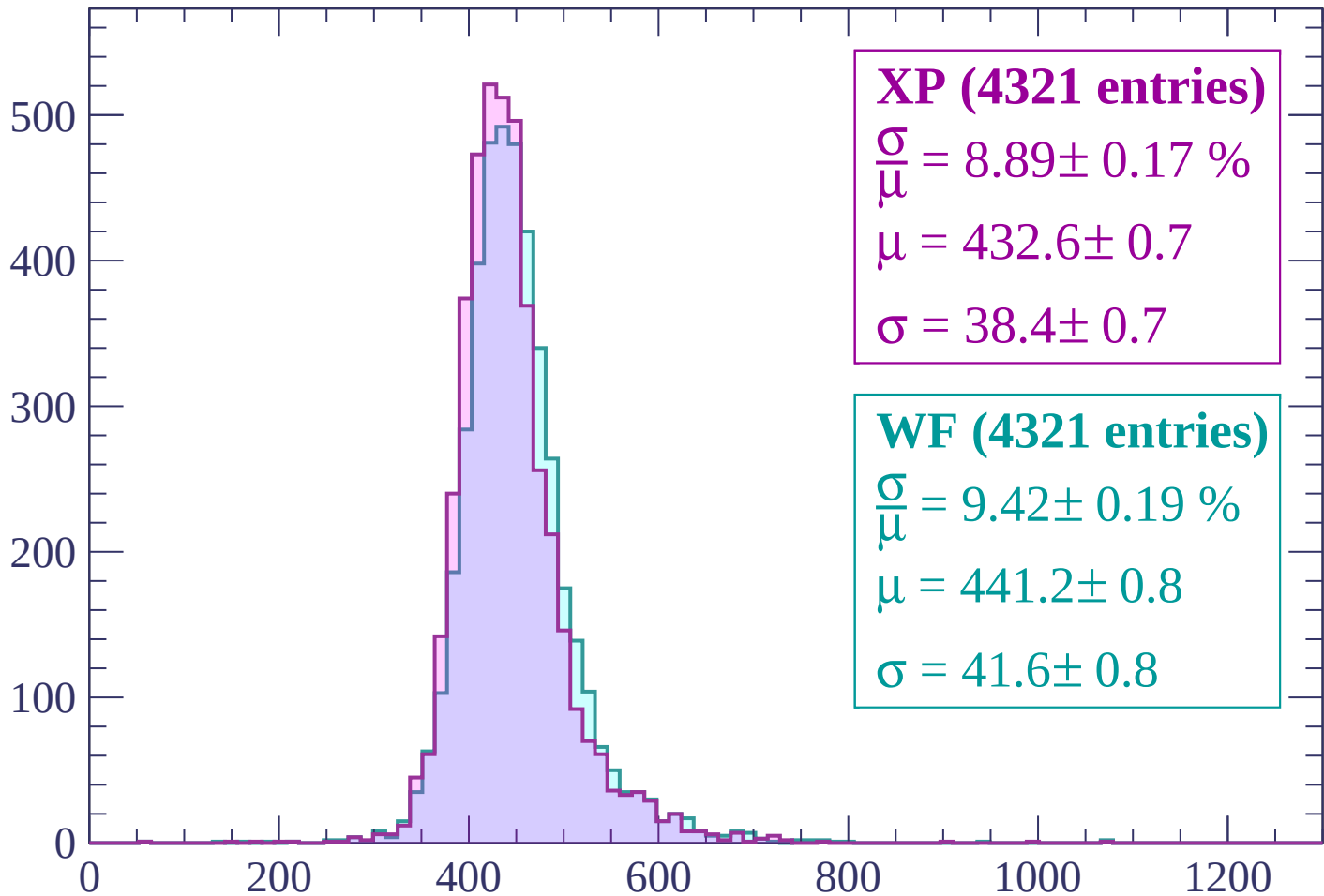
Energy loss | $1080 < p < 1140$

Count



Energy loss | $1140 < p < 1200$

Count



XP (4321 entries)

$$\frac{\sigma}{\mu} = 8.89 \pm 0.17 \%$$

$$\mu = 432.6 \pm 0.7$$

$$\sigma = 38.4 \pm 0.7$$

WF (4321 entries)

$$\frac{\sigma}{\mu} = 9.42 \pm 0.19 \%$$

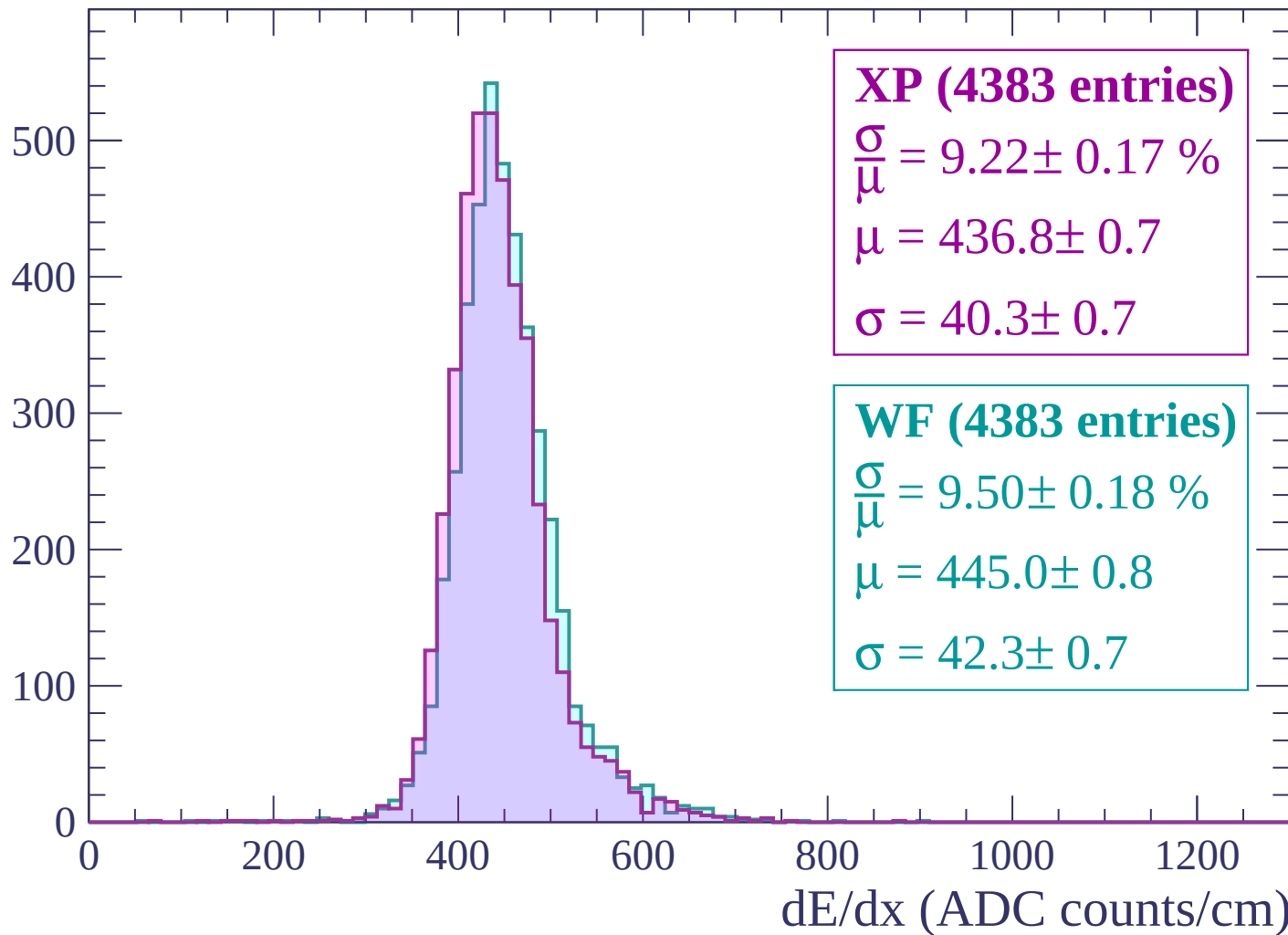
$$\mu = 441.2 \pm 0.8$$

$$\sigma = 41.6 \pm 0.8$$

dE/dx (ADC counts/cm)

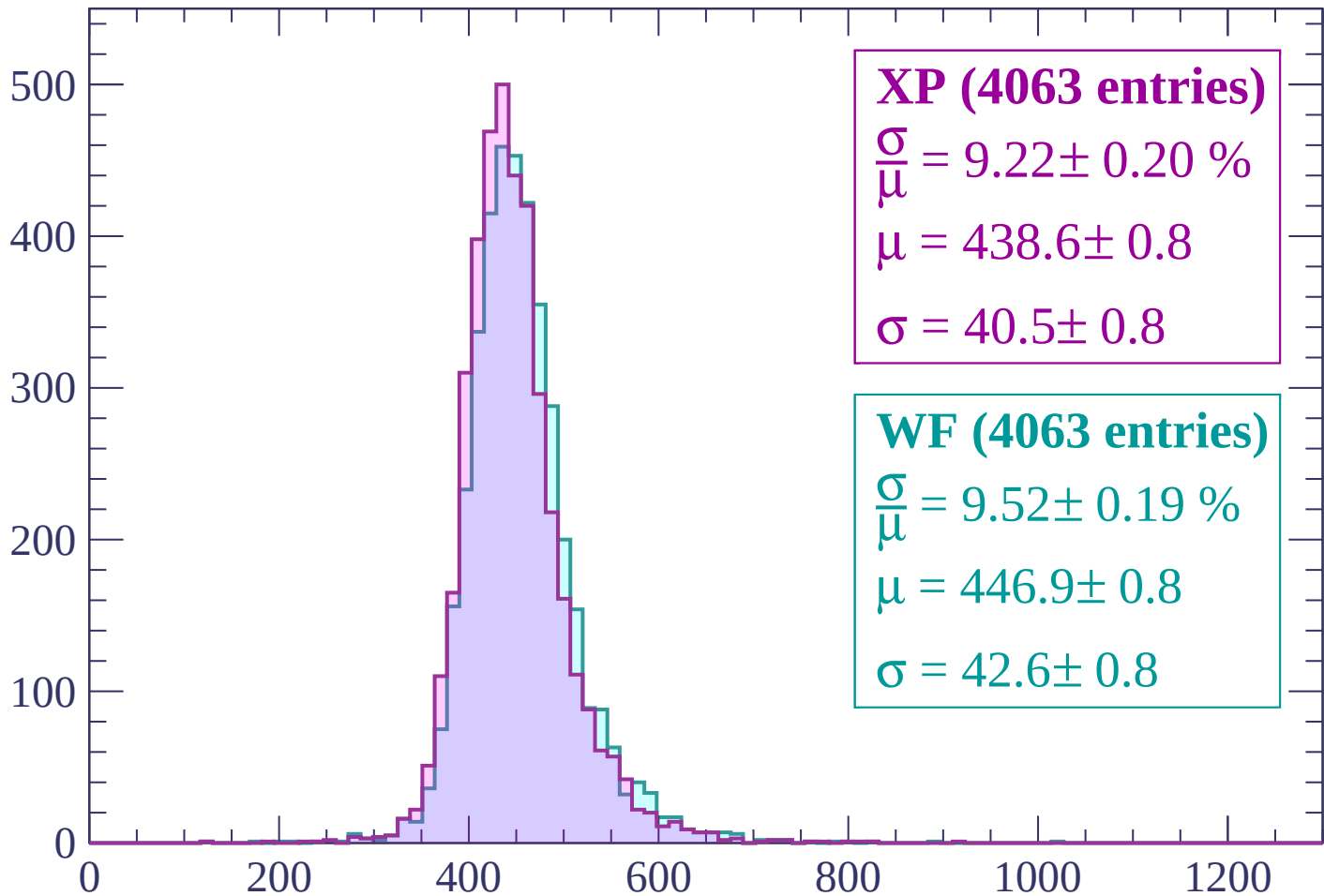
Energy loss | $1200 < p < 1260$

Count



Energy loss | $1260 < p < 1320$

Count



XP (4063 entries)

$\frac{\sigma}{\mu} = 9.22 \pm 0.20 \%$

$\mu = 438.6 \pm 0.8$

$\sigma = 40.5 \pm 0.8$

WF (4063 entries)

$\frac{\sigma}{\mu} = 9.52 \pm 0.19 \%$

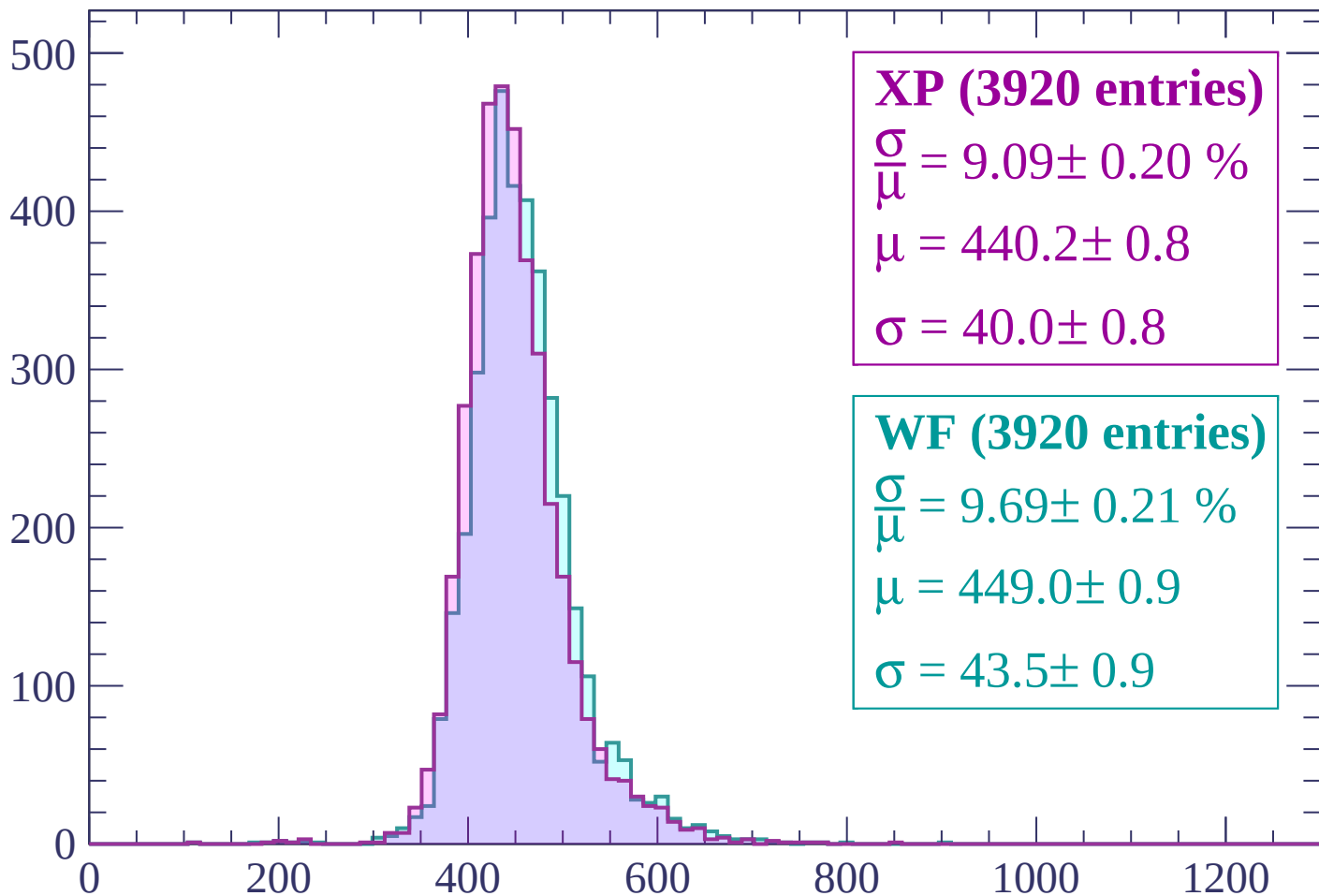
$\mu = 446.9 \pm 0.8$

$\sigma = 42.6 \pm 0.8$

dE/dx (ADC counts/cm)

Energy loss | $1320 < p < 1380$

Count



XP (3920 entries)

$$\frac{\sigma}{\mu} = 9.09 \pm 0.20 \%$$

$$\mu = 440.2 \pm 0.8$$

$$\sigma = 40.0 \pm 0.8$$

WF (3920 entries)

$$\frac{\sigma}{\mu} = 9.69 \pm 0.21 \%$$

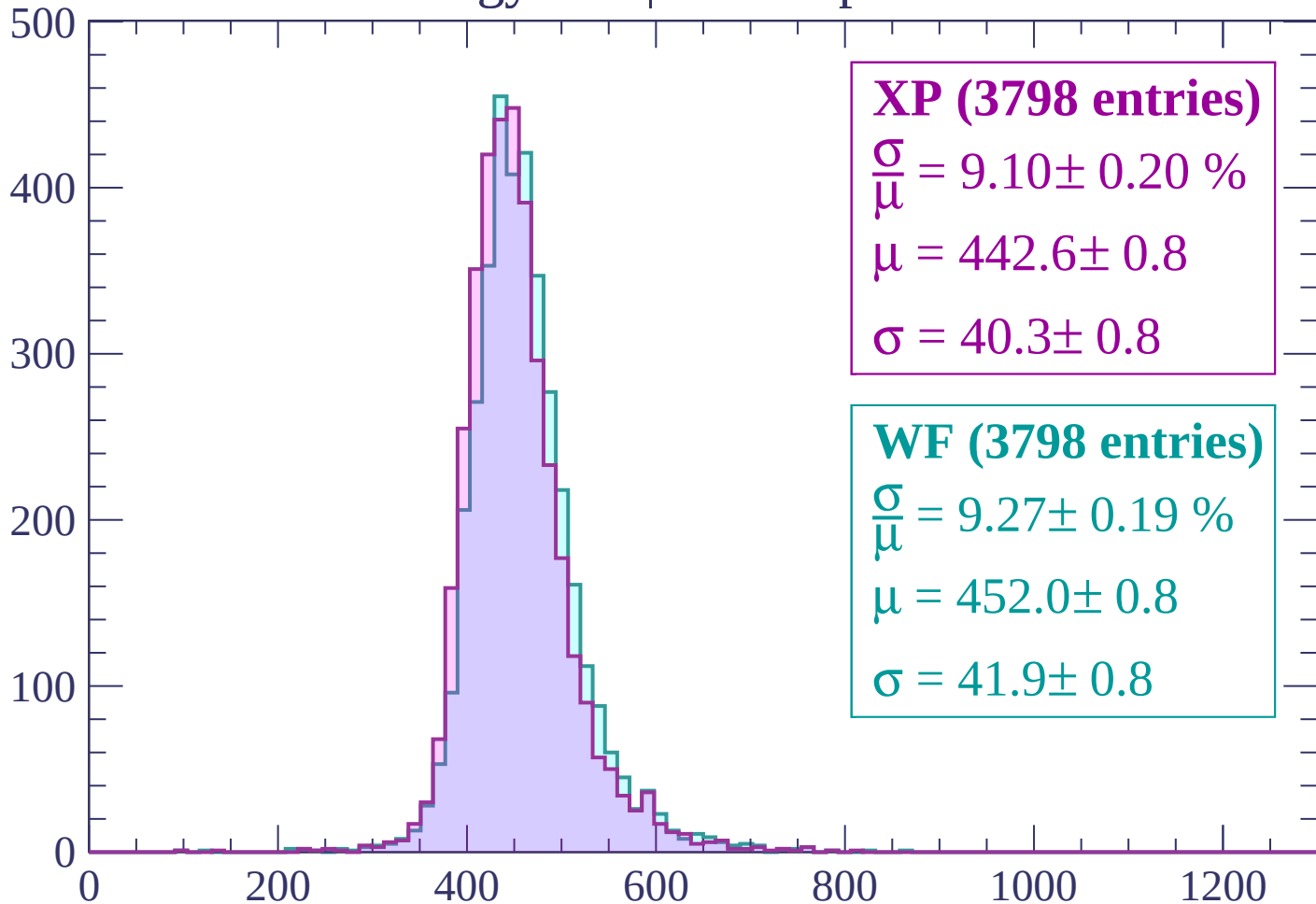
$$\mu = 449.0 \pm 0.9$$

$$\sigma = 43.5 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $1380 < p < 1440$

Count



XP (3798 entries)

$\frac{\sigma}{\mu} = 9.10 \pm 0.20 \%$

$\mu = 442.6 \pm 0.8$

$\sigma = 40.3 \pm 0.8$

WF (3798 entries)

$\frac{\sigma}{\mu} = 9.27 \pm 0.19 \%$

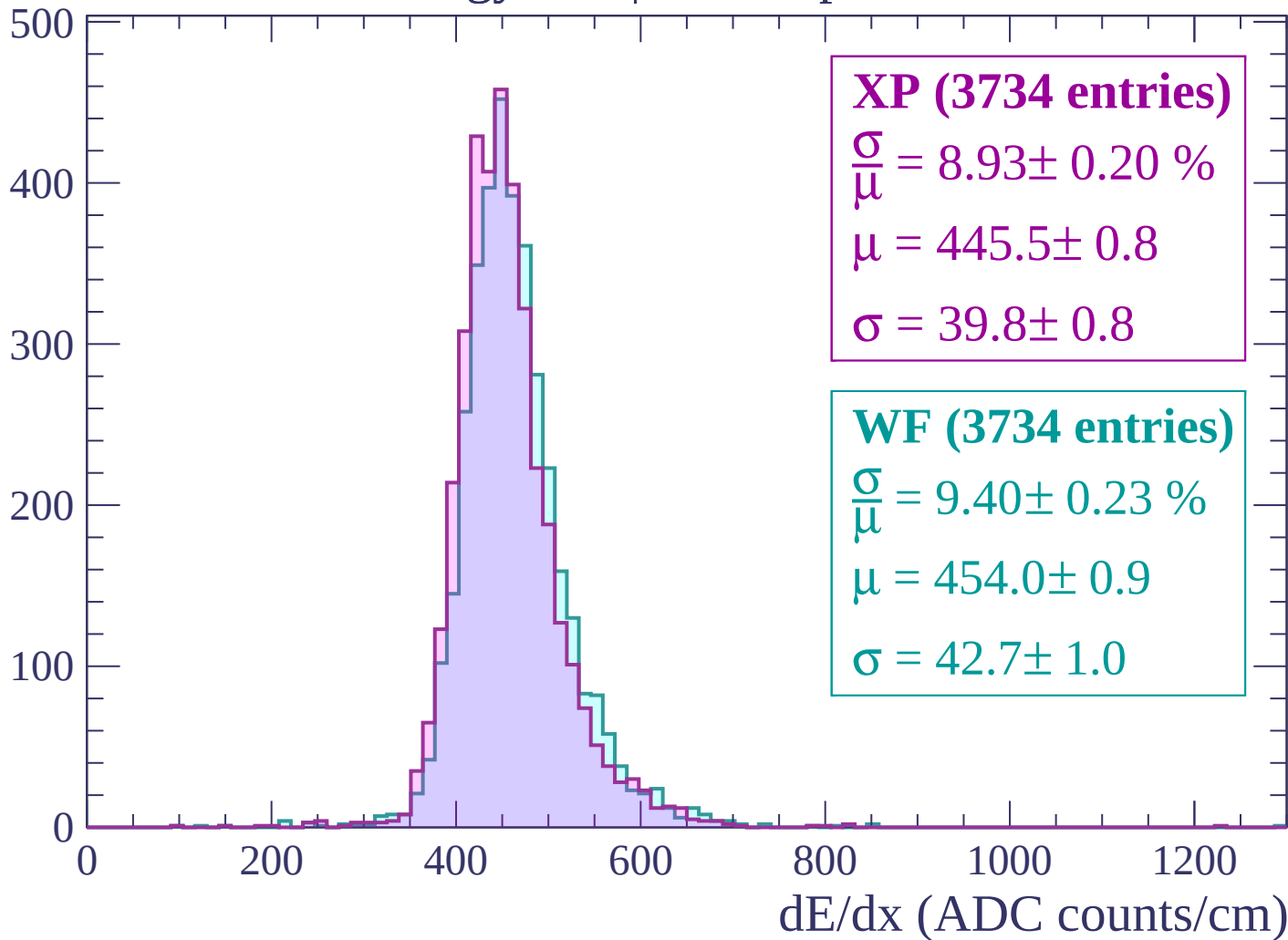
$\mu = 452.0 \pm 0.8$

$\sigma = 41.9 \pm 0.8$

dE/dx (ADC counts/cm)

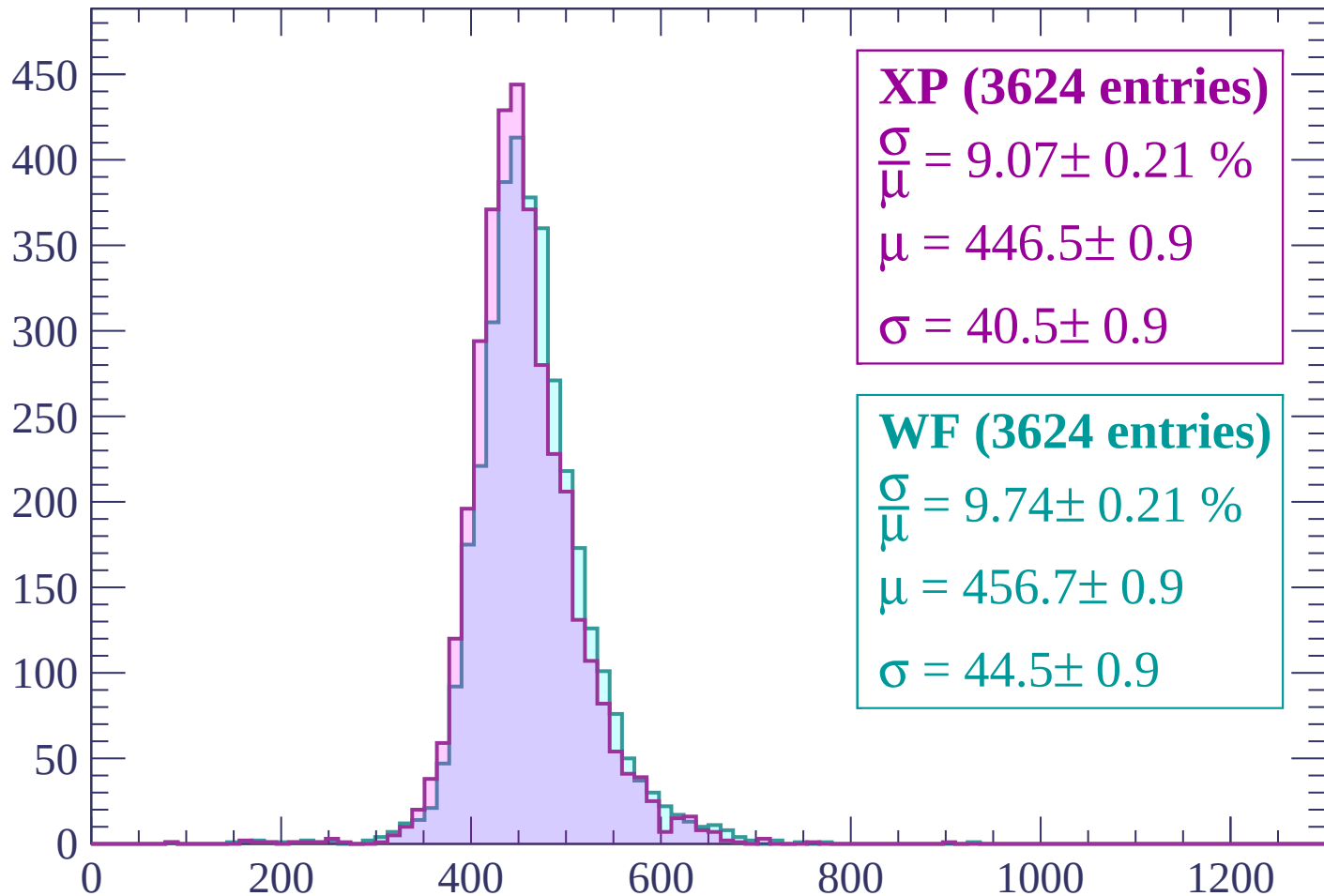
Energy loss | $1440 < p < 1500$

Count



Energy loss | $1500 < p < 1560$

Count



XP (3624 entries)

$$\frac{\sigma}{\mu} = 9.07 \pm 0.21 \%$$

$$\mu = 446.5 \pm 0.9$$

$$\sigma = 40.5 \pm 0.9$$

WF (3624 entries)

$$\frac{\sigma}{\mu} = 9.74 \pm 0.21 \%$$

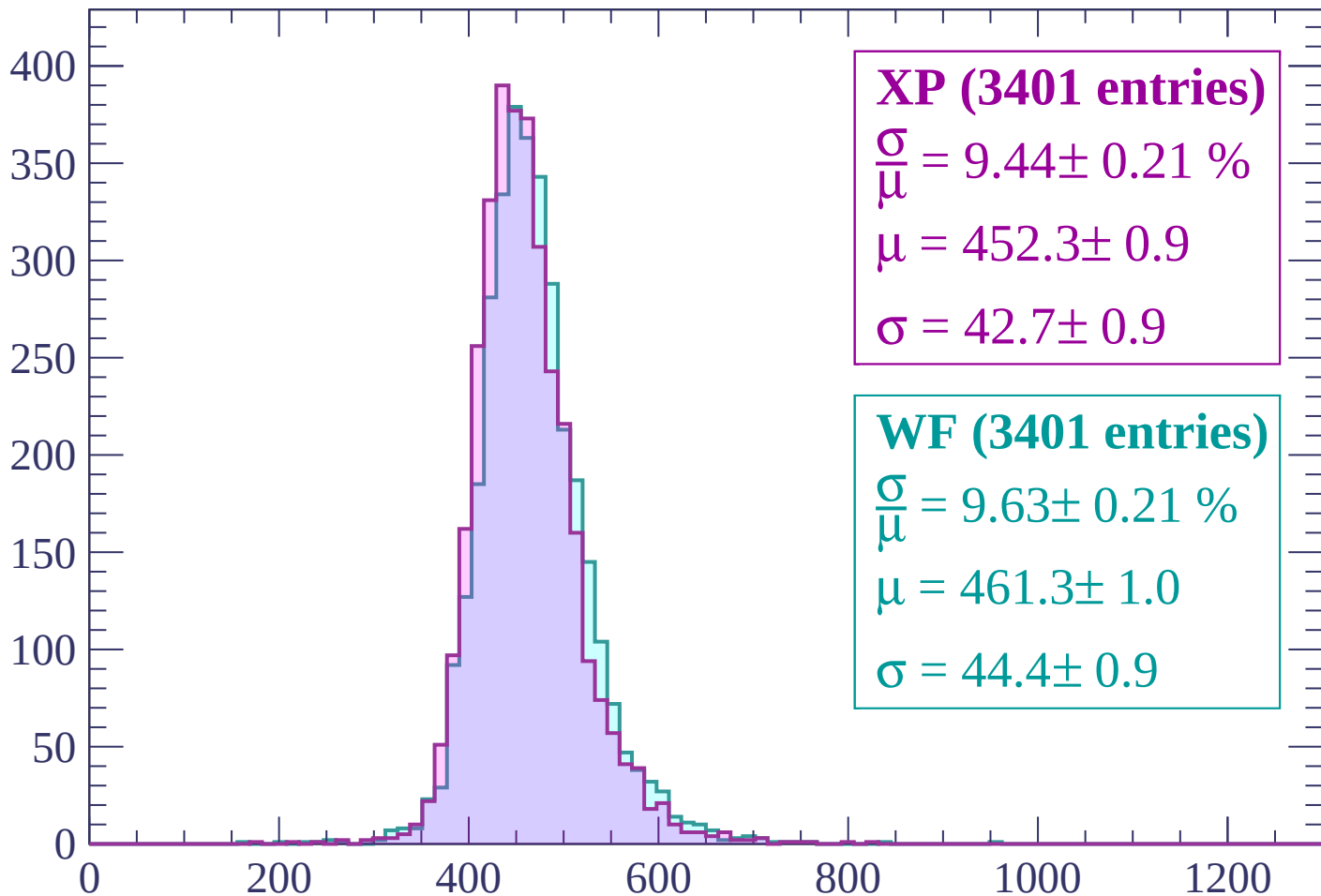
$$\mu = 456.7 \pm 0.9$$

$$\sigma = 44.5 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $1560 < p < 1620$

Count



XP (3401 entries)

$$\frac{\sigma}{\mu} = 9.44 \pm 0.21 \%$$

$$\mu = 452.3 \pm 0.9$$

$$\sigma = 42.7 \pm 0.9$$

WF (3401 entries)

$$\frac{\sigma}{\mu} = 9.63 \pm 0.21 \%$$

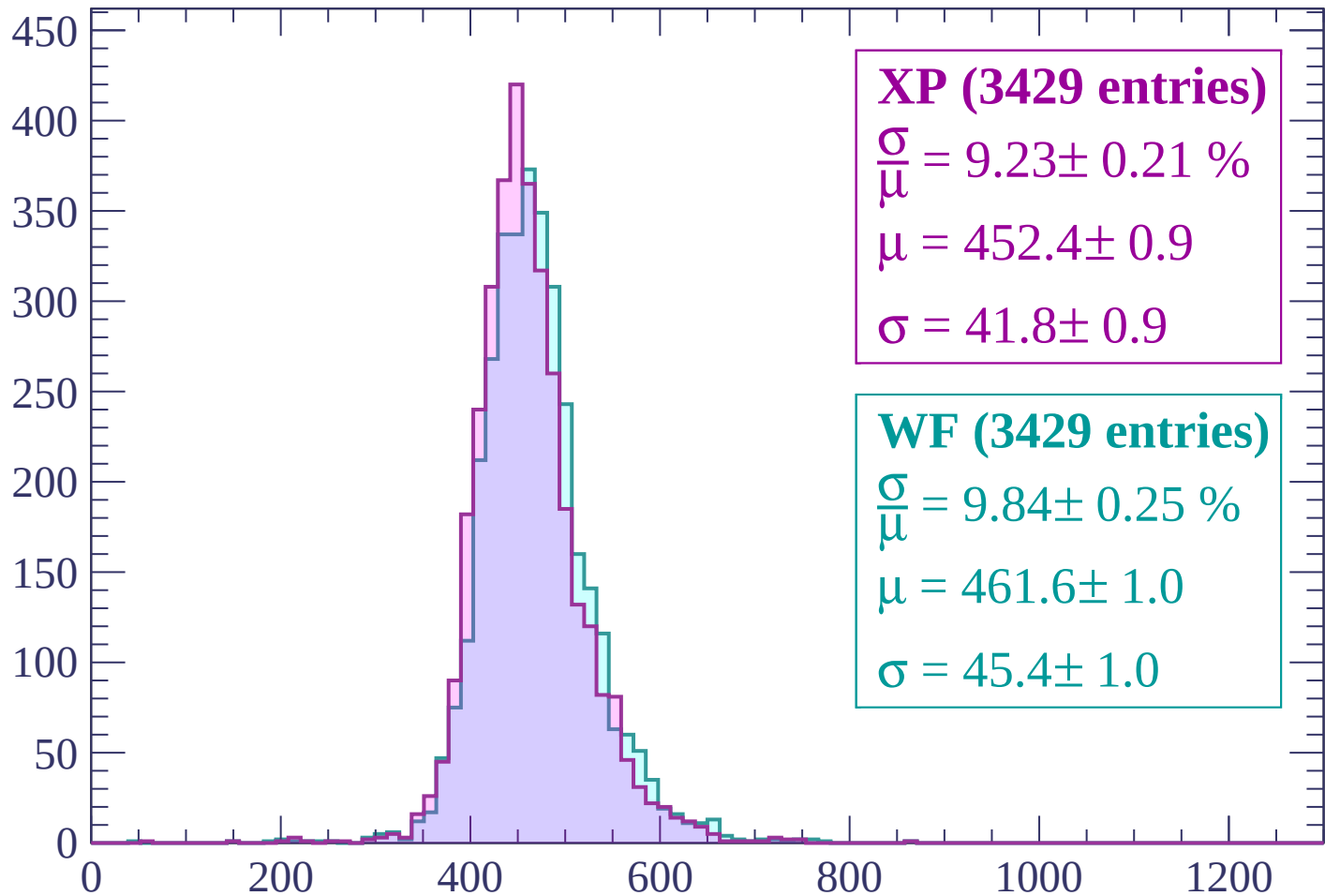
$$\mu = 461.3 \pm 1.0$$

$$\sigma = 44.4 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $1620 < p < 1680$

Count



XP (3429 entries)

$\frac{\sigma}{\mu} = 9.23 \pm 0.21 \%$

$\mu = 452.4 \pm 0.9$

$\sigma = 41.8 \pm 0.9$

WF (3429 entries)

$\frac{\sigma}{\mu} = 9.84 \pm 0.25 \%$

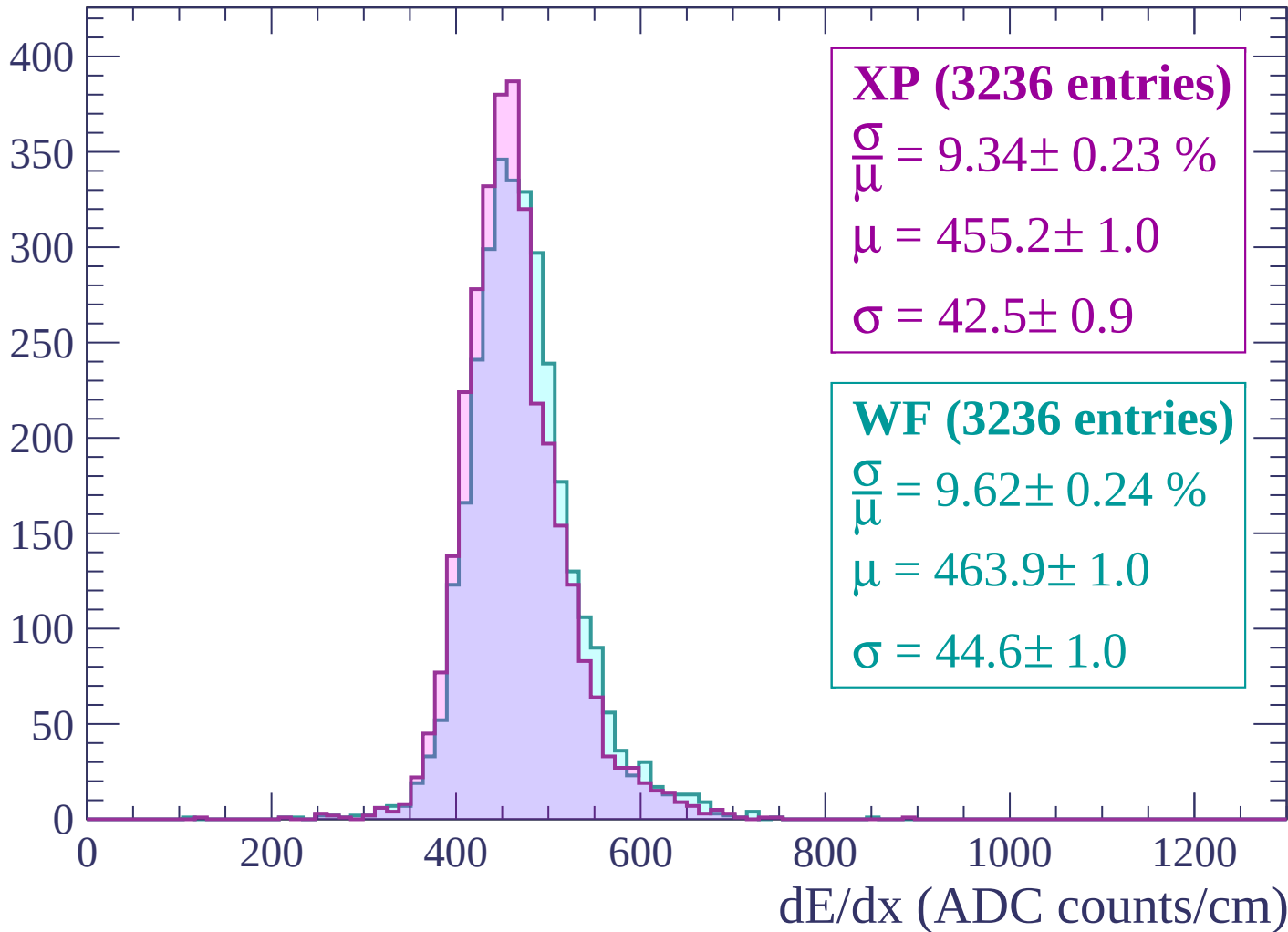
$\mu = 461.6 \pm 1.0$

$\sigma = 45.4 \pm 1.0$

dE/dx (ADC counts/cm)

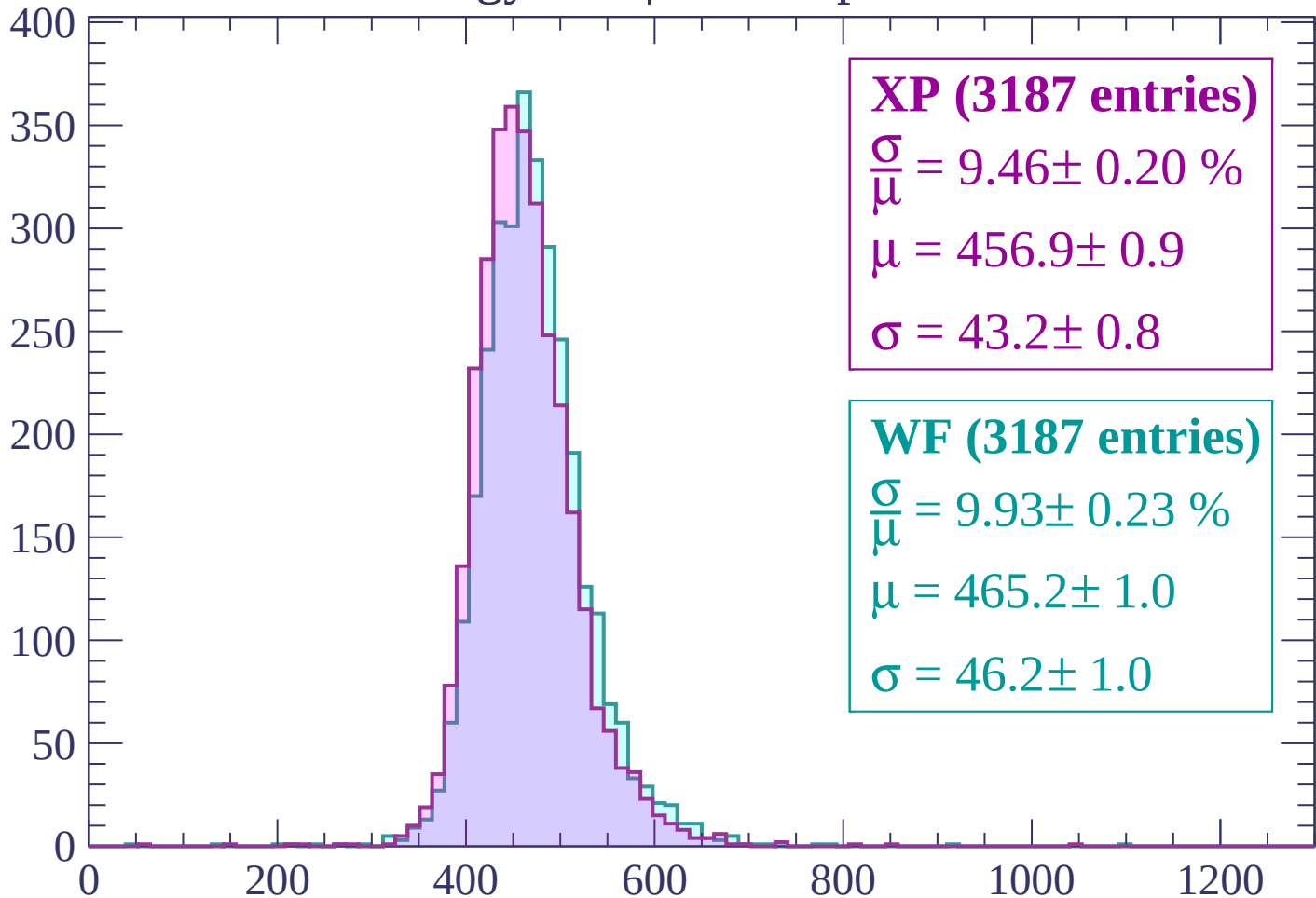
Energy loss | $1680 < p < 1740$

Count



Energy loss | $1740 < p < 1800$

Count



XP (3187 entries)

$$\frac{\sigma}{\mu} = 9.46 \pm 0.20 \%$$

$$\mu = 456.9 \pm 0.9$$

$$\sigma = 43.2 \pm 0.8$$

WF (3187 entries)

$$\frac{\sigma}{\mu} = 9.93 \pm 0.23 \%$$

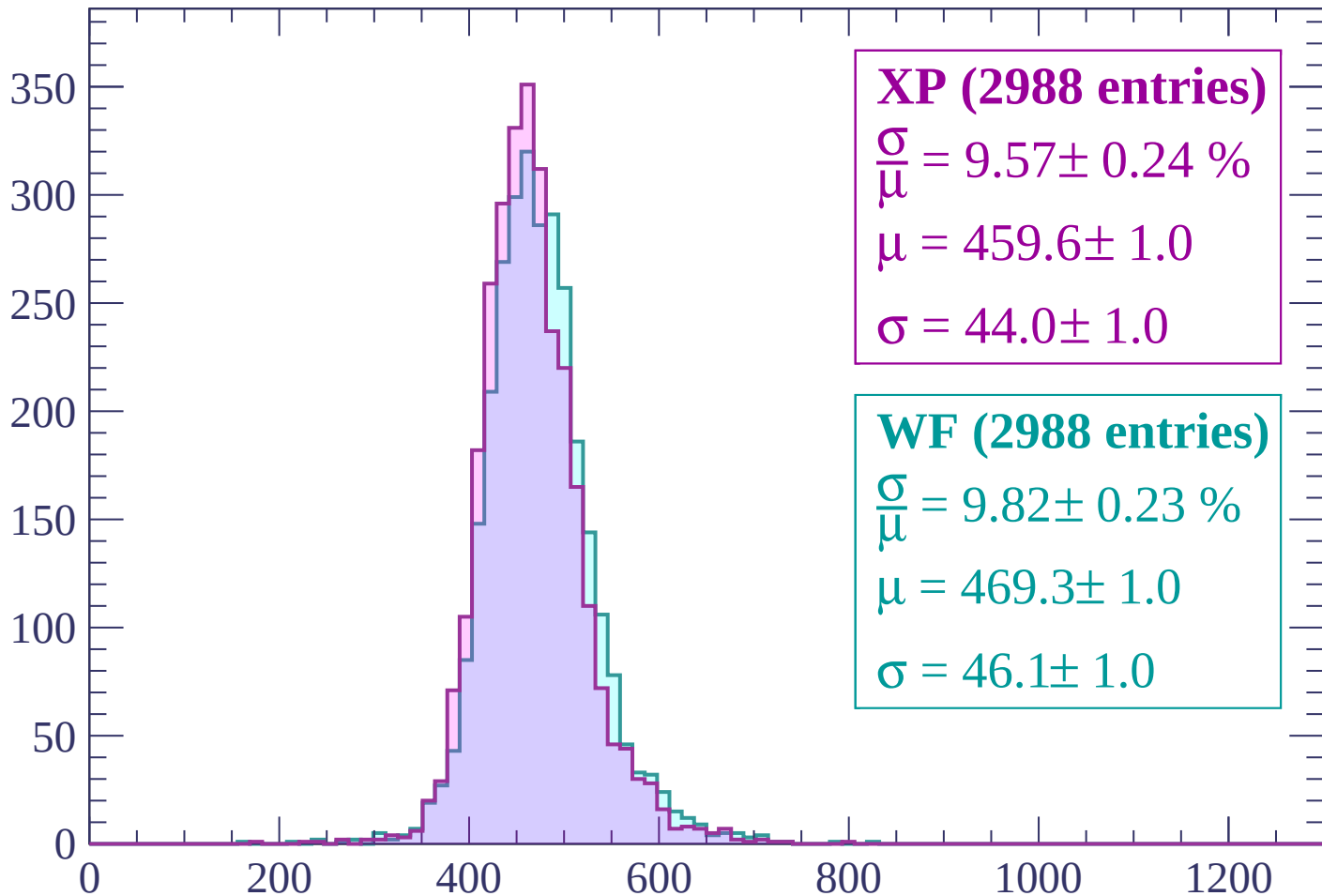
$$\mu = 465.2 \pm 1.0$$

$$\sigma = 46.2 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $1800 < p < 1860$

Count



XP (2988 entries)

$$\frac{\sigma}{\mu} = 9.57 \pm 0.24 \%$$

$$\mu = 459.6 \pm 1.0$$

$$\sigma = 44.0 \pm 1.0$$

WF (2988 entries)

$$\frac{\sigma}{\mu} = 9.82 \pm 0.23 \%$$

$$\mu = 469.3 \pm 1.0$$

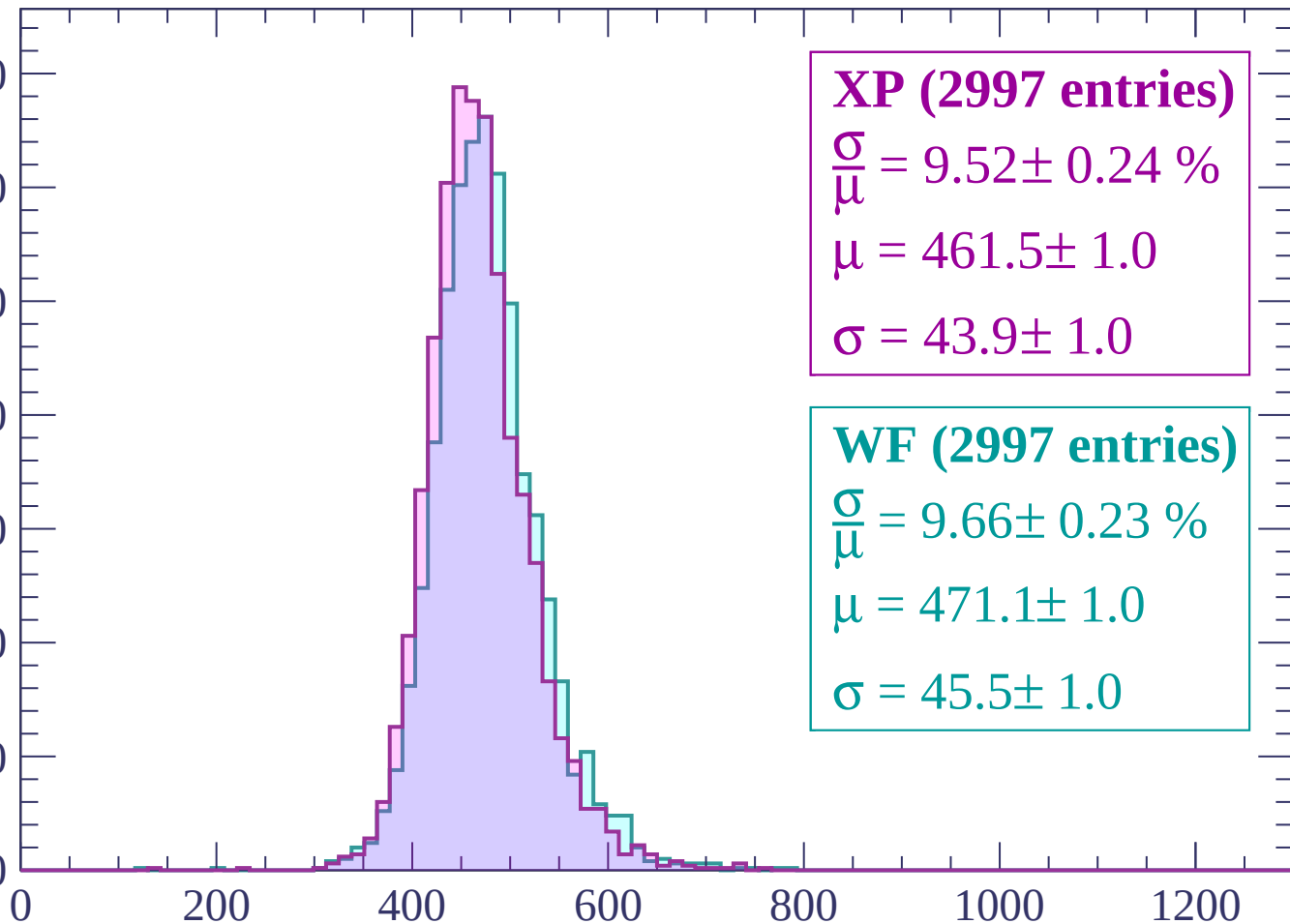
$$\sigma = 46.1 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $1860 < p < 1920$

Count

350
300
250
200
150
100
50
0



XP (2997 entries)

$$\frac{\sigma}{\mu} = 9.52 \pm 0.24 \%$$

$$\mu = 461.5 \pm 1.0$$

$$\sigma = 43.9 \pm 1.0$$

WF (2997 entries)

$$\frac{\sigma}{\mu} = 9.66 \pm 0.23 \%$$

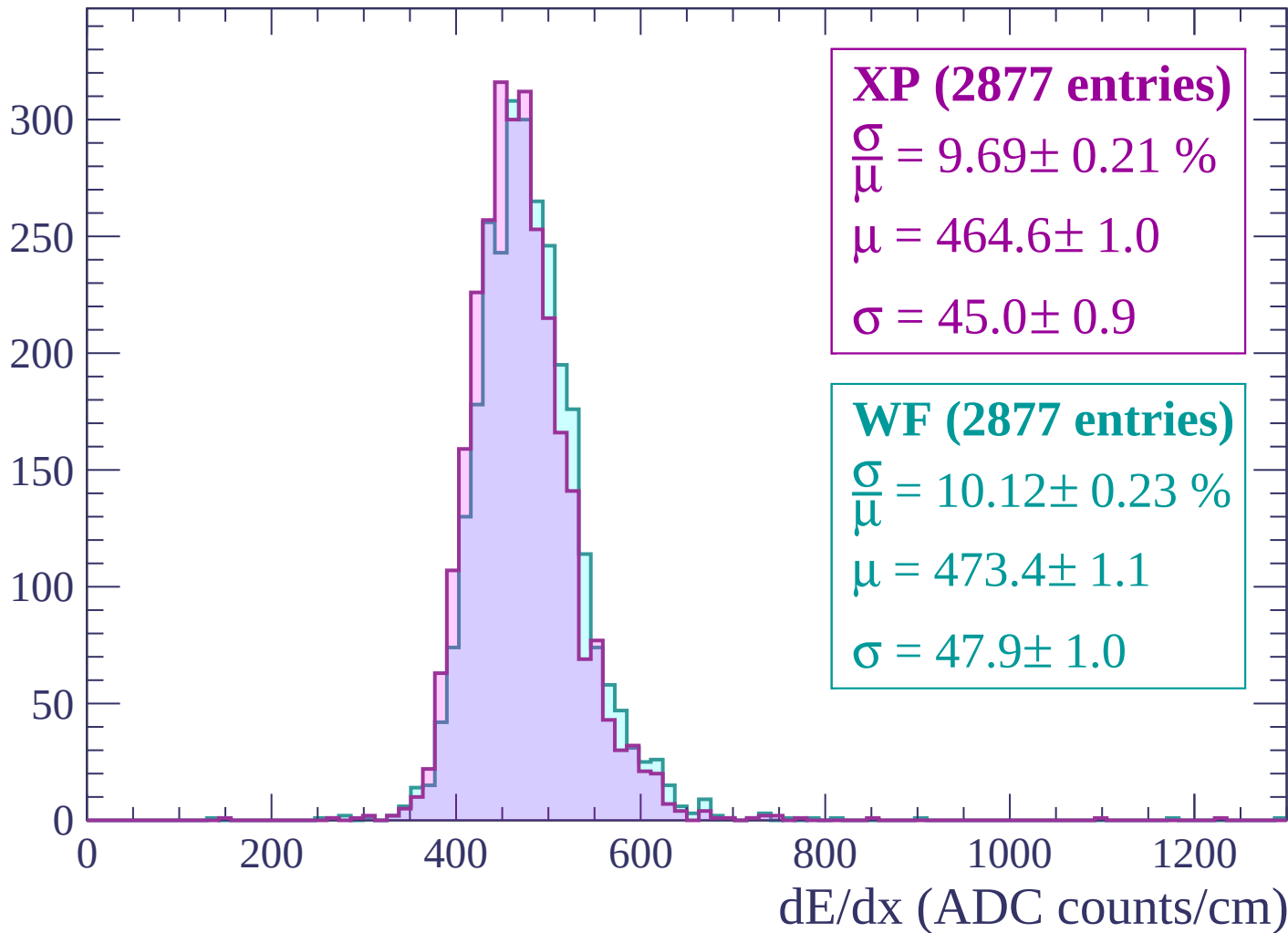
$$\mu = 471.1 \pm 1.0$$

$$\sigma = 45.5 \pm 1.0$$

dE/dx (ADC counts/cm)

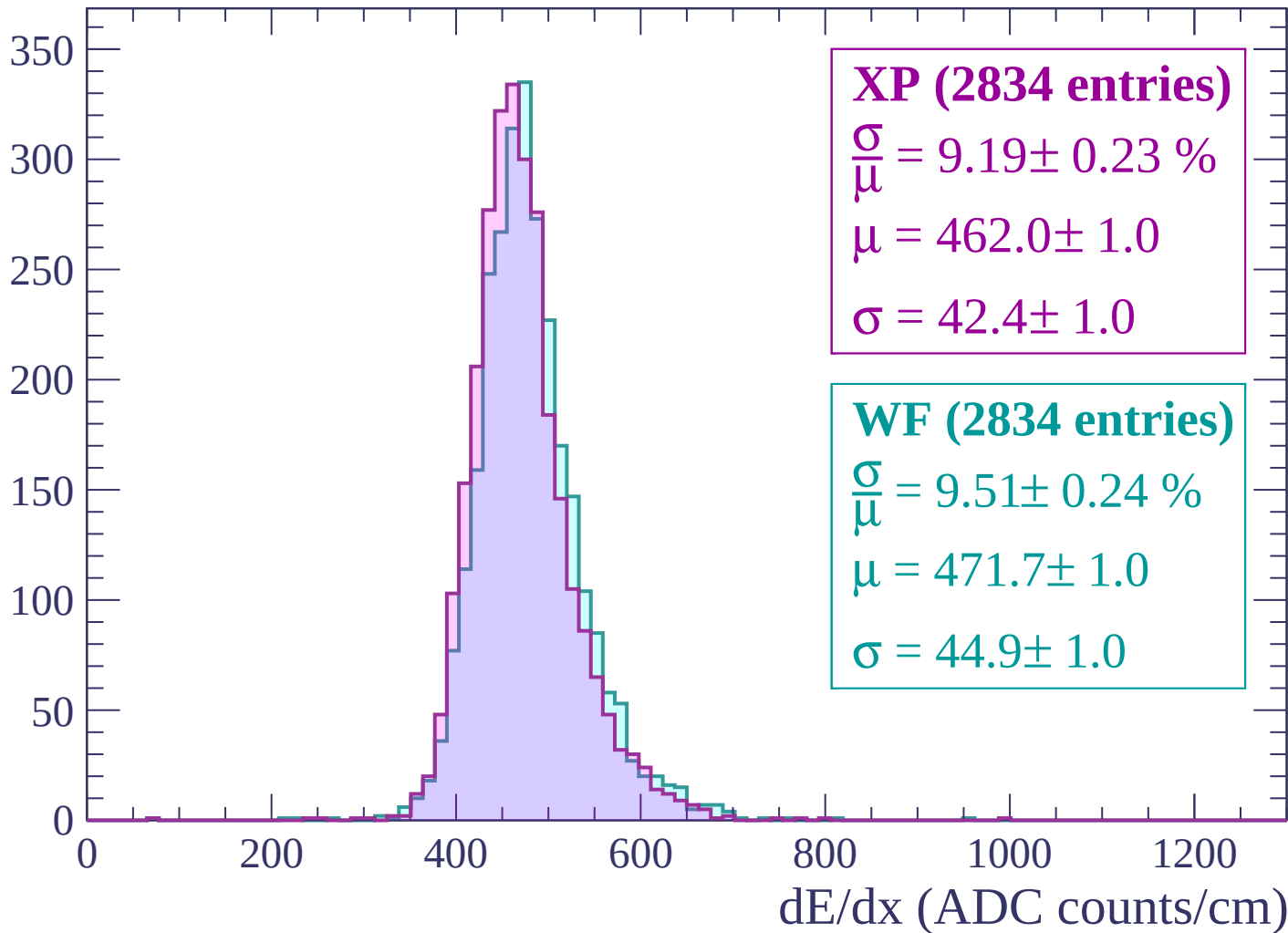
Energy loss | $1920 < p < 1980$

Count



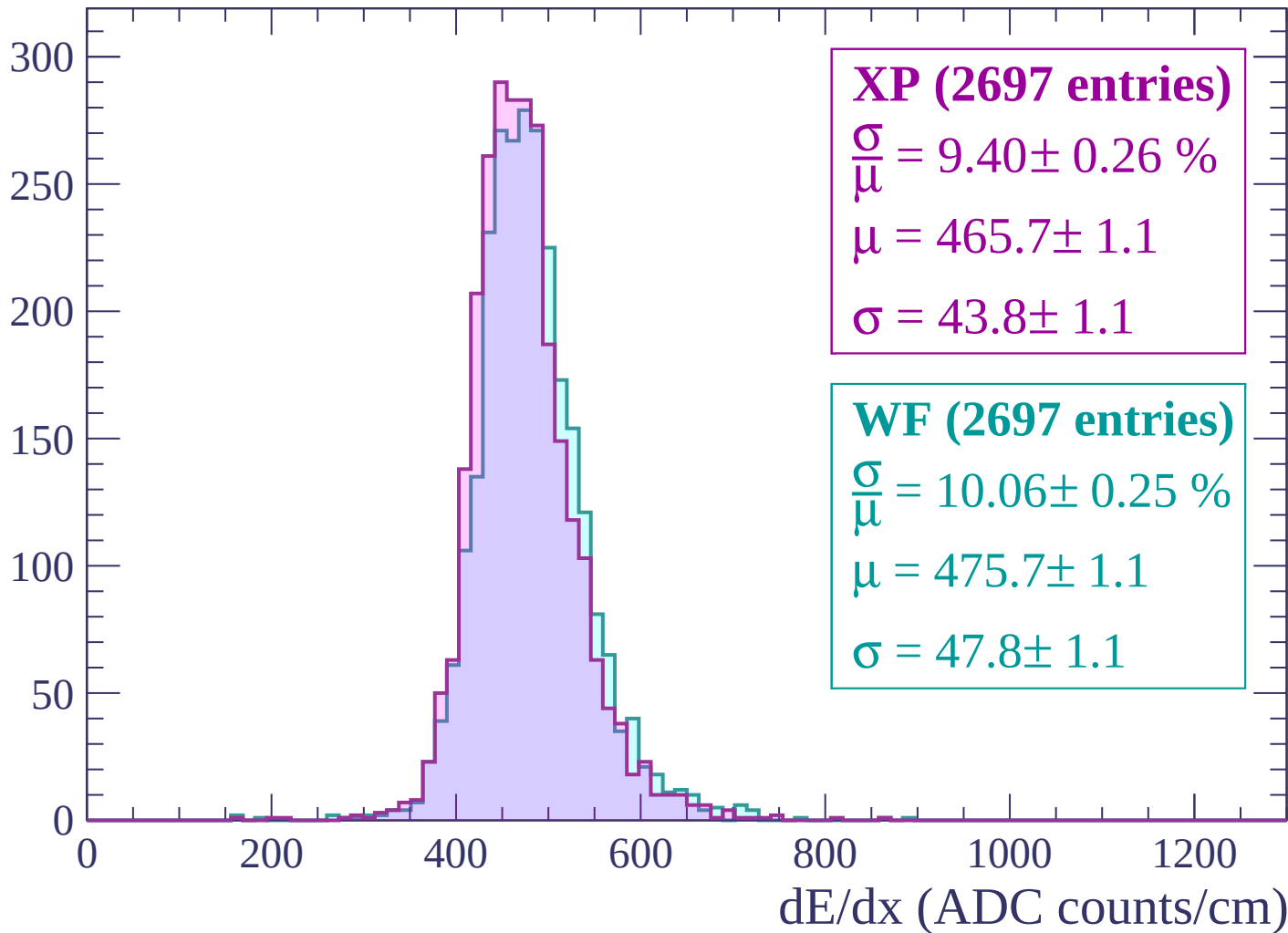
Energy loss | $1980 < p < 2040$

Count



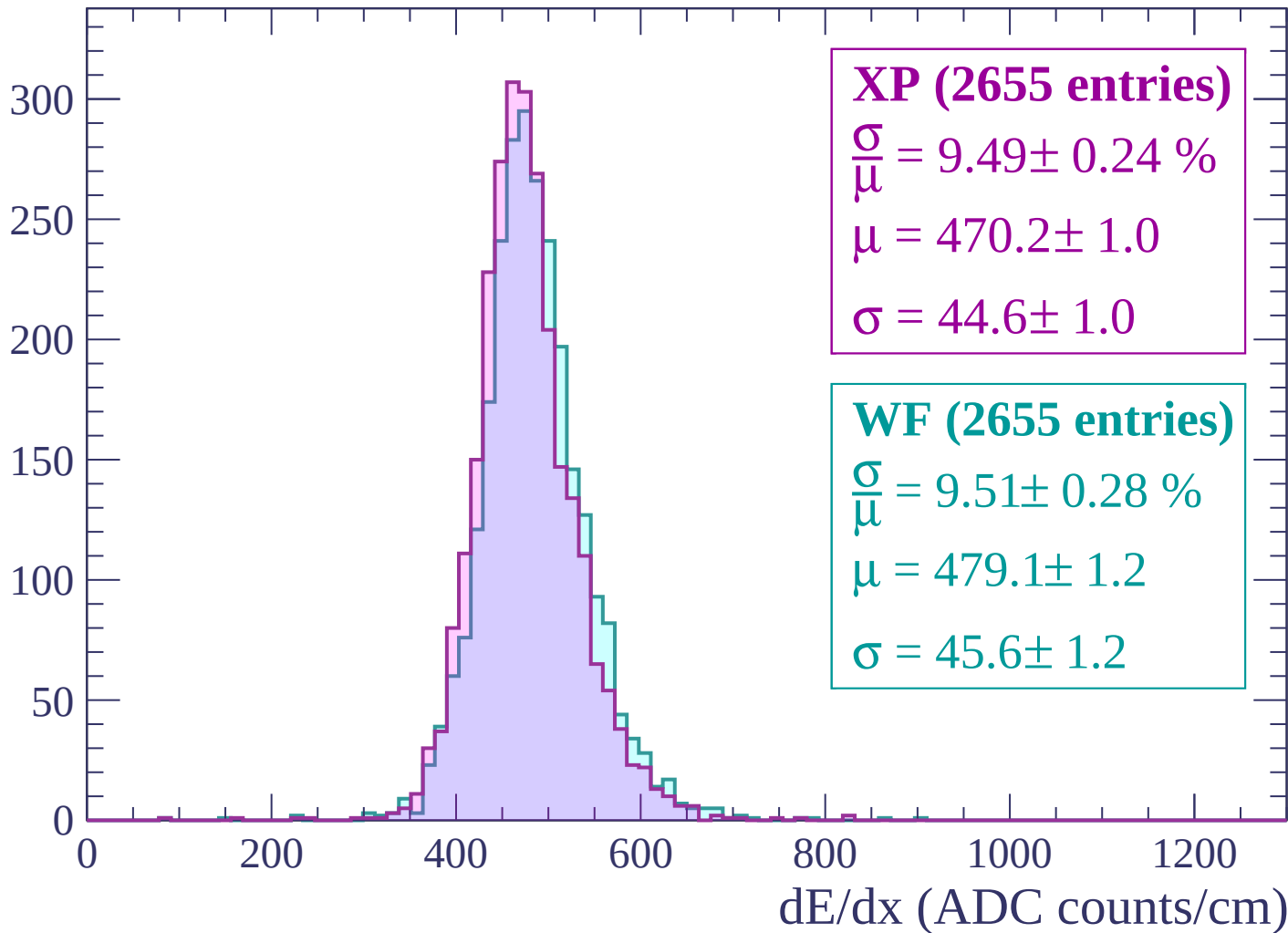
Energy loss | $2040 < p < 2100$

Count



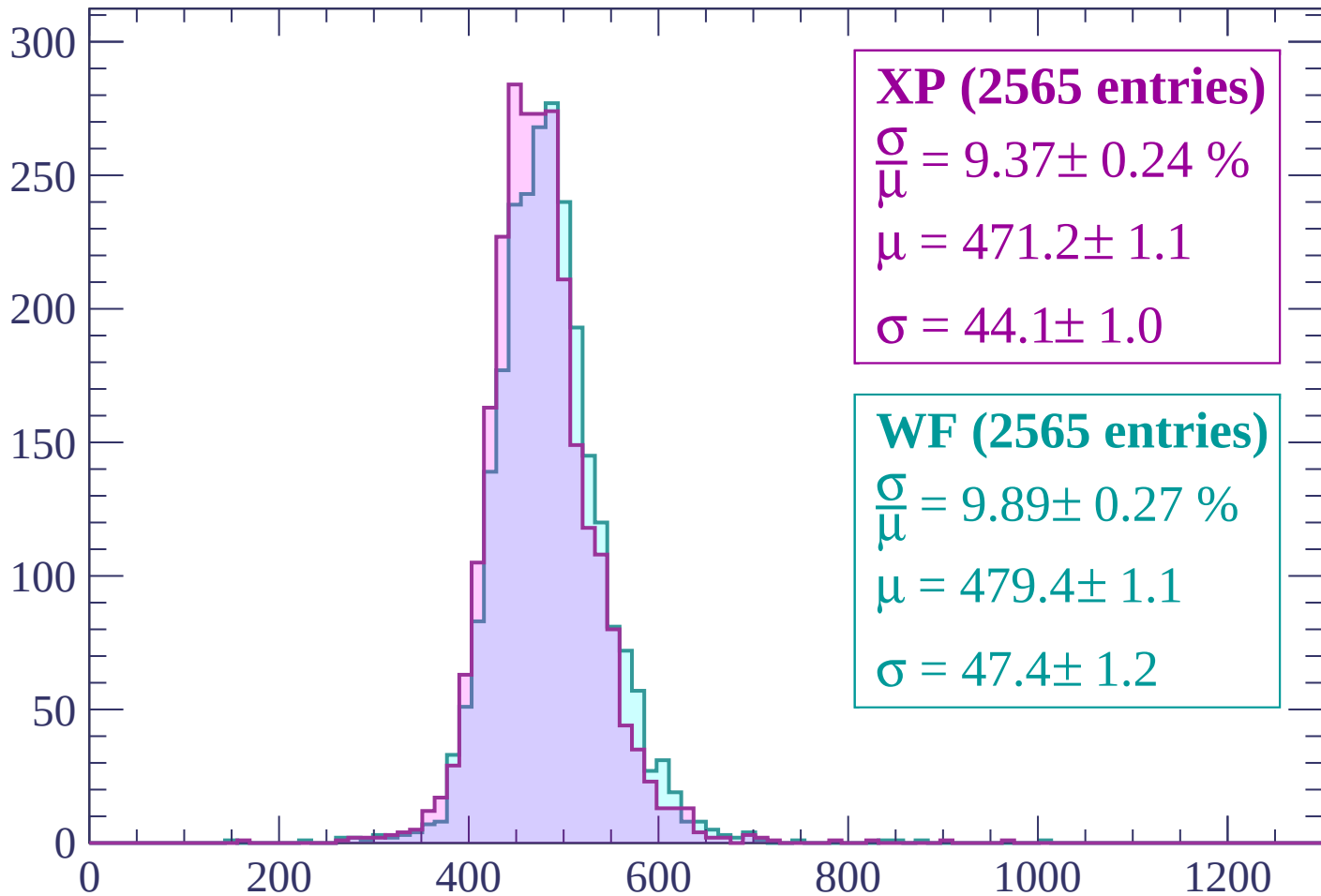
Energy loss | $2100 < p < 2160$

Count



Energy loss | $2160 < p < 2220$

Count



XP (2565 entries)

$$\frac{\sigma}{\mu} = 9.37 \pm 0.24 \%$$

$$\mu = 471.2 \pm 1.1$$

$$\sigma = 44.1 \pm 1.0$$

WF (2565 entries)

$$\frac{\sigma}{\mu} = 9.89 \pm 0.27 \%$$

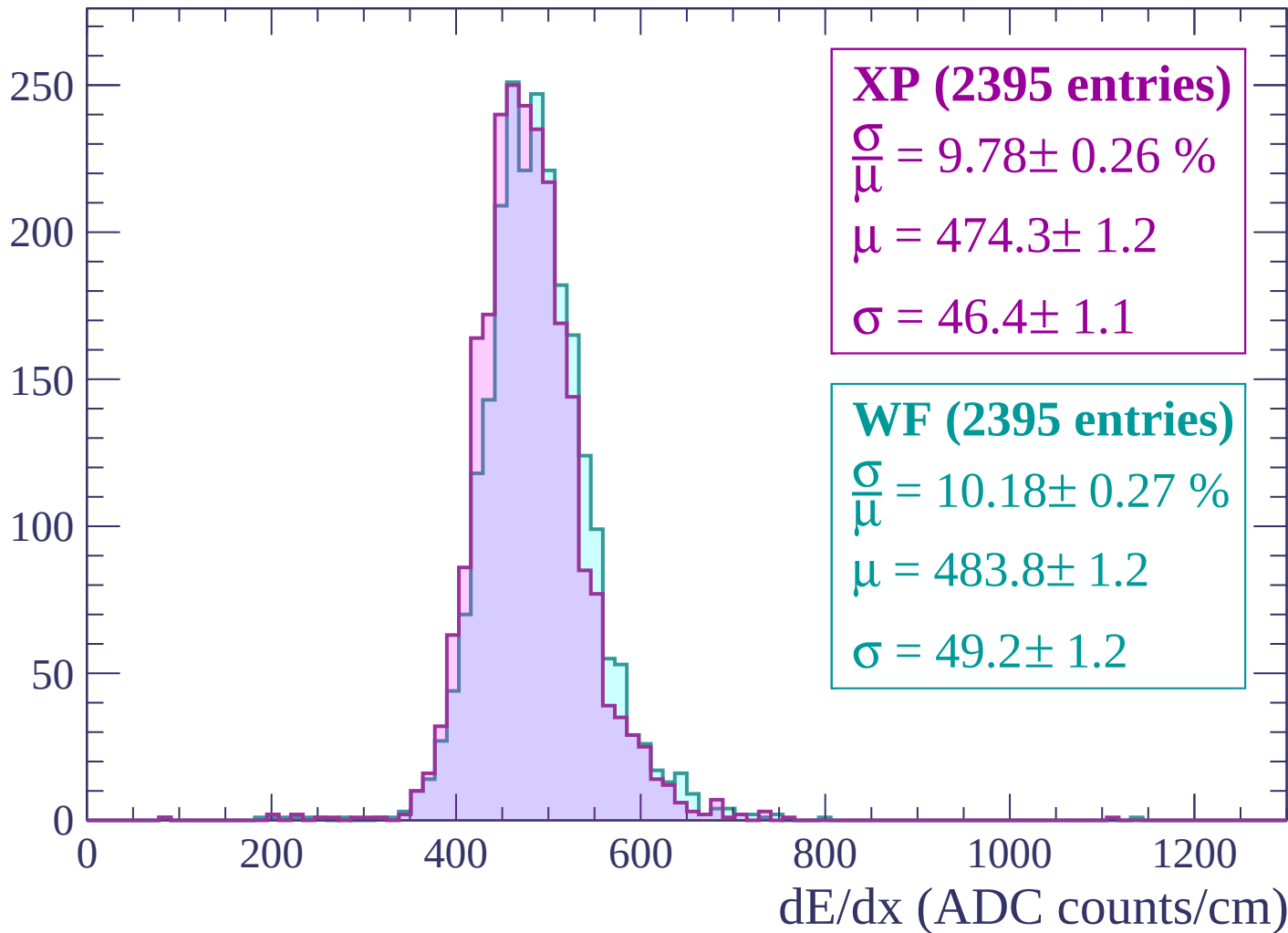
$$\mu = 479.4 \pm 1.1$$

$$\sigma = 47.4 \pm 1.2$$

dE/dx (ADC counts/cm)

Energy loss | $2220 < p < 2280$

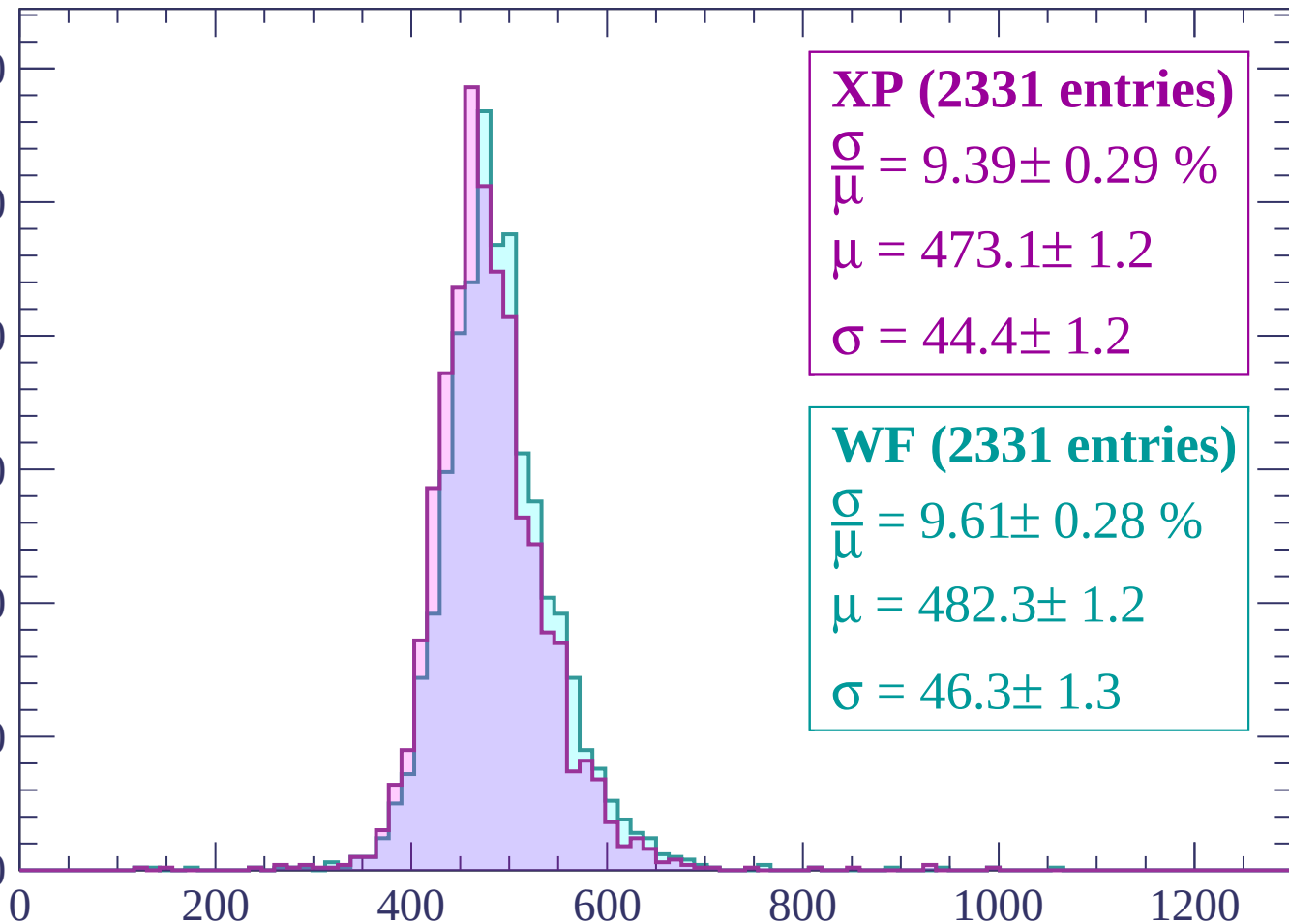
Count



Energy loss | $2280 < p < 2340$

Count

300
250
200
150
100
50
0



XP (2331 entries)

$$\frac{\sigma}{\mu} = 9.39 \pm 0.29 \%$$

$$\mu = 473.1 \pm 1.2$$

$$\sigma = 44.4 \pm 1.2$$

WF (2331 entries)

$$\frac{\sigma}{\mu} = 9.61 \pm 0.28 \%$$

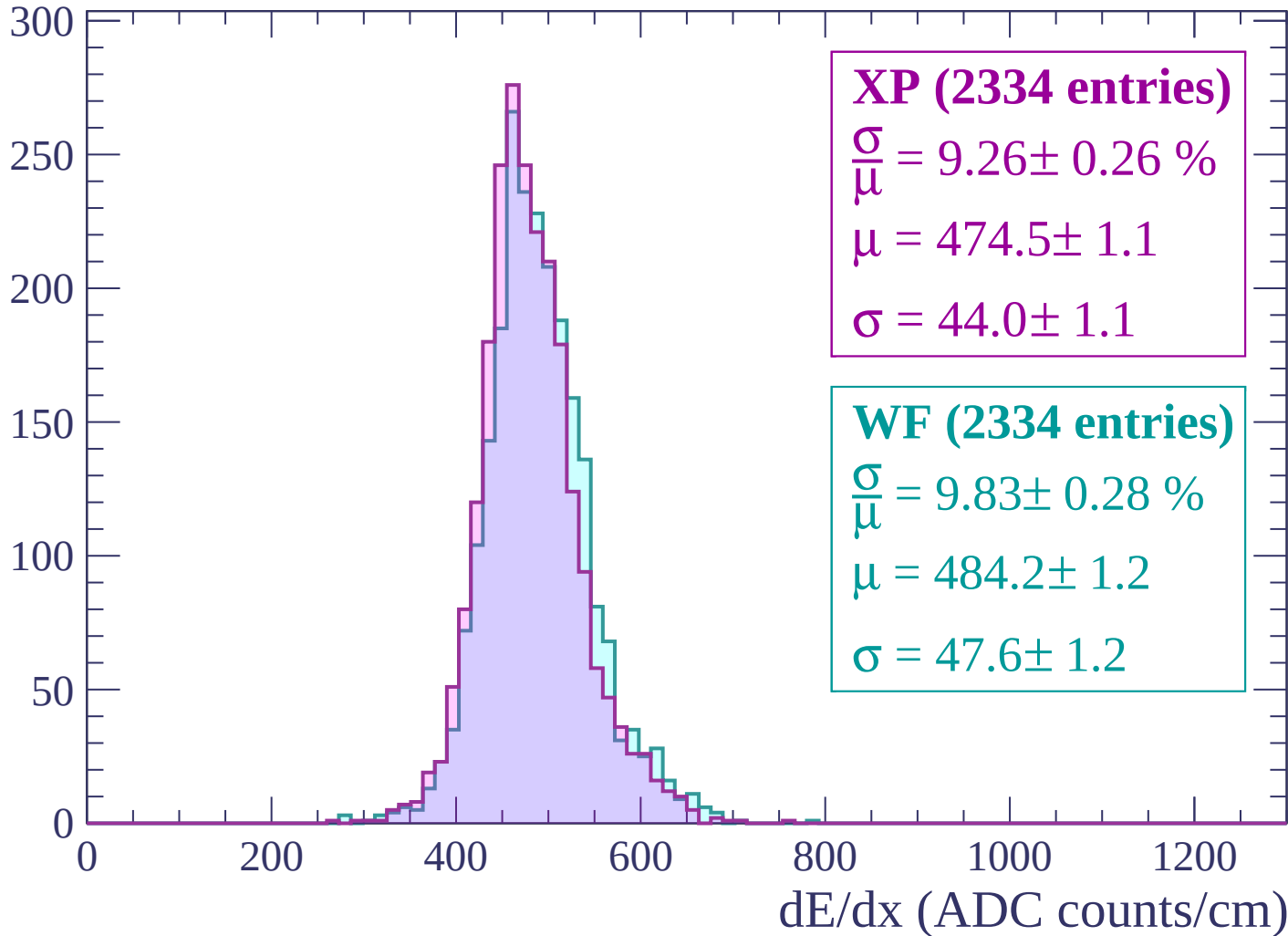
$$\mu = 482.3 \pm 1.2$$

$$\sigma = 46.3 \pm 1.3$$

dE/dx (ADC counts/cm)

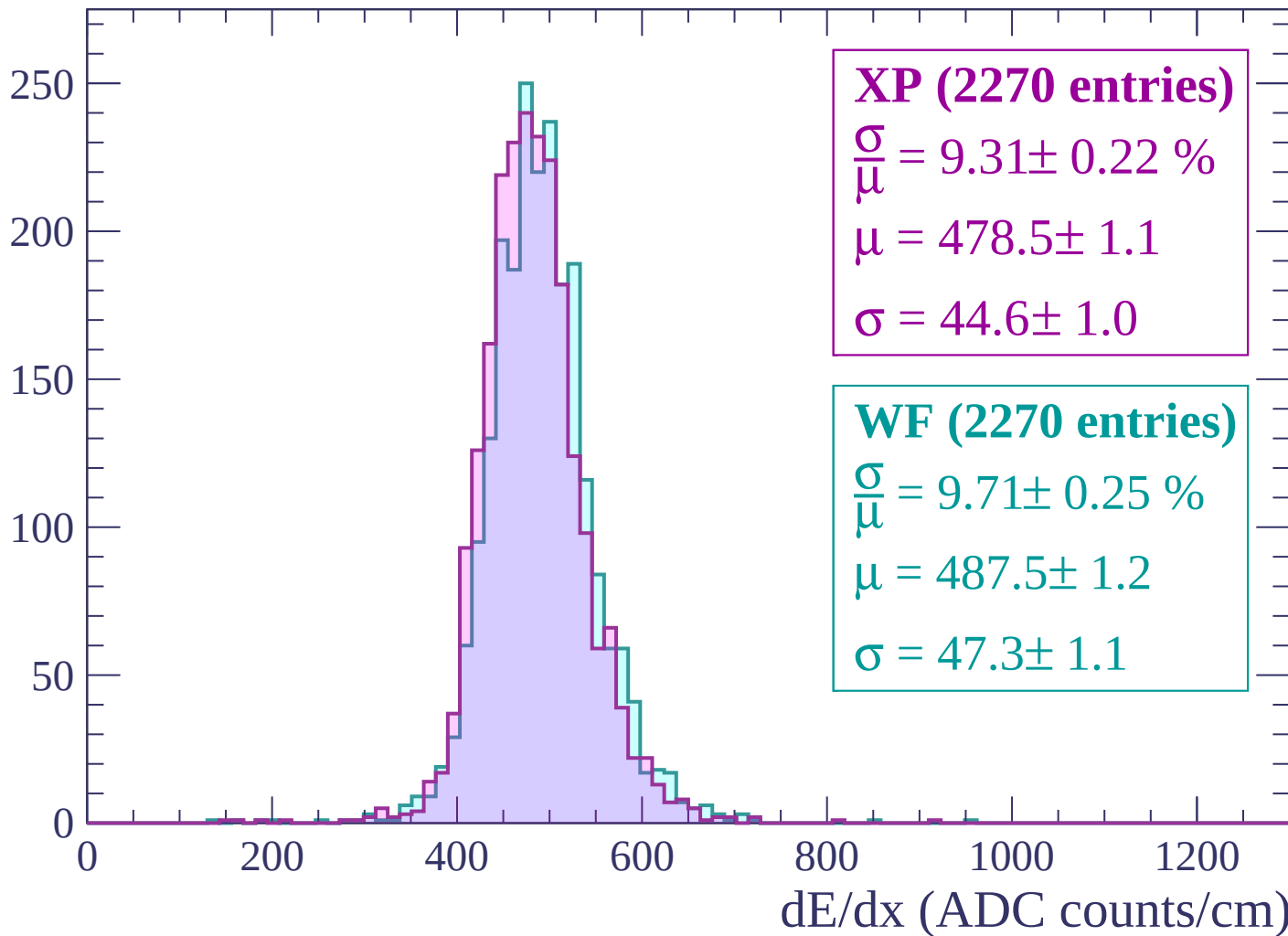
Energy loss | $2340 < p < 2400$

Count



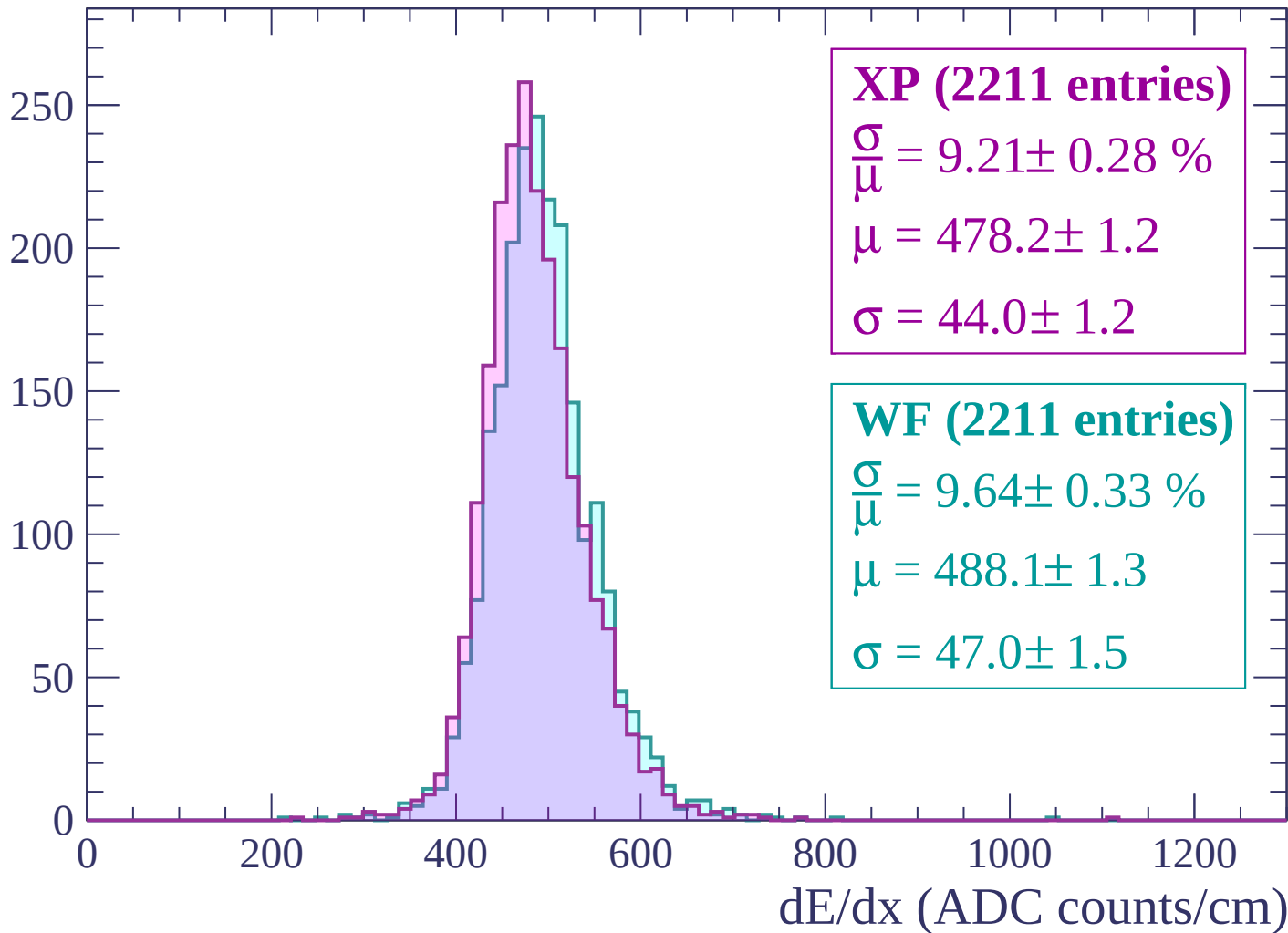
Energy loss | $2400 < p < 2460$

Count



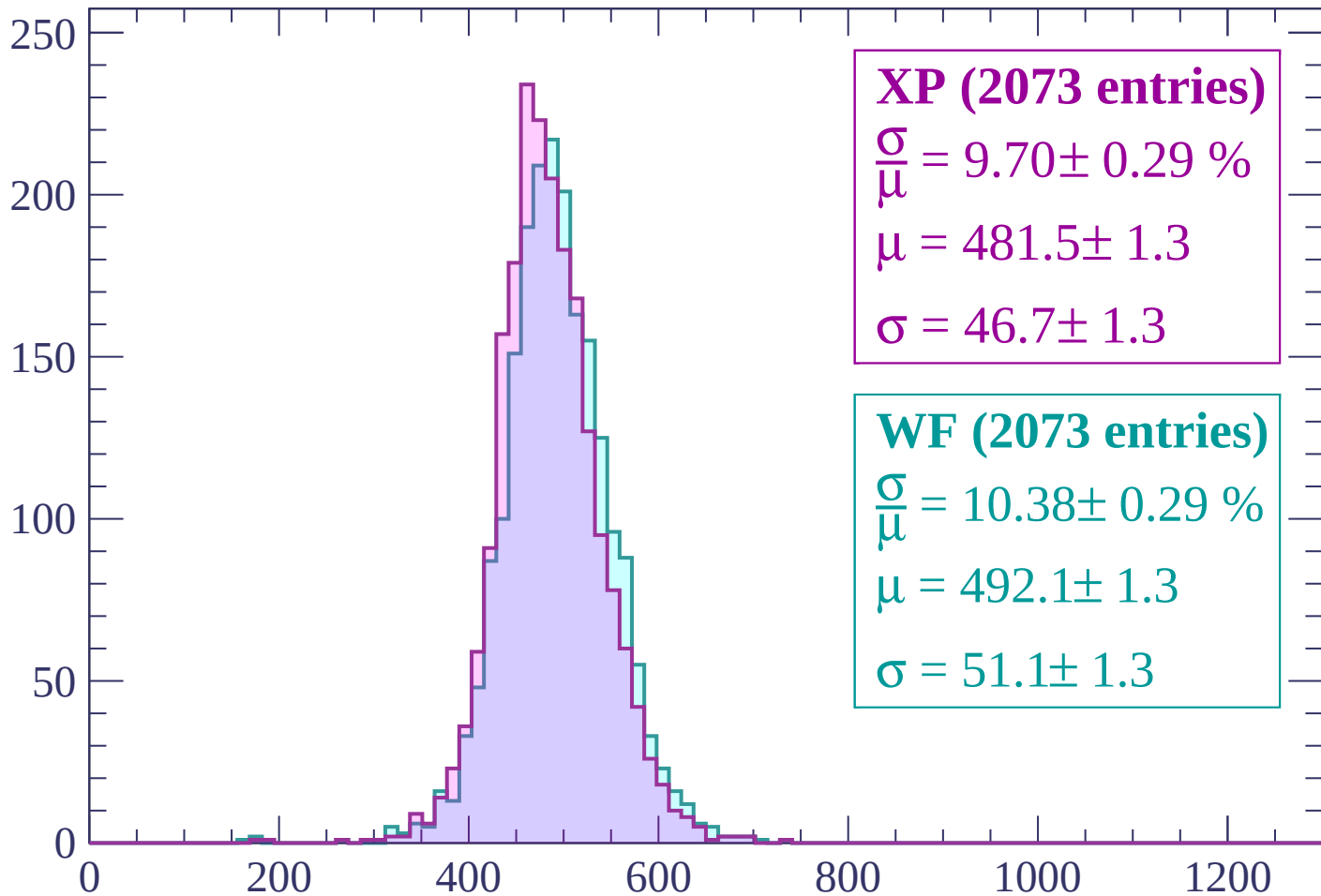
Energy loss | $2460 < p < 2520$

Count



Energy loss | $2520 < p < 2580$

Count



XP (2073 entries)

$$\frac{\sigma}{\mu} = 9.70 \pm 0.29 \%$$

$$\mu = 481.5 \pm 1.3$$

$$\sigma = 46.7 \pm 1.3$$

WF (2073 entries)

$$\frac{\sigma}{\mu} = 10.38 \pm 0.29 \%$$

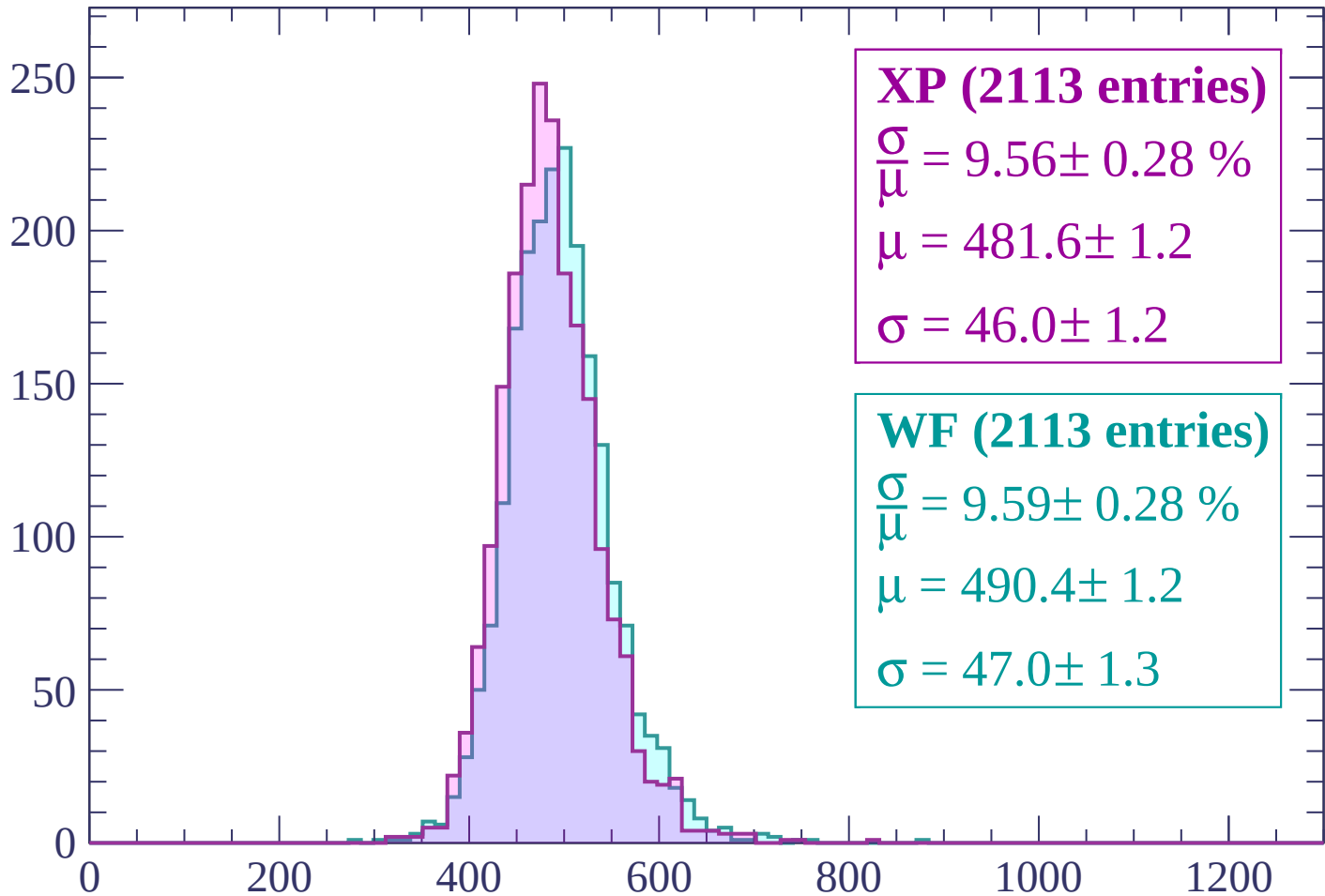
$$\mu = 492.1 \pm 1.3$$

$$\sigma = 51.1 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $2580 < p < 2640$

Count



XP (2113 entries)

$$\frac{\sigma}{\mu} = 9.56 \pm 0.28 \%$$

$$\mu = 481.6 \pm 1.2$$

$$\sigma = 46.0 \pm 1.2$$

WF (2113 entries)

$$\frac{\sigma}{\mu} = 9.59 \pm 0.28 \%$$

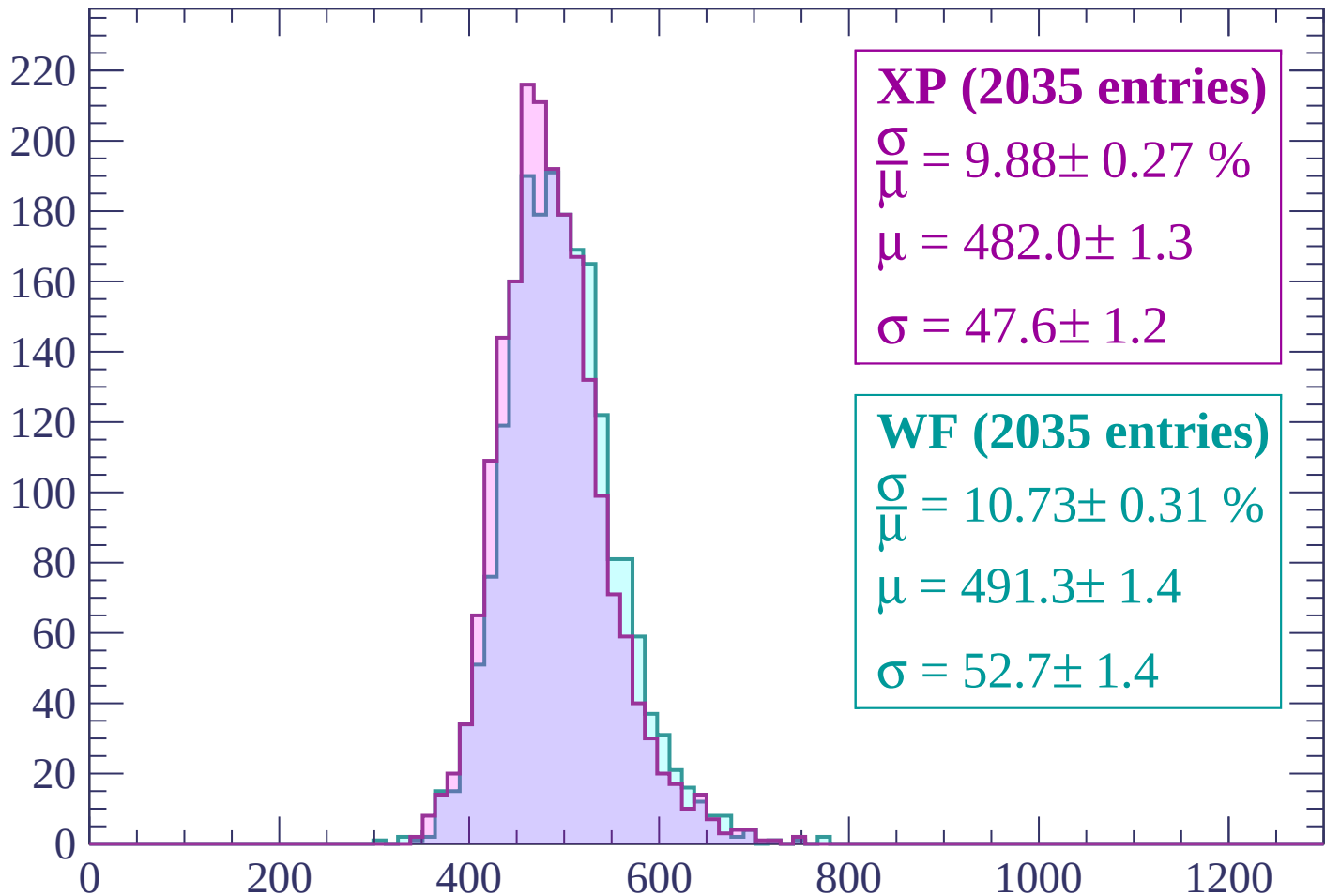
$$\mu = 490.4 \pm 1.2$$

$$\sigma = 47.0 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $2640 < p < 2700$

Count



XP (2035 entries)

$$\frac{\sigma}{\mu} = 9.88 \pm 0.27 \%$$

$$\mu = 482.0 \pm 1.3$$

$$\sigma = 47.6 \pm 1.2$$

WF (2035 entries)

$$\frac{\sigma}{\mu} = 10.73 \pm 0.31 \%$$

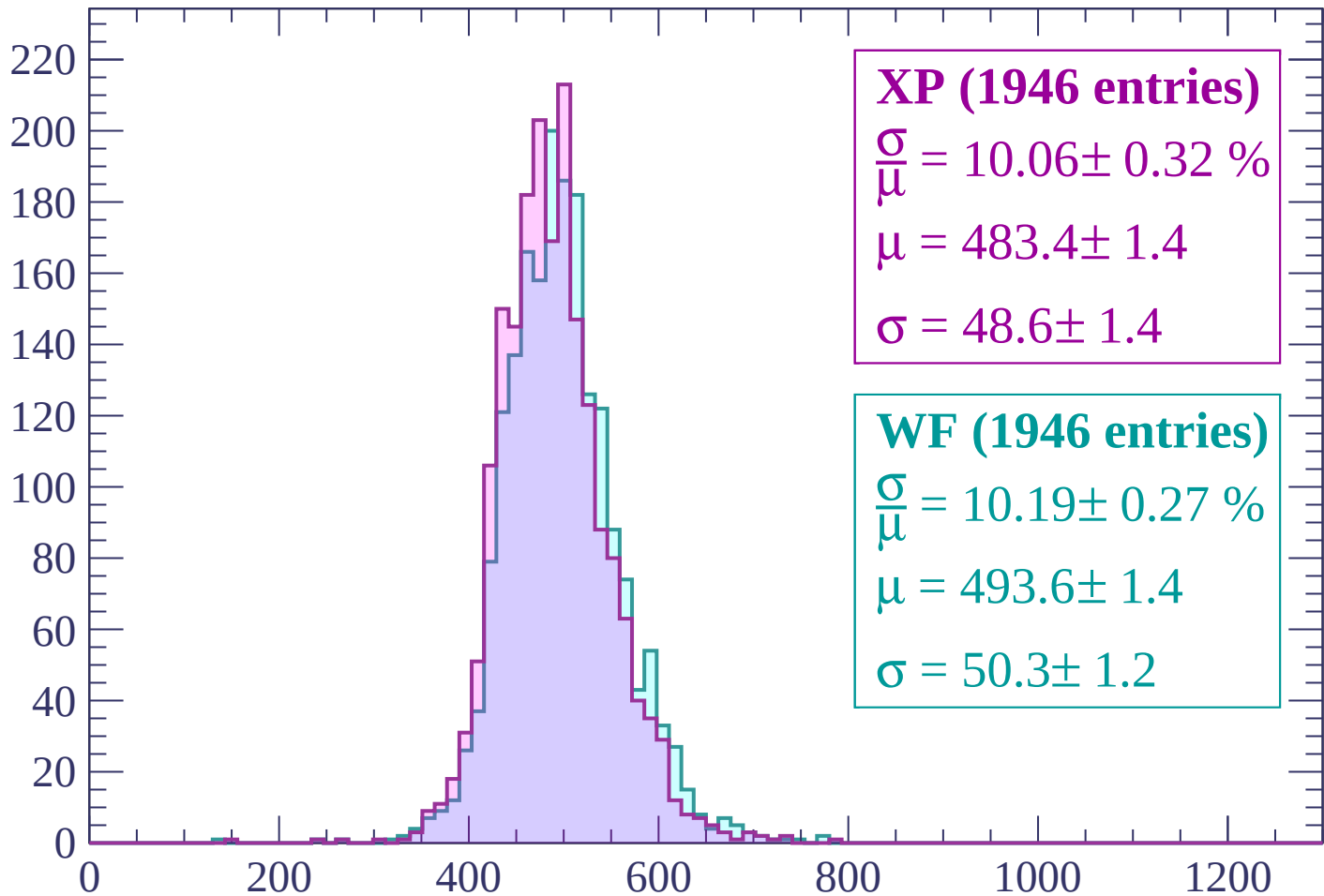
$$\mu = 491.3 \pm 1.4$$

$$\sigma = 52.7 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | $2700 < p < 2760$

Count



XP (1946 entries)

$$\frac{\sigma}{\mu} = 10.06 \pm 0.32 \%$$

$$\mu = 483.4 \pm 1.4$$

$$\sigma = 48.6 \pm 1.4$$

WF (1946 entries)

$$\frac{\sigma}{\mu} = 10.19 \pm 0.27 \%$$

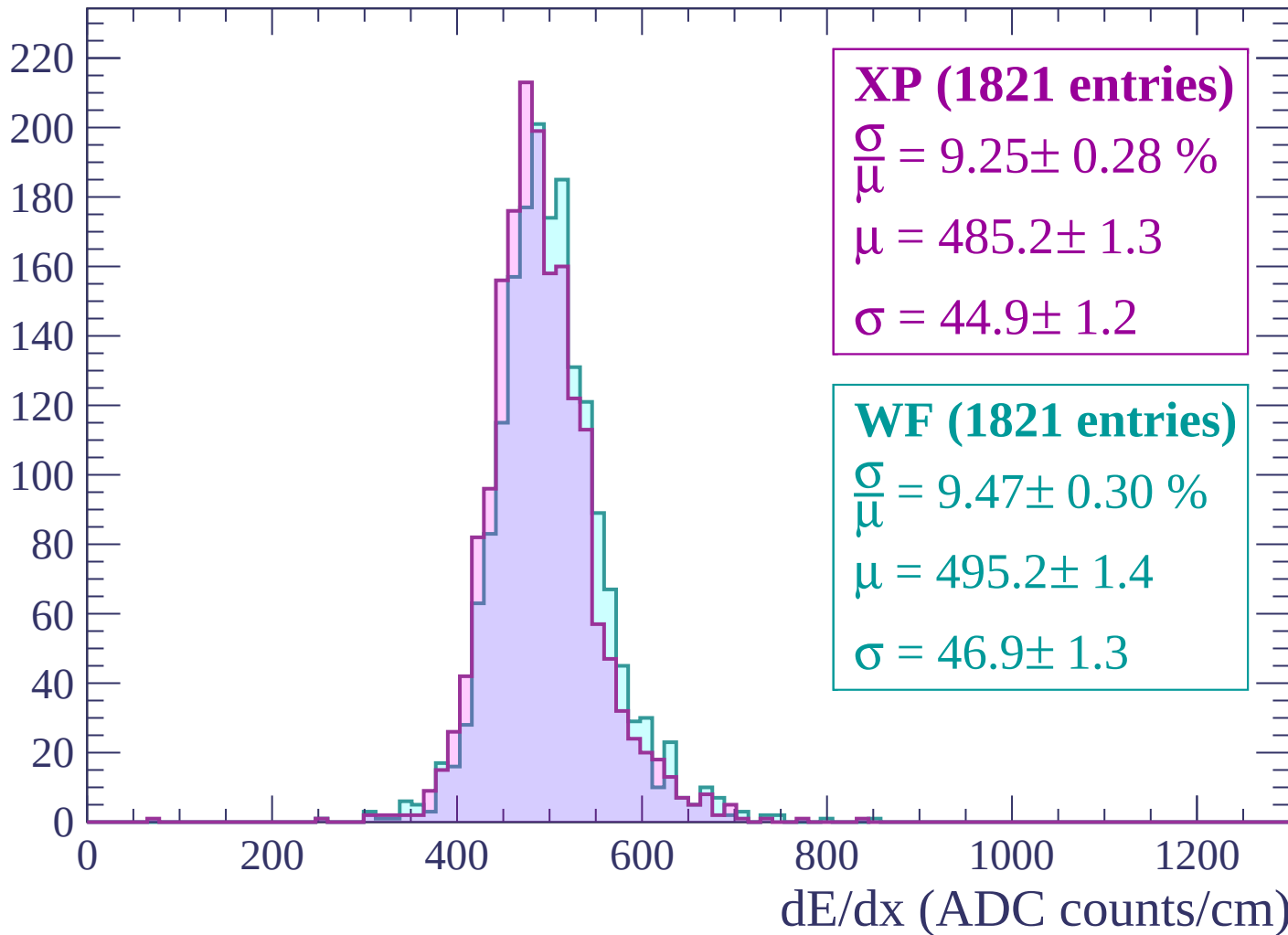
$$\mu = 493.6 \pm 1.4$$

$$\sigma = 50.3 \pm 1.2$$

dE/dx (ADC counts/cm)

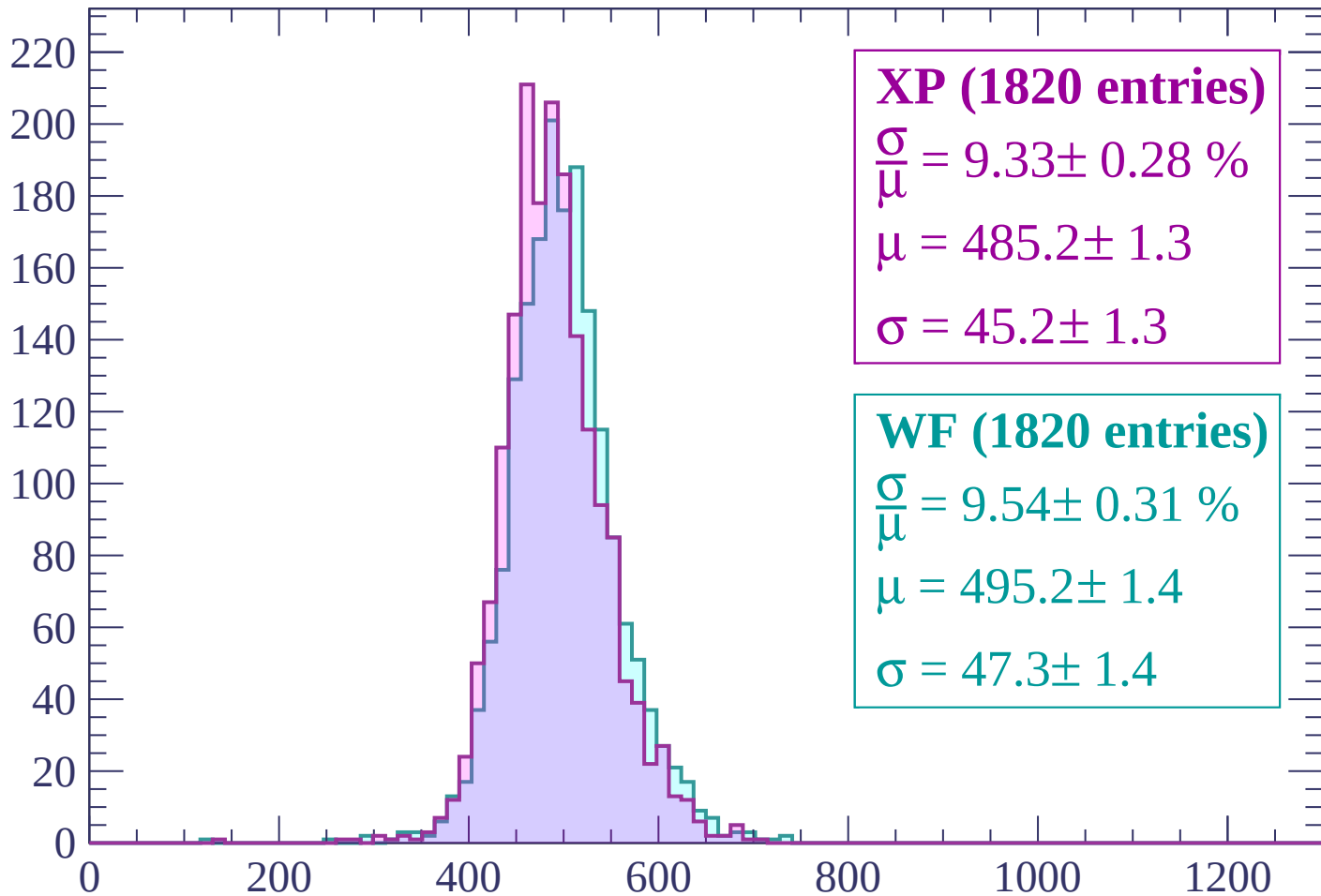
Energy loss | $2760 < p < 2820$

Count



Energy loss | $2820 < p < 2880$

Count



XP (1820 entries)

$$\frac{\sigma}{\mu} = 9.33 \pm 0.28 \%$$

$$\mu = 485.2 \pm 1.3$$

$$\sigma = 45.2 \pm 1.3$$

WF (1820 entries)

$$\frac{\sigma}{\mu} = 9.54 \pm 0.31 \%$$

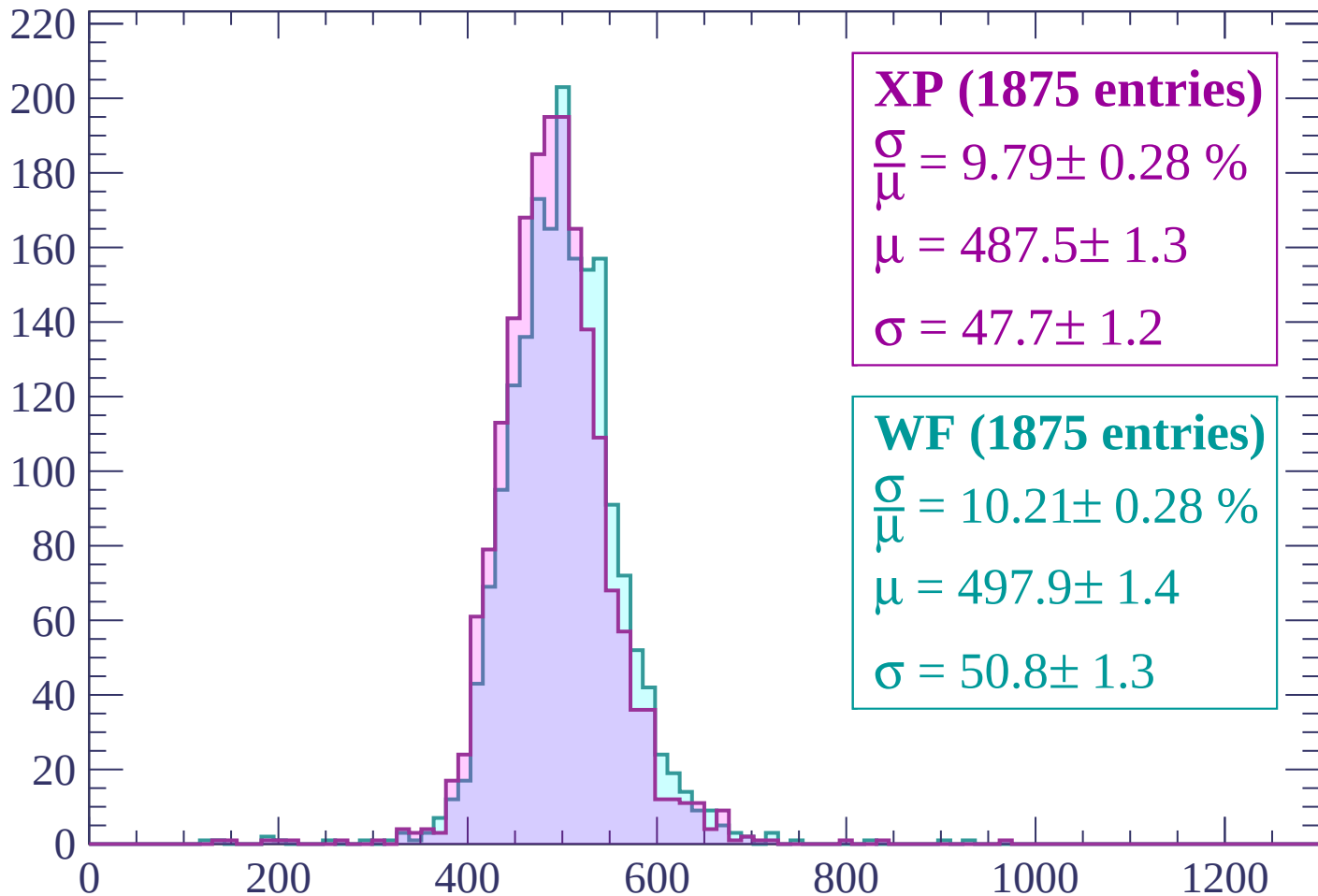
$$\mu = 495.2 \pm 1.4$$

$$\sigma = 47.3 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | $2880 < p < 2940$

Count



XP (1875 entries)

$$\frac{\sigma}{\mu} = 9.79 \pm 0.28 \%$$

$$\mu = 487.5 \pm 1.3$$

$$\sigma = 47.7 \pm 1.2$$

WF (1875 entries)

$$\frac{\sigma}{\mu} = 10.21 \pm 0.28 \%$$

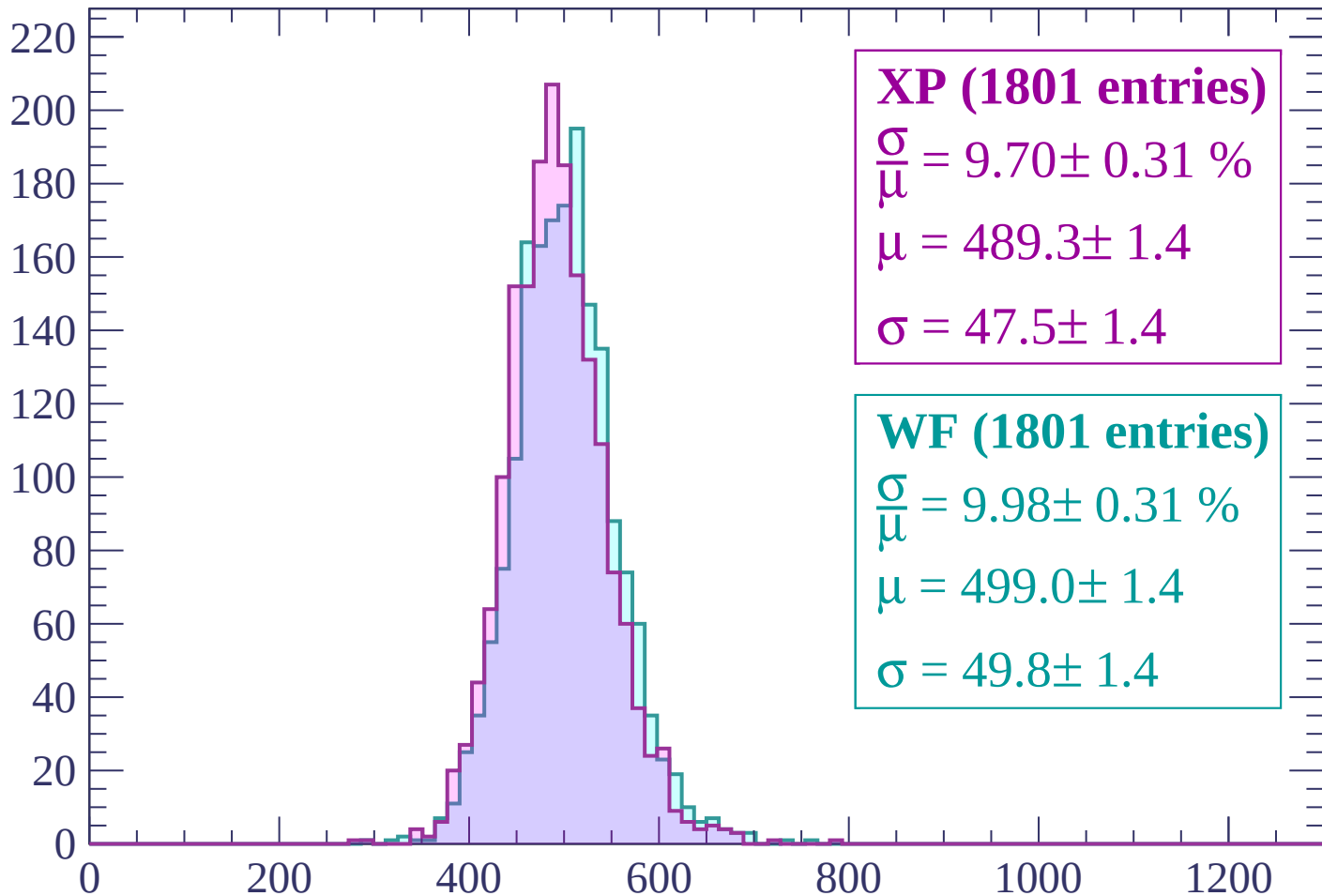
$$\mu = 497.9 \pm 1.4$$

$$\sigma = 50.8 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $2940 < p < 3000$

Count



XP (1801 entries)

$$\frac{\sigma}{\mu} = 9.70 \pm 0.31 \%$$

$$\mu = 489.3 \pm 1.4$$

$$\sigma = 47.5 \pm 1.4$$

WF (1801 entries)

$$\frac{\sigma}{\mu} = 9.98 \pm 0.31 \%$$

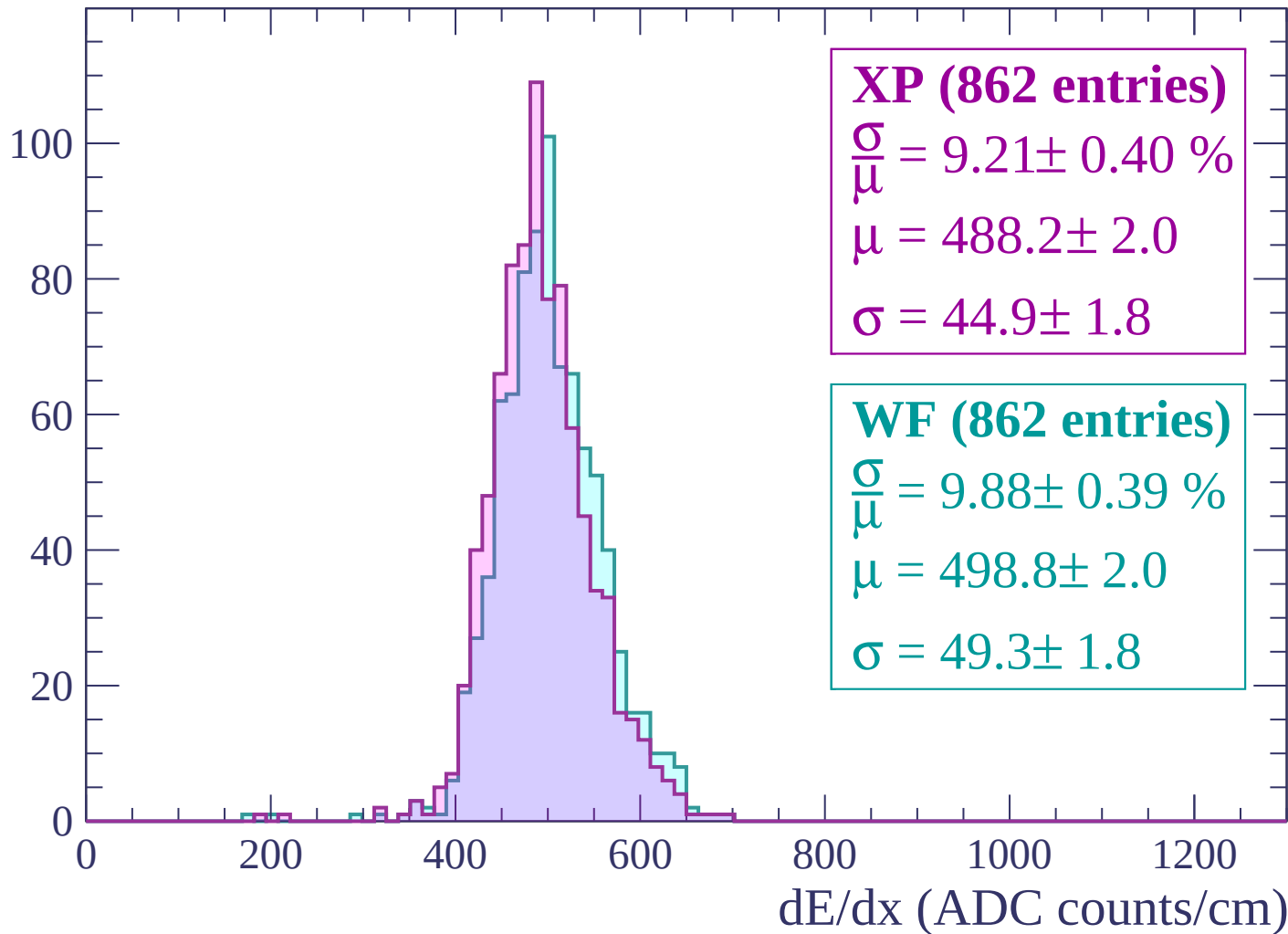
$$\mu = 499.0 \pm 1.4$$

$$\sigma = 49.8 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | $3000 < p < 3060$

Count



Energy loss | $-90 < \theta < -88$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-88 < \theta < -86$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-86 < \theta < -84$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-84 < \theta < -82$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-82 < \theta < -80$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-80 < \theta < -78$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-78 < \theta < -76$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-76 < \theta < -74$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-74 < \theta < -72$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-72 < \theta < -70$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-70 < \theta < -68$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-68 < \theta < -66$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-66 < \theta < -64$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-64 < \theta < -62$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-62 < \theta < -60$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-60 < \theta < -58$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-58 < \theta < -56$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-56 < \theta < -54$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-54 < \theta < -52$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-52 < \theta < -50$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-50 < \theta < -48$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-48 < \theta < -46$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-46 < \theta < -44$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

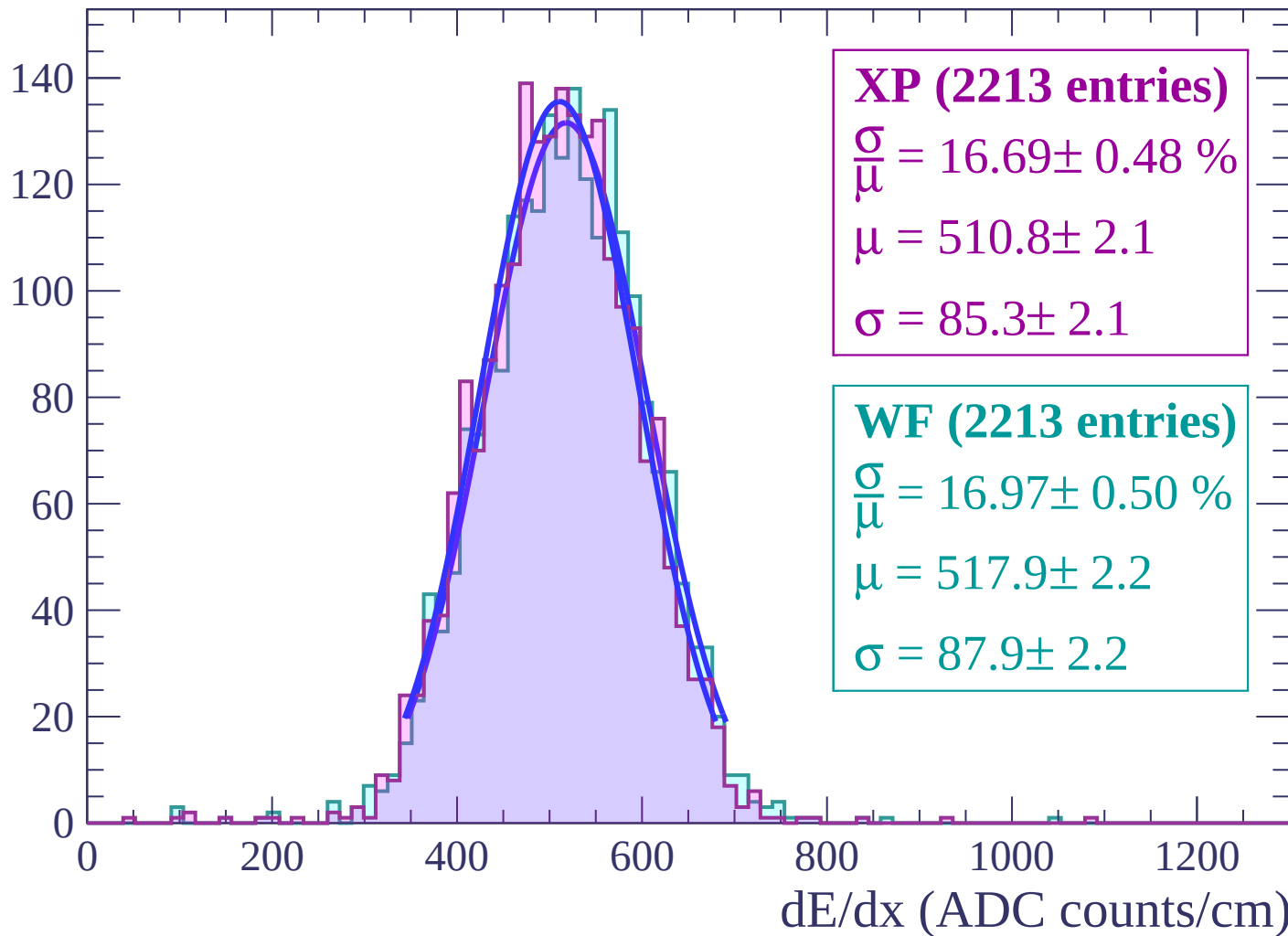
$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

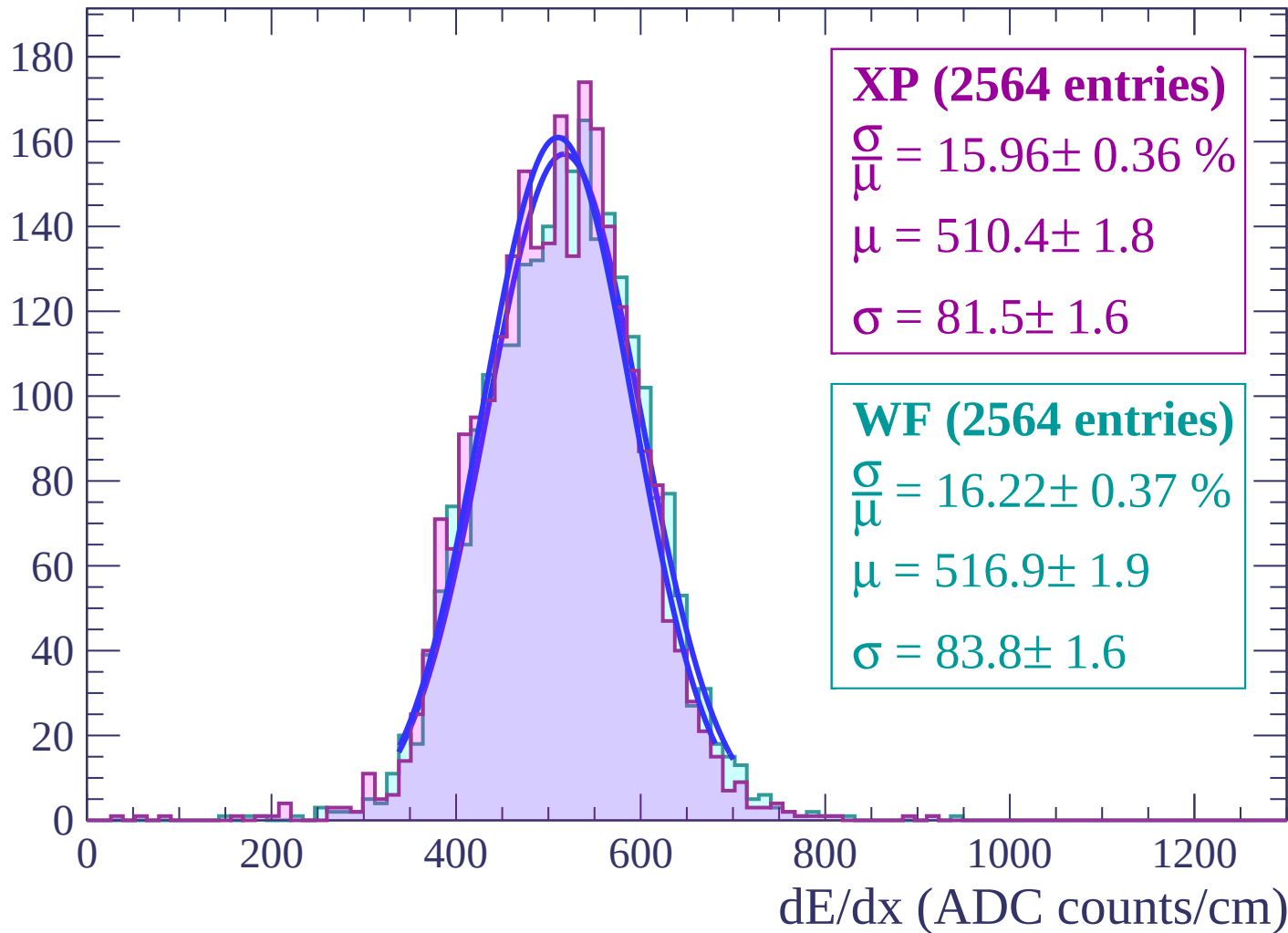
Energy loss | $-44 < \theta < -42$

Count



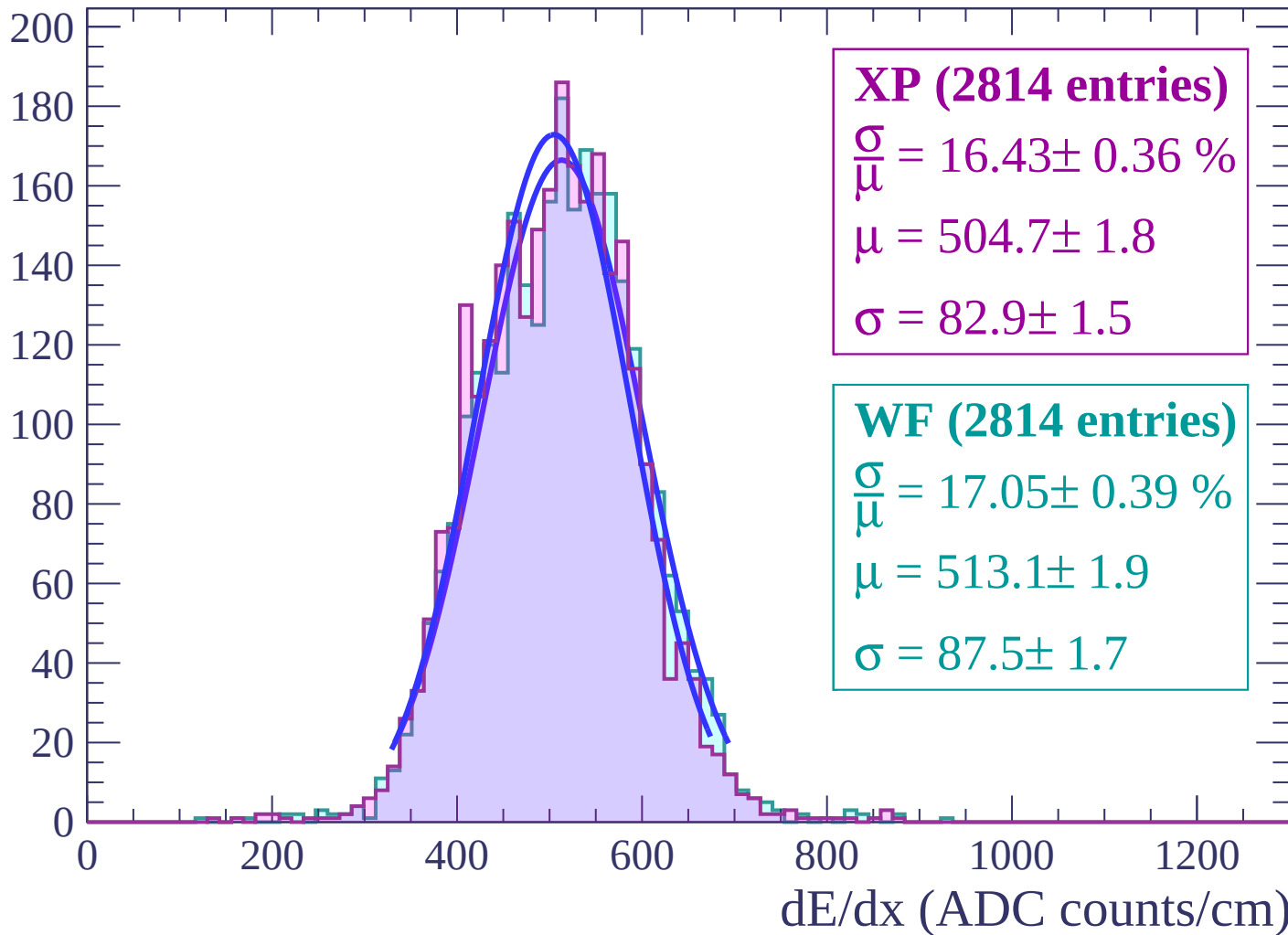
Energy loss | $-42 < \theta < -40$

Count



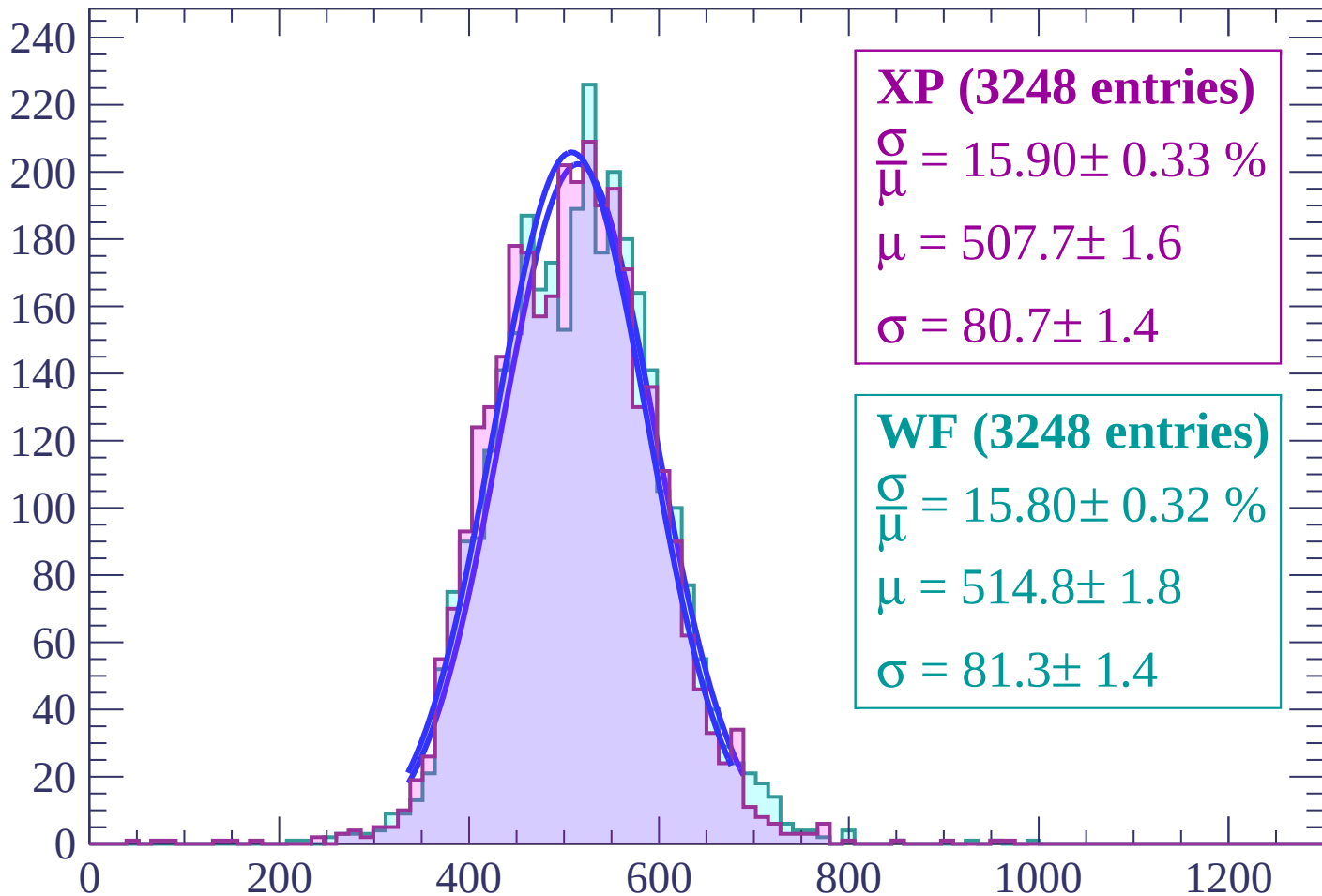
Energy loss | $-40 < \theta < -38$

Count



Energy loss | $-38 < \theta < -36$

Count



XP (3248 entries)

$\frac{\sigma}{\mu} = 15.90 \pm 0.33 \%$

$\mu = 507.7 \pm 1.6$

$\sigma = 80.7 \pm 1.4$

WF (3248 entries)

$\frac{\sigma}{\mu} = 15.80 \pm 0.32 \%$

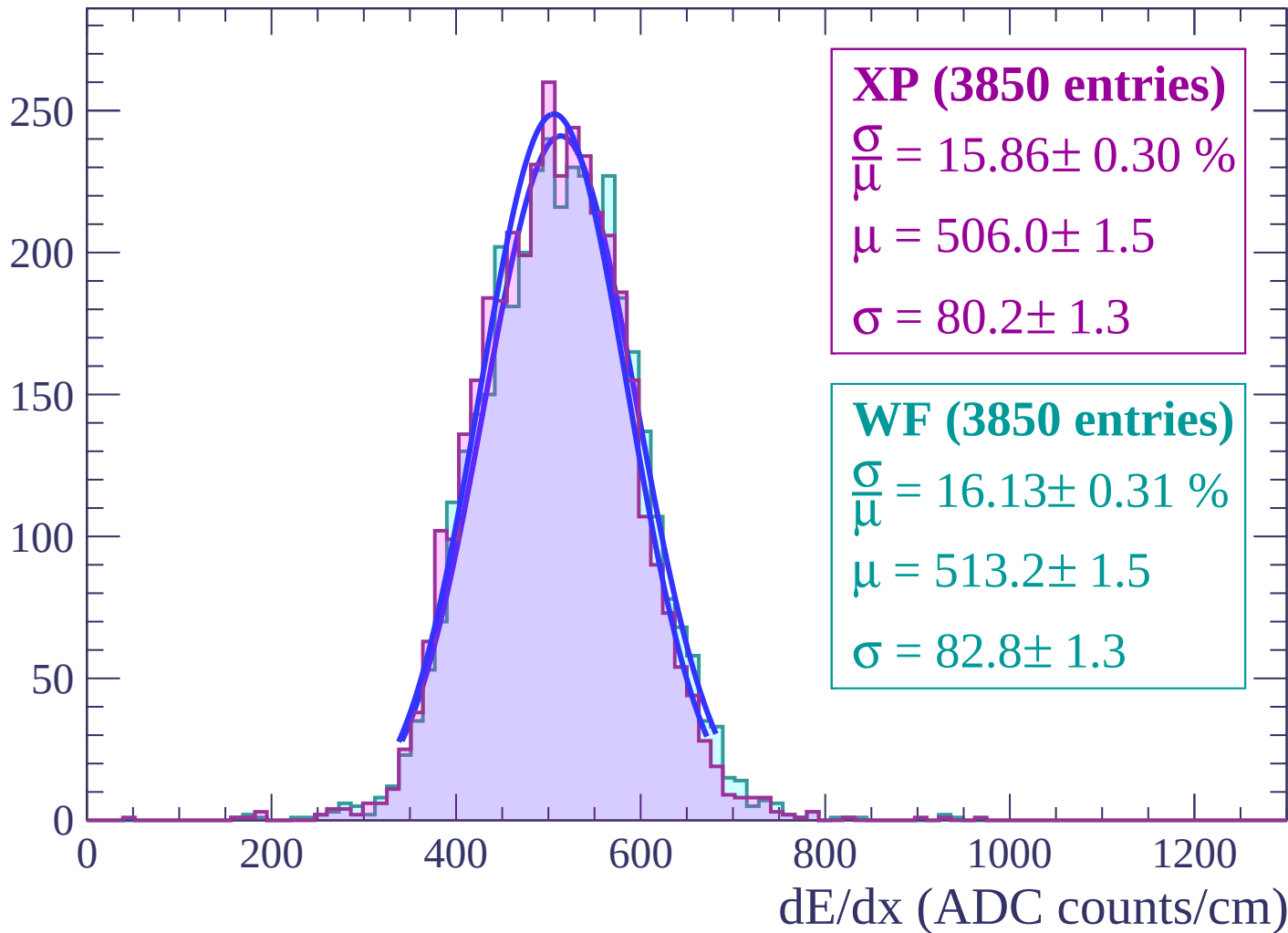
$\mu = 514.8 \pm 1.8$

$\sigma = 81.3 \pm 1.4$

dE/dx (ADC counts/cm)

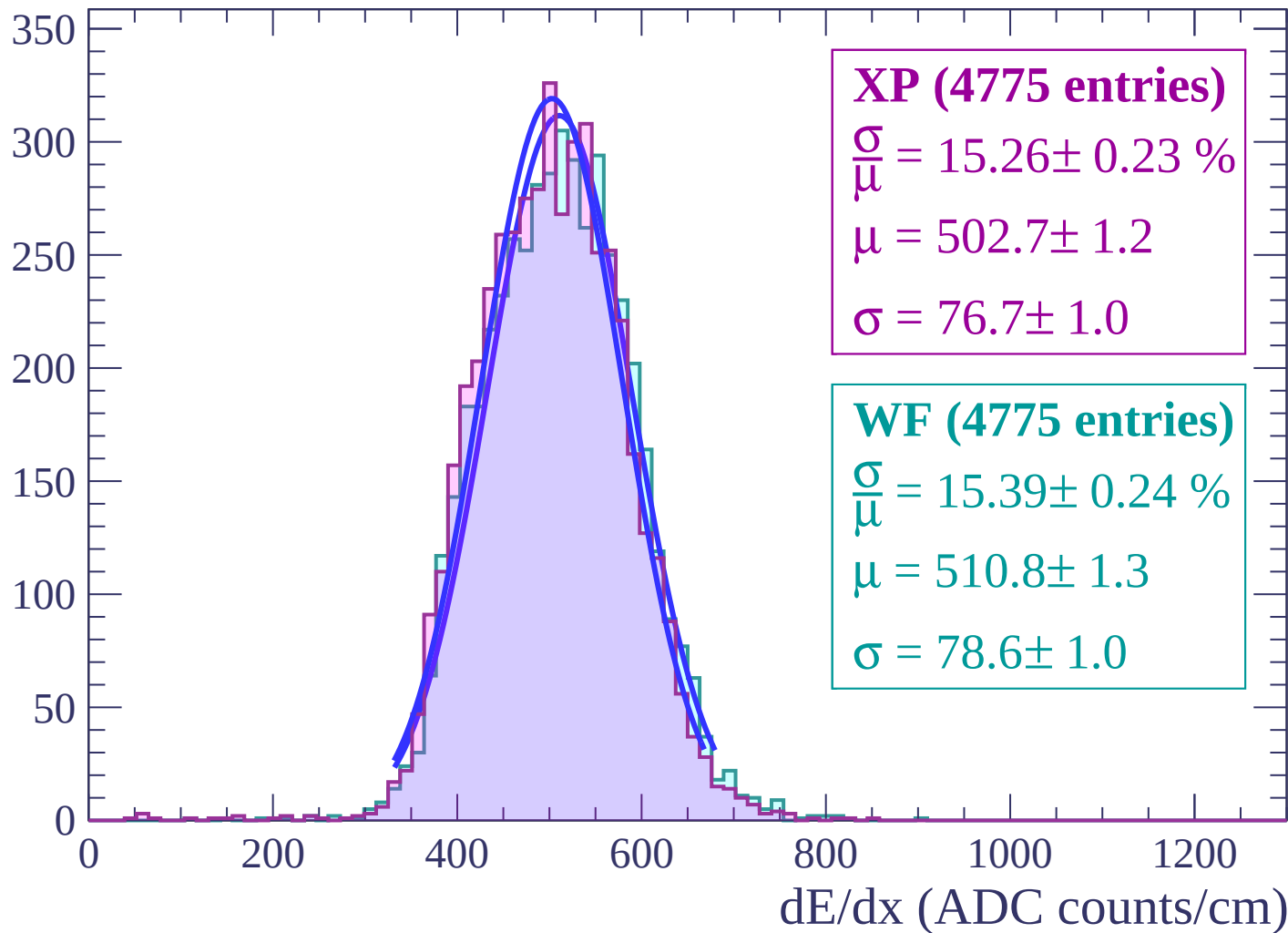
Energy loss | $-36 < \theta < -34$

Count



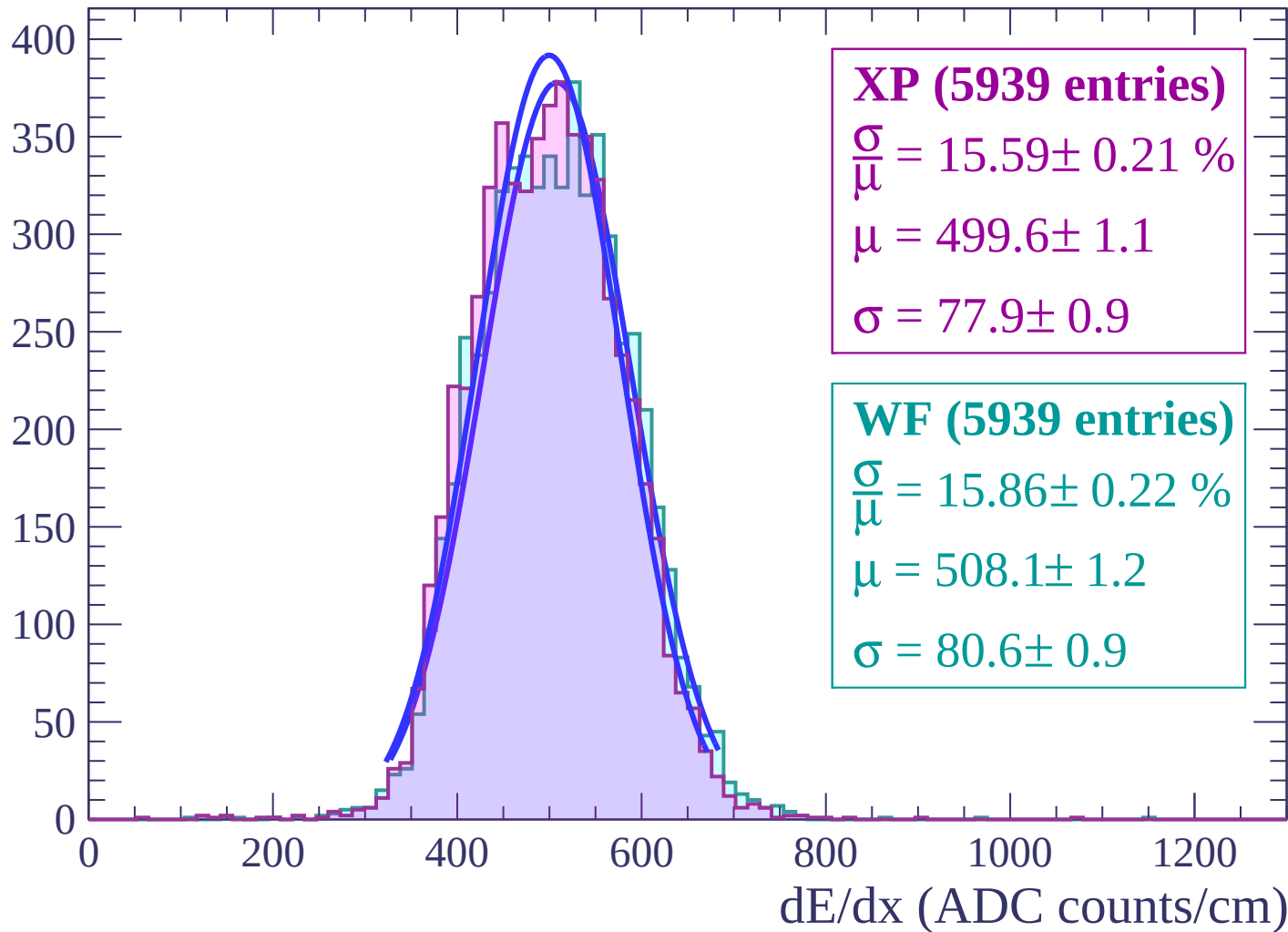
Energy loss | $-34 < \theta < -32$

Count



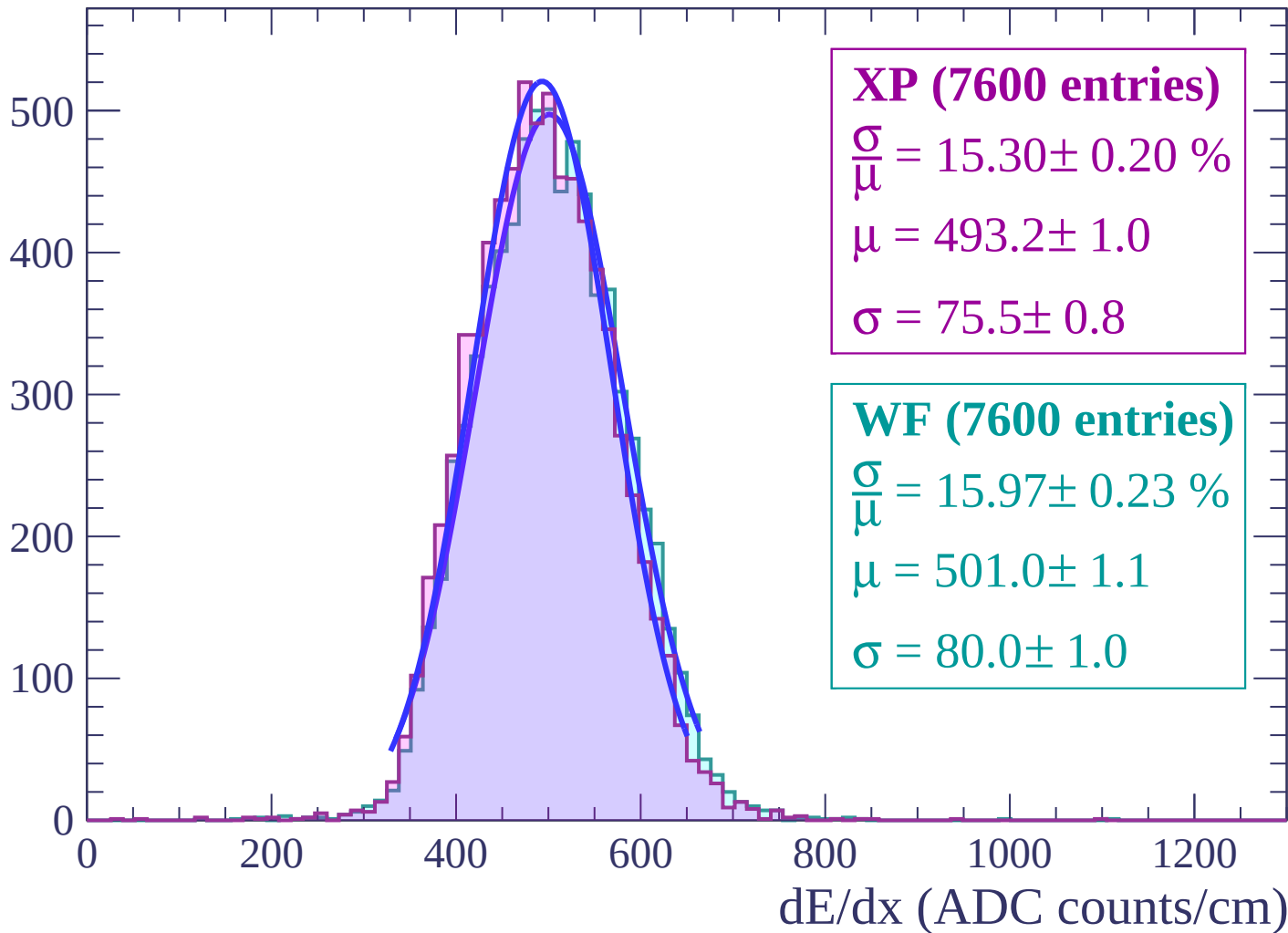
Energy loss | $-32 < \theta < -30$

Count



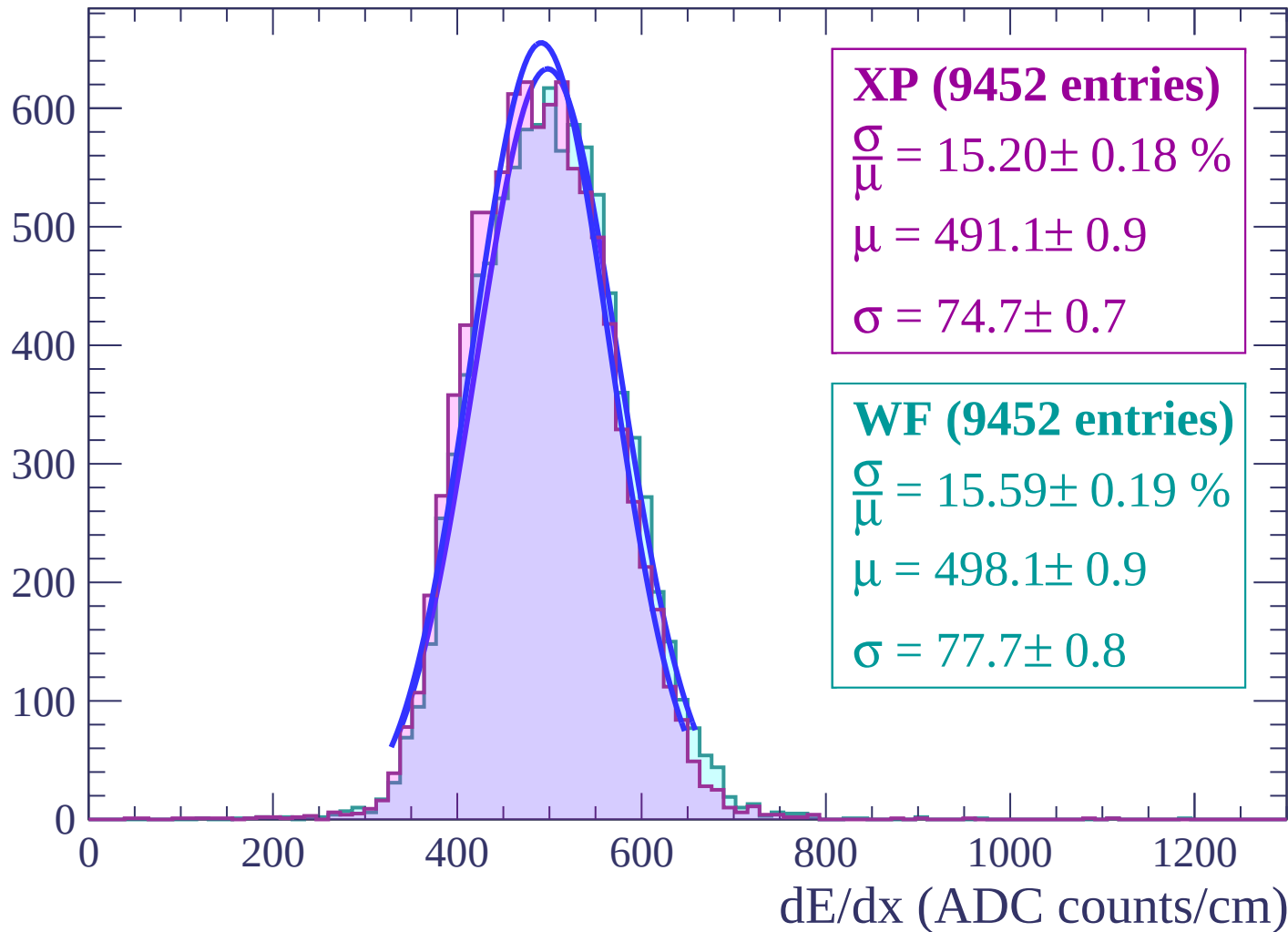
Energy loss | $-30 < \theta < -28$

Count



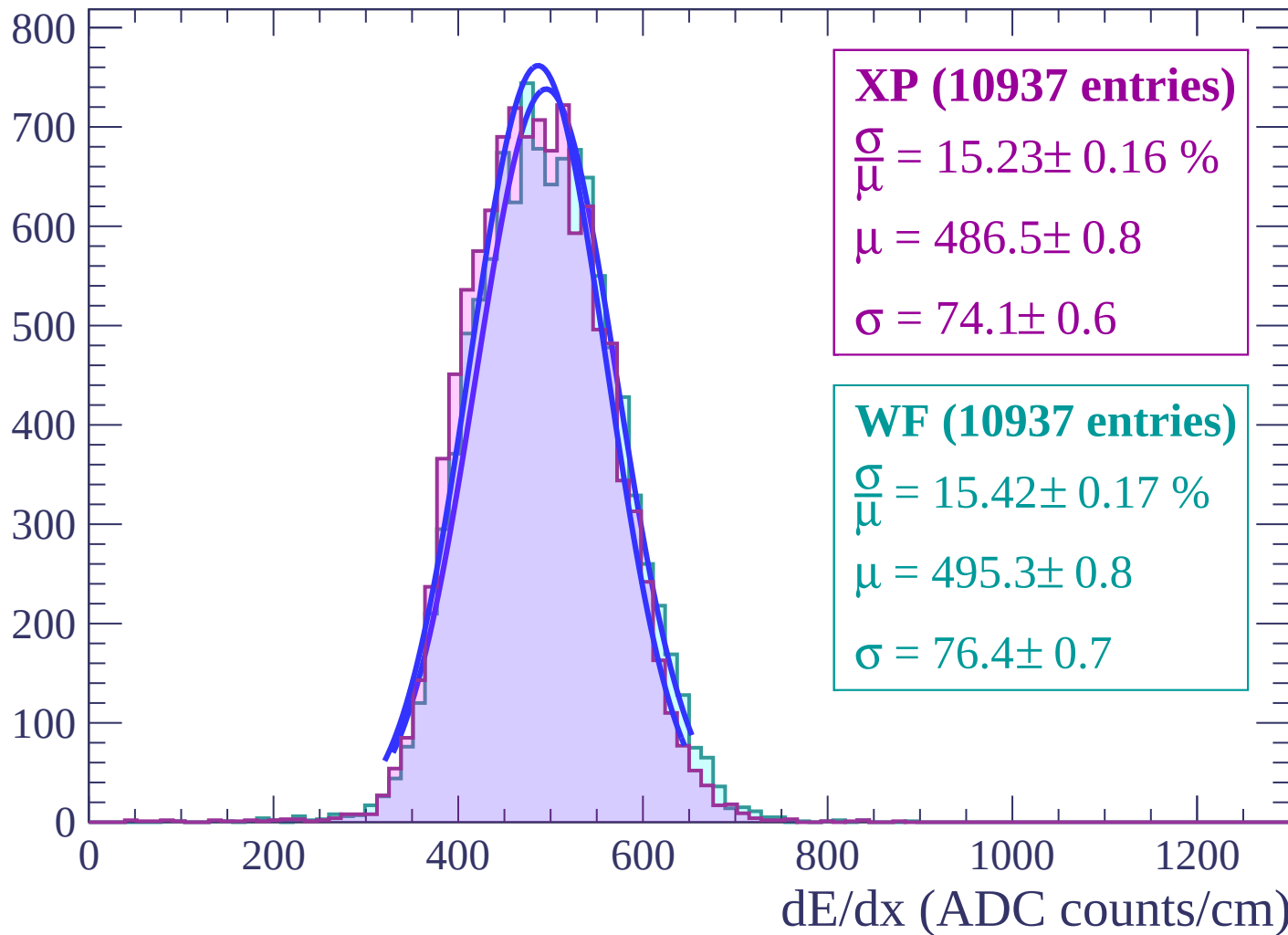
Energy loss | $-28 < \theta < -26$

Count



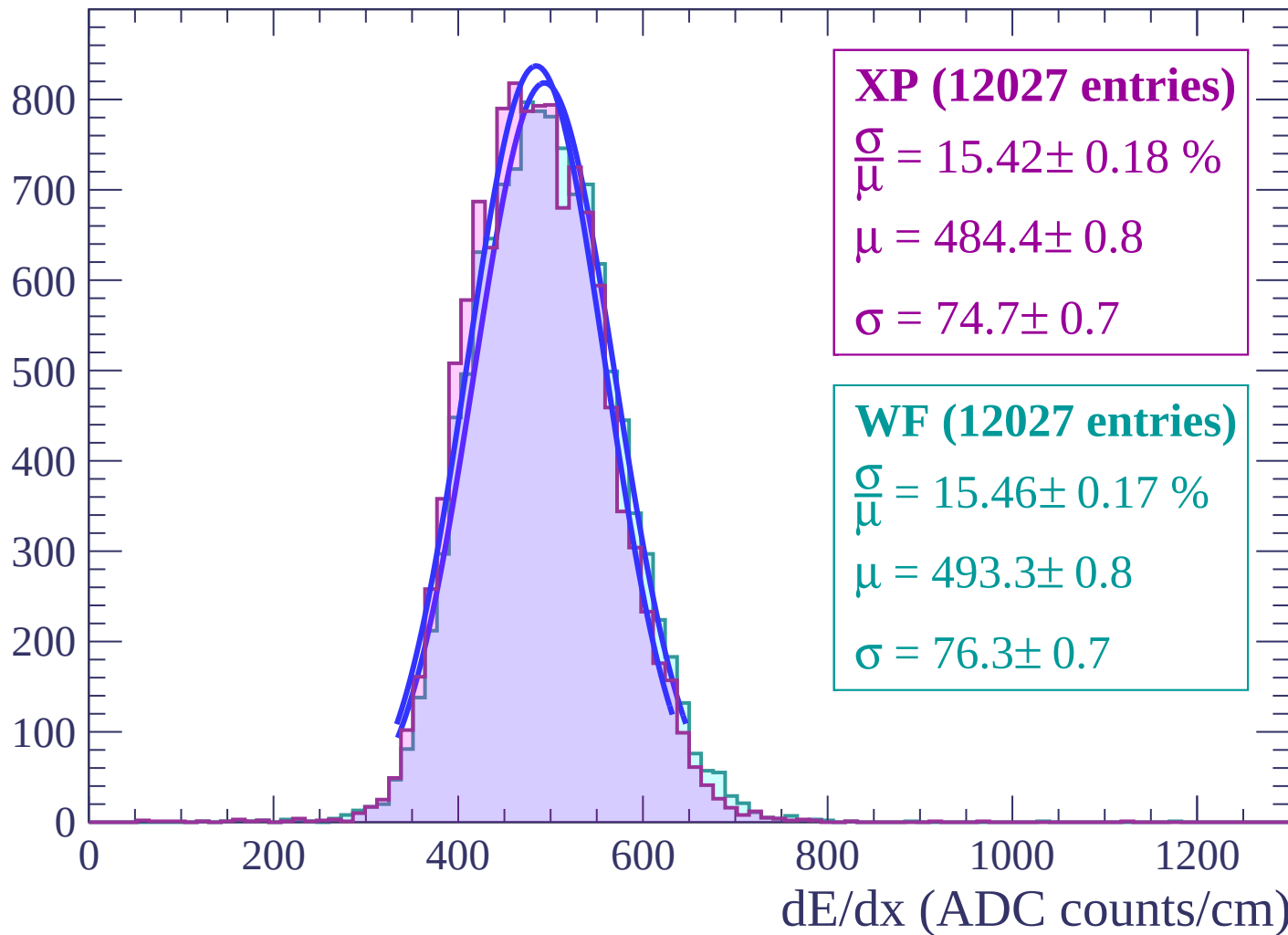
Energy loss | $-26 < \theta < -24$

Count

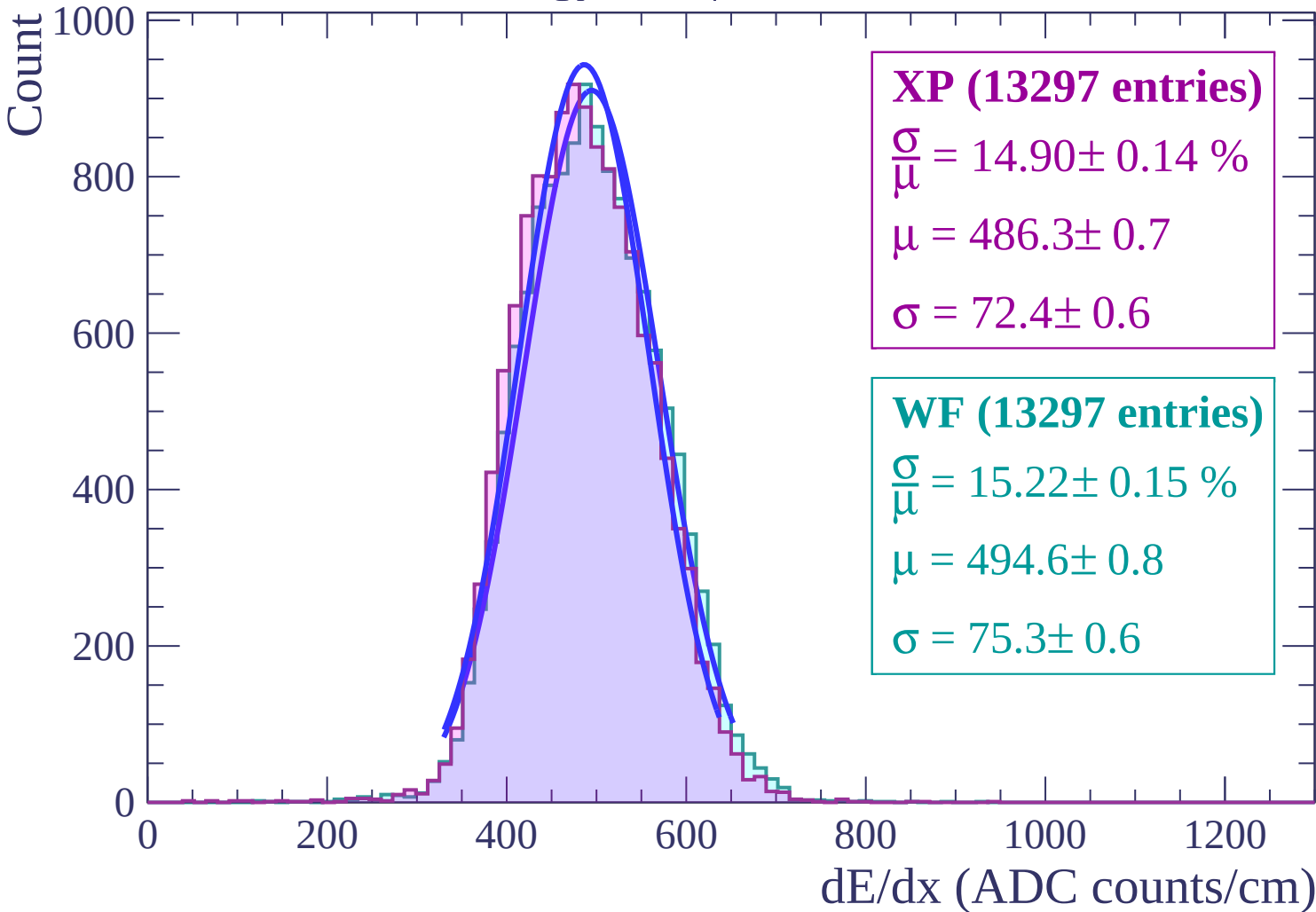


Energy loss | $-24 < \theta < -22$

Count



Energy loss | $-22 < \theta < -20$



Energy loss | $-20 < \theta < -18$

Count

1000

800

600

400

200

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (14611 entries)

$\frac{\sigma}{\mu} = 14.94 \pm 0.14 \%$

$\mu = 485.4 \pm 0.7$

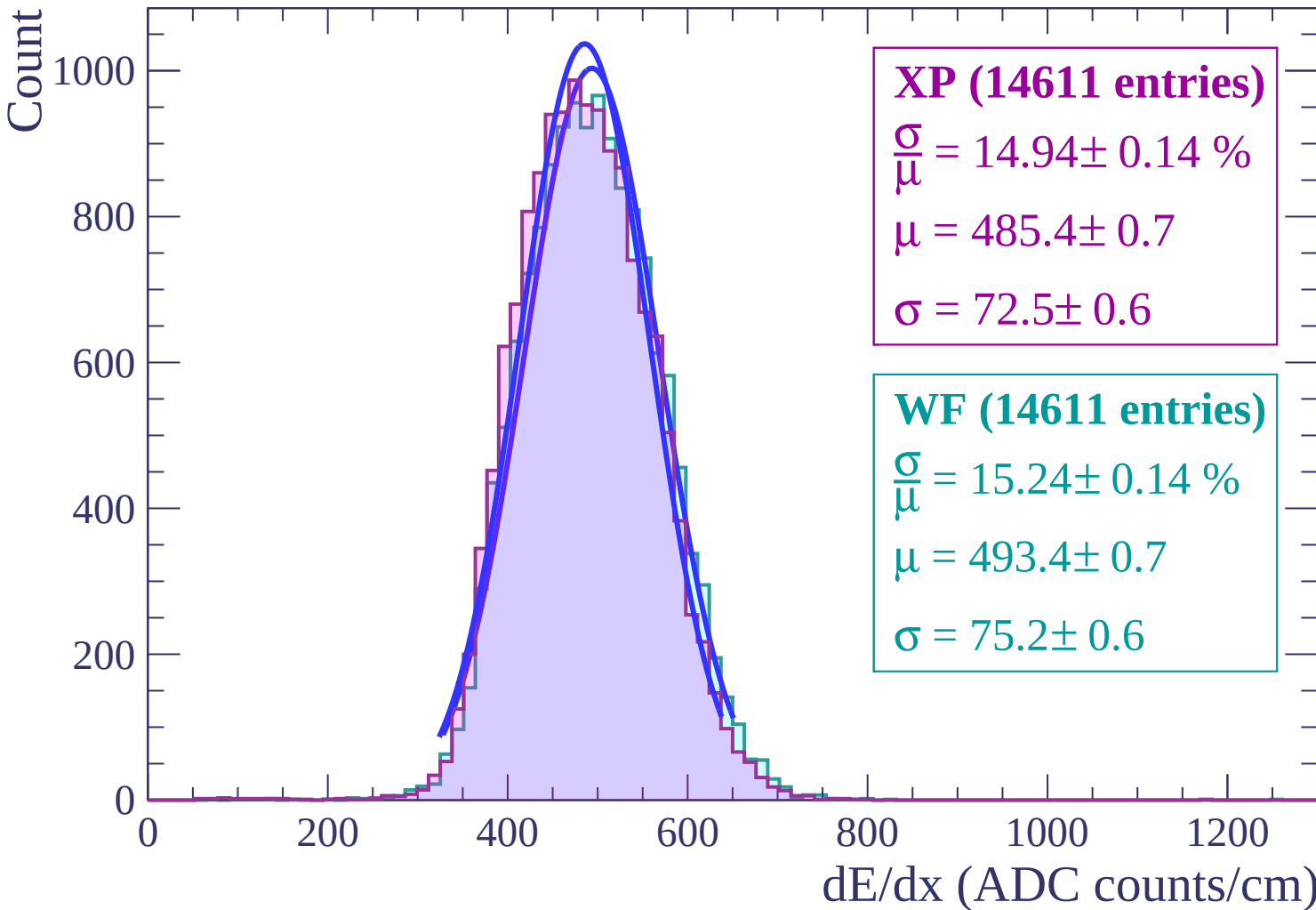
$\sigma = 72.5 \pm 0.6$

WF (14611 entries)

$\frac{\sigma}{\mu} = 15.24 \pm 0.14 \%$

$\mu = 493.4 \pm 0.7$

$\sigma = 75.2 \pm 0.6$



Energy loss | $-18 < \theta < -16$

Count

1000

800

600

400

200

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (16124 entries)

$$\frac{\sigma}{\mu} = 14.99 \pm 0.13 \%$$

$$\mu = 483.8 \pm 0.7$$

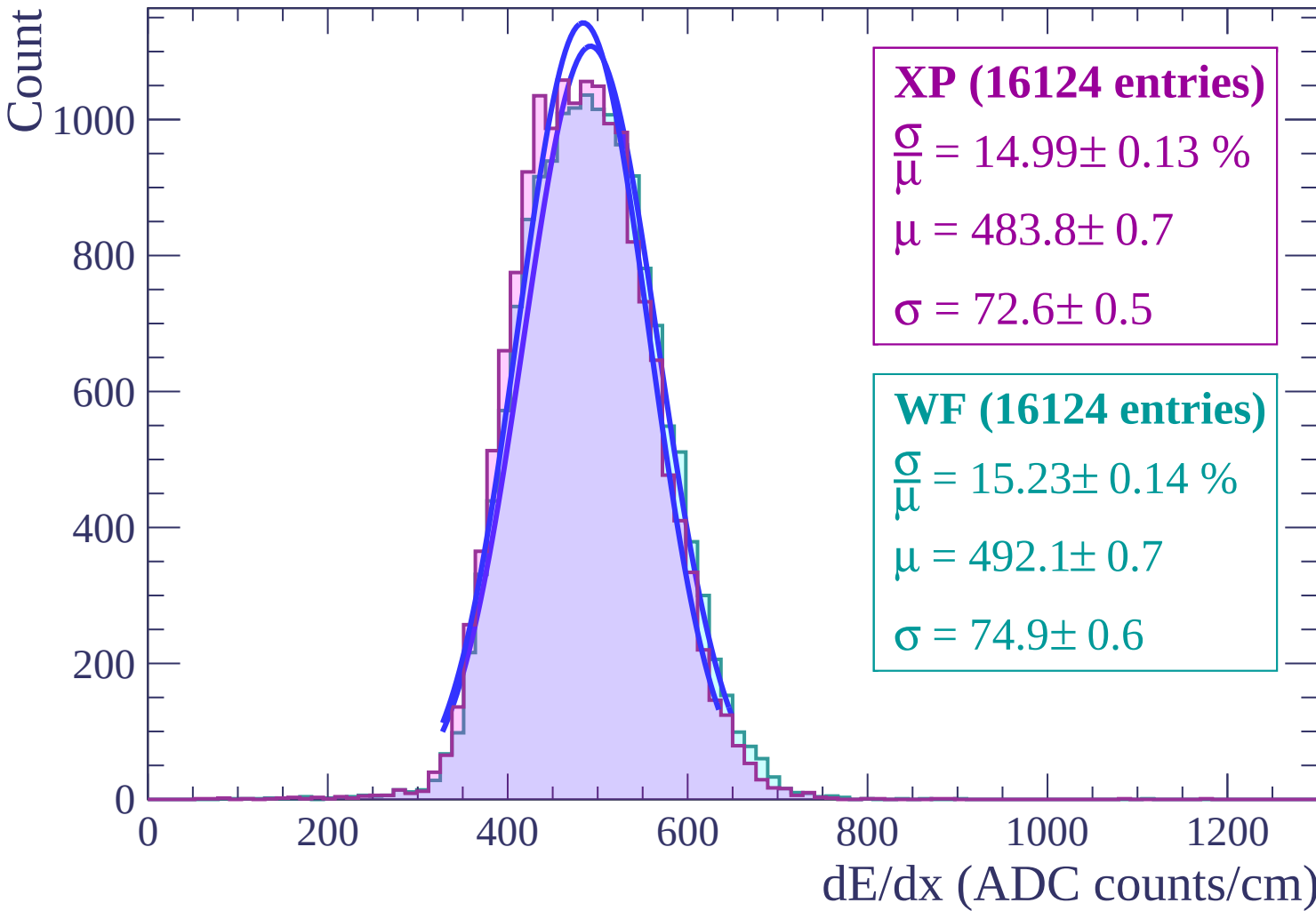
$$\sigma = 72.6 \pm 0.5$$

WF (16124 entries)

$$\frac{\sigma}{\mu} = 15.23 \pm 0.14 \%$$

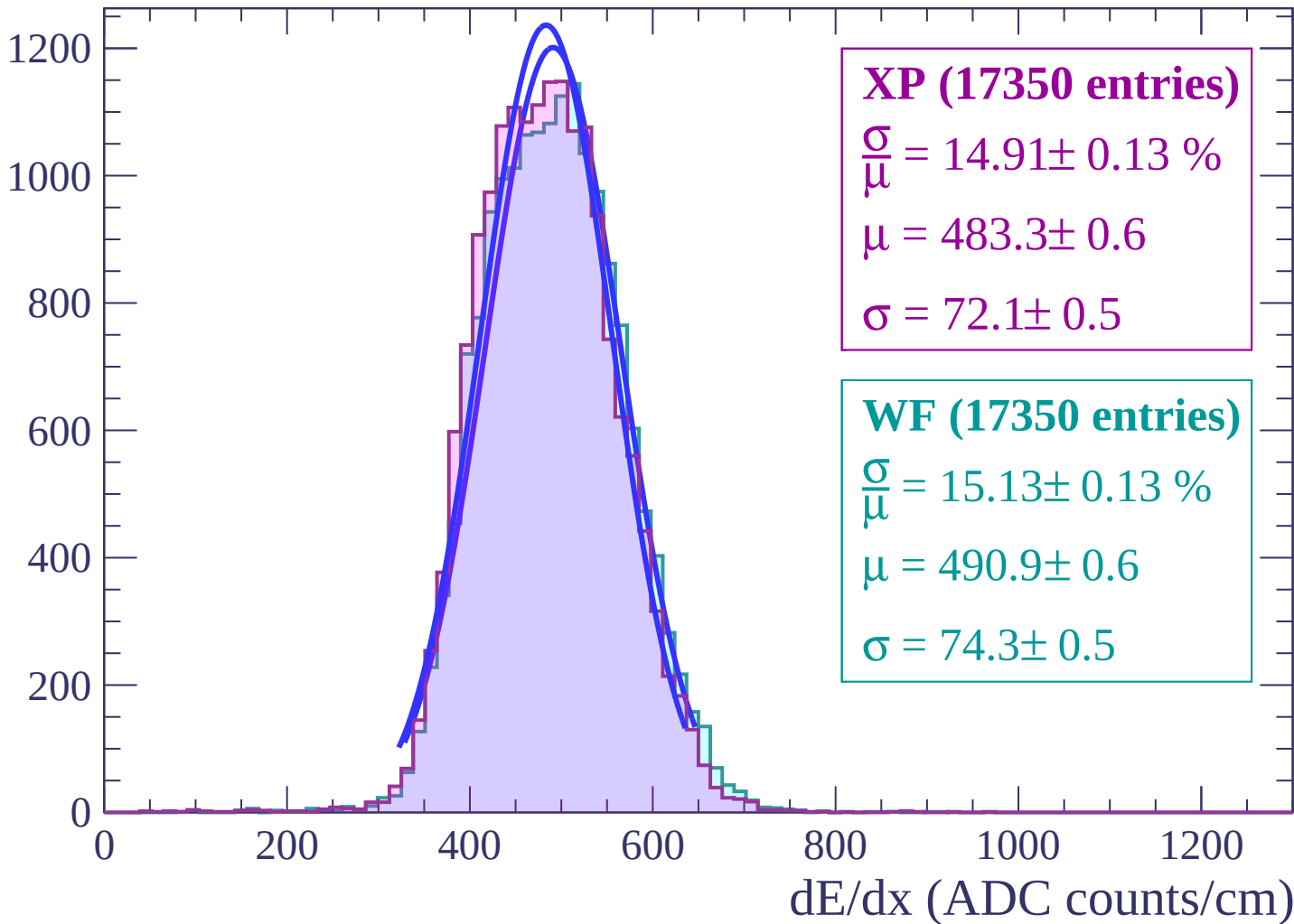
$$\mu = 492.1 \pm 0.7$$

$$\sigma = 74.9 \pm 0.6$$

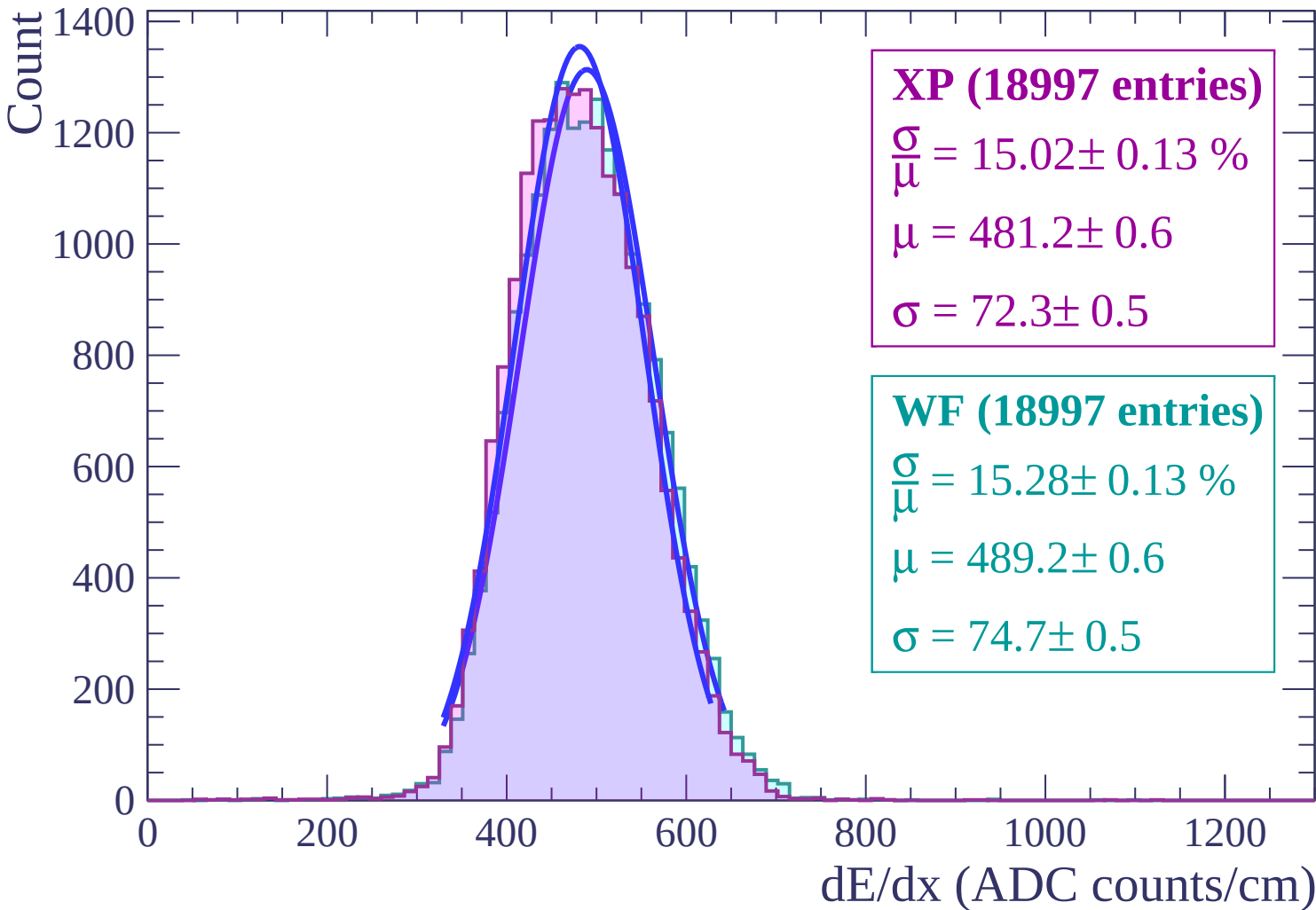


Energy loss | $-16 < \theta < -14$

Count



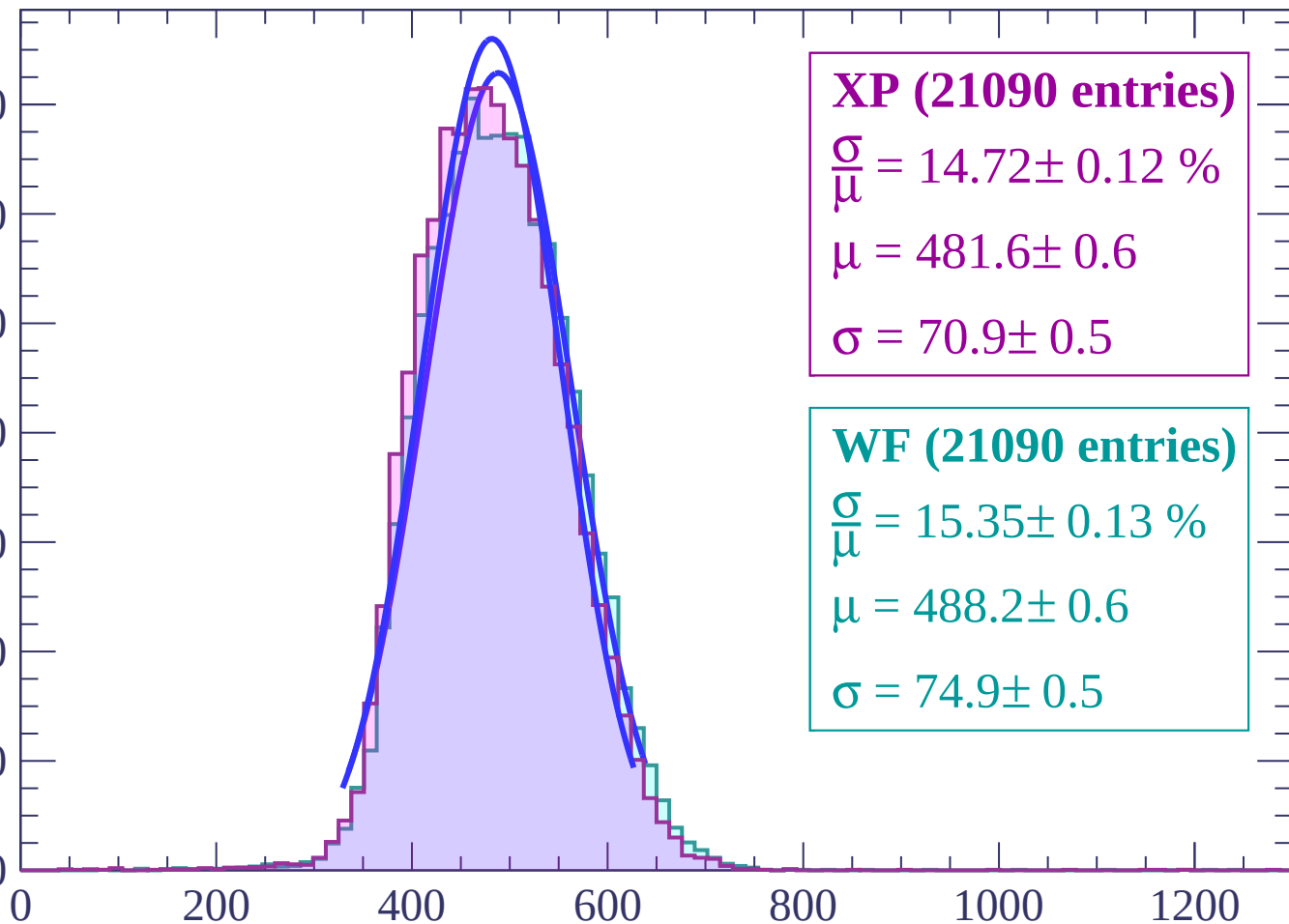
Energy loss | $-14 < \theta < -12$



Energy loss | $-12 < \theta < -10$

Count

1400
1200
1000
800
600
400
200
0



XP (21090 entries)

$$\frac{\sigma}{\mu} = 14.72 \pm 0.12 \%$$

$$\mu = 481.6 \pm 0.6$$

$$\sigma = 70.9 \pm 0.5$$

WF (21090 entries)

$$\frac{\sigma}{\mu} = 15.35 \pm 0.13 \%$$

$$\mu = 488.2 \pm 0.6$$

$$\sigma = 74.9 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss | $-10 < \theta < -8$

Count

1600
1400
1200
1000
800
600
400
200
0

XP (23000 entries)

$$\frac{\sigma}{\mu} = 14.90 \pm 0.11 \%$$

$$\mu = 479.6 \pm 0.6$$

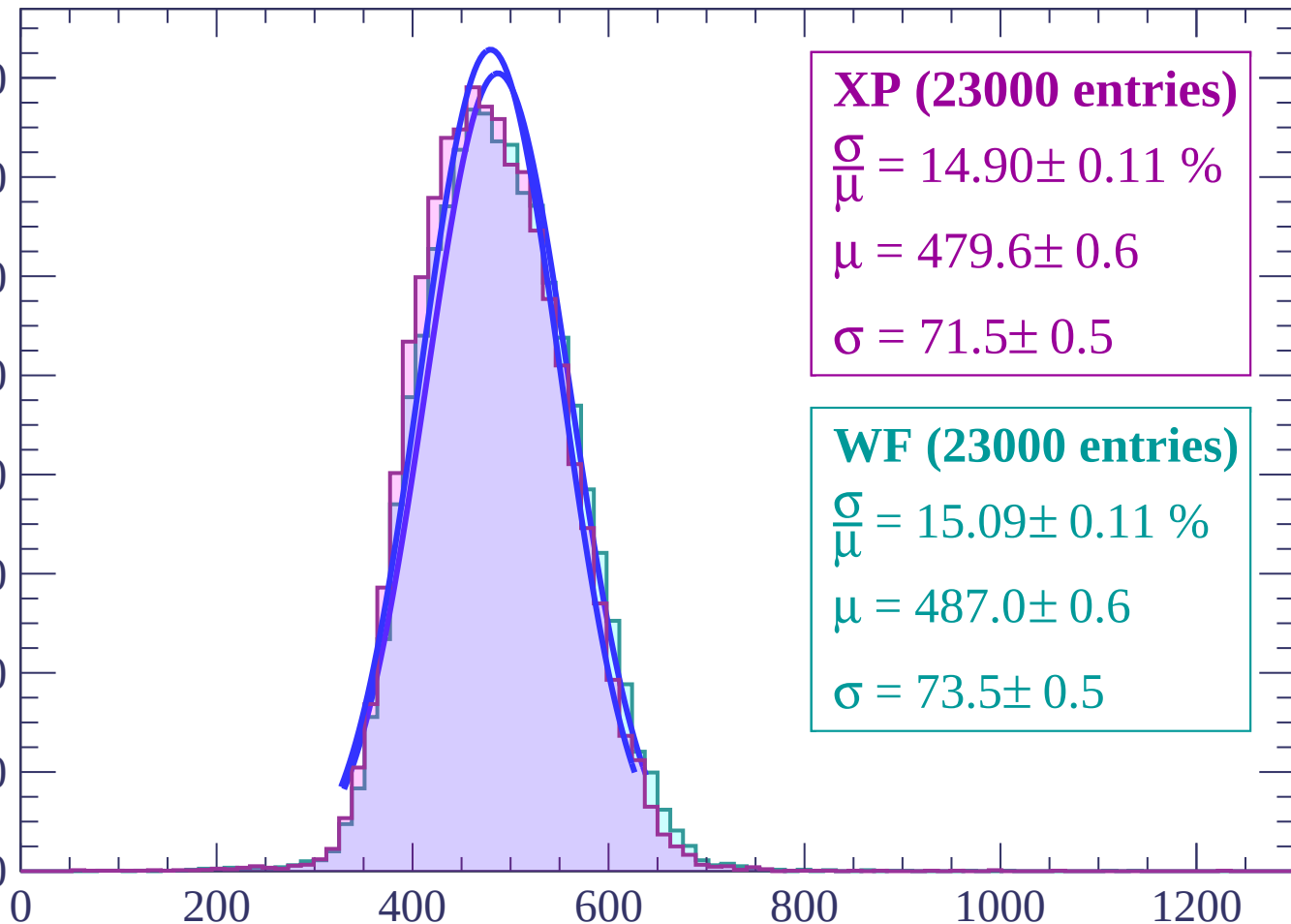
$$\sigma = 71.5 \pm 0.5$$

WF (23000 entries)

$$\frac{\sigma}{\mu} = 15.09 \pm 0.11 \%$$

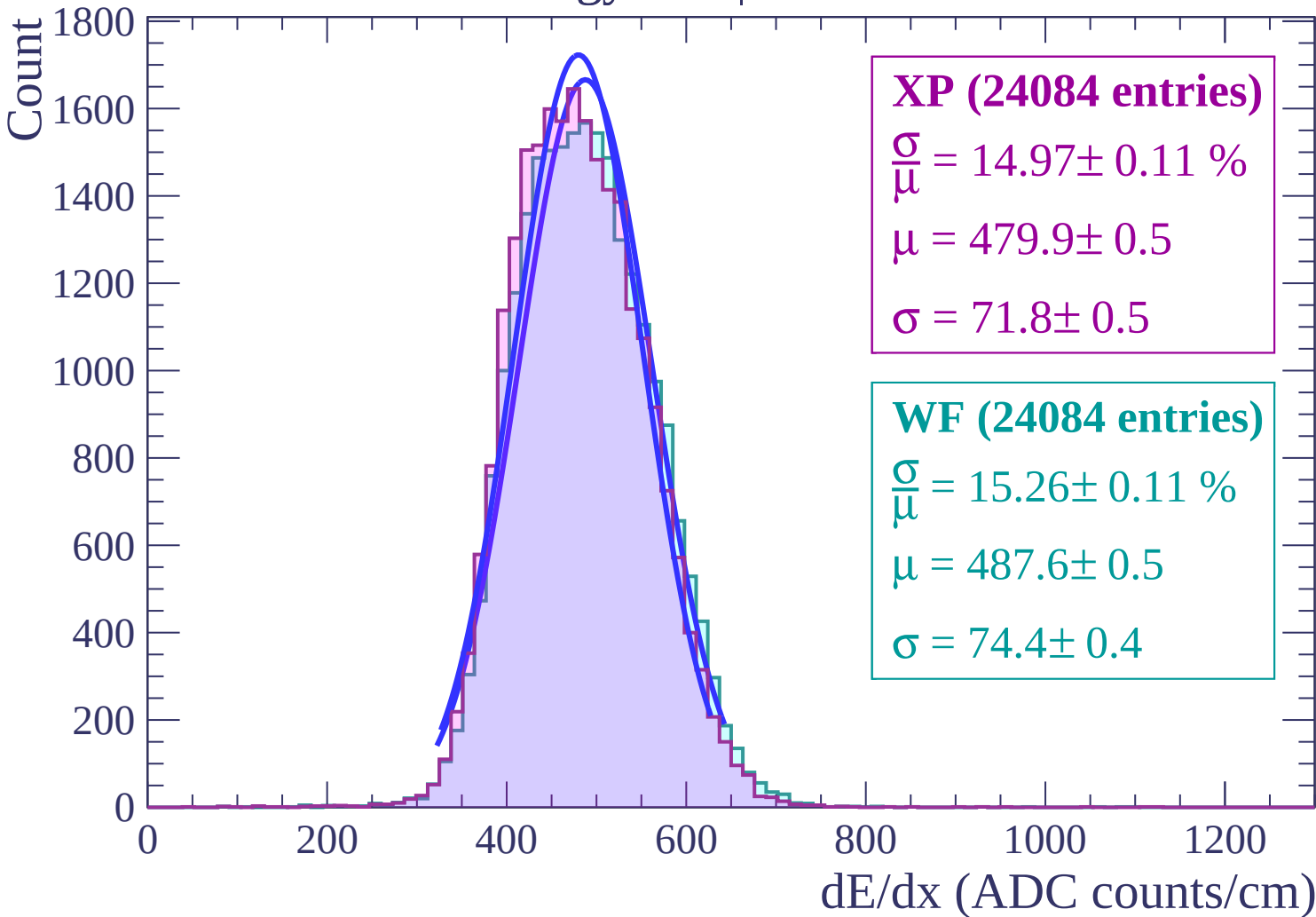
$$\mu = 487.0 \pm 0.6$$

$$\sigma = 73.5 \pm 0.5$$

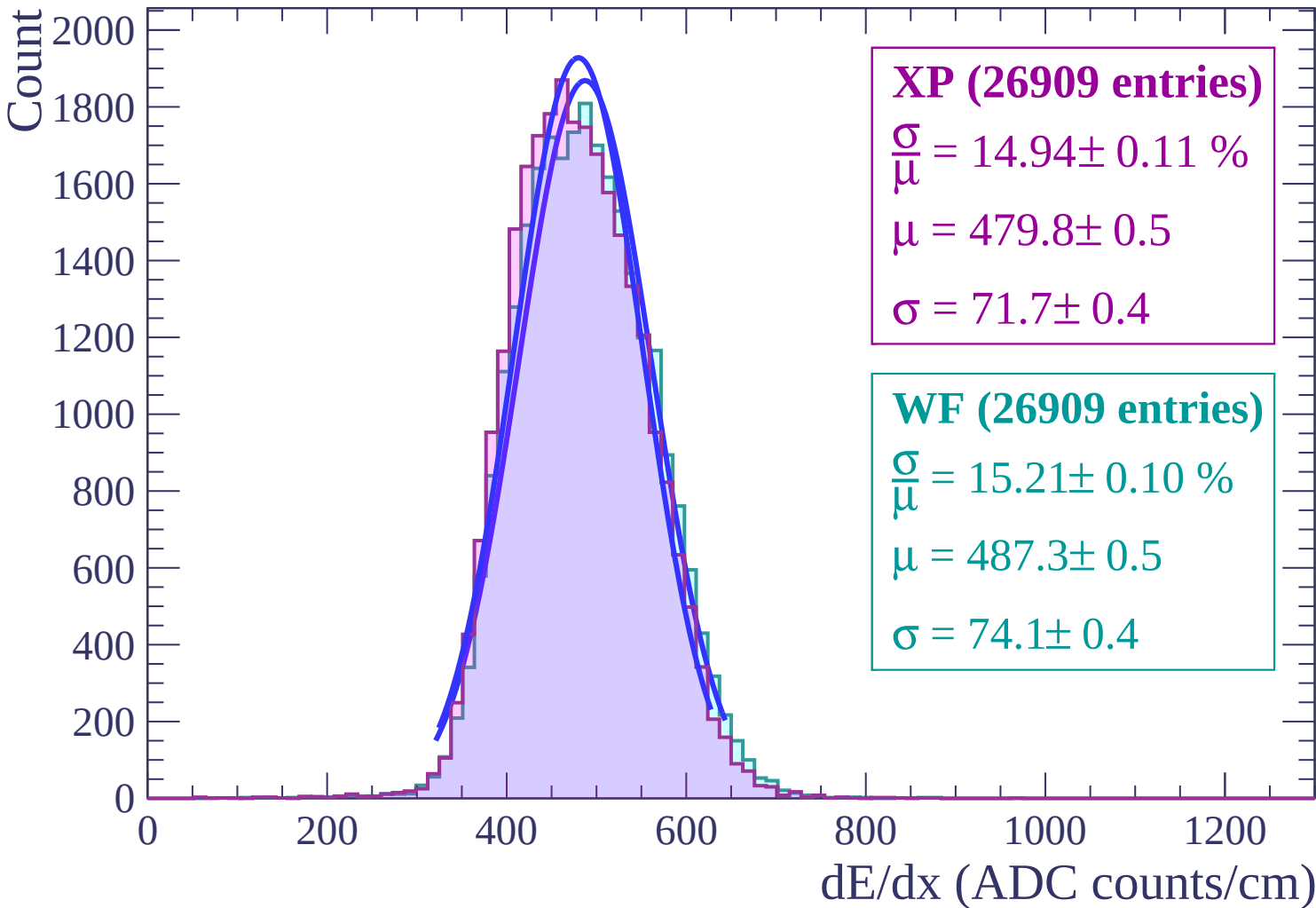


dE/dx (ADC counts/cm)

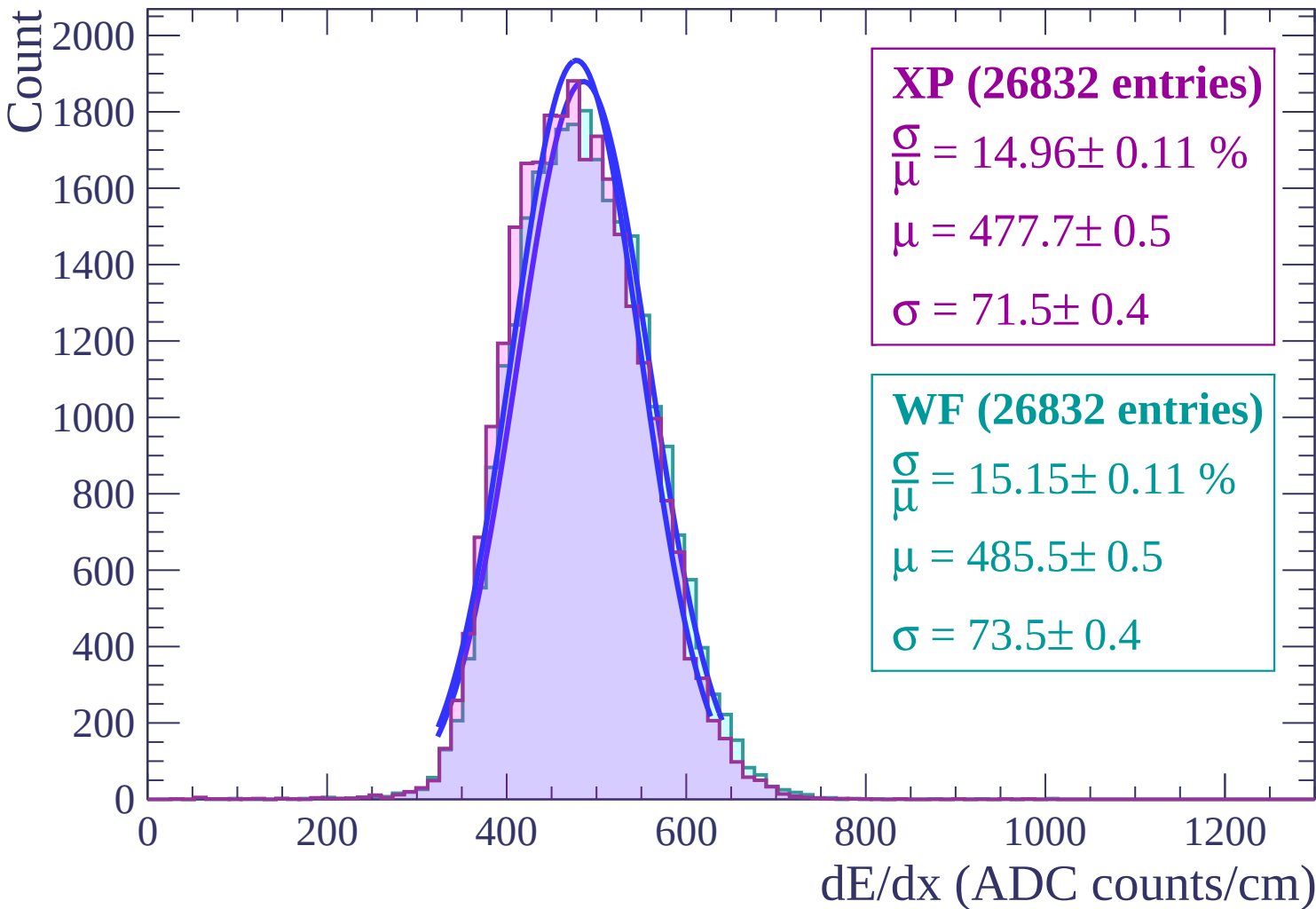
Energy loss | $-8 < \theta < -6$



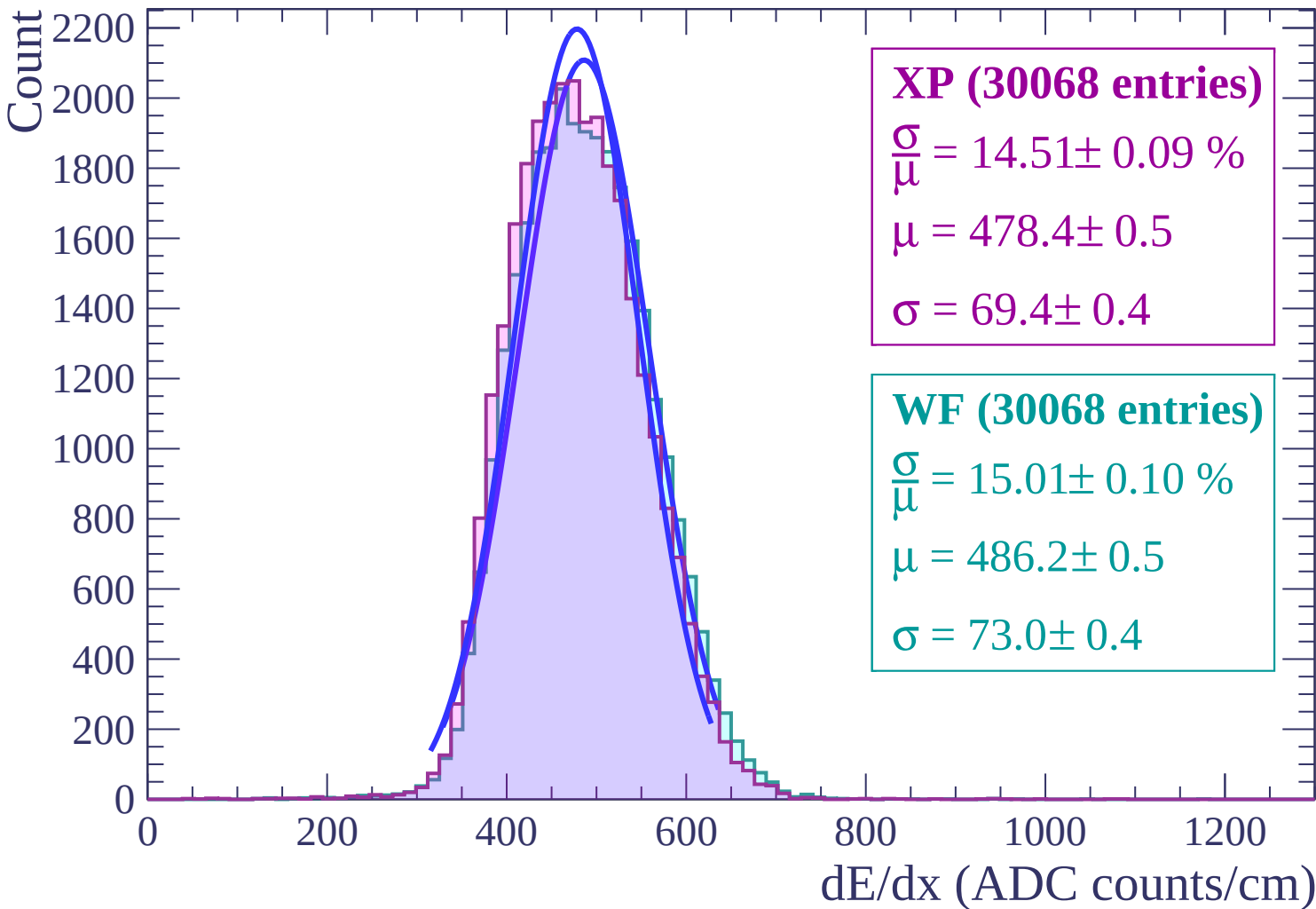
Energy loss | $-6 < \theta < -4$



Energy loss | $-4 < \theta < -2$



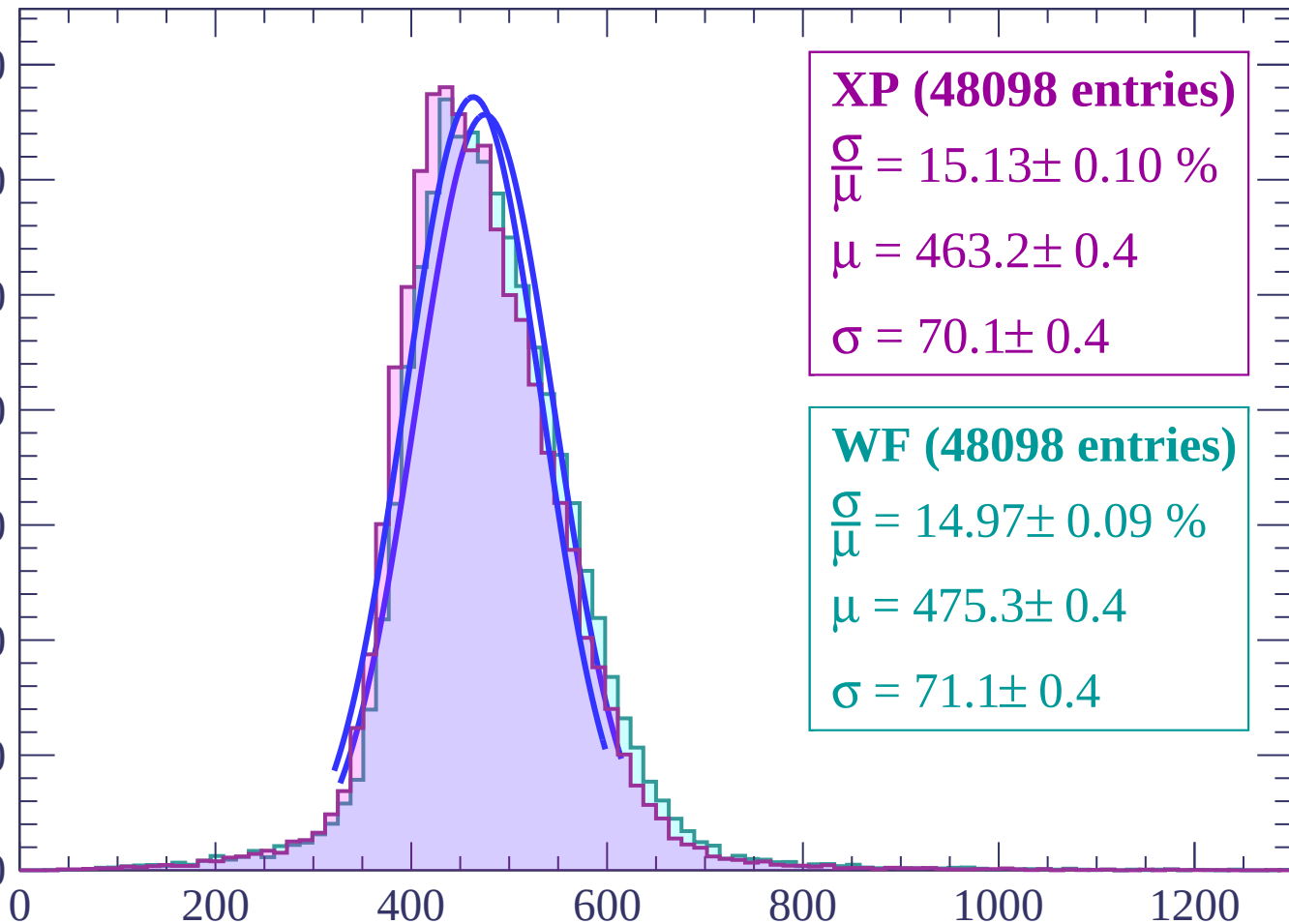
Energy loss $-2 < \theta < 0$



Energy loss | $0 < \theta < 2$

Count

3500
3000
2500
2000
1500
1000
500
0



XP (48098 entries)

$\frac{\sigma}{\mu} = 15.13 \pm 0.10 \%$

$\mu = 463.2 \pm 0.4$

$\sigma = 70.1 \pm 0.4$

WF (48098 entries)

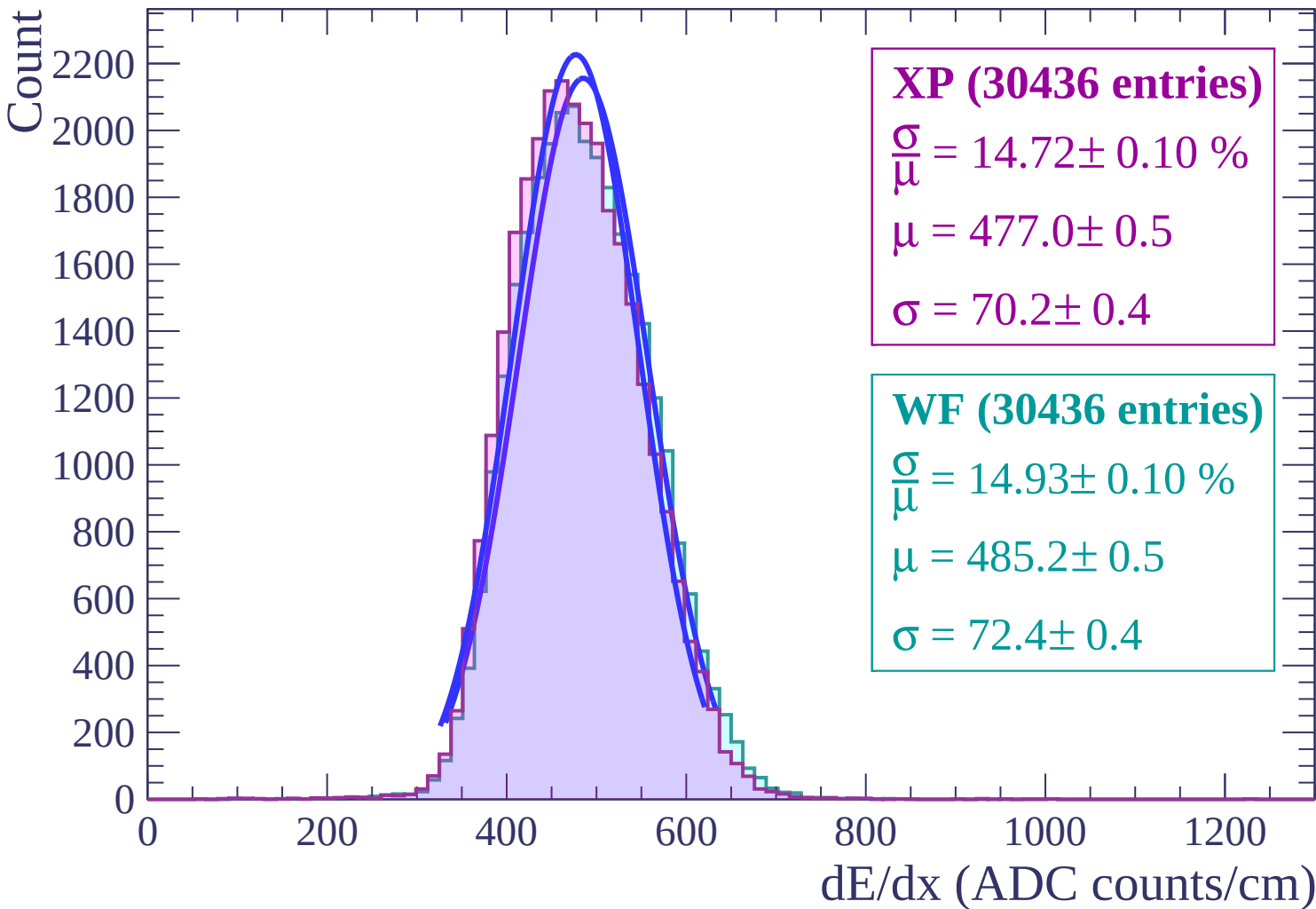
$\frac{\sigma}{\mu} = 14.97 \pm 0.09 \%$

$\mu = 475.3 \pm 0.4$

$\sigma = 71.1 \pm 0.4$

dE/dx (ADC counts/cm)

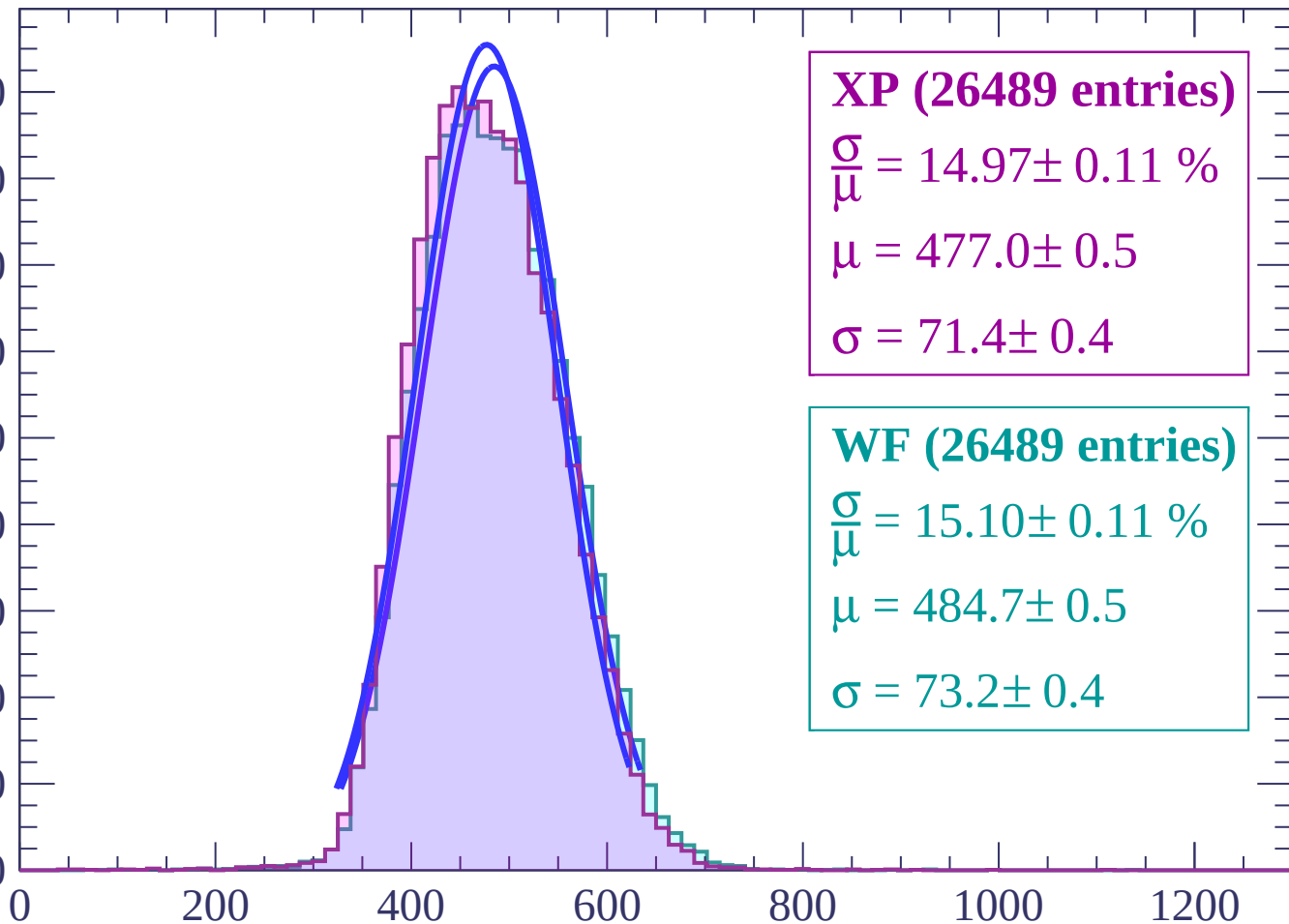
Energy loss | $2 < \theta < 4$



Energy loss | $4 < \theta < 6$

Count

1800
1600
1400
1200
1000
800
600
400
200
0



XP (26489 entries)

$\frac{\sigma}{\mu} = 14.97 \pm 0.11 \%$

$\mu = 477.0 \pm 0.5$

$\sigma = 71.4 \pm 0.4$

WF (26489 entries)

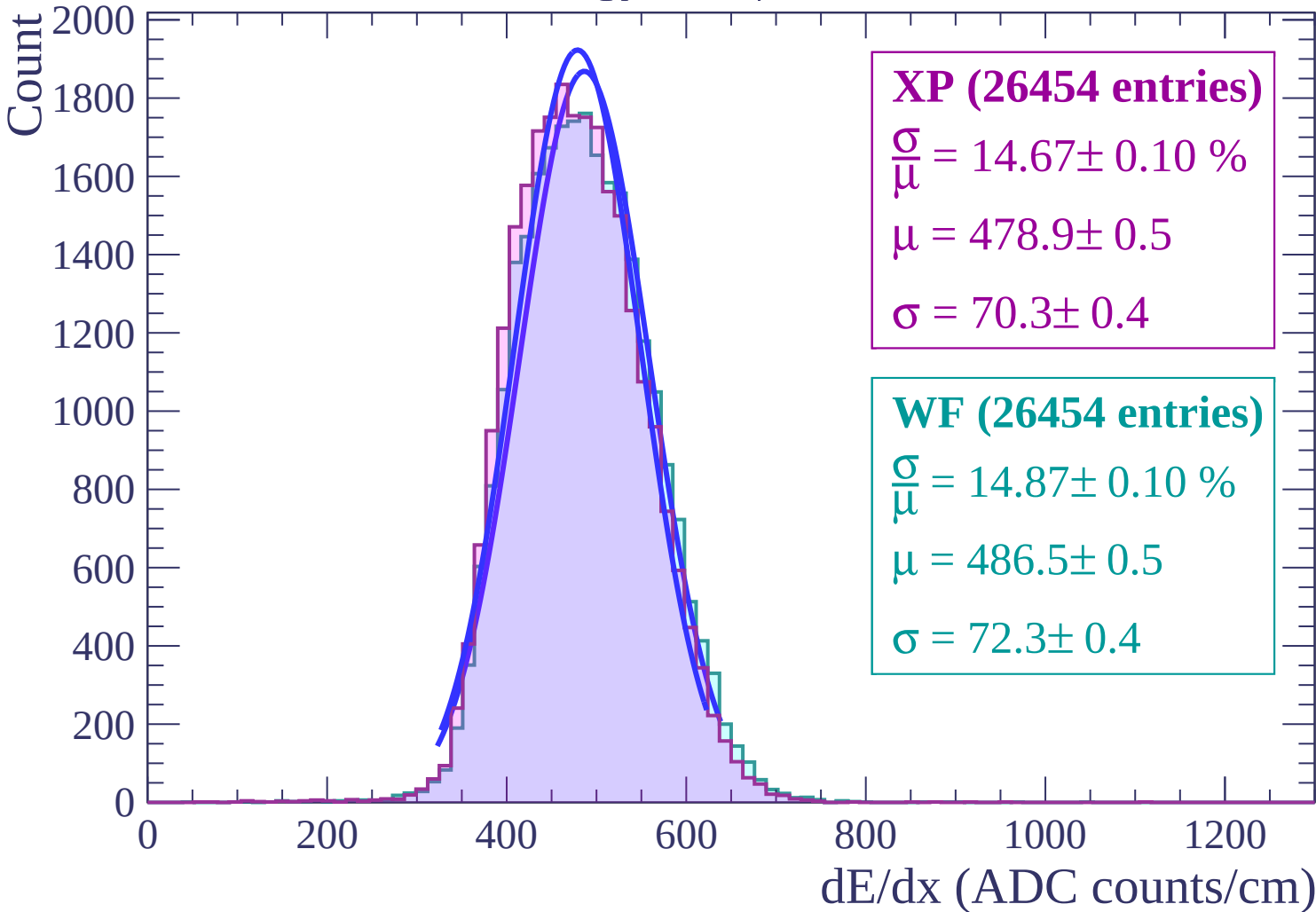
$\frac{\sigma}{\mu} = 15.10 \pm 0.11 \%$

$\mu = 484.7 \pm 0.5$

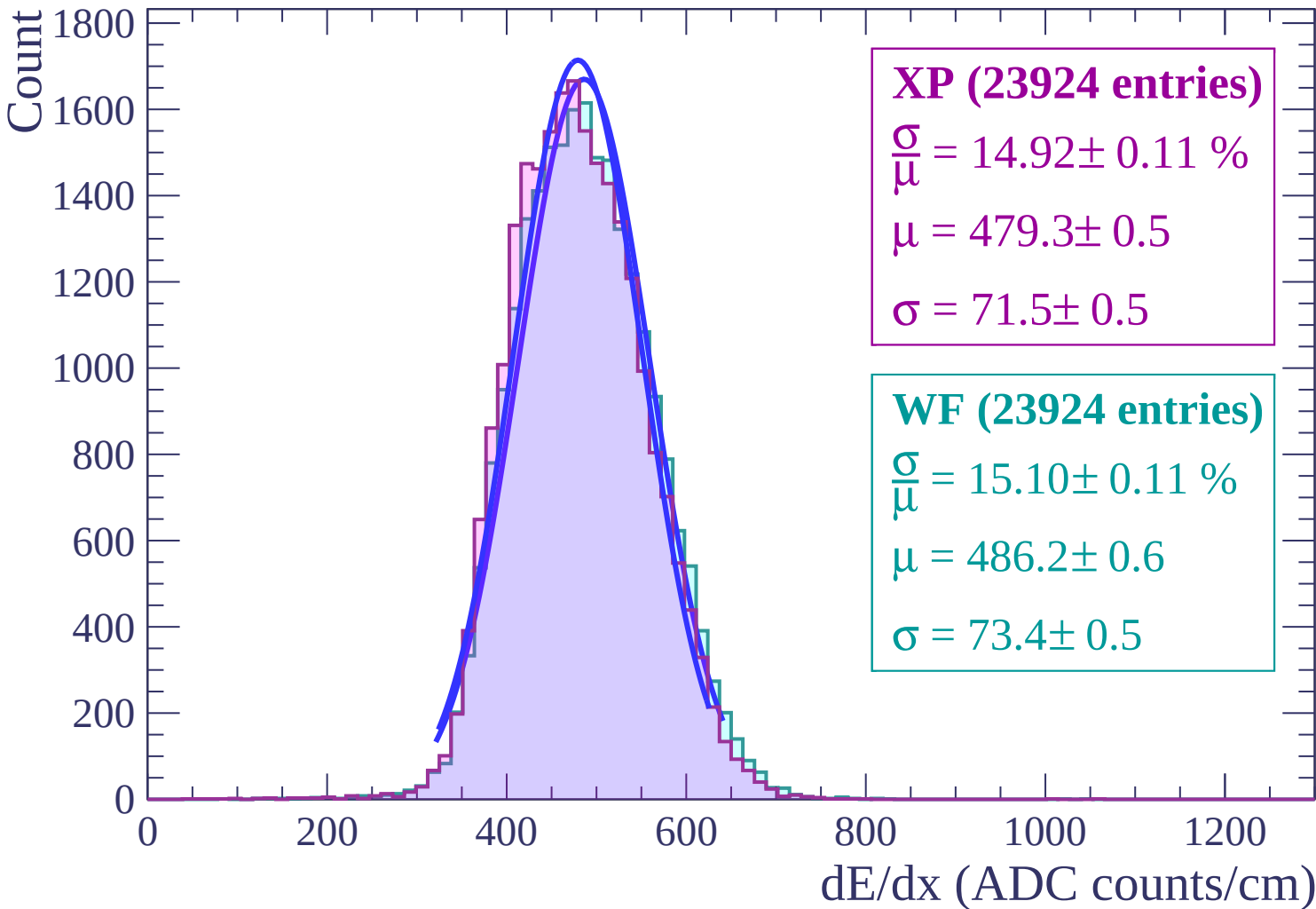
$\sigma = 73.2 \pm 0.4$

dE/dx (ADC counts/cm)

Energy loss | $6 < \theta < 8$



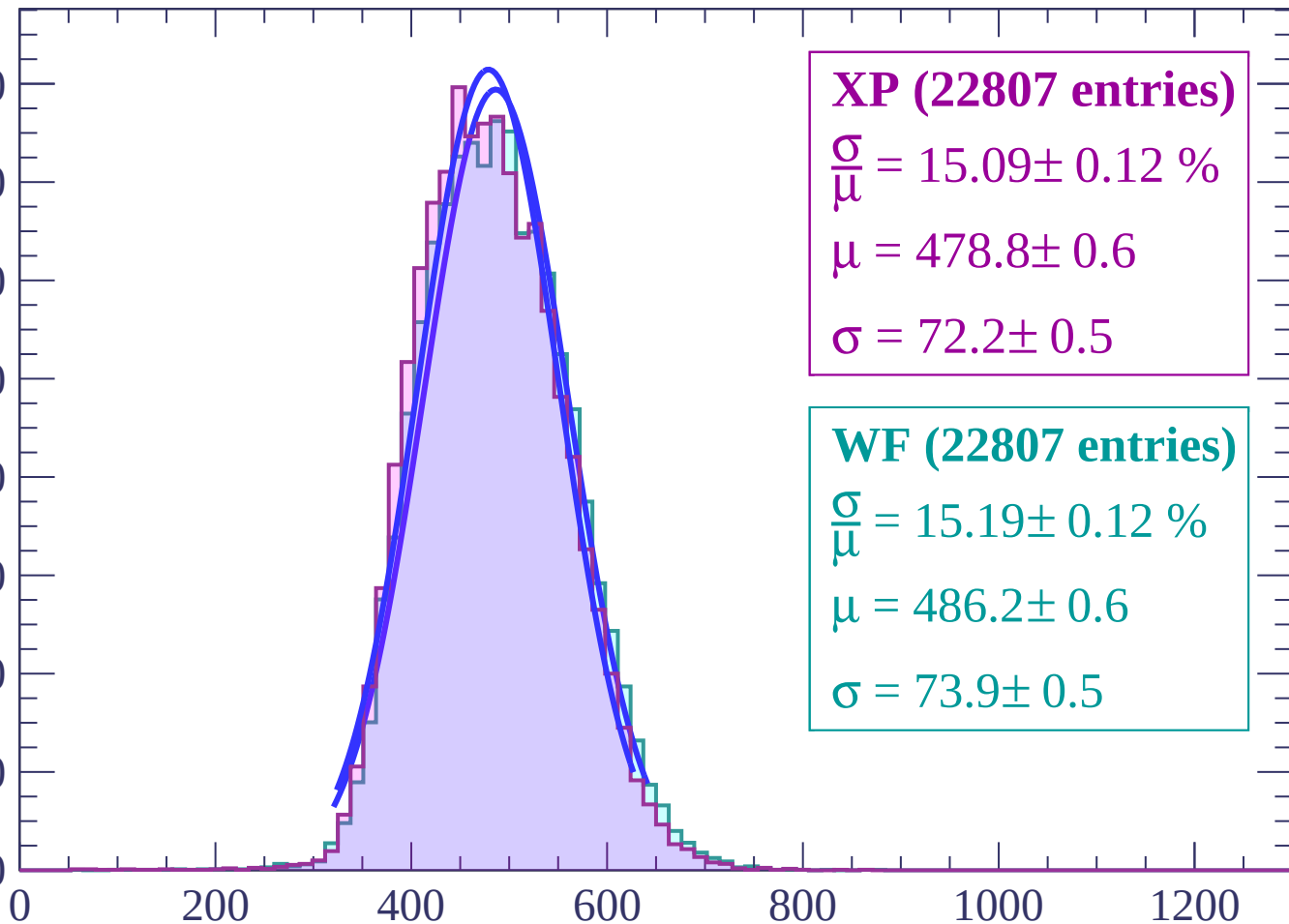
Energy loss | $8 < \theta < 10$



Energy loss | $10 < \theta < 12$

Count

1600
1400
1200
1000
800
600
400
200
0



XP (22807 entries)

$\frac{\sigma}{\mu} = 15.09 \pm 0.12 \%$

$\mu = 478.8 \pm 0.6$

$\sigma = 72.2 \pm 0.5$

WF (22807 entries)

$\frac{\sigma}{\mu} = 15.19 \pm 0.12 \%$

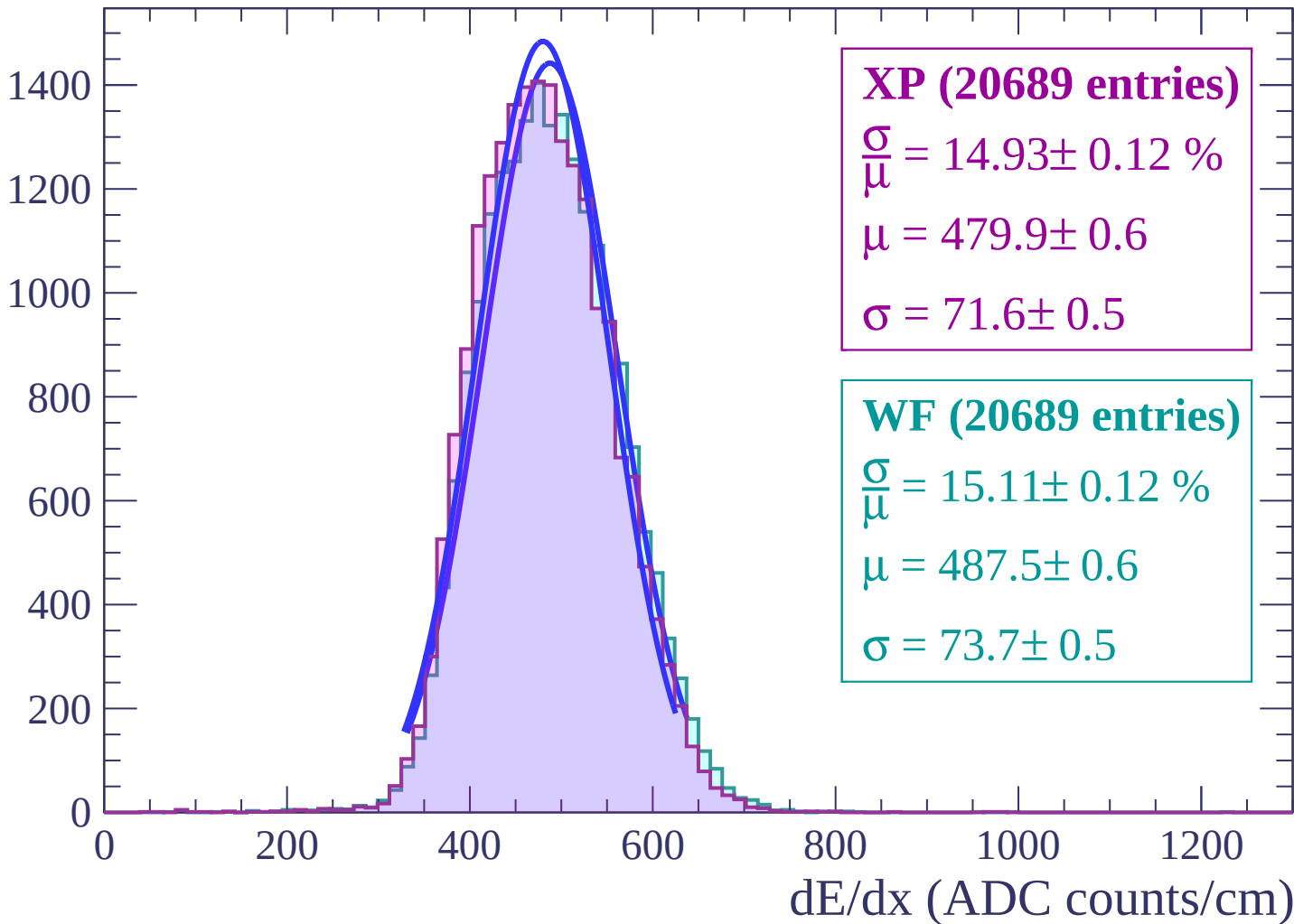
$\mu = 486.2 \pm 0.6$

$\sigma = 73.9 \pm 0.5$

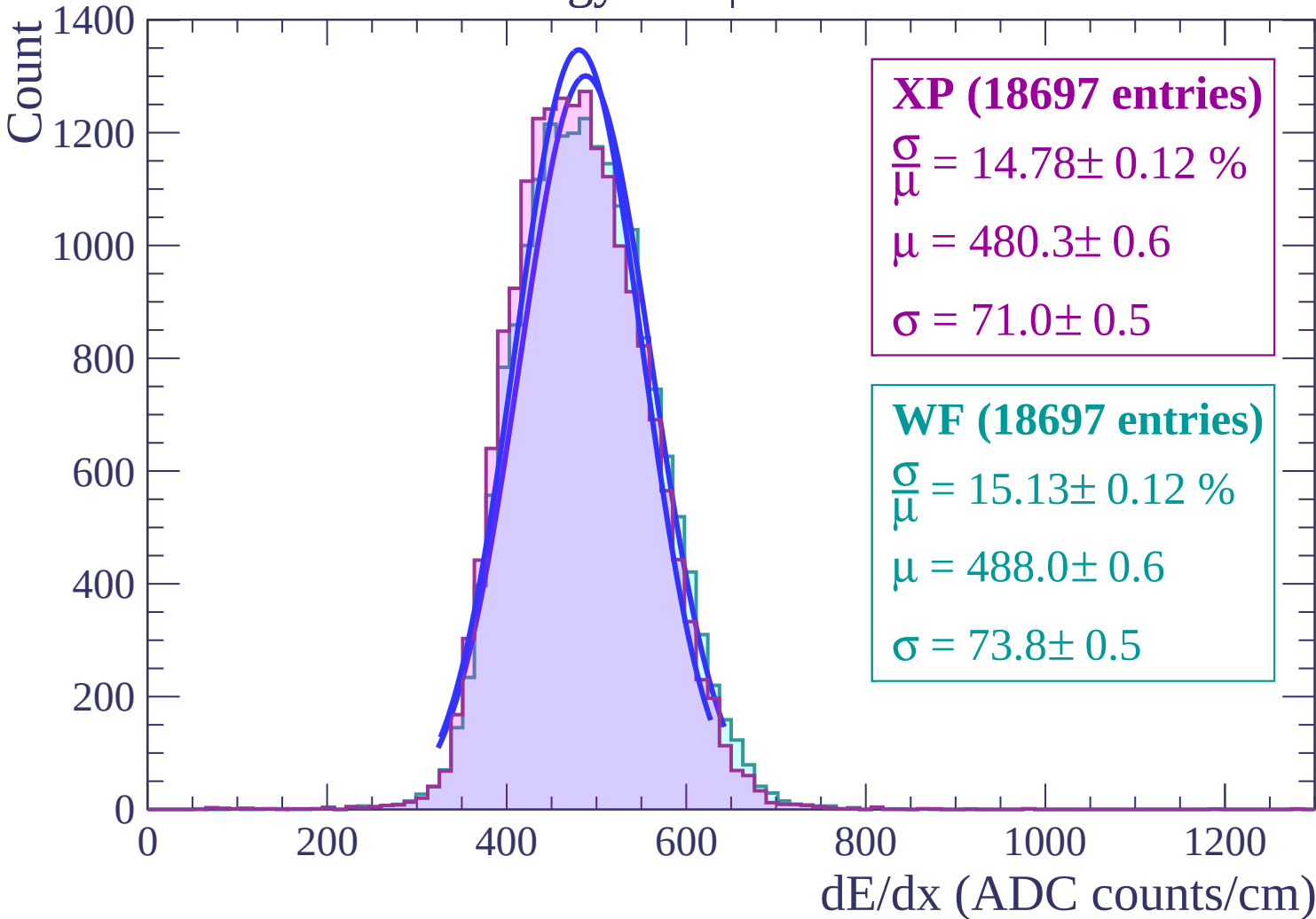
dE/dx (ADC counts/cm)

Energy loss | $12 < \theta < 14$

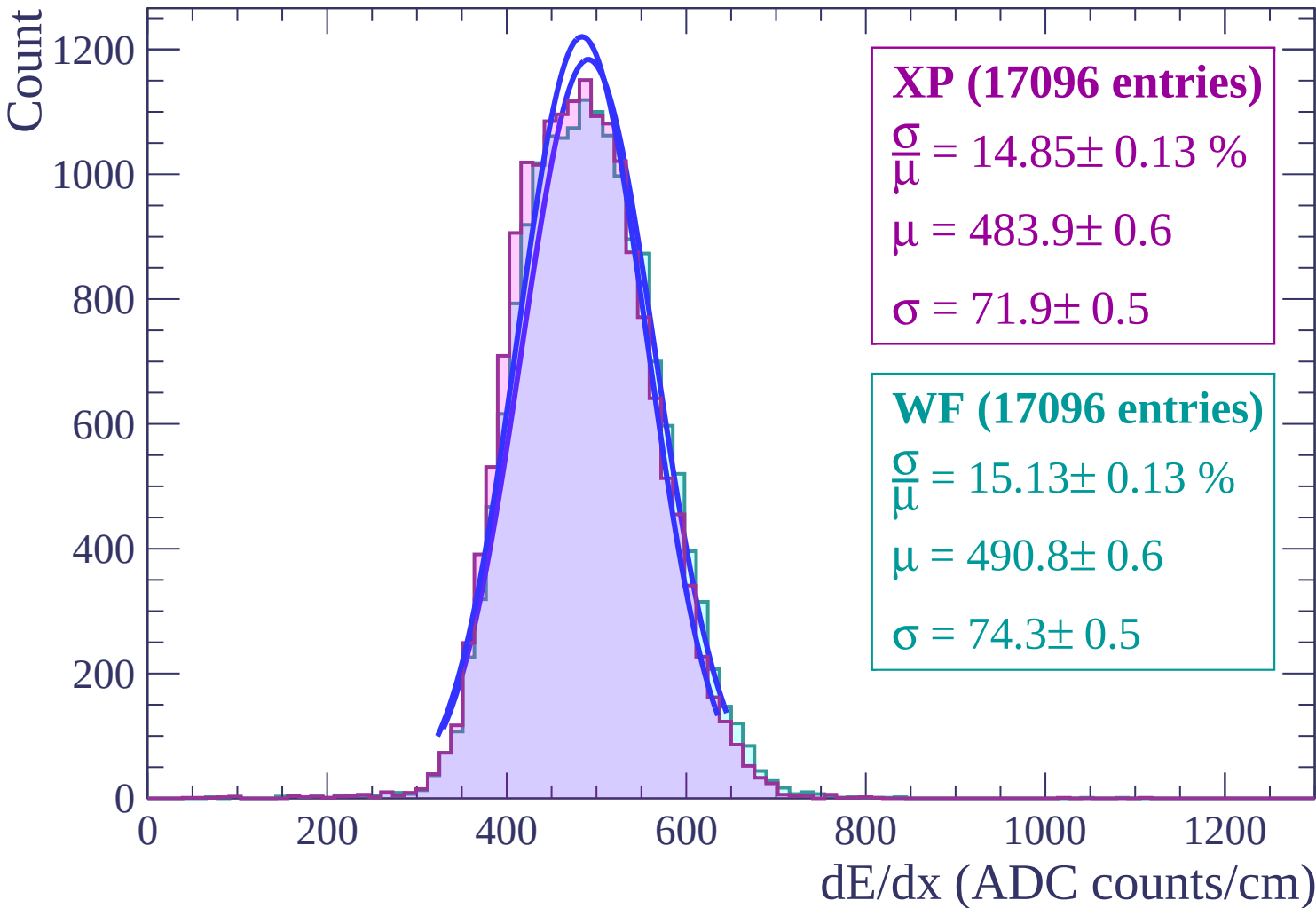
Count



Energy loss | $14 < \theta < 16$



Energy loss | $16 < \theta < 18$



Energy loss | $18 < \theta < 20$

Count

1000

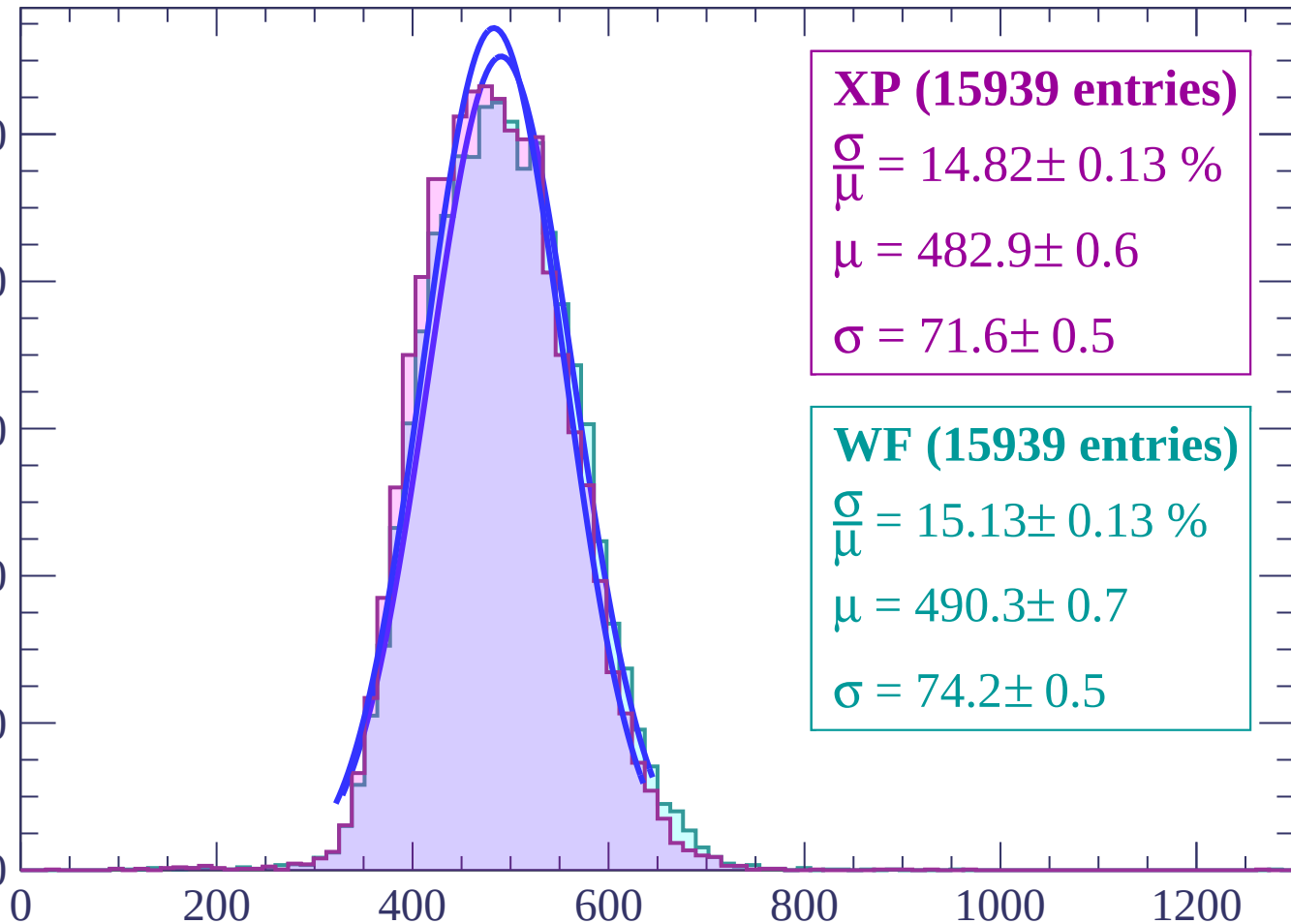
800

600

400

200

0



XP (15939 entries)

$\frac{\sigma}{\mu} = 14.82 \pm 0.13 \%$

$\mu = 482.9 \pm 0.6$

$\sigma = 71.6 \pm 0.5$

WF (15939 entries)

$\frac{\sigma}{\mu} = 15.13 \pm 0.13 \%$

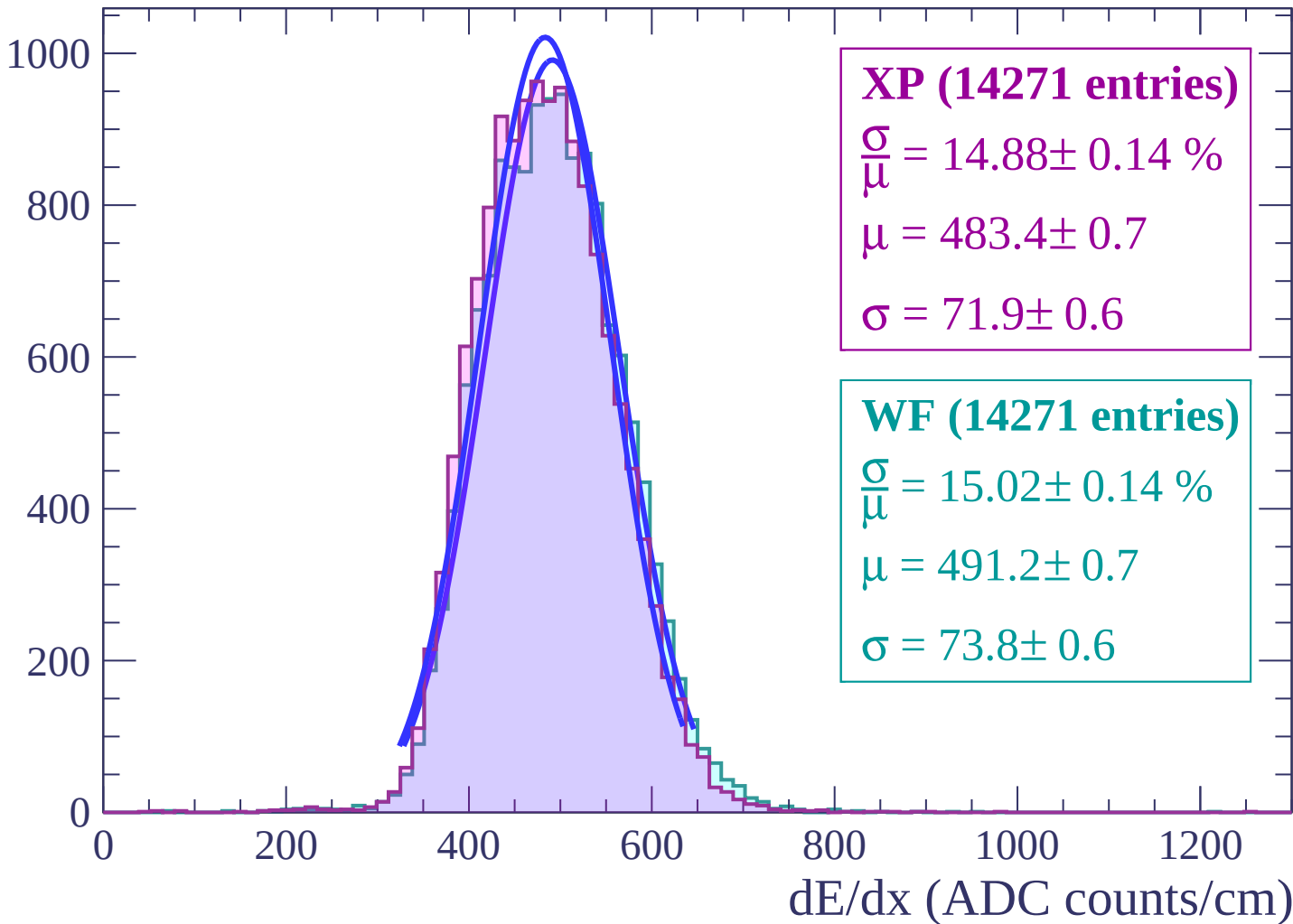
$\mu = 490.3 \pm 0.7$

$\sigma = 74.2 \pm 0.5$

dE/dx (ADC counts/cm)

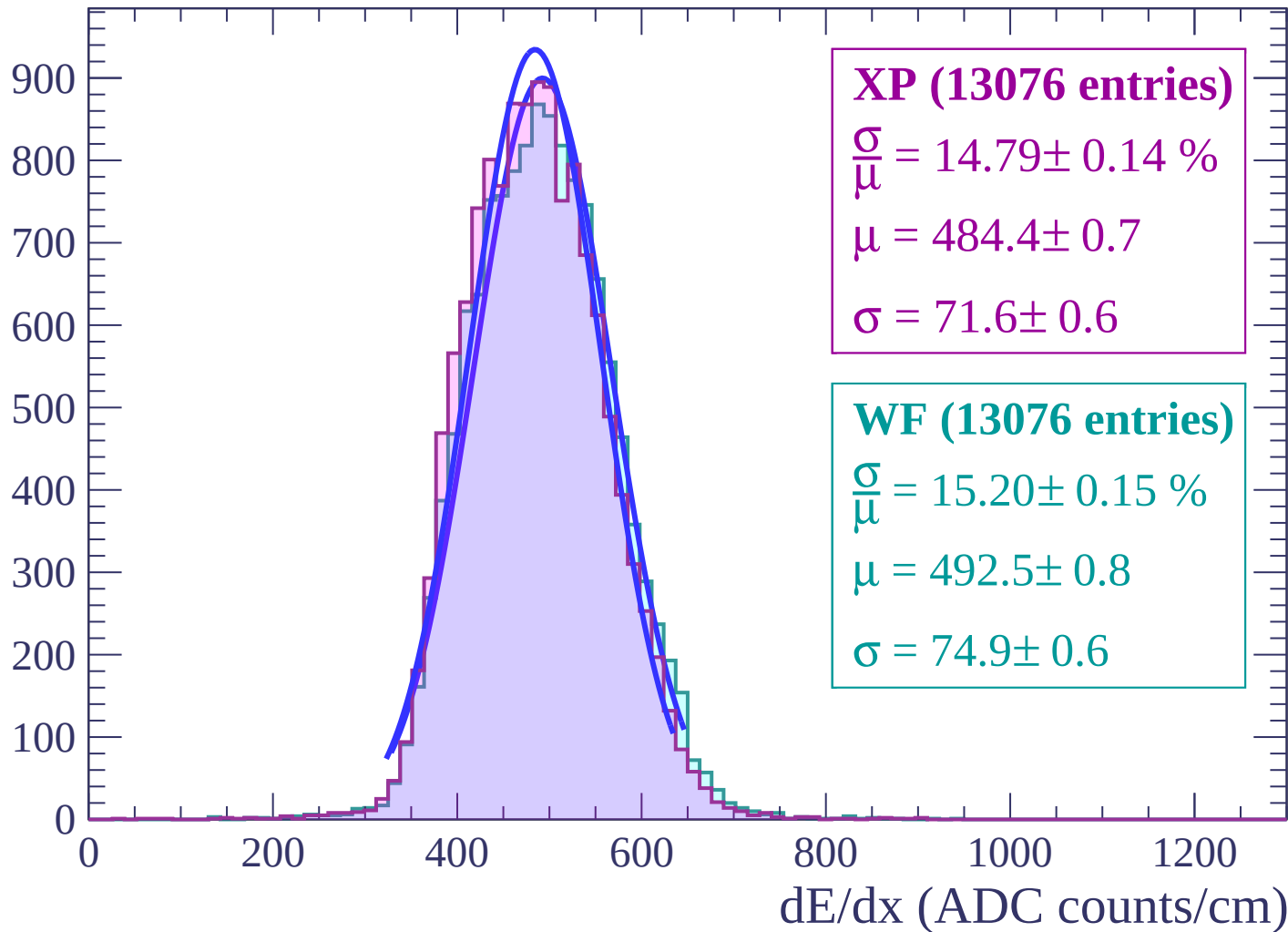
Energy loss | $20 < \theta < 22$

Count



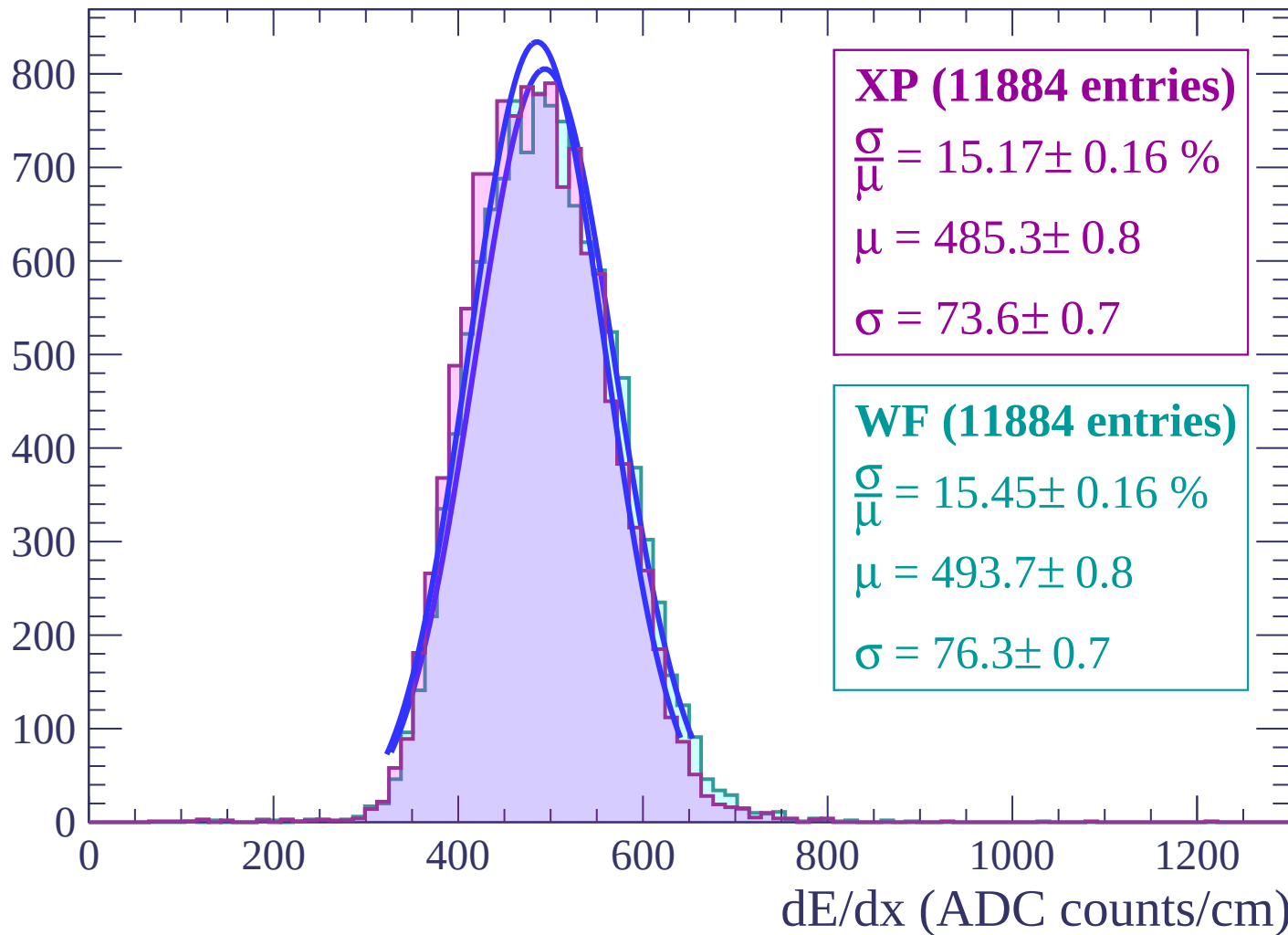
Energy loss | $22 < \theta < 24$

Count



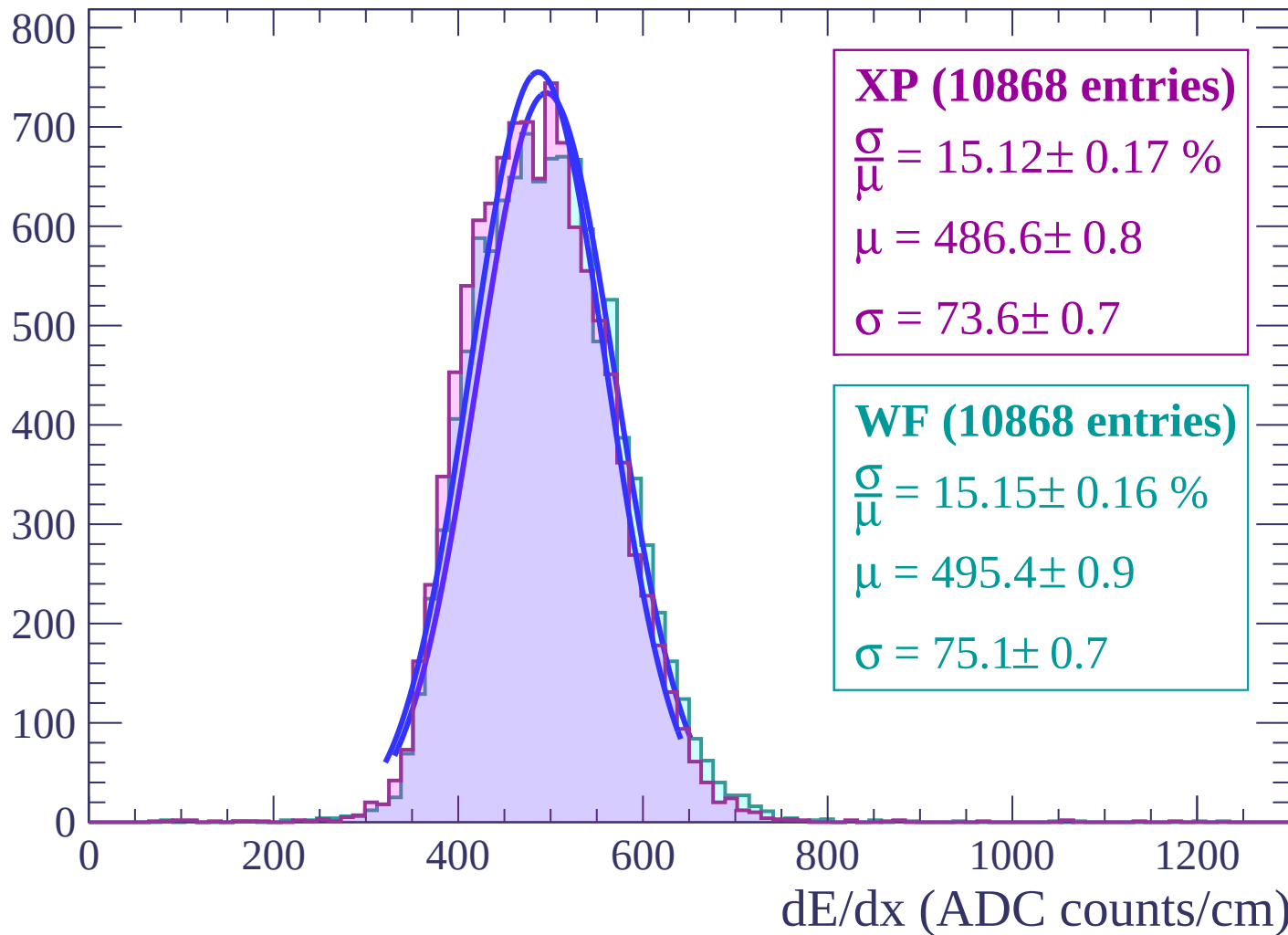
Energy loss | $24 < \theta < 26$

Count



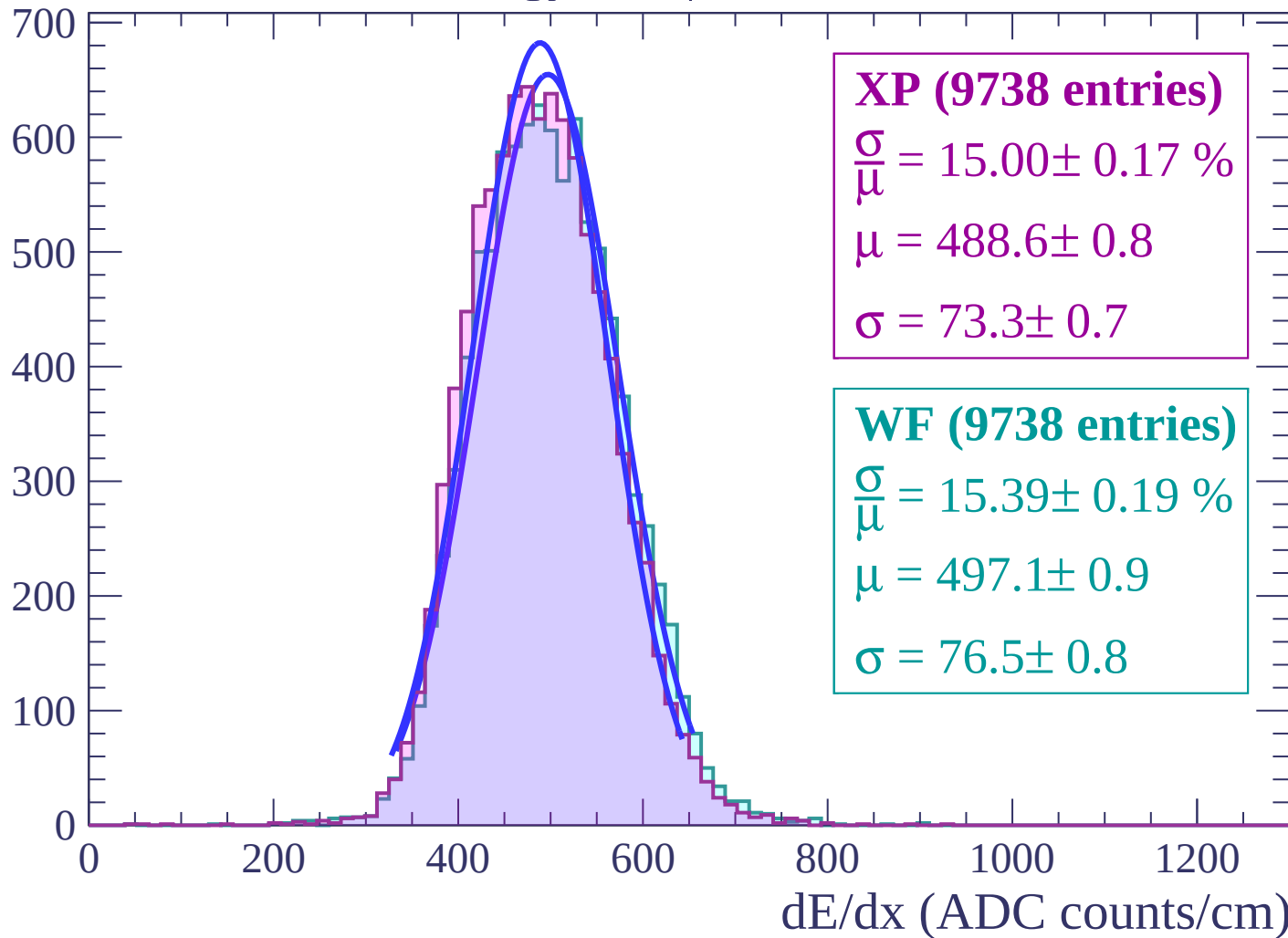
Energy loss | $26 < \theta < 28$

Count



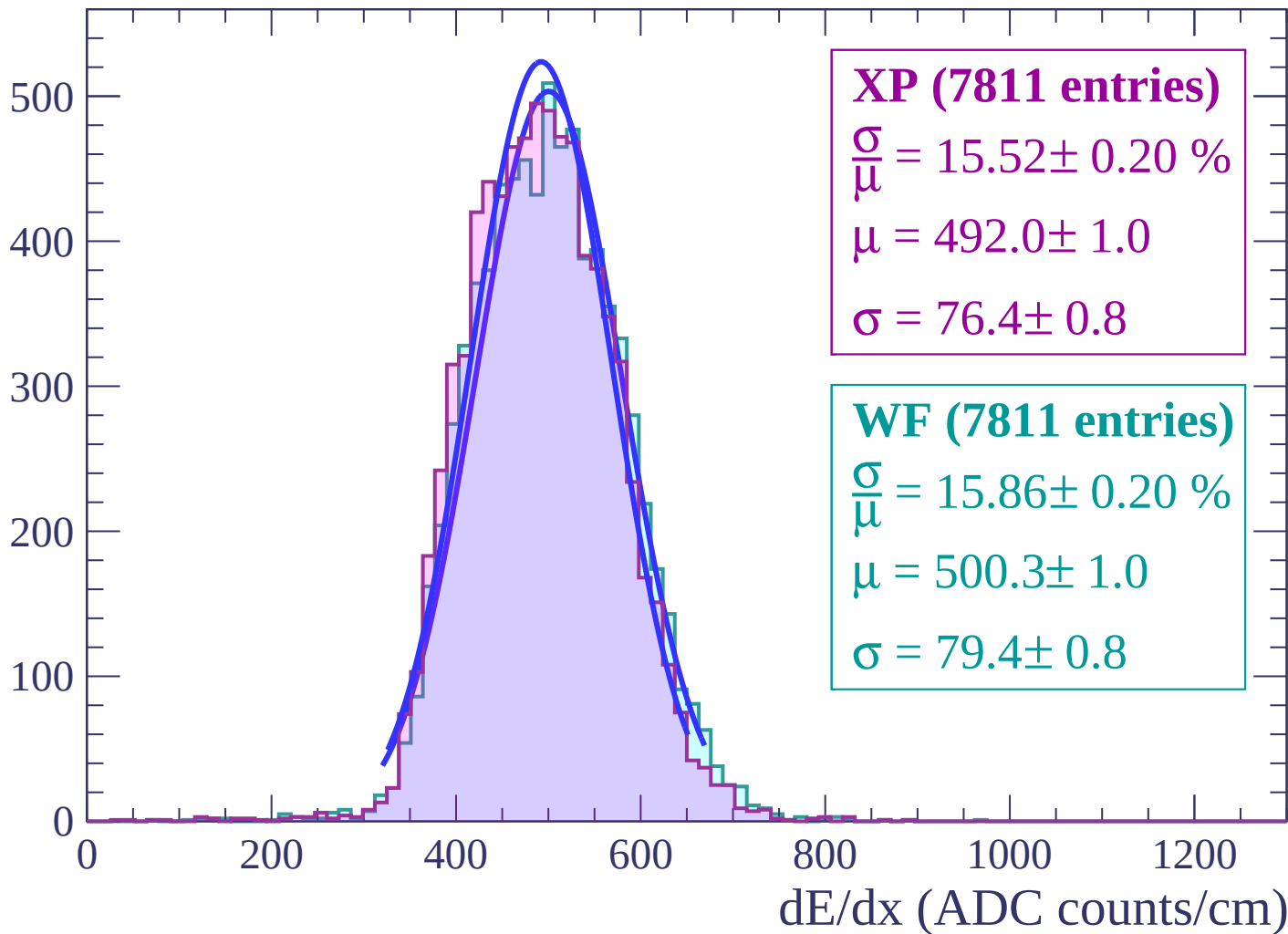
Energy loss | $28 < \theta < 30$

Count



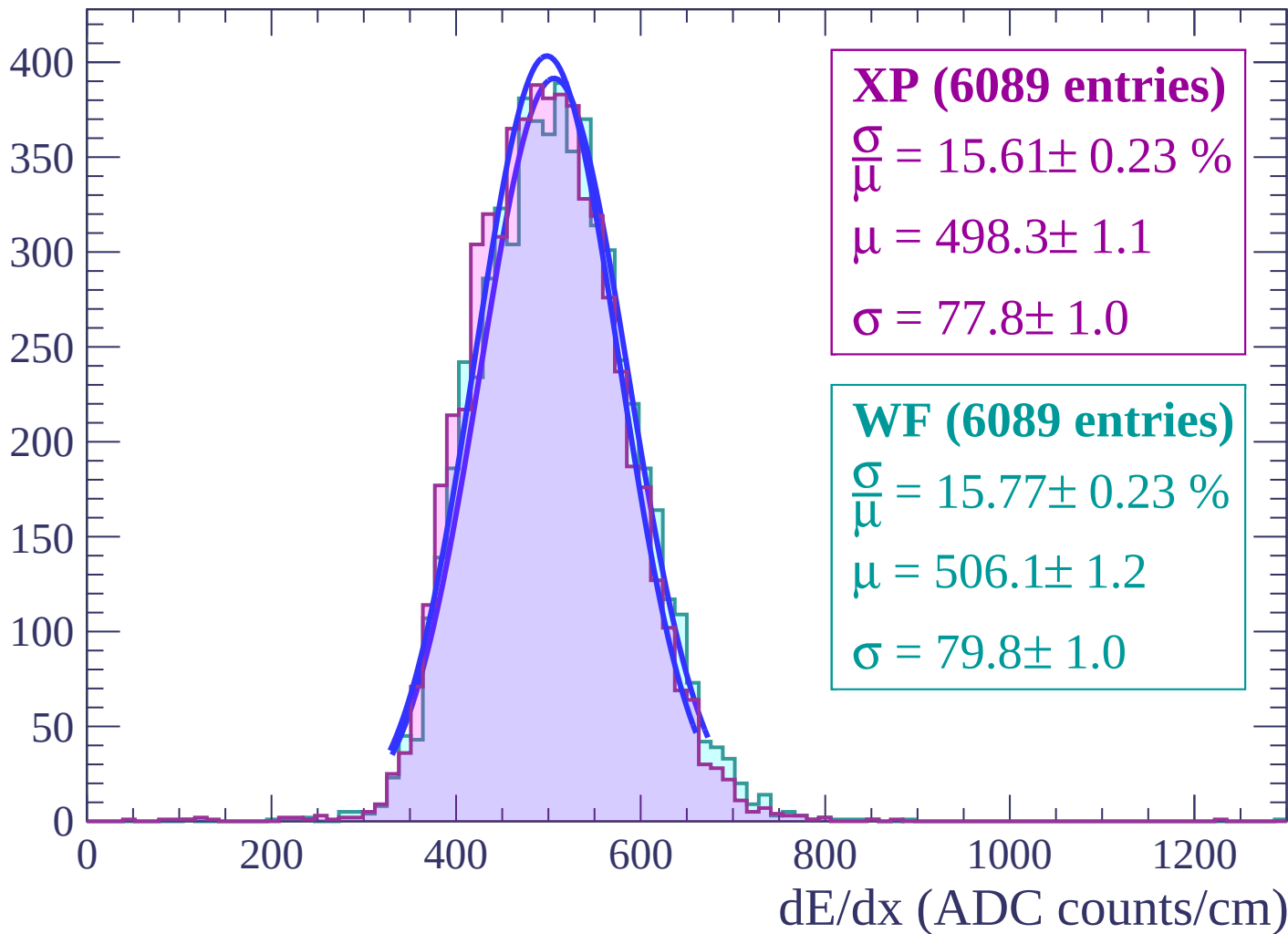
Energy loss | $30 < \theta < 32$

Count



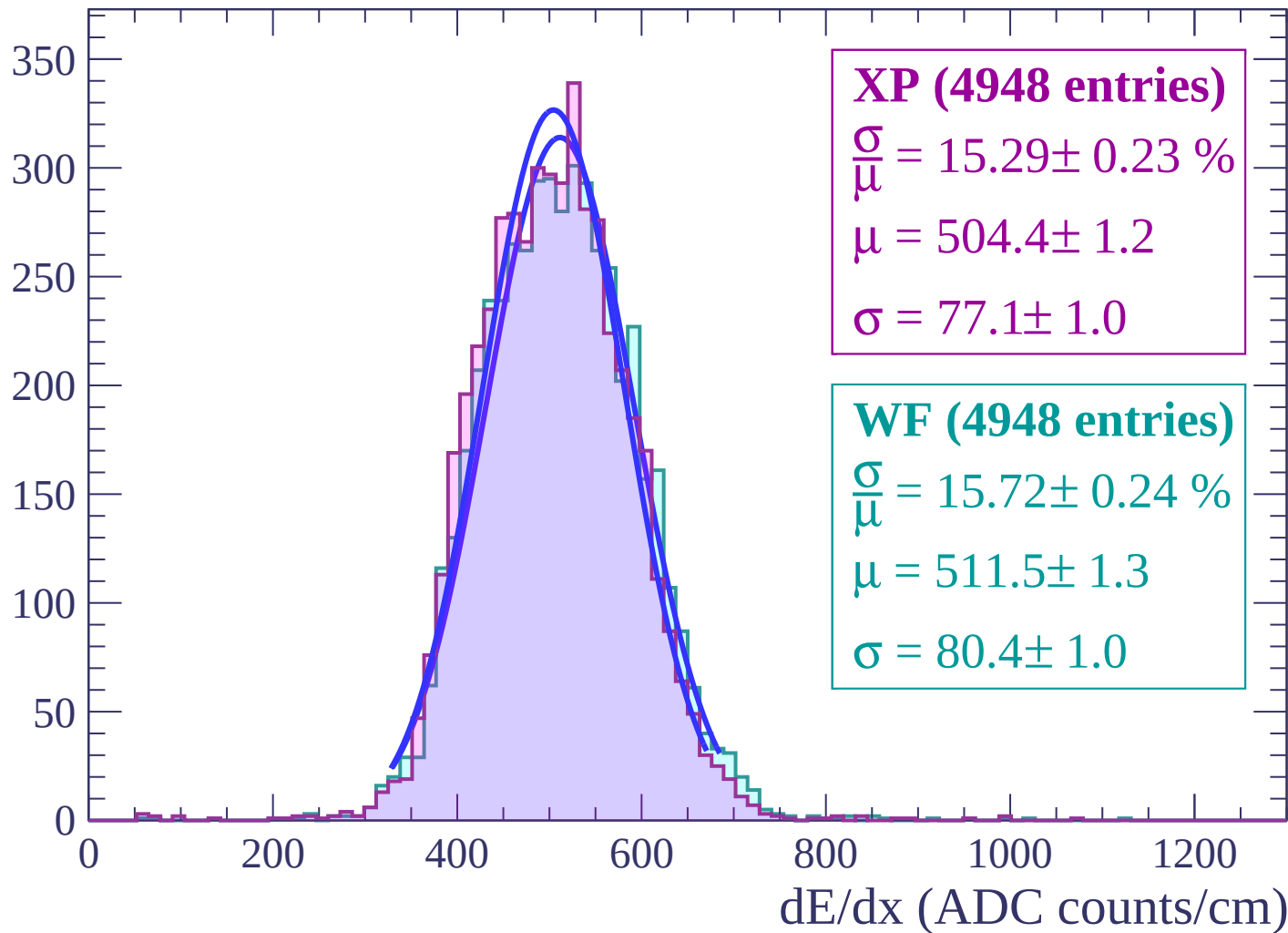
Energy loss | $32 < \theta < 34$

Count



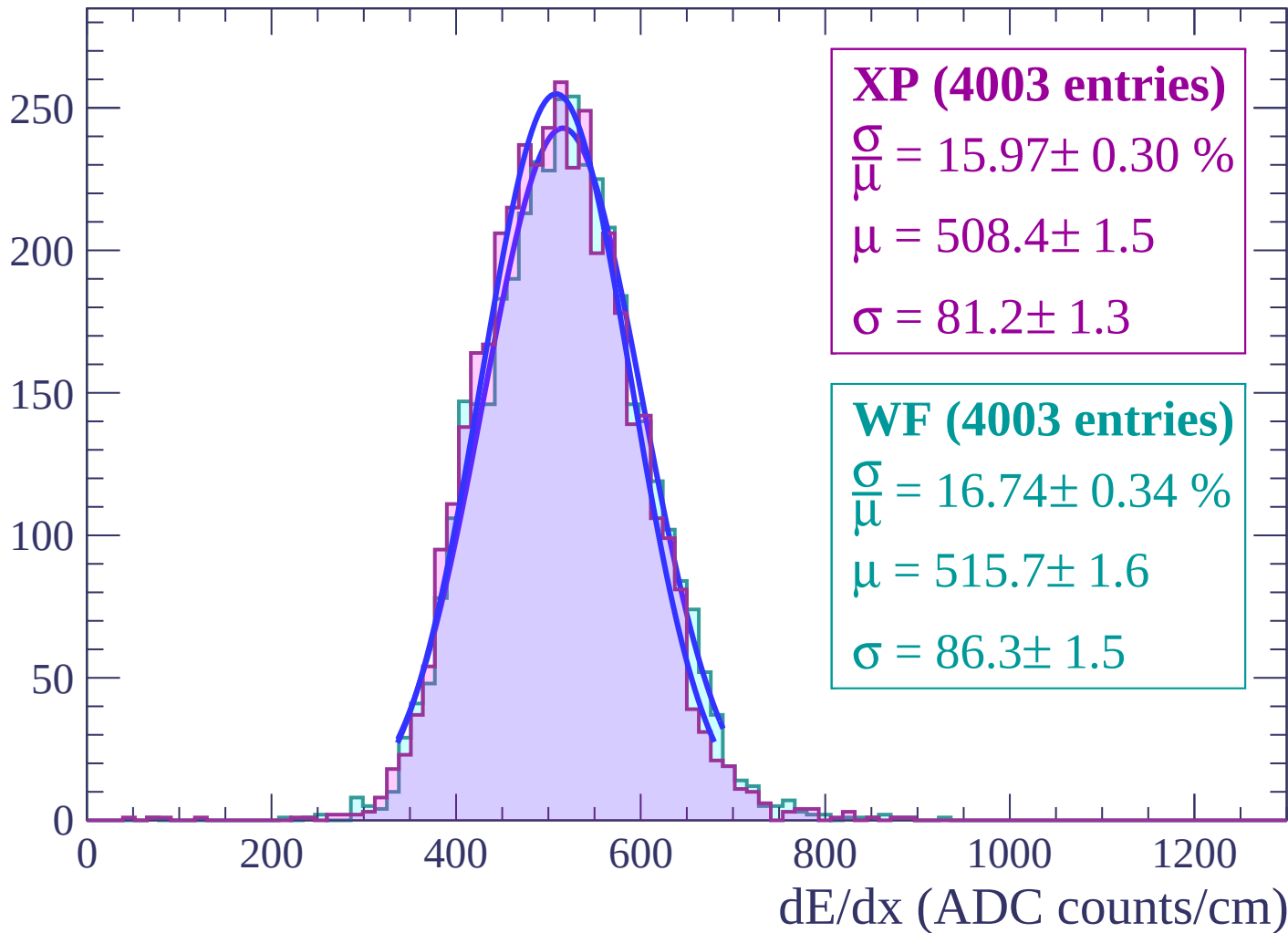
Energy loss | $34 < \theta < 36$

Count



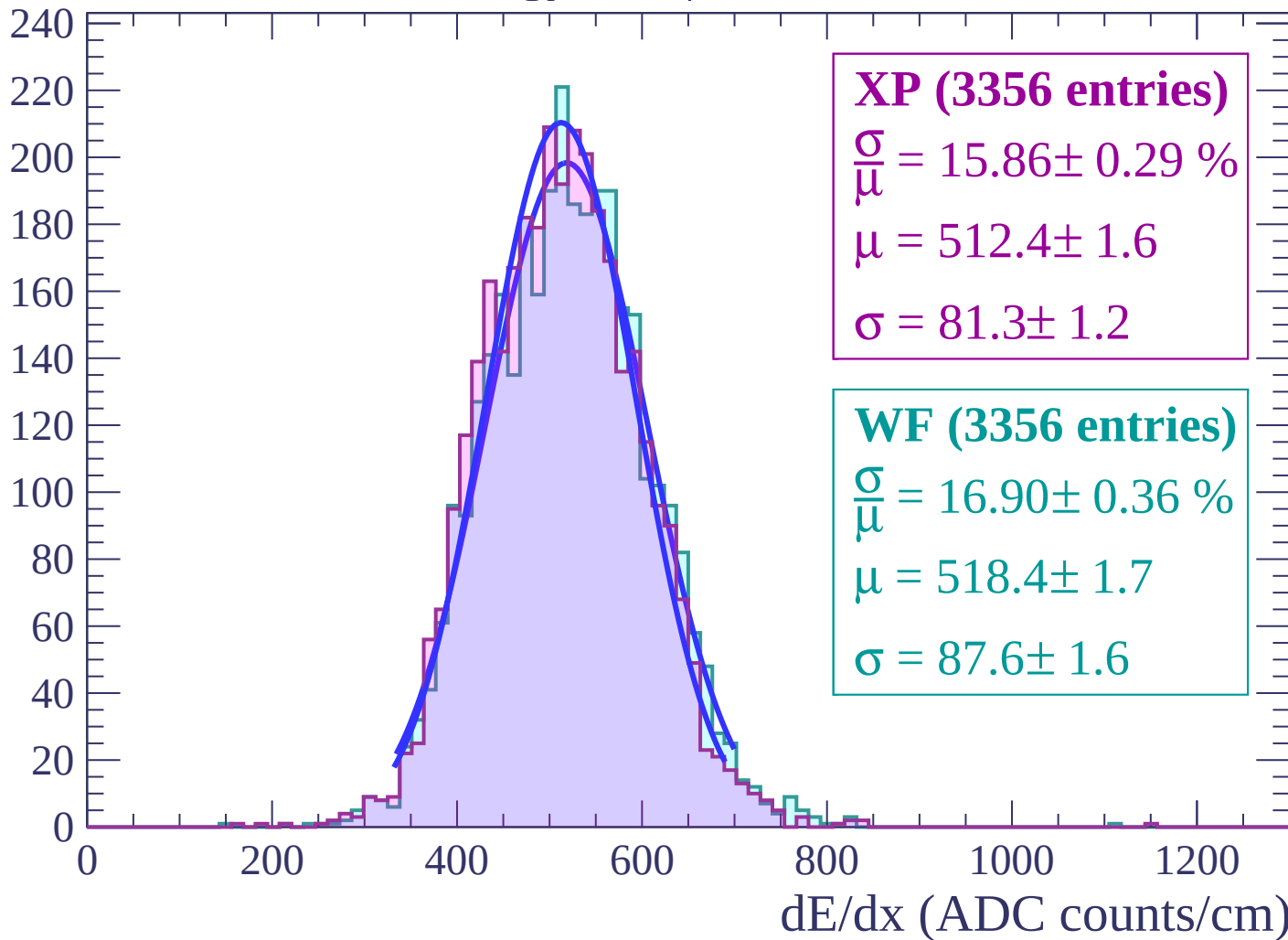
Energy loss | $36 < \theta < 38$

Count



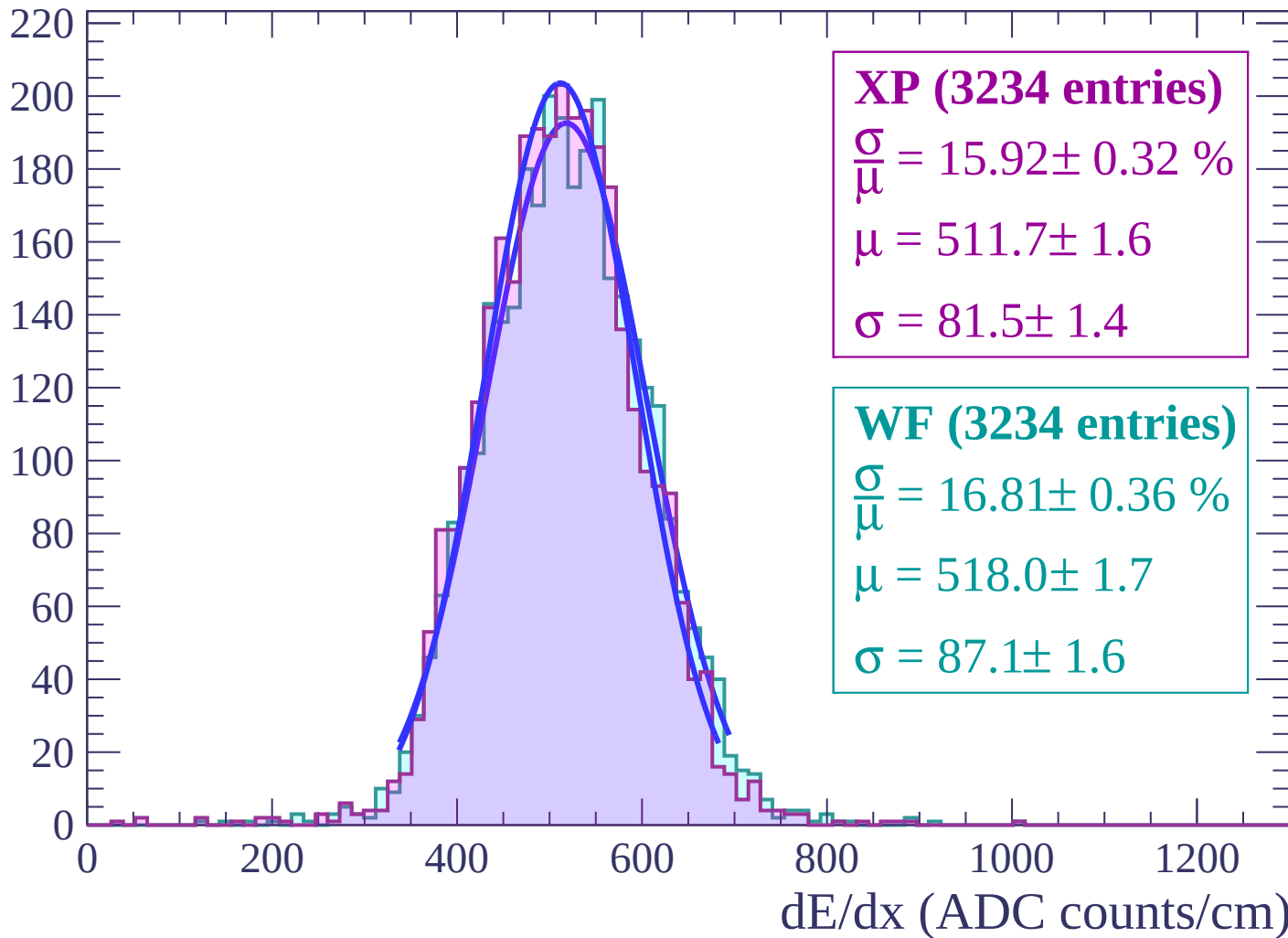
Energy loss | $38 < \theta < 40$

Count



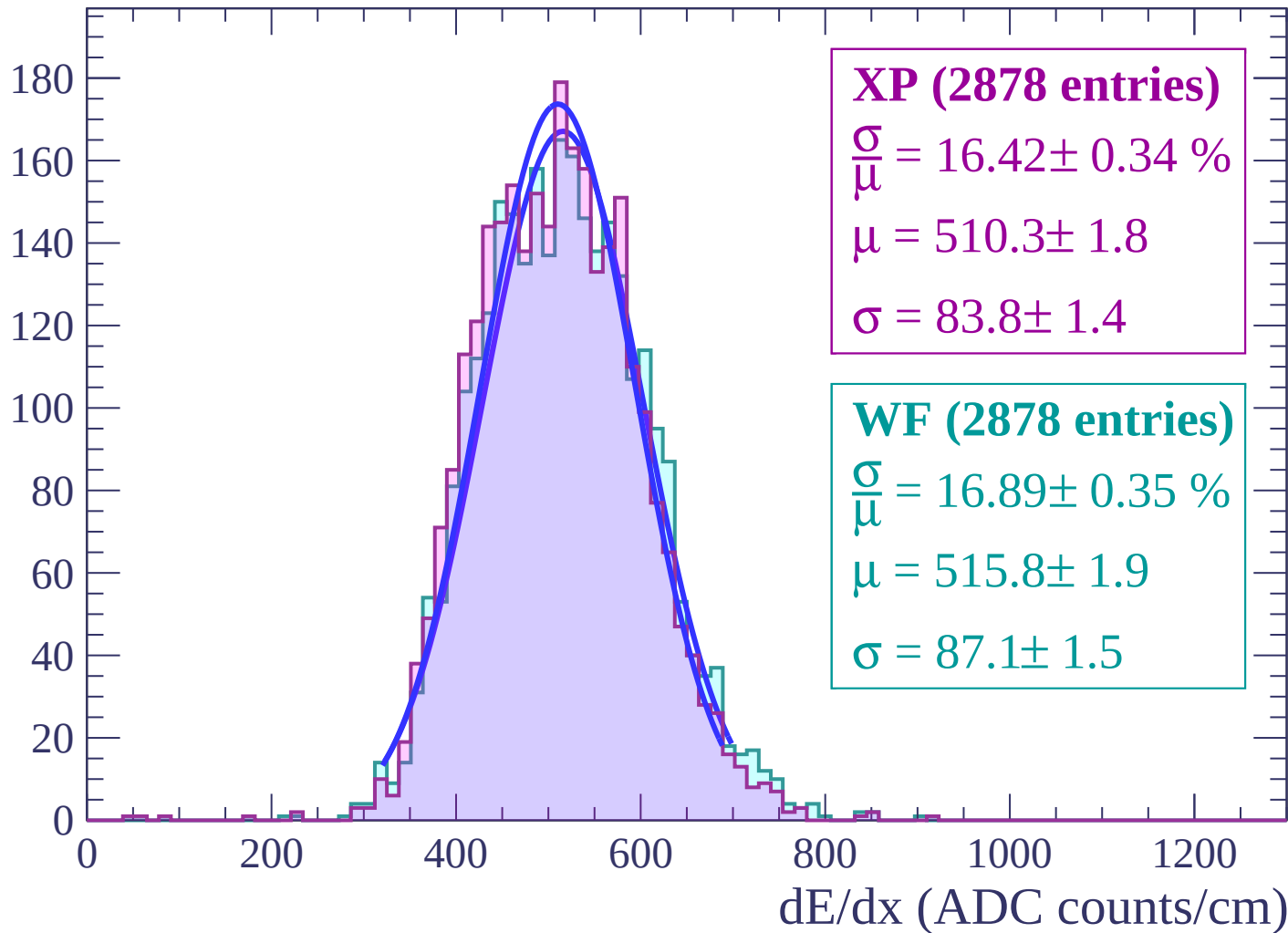
Energy loss | $40 < \theta < 42$

Count



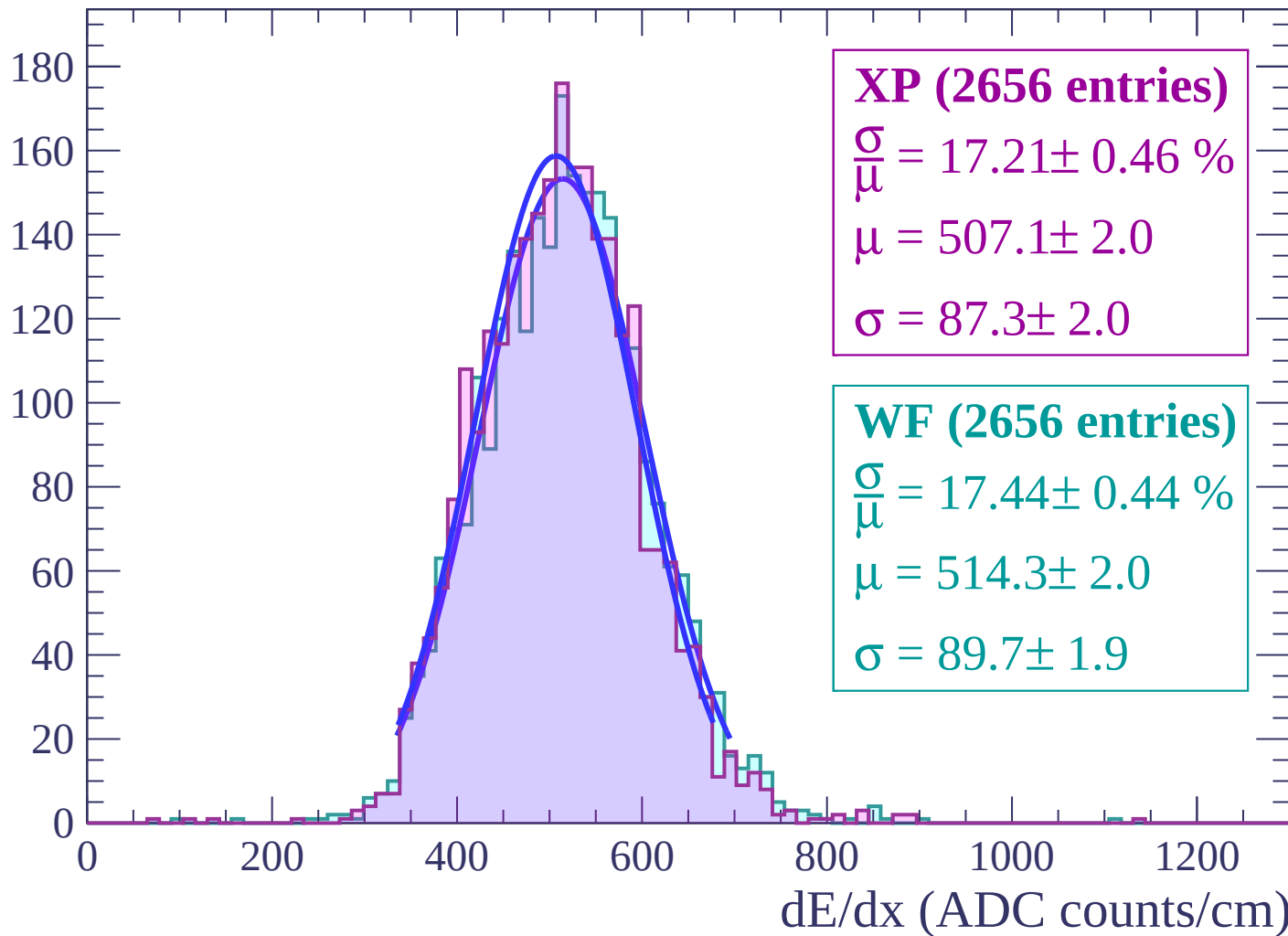
Energy loss | $42 < \theta < 44$

Count



Energy loss | $44 < \theta < 46$

Count



Energy loss | $46 < \theta < 48$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $48 < \theta < 50$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $50 < \theta < 52$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $52 < \theta < 54$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $54 < \theta < 56$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $56 < \theta < 58$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $58 < \theta < 60$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $60 < \theta < 62$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $62 < \theta < 64$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $64 < \theta < 66$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $66 < \theta < 68$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $68 < \theta < 70$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $70 < \theta < 72$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $72 < \theta < 74$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $74 < \theta < 76$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $76 < \theta < 78$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $78 < \theta < 80$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $80 < \theta < 82$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $82 < \theta < 84$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $84 < \theta < 86$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\bar{\mu}} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

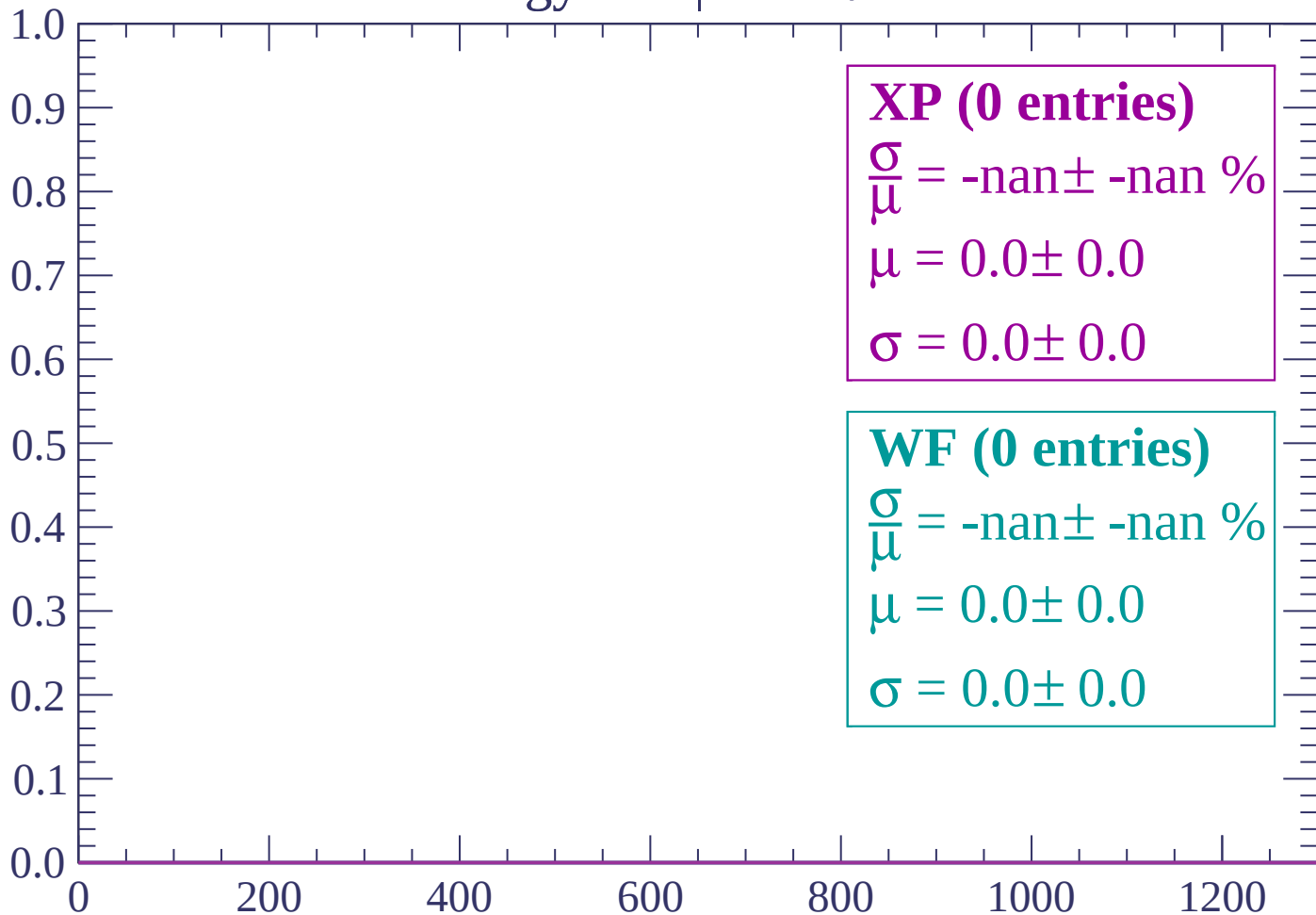
$\frac{\sigma}{\bar{\mu}} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $86 < \theta < 88$

Count



dE/dx (ADC counts/cm)

Energy loss | $88 < \theta < 90$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $90 < \theta < 92$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\bar{\mu}} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\bar{\mu}} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

